
Econometrics Notes - PhD in Economics

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Note to the reader

The first part of these notes is designed to provide a strong theoretical econometrics background. It is based partially on older versions of notes from Nicola Limodio's PhD course at Bocconi, and Taisuke Otsu's EC484 course at LSE. One proof is borrowed from Javier Hidalgo's EC484 course at LSE. Some probability/statistic notions are borrowed from Sandra Fortini's Intro to Probability course at Bocconi. I also use some textbooks, with the main reference being Hansen, 2022.

For the second part, I rely heavily on [Erich Battistin's](#) notes on causal reasoning and design-driven identification in the social sciences. I also use some notes from Michael Gmeiner's EC475 course at LSE.

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Chapter 0: Introduction: Parameters, Estimands, and Estimators

A useful way to organize econometrics is to distinguish carefully between three different objects: *parameters*, *estimands*, and *estimators*. Although these objects are closely related, they belong to different conceptual layers of the econometric enterprise and should not be conflated.

A *parameter* is the object of interest defined by an economic model or a causal framework. It is the quantity we ultimately care about. For example, in a structural model of labor supply, a parameter may measure the elasticity of hours worked with respect to wages; in a causal model, it may be the average effect of an additional year of schooling on earnings. Parameters are therefore theory-driven objects: they arise from an economic question, not directly from the data.

An *estimand*, by contrast, is a population quantity defined in terms of the joint distribution of observables. It is a statistical object. Examples include regression coefficients, differences in conditional means, average treatment effects, or instrumental-variables ratios. The key role of econometric theory is to understand when an estimand can be interpreted as the parameter of interest. This is the problem of *identification*. Identification is therefore a population statement: under appropriate assumptions, it tells us whether the parameter we care about can be recovered from features of the data-generating process.

Finally, an *estimator* is a function of the observed sample. It is the empirical object we compute in practice. Sample averages, OLS coefficients, two-stage least squares estimators, kernel estimators, or matching estimators are all estimators. Since the observed data are random, estimators are random variables as well, and they possess finite-sample and asymptotic distributions. Understanding these distributions allows us to perform *statistical inference*, that is, to learn about the estimand—and, through identification, about the underlying parameter.

This distinction is especially useful because it clarifies that econometrics involves two logically separate tasks. The first task is *identification*: linking a theoretical or causal parameter to a population estimand under explicit assumptions. The second task is *inference*: linking the population estimand to a sample estimator, and quantifying the uncertainty induced by sampling variation. Much confusion in applied work comes from moving too quickly between these layers. A regression coefficient, for instance, is always an estimand and can always be estimated; whether it has a causal interpretation is a distinct question, resolved only by identification assumptions.

A simple example illustrates the point. Consider the linear model

$$Y_i = \beta D_i + \varepsilon_i.$$

The parameter of interest is the causal effect β . A natural estimand is the population

regression coefficient

$$\beta^{OLS} = \frac{\text{Cov}(Y_i, D_i)}{\text{V}(D_i)}$$

Substituting the model into this expression yields

$$\beta^{OLS} = \frac{\text{Cov}(\beta D_i + \varepsilon_i, D_i)}{\text{V}(D_i)} = \beta + \frac{\text{Cov}(\varepsilon_i, D_i)}{\text{V}(D_i)}$$

This decomposition makes the identification problem transparent: the regression estimand equals the causal parameter if and only if $\text{Cov}(\varepsilon_i, D_i) = 0$. Even in very large samples, ordinary least squares only recovers the regression estimand. Whether that estimand coincides with the causal parameter depends entirely on the economic assumptions imposed on the model. In contrast, once identification holds, the problem becomes one of inference: how precisely can the sample OLS estimator recover the population regression coefficient?

The perspective adopted in these notes follows this logic. The first part focuses mainly on estimation and inference for standard econometric procedures, while the second part studies identification and causal interpretation in modern microeconometrics. Keeping separate the roles of parameters, estimands, and estimators helps unify these topics and makes clear what each econometric method is designed to accomplish.

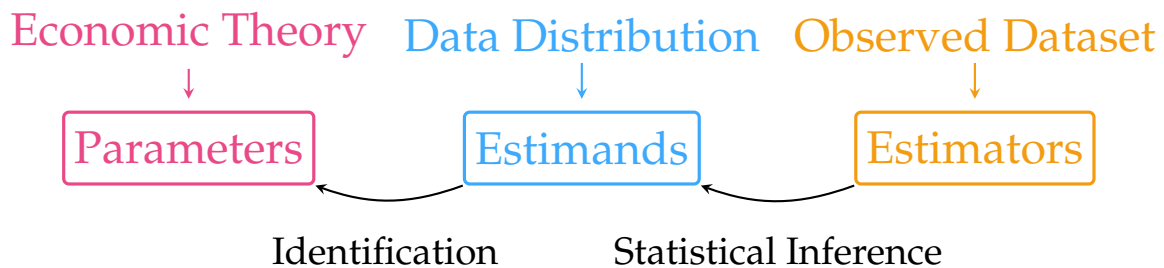


Figure 1: A schematic representation of the relationship between parameters, estimands, and estimators, adapted from lecture slides by Peter Hull.

Chapter 1: Econometrics Foundations

1 Classical Linear Regression Model

1.1 Model and Gauss-Markov Assumptions

Let $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times k}$ and $\beta \in \mathbb{R}^k$ the parameter of interest. The general form of the multiple linear regression model is

$$y = X\beta + \varepsilon. \quad (1.1)$$

Assumption 1.1: Gauss Markov Assumptions + Normality

The Gauss-Markov Linear Regression Assumptions are given below

- (i) Linearity: The true model is $y = X\beta + \varepsilon$, with $\mathbb{E}(\varepsilon) = 0$
- (ii) Full rank: $\text{rank}(X) = k \leq n \implies X'X$ is invertible
- (iii) Exogeneity: $\mathbb{E}[\varepsilon | X] = 0$
- (iv) Spherical error: $\mathbb{V}(\varepsilon | X) = \sigma^2 I_n$
- (v) Normality: $\varepsilon | X \sim \mathcal{N}(0, \sigma^2 I_n)$

We quickly discuss the intuition and implications of some of the assumptions below.

The *full rank* assumption just states that it is impossible to estimate a k number of β with n observations if $k > n$. This assumption also prevents from perfect collinearity.

The *exogeneity* assumption deserves a longer treatment. Consider the case where X are treated as random variables. First, notice that full independence, $X \perp\!\!\!\perp \varepsilon$ implies that for any measurable $g(\cdot)$ with finite moments, $\mathbb{E}(g(X)\varepsilon) = \mathbb{E}(\varepsilon)\mathbb{E}(g(X))$. Specifically, $\mathbb{E}(X'\varepsilon) = \mathbb{E}(X')\mathbb{E}(\varepsilon) = 0$ in this context. Mean independence only implies that $\mathbb{E}(\varepsilon | X) = \mathbb{E}(\varepsilon) = 0$. For our next argument, we introduce the following proposition.

Proposition 1.1: Law of Iterated Expectations (LIE)

Let Z be an integrable random variable and let $\mathcal{G}$ a sigma-algebra generated by a partition. Then, $\mathbb{E}(\mathbb{E}(Z | \mathcal{G})) = \mathbb{E}(Z)$

Finally, notice that in this context, we can write the covariance $\text{Cov}(\varepsilon_i, x_i) = \mathbb{E}(\varepsilon_i x_i) - \mathbb{E}(\varepsilon_i)\mathbb{E}(x_i) = \mathbb{E}(\mathbb{E}(\varepsilon_i x_i | x_i)) = \mathbb{E}(x_i \mathbb{E}(\varepsilon_i | x_i)) = 0$, where the second equality is an application of LIE.

An interesting observation is hence that independence $\implies$ mean independence $\implies$ uncorrelatedness.

The *spherical errors* assumption can be rewritten as follows:

$$\mathbb{V}(\varepsilon \mid X) = \sigma^2 I_n \Leftrightarrow \mathbb{V}(\varepsilon_i \mid X) = \sigma^2, \forall i \text{ and } \text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \forall i \neq j$$

Therefore, we can see that this is an assumption about homoskedasticity and autocorrelation. Notice that iid assumption for $\varepsilon_i \mid X$ implies spherical errors, but the other way around is not true.

1.2 Estimation procedures

1.2.1 Minimum Distance Estimation - OLS

The problem is

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - x_i' \beta)^2 = \arg \min_{\beta} (y - X\beta)'(y - X\beta) = \arg \min_{\beta} S(\beta)$$

With $S(\beta)$ being the sum of squared residuals. The necessary condition for a minimum is a first order condition (FOC):

$$\left. \frac{\partial S(\beta)}{\partial \beta} \right|_{\hat{\beta}} = 0 - 2X'y + 2X'X\hat{\beta} = 0$$

This is equivalent to $X'\hat{\varepsilon} = 0$, since $\hat{\varepsilon} = y - X\hat{\beta}$ are the residuals.

We solve this as follows, and these equations are also called *the normal equation*:

$$-2X'y + 2X'X\hat{\beta} = 0 \Rightarrow \hat{\beta}_{OLS} = (X'X)^{-1}X'y$$

Notice that by assumption (ii), the inverse of exists $(X'X)$, which ensures that $\hat{\beta}_{OLS}$ is unique. This estimator is a minimum, by second order conditions:

$$\left. \frac{\partial^2 S(\beta)}{\partial \beta \partial \beta'} \right|_{\hat{\beta}} = 2X'X$$

This matrix must be positive definite.

1.2.2 Method of Moments (MM)

The MM estimator of β is based on the following moment conditions

$$E(x_{ij}\varepsilon_i) = 0 \quad j = 1 \dots k$$

These are the orthogonality conditions: errors are uncorrelated with the regressors.

$\hat{\beta}_{MM}$ ensures that the sample analogue moment conditions are satisfied. To find the estimator, replace the expectation with the sample average

$$\frac{1}{n} \sum_{i=1}^n x_{ij} \hat{\varepsilon}_i = 0 \quad \forall j \quad \text{or} \quad \frac{1}{n} \sum_{i=1}^n x_{ij} (y_i - x_i' \hat{\beta}_{MM}) = 0$$

In matrix notations this implies that

$$\frac{1}{n} X'(y - X\hat{\beta}_{MM}) = \frac{1}{n} X'\hat{\varepsilon} = 0 \Rightarrow \hat{\beta}_{MM} = \hat{\beta}_{OLS}$$

The MM estimator of σ^2 is based on the moment condition $\mathbb{E}(\varepsilon_i^2) = \sigma^2$ using the sample analogue moment

$$\hat{\sigma}_{MM}^2 = \frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i^2.$$

1.2.3 Maximum Likelihood Estimation (MLE)

The MLE requires us to have a sample of N independently drawn observations $(y_1, x_1), \dots, (y_N, x_N)$ with conditional distribution $f(y|x, \beta)$ where the functional form f is known but parameters β are not. Given assumptions (i) and (v), we have that $y|X \sim N(X\beta, \sigma^2 I_n)$ and that $y_1 \dots y_n$ are independent. We want to estimate β , and hence we choose $\hat{\beta}_{MLE}$ to maximize $\mathbb{P}(y_1, y_2, \dots, y_N | \hat{\beta}_{MLE}, x_1, x_2, \dots, x_N)$, that is the probability of observing the data if $\hat{\beta}_{MLE}$ was true.

$$\begin{aligned} f(y_1 \dots y_n | X; \beta, \sigma^2) &= \prod_{i=1}^n f(y_i | X; \beta, \sigma^2) \quad - \text{independence} \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma}} \cdot \exp \left\{ -\frac{1}{2\sigma^2} (y_i - x_i' \beta)^2 \right\} \quad - \text{normality} \\ &= \left(\frac{1}{\sqrt{2\pi\sigma}} \right)^n \cdot \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i' \beta)^2 \right\} = \left(\frac{1}{\sqrt{2\pi\sigma}} \right)^n \cdot \exp \left\{ -\frac{1}{2\sigma^2} S(\beta) \right\} \end{aligned}$$

The log likelihood function is

$$l(\beta, \sigma^2; y_1 \dots y_n, x_1 \dots x_n) = -\frac{n}{2} \log(2\pi\sigma) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} S(\beta)$$

We take two FOCs wrt to each unknown parameter (β and σ^2).

$$\left. \frac{\partial l(\beta, \sigma^2)}{\partial \beta} \right|_{\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2} = -\frac{1}{2\hat{\sigma}_{MLE}^2} \frac{\partial S(\hat{\beta}_{MLE})}{\partial \beta} = 0 \Leftrightarrow \frac{\partial S(\hat{\beta}_{MLE})}{\partial \beta} = 0 \quad (1.2)$$

This shows that maximizing the likelihood under normality with respect to β leads to $\hat{\beta}_{MLE} = \hat{\beta}_{MM} = \hat{\beta}_{OLS}$.

$$\left. \frac{\partial l(\beta, \sigma^2)}{\partial \sigma^2} \right|_{\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2} = -\frac{n}{2\hat{\sigma}_{MLE}^2} + \frac{1}{2\hat{\sigma}_{MLE}^4} S(\hat{\beta}_{MLE}) = 0 \quad (1.3)$$

1.3 Goodness of Fit

We define different objects required to study goodness of fit.

Definition 1.1: Total sum of squares

The Total Sum of Squares is given by

$$TSS = \sum_i (y_i - \bar{y})^2$$

Definition 1.2: Explained sum of squares

The Explained Sum of Squares is given by

$$ESS = \sum_i (\hat{y}_i - \bar{y})^2$$

Definition 1.3: Residual sum of squares

The Residuals Sum of Squares is given by

$$RSS = \sum_i (y_i - \hat{y}_i)^2$$

*Some intuition:*¹ The TSS represents the total variability in the dependent variable, including the portions of variability both explained and unexplained by the regression model. By its nature, the TSS reflects the overall dispersion of individual values of the observed dependent variable from its mean value. The ESS captures the portion of the total variability in the dependent variable explained by the regression model. Finally, the RSS reflects the remaining error, or unexplained variability, which is the portion of the total variability in the dependent variable that can't be explained by the regression model.

With these definitions in mind, we can now define three alternative definitions of the R^2 .

Definition 1.4: Three alternative definitions of R^2

- (i) $R_1^2 = 1 - \frac{RSS}{TSS}$
- (ii) $R_2^2 = \frac{[\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{y})]^2}{[\sum_i (y_i - \bar{y})]^2 [\sum_i (\hat{y}_i - \bar{y})]^2}$
- (iii) $R_3^2 = \frac{ESS}{TSS}$

¹This [website](#) provides cool intuition.

Proposition 1.2: Equivalence between different definitions of R^2

When an intercept is included in the model, all three measures are the same and $R^2 \in [0, 1]$.

Proof. First, we can show that $R_1^2 = R_3^2$.

$$\begin{aligned} TSS &= \sum (y_i - \bar{y})^2 = \sum [(\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)]^2 \\ &= ESS + RSS + 2 \sum (\hat{y}_i - \bar{y})(y_i - \hat{y}_i) \\ &= ESS + RSS + 2 \sum (\hat{y}_i - \bar{y})\varepsilon_i \end{aligned}$$

The last term is 0 since residuals are orthogonal to the fitted values. Therefore, we have that $\frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$ and therefore $R_3^2 = R_1^2$.

$$R_2^2 = \frac{[\sum (y_i - \bar{y})(\hat{y}_i - \bar{y})]^2}{[\sum (y_i - \bar{y})^2][\sum (\hat{y}_i - \bar{y})^2]} = \frac{[\sum (y_i - \bar{y})(\hat{y}_i - \bar{y})]^2}{TSS \cdot ESS}$$

The numerator can be re-written as follows:

$$\begin{aligned} (\sum (y_i - \bar{y})(\hat{y}_i - \bar{y}))^2 &= (\sum [(\hat{y}_i + e_i - \bar{y})(\hat{y}_i - \bar{y})])^2 \\ &= (\sum [(\hat{y}_i - \bar{y}) + e_i](\hat{y}_i - \bar{y}))^2 \\ &= \left(\sum (\hat{y}_i - \bar{y})^2 + \sum e_i(\hat{y}_i - \bar{y}) \right)^2 \\ &= ESS^2 \end{aligned}$$

Therefore, we have

$$R_2^2 = \frac{ESS^2}{ESS \cdot TSS} = \frac{ESS}{TSS} = R_3^2$$

□

The main drawback of the R^2 is that it never decreases as we increase the number of regressors, even if the new variables have no explanatory power. The adjusted R^2 allows to correct for the number of explanatory variables:

$$\bar{R}^2 = 1 - \frac{RSS/(n-k)}{TSS/(n-1)}$$

Suppose that we are in a case where we add an additional regressor with no explanatory power. RSS does not change (specifically it does not decrease, what we desire), but $(n-k)$ goes down and therefore the $\bar{R}^2$ goes down as well.

1.4 Geometric interpretation of OLS and Frisch–Waugh–Lovell (FWL)

1.4.1 Quick linear algebra recap

We provide some basic definitions and one proposition that are useful in our application.

Definition 1.5: Span

Consider the vector space V and a set of vectors $A \subseteq V$. The span of A , denoted $\text{span}A$, is the set of all linear combinations of vectors in A . This is given by

$$\text{span}A = \left\{ \sum_{i=1}^n \alpha_i v_i \mid \forall i \in \mathbb{N}, \alpha_i \in \mathbb{R}, v_i \in A \right\}$$

Definition 1.6: Column Space (Range)

Consider an $n \times K$ matrix X with columns $x_1, x_2, \dots, x_K$. The column space of X , denoted $\mathcal{R}(X)$ or $\text{col}(X)$, is the set of all possible linear combinations of the column vectors of X . Equivalently, the column space is the span of the columns of X :

$$\mathcal{R}(X) = \text{span}\{x_1, x_2, \dots, x_K\} = \left\{ \sum_{j=1}^K \beta_j x_j \mid \beta_j \in \mathbb{R} \right\}$$

Geometrically, $\mathcal{R}(X)$ is the subspace of $\mathbb{R}^n$ generated by the columns of X .

Definition 1.7: Linear Independence

Given a vector space V , a set of vectors $\{v_1, \dots, v_n\}$ is linearly *dependent* if there exists scalars $\{\alpha_1, \dots, \alpha_n\}$, not all zero, such that $\sum_i \alpha_i v_i = 0$. A set of vectors $\{v_1, \dots, v_n\}$ is linearly *independent* if it is not linearly dependent.

Proposition 1.3: Equivalence

Consider an $n \times n$ matrix M . The following are equivalent.

1. M is invertible
2. The row vectors of M are linearly independent
3. The column vectors of M are linearly independent

1.4.2 Geometric interpretation of OLS

Consider the model in matrix form. The fitted value is $\hat{y} = X\hat{\beta}$, the residual is $\hat{e} = y - X\hat{\beta}$, and the orthogonality condition is given by $X'\hat{e} = 0$. We first need to introduce two important matrices in linear algebra.

Definition 1.8: Projection (with properties)

The projection matrix is defined as $P_X = X(X'X)^{-1}X'$. It satisfies the following properties:

1. $P_X = P_X'$ (symmetry)
2. $P_X^2 = P_X$ (idempotent)
3. $P_X X = X$
4. $P_X y = \hat{y}$ (projection)
5. $P_X \hat{e} = 0$
6. $\text{tr}(P_X) = k$

Proof. We prove properties 4 to 6 here by inspection of the definitions, for illustration. For 4,

$$P_X y = X(X'X)^{-1}X'y = X\hat{\beta} = \hat{y}$$

For 5,

$$\begin{aligned} P_X \hat{e} &= X(X'X)^{-1}X'(y - X\hat{\beta}) = X(X'X)^{-1}X'y - X(X'X)^{-1}X'X(X'X)^{-1}X'y \\ &= X(X'X)^{-1}X'y - X(X'X)^{-1}X'y = 0 \end{aligned}$$

Now, for 6,

$$\text{tr}(P_X) = \text{tr}(X(X'X)^{-1}X') = \text{tr}((X'X)^{-1}X'X) = \text{tr}(I_k) = k$$

where the second equality comes from the fact that $\text{tr}(AB) = \text{tr}(BA)$. □

The key insight from the 5th property is that the projection on a vector orthogonal to X is equal to 0. OLS chooses the fitted vector $\hat{y}$ as the point in the column space $\text{col}(X)$ that is *closest* to y in Euclidean distance, so the residual $\hat{e} = y - \hat{y}$ is the perpendicular from y to that subspace (hence $X'\hat{e} = 0$). The projection P_X “keeps” components along $\text{col}(X)$ and “kills” components orthogonal to it, which is why $P_X y = \hat{y}$ and $P_X \hat{e} = 0$.

Definition 1.9: Annihilator matrix (with properties)

The annihilator matrix (or orthogonal projection matrix) is defined as $M_X = I_k - P_X$. It satisfies the following properties:

1. $M_X = M_X'$ (symmetry)
2. $M_X^2 = M_X$ (idempotent)
3. $M_X X = 0$ (annihilator matrix)
4. $M_X y = \hat{e}$ (orthogonal projection)

$$5. M_X \hat{e} = 0$$

$$6. \text{tr}(M_X) = n - k$$

Equivalently, the residual-maker $M_X = I - P_X$ removes everything explainable by X since $M_X y = \hat{e}$ and $M_X X = 0$. Note that by definition, the RSS is just $\hat{e}'\hat{e} = y' M_X' M_X y = y' M_X y$.

Before moving to the Frisch–Waugh–Lovell Theorem, I construct a simple low-dimensional example to understand better what we just discussed. Figure 1.4.2 helps visualize.

Example 1.1: Geometric OLS in $\mathbb{R}^3$

Consider a case where the data space is $\mathbb{R}^3$ ($n = 3$), and we have two explanatory variables, x_1 and x_2 ($K = 2$). The matrix $\mathbf{X}$ is 3×2 :

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \\ x_{13} & x_{23} \end{pmatrix} = (\mathbf{x}_1 \quad \mathbf{x}_2)$$

The column space $\mathcal{R}(\mathbf{X})$ is the span of the two columns, $\mathbf{x}_1$ and $\mathbf{x}_2$.

$$\mathcal{R}(\mathbf{X}) = \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$$

Since $K = 2$, $\mathcal{R}(\mathbf{X})$ is a 2-dimensional plane (the shaded gray area) embedded within the 3-dimensional space $\mathbb{R}^3$. Now, the vector $\mathbf{y}$ lies somewhere in $\mathbb{R}^3$, generally off the plane $\mathcal{R}(\mathbf{X})$.

Now, the expression $\mathbf{X}\hat{\beta}$ is, by definition, a linear combination of the columns of $\mathbf{X}$:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = (\mathbf{x}_1 \quad \mathbf{x}_2) \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \hat{\beta}_1 \mathbf{x}_1 + \hat{\beta}_2 \mathbf{x}_2$$

The OLS estimator finds $\hat{\mathbf{y}}$ on the plane $\mathcal{R}(\mathbf{X})$ that is closest to $\mathbf{y}$. This closest point is found by dropping a perpendicular line from $\mathbf{y}$ to the plane. In other words, we find the vector $\hat{\mathbf{y}}$ on the $\mathbf{x}_1, \mathbf{x}_2$ plane that makes the residual $\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}}$ orthogonal to the plane.

Since $\hat{\mathbf{y}}$ is formed by scaling $\mathbf{x}_1$ by $\hat{\beta}_1$ and $\mathbf{x}_2$ by $\hat{\beta}_2$ and summing them up, $\hat{\mathbf{y}}$ must lie on the plane (the column space $\mathcal{R}(\mathbf{X})$). The key is that the orthogonality condition ($\mathbf{X}'\hat{\mathbf{e}} = \mathbf{0}$) is what forces the solution $\hat{\beta}$. If you choose any arbitrary β , the resulting $\mathbf{X}\beta$ will be a vector on the plane, but the residual $\mathbf{y} - \mathbf{X}\beta$ will not be perpendicular to the plane (it will just be some other vector).

Finally, $\hat{\mathbf{y}} = \mathbf{P}_X \mathbf{y}$ means $\mathbf{P}_X$ is the matrix that transforms the raw data vector $\mathbf{y}$ into the best possible linear combination of $\mathbf{X}$ (the predicted vector $\hat{\mathbf{y}}$). It collapses the n -dimensional vector $\mathbf{y}$ onto the K -dimensional subspace $\mathcal{R}(\mathbf{X})$.

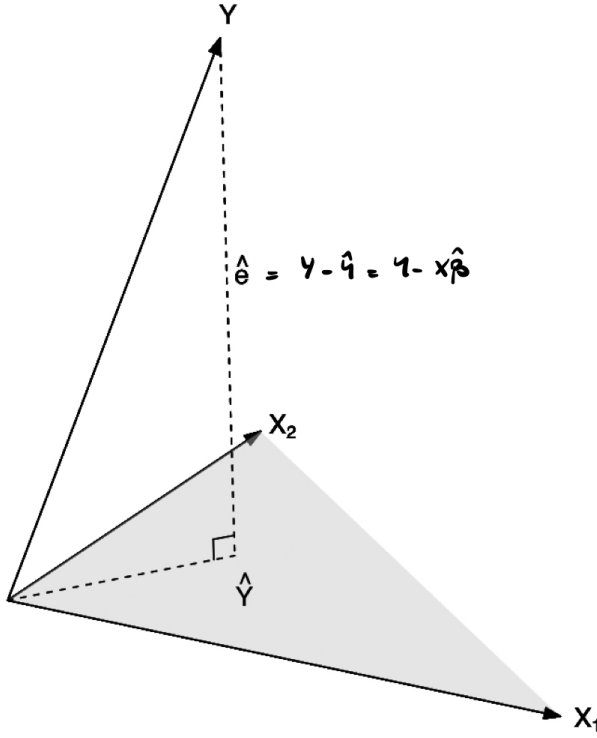


Figure 2: Projection of Y onto X_1 and X_2 (Figure 3.3 from Hansen, 2022)

1.4.3 Frisch–Waugh–Lovell (FWL)

We are now well-equipped to study the Frisch–Waugh–Lovell (FWL) theorem.

Consider the regression $y = X\beta + \varepsilon$, which can be rewritten, partitioning matrix X as $X = [X_1 : X_2]$. We can hence rewrite the regression as $y = X_1\beta_1 + X_2\beta_2 + \varepsilon$. Notice that $\dim(X_1) = n \times k_1, \dim(X_2) = n \times (k - k_1)$. Additionally, recall that $P_j = X_j(X_j'X_j)^{-1}X_j', M_j = I - P_j, \forall j$

Our OLS estimator is as follows:

$$\begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = (X'X)^{-1}X'y = \begin{bmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{bmatrix}^{-1} \begin{bmatrix} X_1'y \\ X_2'y \end{bmatrix}$$

Notice that by the inverse formula for partitioned matrix, we can rewrite

$$\begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} S_1^{-1} & -S_1^{-1}X_1'X_2(X_2'X_2)^{-1} \\ -S_2^{-1}X_2'X_1(X_1'X_1)^{-1} & S_2^{-1} \end{bmatrix} \begin{bmatrix} X_1'y \\ X_2'y \end{bmatrix},$$

where $S_1 := X_1'X_1 - X_1'X_2(X_2'X_2)^{-1}X_2'X_1$, $S_2 := X_2'X_2 - X_2'X_1(X_1'X_1)^{-1}X_1'X_2$.

We can hence find an expression for $\hat{\beta}_1, \hat{\beta}_2$.

$$\begin{aligned}\hat{\beta}_1 &= S_1^{-1} \left(X_1'y - X_1'X_2(X_2'X_2)^{-1}X_2'y \right) \\ &= \left(X_1'(I - X_2(X_2'X_2)^{-1}X_2')X_1 \right)^{-1} X_1'(I - X_2(X_2'X_2)^{-1}X_2')y \\ &= (X_1'M_2X_1)^{-1}(X_1'M_2y), \\ \hat{\beta}_2 &= S_2^{-1} \left(X_2'y - X_2'X_1(X_1'X_1)^{-1}X_1'y \right) \\ &= \left(X_2'(I - X_1(X_1'X_1)^{-1}X_1')X_2 \right)^{-1} X_2'(I - X_1(X_1'X_1)^{-1}X_1')y \\ &= (X_2'M_1X_2)^{-1}(X_2'M_1y)\end{aligned}$$

Now, since M_2 is symmetric and idempotent, we have:

$$\hat{\beta}_1 = (X_1'M_2M_2X_1)^{-1}(X_1'M_2M_2y) = (\tilde{X}_1'\tilde{X}_1)^{-1}(\tilde{X}_1'\tilde{\varepsilon}_2),$$

where $\tilde{X}_1 = M_2X_1$ and $\tilde{\varepsilon}_2 = M_2y$.

Therefore, the coefficient estimator $\hat{\beta}_1$ is algebraically equal to the least squares regression of $\tilde{\varepsilon}_2$ on $\tilde{X}_1$. The conclusion follows since M_2X_1 are the residuals from a regression of X_2 on X_1 , and M_2y are the residuals of a regression of X_2 on y .

Now, for the residuals, in the case of OLS, we have that $\hat{\varepsilon} = y - X\hat{\beta} = y - P_Xy = M_Xy$. Notice that it can be written as $y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2$. Similarly, we have that $\tilde{\varepsilon} = \tilde{\varepsilon}_2 - \tilde{X}_1\hat{\beta}_1 = M_2y - M_2X_1\hat{\beta}_1$. If we pre-multiply $\hat{\varepsilon}$ by M_2 , we get

$$M_2\hat{\varepsilon} = M_2y - M_2X_1\hat{\beta}_1 - M_2X_2\hat{\beta}_2$$

Notice that by properties of the annihilator matrix, for any matrix X , $M_XX = 0$. It is easy to see since $M_XX = (I - P_X)X = 0$ as $P_XX = X$. Therefore, we have that $M_2X_2\hat{\beta}_2 = 0$, which implies that $M_2\hat{\varepsilon} = M_2y - M_2X_1\hat{\beta}_1$. The conclusion follows by orthogonality condition $X'\hat{\varepsilon} = 0 \implies X_2'\hat{\varepsilon} = 0$:

$$M_2\hat{\varepsilon} = \hat{\varepsilon} - X_2(X_2'X_2)^{-1}X_2'\hat{\varepsilon} = \hat{\varepsilon}$$

We have hence proven the following theorem.

Theorem 1.1: Frisch–Waugh–Lovell Theorem (FWL)

The OLS estimator of β_1 and the OLS residuals can be computed by either the OLS regression or using the following algorithm:

1. Regress y on X_2 , obtain $\tilde{\varepsilon}_2$
2. Regress X_1 on X_2 , obtain residuals $\tilde{X}_1$
3. Regress $\tilde{\varepsilon}_2$ on $\tilde{X}_1$, and obtain OLS estimator $\hat{\beta}_1$ and residuals $\hat{\varepsilon}$

Additionally, the residual-based regression of $M_2 y$ on $M_2 X_1$ yields residuals that are numerically equivalent to those obtained from regressing y on $(X_1 : X_2)$.

We are giving some intuition about the theorem below.

1. Regress y on X_2 , obtain $\tilde{\varepsilon}_2$. This means that we are calculating the part of y which is not explained by X_2 . In other words, we are projecting y onto the space which is orthogonal to X_2 , so we are projecting X_2 out of y .
2. Regress X_1 on X_2 , obtain residuals $\tilde{X}_1$. Again, we are computing the part of X_1 which is not explained by X_2 , or we are projecting X_1 onto the space which is orthogonal to X_2 .
3. Regress $\tilde{\varepsilon}_2$ on $\tilde{X}_1$, and obtain OLS estimator $\hat{\beta}_1$ and residuals $\hat{\varepsilon}$. The OLS estimator for the final regression is $\hat{\beta} = (\tilde{X}_1' \tilde{X}_1)^{-1} \tilde{X}_1' \tilde{\varepsilon}_2 = (X_1' M_2' M_2 X_1)^{-1} (X_1' M_2' M_2 y)$

1.5 Finite-Sample Properties

1.5.1 Unbiasedness

Recall our model $y = X\beta + \varepsilon$, which satisfies properties (i) to (v) (Gauss Markov + normality). Our estimator is therefore $\hat{\beta} = (X'X)^{-1}X'y$. It is important to understand that $\hat{\beta}$ is a random variable. We can now compute the conditional mean given X .

$$\mathbb{E}[\hat{\beta} | X] = (X'X)^{-1}X'\mathbb{E}[y | X] = (X'X)^{-1}X'X\hat{\beta} + (X'X)^{-1}X'\mathbb{E}[\varepsilon | X] = \beta \quad (1.4)$$

The last equality comes from assumption (iii). I like the interpretation Hansen has of this result. He says that the estimator $\hat{\beta}$ is unbiased for β , conditional on X means that the conditional distribution of $\hat{\beta}$ is centered at β . Conditional on X means that the distribution is unbiased for any realisation on the regressor matrix X . Additionally, notice that conditional unbiasedness implies unconditional unbiasedness by LIE:

$$\mathbb{E}(\hat{\beta}) = \mathbb{E}[\mathbb{E}[\hat{\beta} | X]] = \beta$$

Given (i) to (iv), we can define an estimator that is an unbiased estimator of σ^2 : $s^2 = \frac{\hat{\varepsilon}'\hat{\varepsilon}}{n-k}$. This estimator gives rise to the standard error of the regression, s , which is the square root of s^2 . Recall that $\hat{\varepsilon} = y - \hat{y}$. (i) and (ii) yield

$$\hat{\varepsilon} = M_X y = M_X (X\beta + \varepsilon) = M_X \varepsilon, \text{ hence}$$

$$\hat{\varepsilon}'\hat{\varepsilon} = \varepsilon' M_X \varepsilon \quad \text{with} \quad M_X = I - X(X'X)^{-1}X'$$

then: a) first equality, we divide by $n - k$ to adjust biases (see at the end), b) second equality, we introduce the trace operator, c) third equality exploit the linearity of tr and E

$$\mathbb{E}(s^2|X) = \frac{1}{n-k} \mathbb{E}(\varepsilon' M_X \varepsilon | X) = \frac{1}{n-k} E[tr(\varepsilon' M_X \varepsilon) | X] = \frac{1}{n-k} tr[M_X (\mathbb{E}(\varepsilon' \varepsilon | X))]$$

recalling A.3 and A.4

$$\mathbb{E}(s^2|X) = \frac{1}{n-k} tr[M_X \sigma^2 I] = \frac{\sigma^2}{n-k} tr(M_X)$$

We can show that

$$\begin{aligned} tr(M_X) &= tr(I_n - X(X'X)^{-1}X') \\ tr(I_n) - tr(X(X'X)^{-1}X') &= tr(I_n) - tr(I_k) = n - k \end{aligned}$$

then we can simplify

$$E(s^2|X) = \frac{1}{n-k} tr[M_X \sigma^2 I] = \frac{\sigma^2}{n-k} n - k = \sigma^2$$

s^2 is conditionally unbiased. Using the law of iterated expectations, we can show that s^2 is unconditionally unbiased too.

1.5.2 Variance

Given observations (i) to (iii), we have that the conditional variance is given by:

$$\begin{aligned} \mathbb{V}(\hat{\beta}|X) &= \mathbb{E}[(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])' | X] \\ &= \mathbb{E}\left[(\beta + (X'X)^{-1}X'\varepsilon - \beta)((\beta + (X'X)^{-1}X'\varepsilon - \beta))' | X\right] \\ &= \mathbb{E}\left[(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1}\right] \end{aligned}$$

This is the general variance-covariance matrix where the diagonal elements are $\mathbb{V}(\hat{\beta}_k|X)$ and the off-diagonal elements are $\text{Cov}(\hat{\beta}_k, \hat{\beta}_\ell|X)$. If we assume spherical errors (iv), $\varepsilon\varepsilon' = \sigma^2$, and the variance becomes

$$\mathbb{V}(\hat{\beta}|X) = \sigma^2(X'X)^{-1}$$

For X non-stochastic, $\mathbb{V}(\hat{\beta}|X) = \mathbb{V}(\hat{\beta})$. If X is stochastic, we have

$$\mathbb{V}(\hat{\beta}) = \mathbb{E}(\mathbb{V}(\hat{\beta}|X)) + \mathbb{V}(\mathbb{E}(\hat{\beta}|X)) = \sigma^2 \mathbb{E}\left[(X'X)^{-1}\right]$$

The last equality comes from the fact that β is just a number, so has variance 0.

The variance can be estimated as follows

$$\hat{\mathbb{V}}(\hat{\beta}) = s^2(X'X)^{-1}, \text{ where } s^2 = \frac{\hat{\varepsilon}'\hat{\varepsilon}}{n-k}$$

The standard error of the regression is simply the square root of s^2 . The standard of $\hat{\beta}_j$ is

$$SE(\hat{\beta}_j) = \sqrt{\hat{\mathbb{V}}(\hat{\beta})} = \sqrt{s^2(X'X)^{-1}_{jj}}$$

Example 1.2: Variance in simple regression model

Consider the simple regression

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X] = 0, \quad \mathbb{V}(\varepsilon_i | X) = \sigma^2.$$

Conditional on X ,

$$\mathbb{V}(\hat{\beta} | X) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} := \frac{\sigma^2}{(n-1)S_x^2},$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $S_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is the sample variance of x .

The key takeaways here are:

1. The $(n-1)$ in the denominator highlights that larger samples reduce the variance.
2. The S_x^2 in the denominator highlights that greater variability of the regressors reduce the variability of our estimators.

We can now state one of the most important theorems so far.

Theorem 1.2: Gauss Markov Theorem

Let assumptions (i) - (iv) to hold. Then,

- The least square estimator $\hat{\beta}$ is the minimum variance linear unbiased estimator.
- For any vector of constants w , the minimum variance linear unbiased estimator of $w'\beta$ is $w'\hat{\beta}$.

Proof. Let $\tilde{\beta}$ a linear estimator where $\tilde{\beta} = Cy$. Unbiasedness requires

$$\mathbb{E}[\tilde{\beta} | X] = CX\beta + \mathbb{E}(\varepsilon|X) = CX\beta \stackrel{!}{=} \beta \implies CX = I_k.$$

Under spherical errors, $\mathbb{V}(y | X) = \sigma^2 I_n$, so

$$\mathbb{V}(\tilde{\beta} | X) = \sigma^2 CC', \quad \mathbb{V}(\hat{\beta} | X) = \sigma^2 (X'X)^{-1}.$$

Write the residual-maker $M_X := I_n - P_X$ with $P_X := X(X'X)^{-1}X'$. Note M_X is symmetric, idempotent, and positive semidefinite, and $M_X X = 0$.

Use the decomposition

$$C = (X'X)^{-1}X' + CM_X$$

(which holds because multiplying both sides by X yields I_k). Then

$$\begin{aligned}\mathbb{V}(\tilde{\beta} | X) &= \sigma^2 [(X'X)^{-1}X' + CM_X] [(X'X)^{-1}X' + CM_X]' \\ &= \sigma^2 \left\{ (X'X)^{-1}X'X(X'X)^{-1} + CM_XM_XC' \right\} \quad (\text{cross terms vanish since } X'M_X = 0) \\ &= \sigma^2 (X'X)^{-1} + \sigma^2 CM_XC' .\end{aligned}$$

Hence

$$\mathbb{V}(\tilde{\beta} | X) - \mathbb{V}(\hat{\beta} | X) = \sigma^2 CM_XC' \geq_{psd} 0,$$

because M_X is positive semidefinite. Therefore OLS has the smallest variance among all linear unbiased estimators.

Finally, for any fixed $w \in \mathbb{R}^k$,

$$\mathbb{V}(w'\tilde{\beta} | X) - \mathbb{V}(w'\hat{\beta} | X) = w'[\mathbb{V}(\tilde{\beta} | X) - \mathbb{V}(\hat{\beta} | X)]w = \sigma^2 (C'w)'M_X(C'w) \geq_{psd} 0,$$

so every linear combination $w'\hat{\beta}$ is minimum-variance among linear unbiased estimators. $\square$

1.6 Asymptotic Properties of OLS

1.6.1 Probability Theory foundations

We start by defining concepts that will be useful in our treatment of asymptotic theory. First, we need to define a (non-exhaustive) list of modes of convergence for random variables.

Definition 1.10: Convergence of random variables

Consider a sequence X_n of random variables.

- The strongest mode of convergence we mention here is almost sure convergence. X_n is said to convergence a.s. to X (or $X_n \xrightarrow{a.s.} X$) if for every $\epsilon > 0$,

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} (|X_n - X|) > \epsilon \right) = 0$$

- X_n is said to converge in probability to X (or $X_n \xrightarrow{p} X$) if for every $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} (|X_n - X| > \epsilon) = 0$$

- Let $F_{X_n}(x)$ the CDF of X_n and $F_X(x)$ the CDF of X . X_n is said to converge in probability to X (or $X_n \xrightarrow{d} X$) if for every point of continuity of F ,

$$F_{X_n}(x) \rightarrow F_X(x)$$

Now, we need to introduce two of the most important concepts in probability theory: laws of large numbers (LLN) and central limit theorems (CLT). We start by three LLNs. The first one is the one we will mostly use. It has the weakest assumptions, but turns out to be hard to prove. I will therefore prove Chebyshev's (like) WLLN, which will be useful later once we relax the spherical error assumption.

Theorem 1.3: Khinchin's Weak Law of Large Number

Suppose that $X_i, i \geq 1$ is an i.i.d. sequence of random variables with a finite first moment (i.e., $\mathbb{E}(|X_i|) < \infty$). Let $\mu = \mathbb{E}(X_i)$. Then, the sample mean converges in probability to the expected value:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

Now, we can state a stronger statement.

Theorem 1.4: Kolmogorov's Law of Large Numbers

Suppose that $X_i, i \geq 1$ is an i.i.d. sequence of random variables with common mean $\mu = \mathbb{E}(X_1)$. The sample mean converges almost surely to the expected value,

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu$$

if and only if the expected value is finite:

$$\mathbb{E}(|X_1|) < \infty$$

Finally, we state the Chebyshev's WLLN, which uses assumption on the second moment.

Theorem 1.5: Chebyshev's WLLN

Let $\{X_n\}$ be a sequence of uncorrelated random variables with finite means $\mu_i = \mathbb{E}[X_i]$ and such that

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i) = 0.$$

Then

$$\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mu_i \xrightarrow{p} 0.$$

Equivalently,

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i].$$

Proof. Let $\bar{X}_n = \frac{1}{n} \sum_i X_i$. First, notice that

$$\mathbb{E}(\bar{X}_n) = \frac{1}{n} \sum_i \mathbb{E}(X_i)$$

Additionally, we can write the variance as

$$\mathbb{V}(\bar{X}_n) = \mathbb{V}\left(\frac{1}{n} \sum_i X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) = \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i) \rightarrow 0$$

where the last equality comes from uncorrelatedness and convergence comes by assumptions. Now, by Chebyshev's Inequality, we know that for some $\varepsilon > 0$

$$\mathbb{P}\left(\left|\bar{X}_n - \frac{1}{n} \sum_i \mathbb{E}(X_i)\right| > \varepsilon\right) \leq \frac{\mathbb{V}(\bar{X}_n)}{\varepsilon^2} \rightarrow 0$$

□

Corollary 1.1: Chebyshev's WLLN for Identical Means

Under the assumptions of Theorem 1.5, suppose in addition that

$$\mathbb{E}[X_i] = \mu \quad \text{for all } i.$$

Then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu.$$

Proof. By Theorem 1.5,

$$\frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] + o_p(1).$$

If $\mathbb{E}[X_i] = \mu$ for all i , then

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mu,$$

so the conclusion follows.

□

The condition on the second moments stated in 1.5 is actually much stronger than what we need. We can introduce the concept of uniform integrability to find weaker sufficient conditions. This is not directly relevant for the course, so I show that in Appendix A.1.

We can now state the most standard CLT, which applies only to iid data.

Theorem 1.6: Lindeberg-Levy's Central Limit Theorem

Suppose that $X_i, i \geq 1$ is an i.i.d. sequence of random variables with common finite mean $\mu = \mathbb{E}(X_1)$ and common finite, positive variance $\sigma^2 = \mathbb{V}(X_1) < \infty$. Then,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Now, if we face a case where we don't have iid data (for instance if we have heteroskedasticity), we need to use another CLT.

Theorem 1.7: Lindeberg-Feller's Central Limit Theorem

Let $\{X_n\}$ be an integrable sequence of independent random variables with $0 < \mathbb{V}(X_i) < \infty$ for all n . Let $\mathbb{E}(X_i) = \mu_i$ and $\mathbb{V}(X_i) = \sigma_i^2$. Now, let

$$C_n := \sqrt{\sum_i \sigma_i^2}, \text{ where } \lim_{n \rightarrow \infty} \max_{i \in [1, n]} \frac{\sigma_i^2}{C_n^2} = 0$$

Then, for any $\varepsilon > 0$,

$$\frac{1}{C_n} \sum_i (X_i - \mu_i) \xrightarrow{d} \mathcal{N}(0, 1) \iff \lim_{n \rightarrow \infty} \frac{1}{C_n^2} \sum_i \mathbb{E} \left[(X_i - \mu_i)^2 \mathbb{1}_{\{|X_i - \mu_i| \geq \varepsilon C_n\}} \right] = 0$$

The condition on the right-hand side of the equivalence operator is known as the **Lindeberg Condition**. This condition requires that the sum of the extreme-value variances (the numerator) becomes negligible compared to the total variance C_n^2 (the denominator) as $n \rightarrow \infty$.

While the theorems above describe the limiting behavior of a single sum, many problems, particularly in time series and non-parametric statistics, require understanding the behavior of an entire cumulative process. This extension is known as the Functional Central Limit Theorem, which I introduce in Appendix A.2 in the context of Brownian Motions.

Now, we can state two other important results that will come handy in our applications.

Theorem 1.8: Continuous Mapping Theorem

Let $\{X_n\}$ be a sequence of random variables (or vectors) and X be a random variable. Let g be a function that is continuous. Then the following hold:

- If $X_n \xrightarrow{a.s.} X$ then $g(X_n) \xrightarrow{a.s.} g(X)$
- If $X_n \xrightarrow{p} X$ then $g(X_n) \xrightarrow{p} g(X)$

- If $X_n \xrightarrow{d} X$ then $g(X_n) \xrightarrow{d} g(X)$

Theorem 1.9: Slutsky's Theorem

Let $\{X_n\}$ and $\{Y_n\}$ be two sequences of random variables. Suppose X_n converges in distribution to a random variable X , and Y_n converges in probability to a constant c :

$$X_n \xrightarrow{d} X \quad \text{and} \quad Y_n \xrightarrow{p} c$$

Then the following hold:

- Sum: $X_n + Y_n \xrightarrow{d} X + c$
- Product: $X_n Y_n \xrightarrow{d} cX$
- Quotient (if $c \neq 0$): $\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}$

Finally, we introduce objects called (stochastic) orders of magnitude, which can be useful in some applications.

Definition 1.11: Big and small oh

Consider (X_n) and (f_n) two sequences of real numbers.

1. We say that $X_n = O(f_n)$ if $\left| \frac{X_n}{f_n} \right| \rightarrow c < \infty$ (i.e. $\left| \frac{X_n}{f_n} \right|$ is bounded for all sufficiently large n)
2. We say that $X_n = o(f_n)$ if $\left| \frac{X_n}{f_n} \right| \rightarrow 0$ as $n \rightarrow \infty$

Notice that $X_n = O(1) \Leftrightarrow X_n$ is bounded uniformly in n , that is there exists $M < \infty$ such that $|X_n| \leq M$ for all n . Naturally, $X_n = o(1) \Leftrightarrow X_n \rightarrow 0$ as $n \rightarrow \infty$

Definition 1.12: Big and small oh-P

Consider (X_n) be a sequence of random variable and (f_n) a sequence of real numbers.

1. We say that $X_n = O_p(f_n)$ if $\forall \varepsilon > 0, \exists c \geq 0$ and $n_0 \in \mathbb{N}$ such that $\forall n \geq n_0$,

$$\mathbb{P}(|X_n| > cf_n) < \varepsilon$$

2. We say that $X_n = o_p(f_n)$ if $\frac{X_n}{f_n} \xrightarrow{p} 0$

Notice that the definition of $O_p(\cdot)$ is equivalent to the random variable being bounded in probability. If we have $X_n = O_p(1)$, we can equivalently write that for any $\varepsilon > 0$ there

exists a constant $c < \infty$ such that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(|X_n| > c) < \varepsilon$$

Finally, $X_n = o_p(1)$ is equivalent to $X_n \xrightarrow{p} 0$ as $n \rightarrow \infty$.

1.6.2 Consistency of the least squares estimator

Define the OLS estimator $\hat{\beta}$ and the estimation error:

$$\hat{\beta} = \beta + \left(\frac{X'X}{n} \right)^{-1} \left(\frac{X'\varepsilon}{n} \right)$$

To show that $\hat{\beta} \xrightarrow{p} \beta$ (Consistency), we need to show that $\left(\frac{X'X}{n} \right) \xrightarrow{p} Q$, where Q is finite and invertible, and $\left(\frac{X'\varepsilon}{n} \right) \xrightarrow{p} 0$.

First, consider the case where the data $\{(x_i, \varepsilon_i)\}_{i=1}^n$ is iid. Additionally, we assume contemporaneous exogeneity: $\mathbb{E}(x_i \varepsilon_i) = 0$, and we impose some moment conditions: $\mathbb{E}(x_i x_i') < \infty$ and $\mathbb{E}|x_i \varepsilon_i| < \infty$. Notice that $\mathbb{E}(x_i x_i')$ is invertible by assumption ii) (full rank). Therefore, by Khinchine's WLLN, we obtain that

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} \mathbb{E}(x_i x_i') \equiv Q \text{ and } \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}(x_i \varepsilon_i) = 0$$

Note that Slutsky's Theorem guarantees

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \xrightarrow{p} Q^{-1}$$

Therefore, by Continuous Mapping Theorem, we obtain

$$\hat{\beta} \xrightarrow{p} \beta + \mathbb{E}(x_i x_i')^{-1} \mathbb{E}(x_i \varepsilon_i) = \beta + O_p(1) \cdot o_p(1) = \beta$$

Notice that the assumptions imposed here are stronger than what we need to establish consistency. Indeed, only assuming that $\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} Q$ and imposing weak exogeneity is enough to guarantee consistency.

Weak exogeneity (or contemporaneous uncorrelation), written as $E(x_i \varepsilon_i) = 0$, is the necessary population moment condition required to ensure OLS consistency. Strict exogeneity, written as $E(\varepsilon_i | X) = 0$ is a much stronger assumption, requiring the error term to be uncorrelated with all regressors in the sample, and is the condition needed to guarantee OLS unbiasedness. Since Strict Exogeneity implies Mean Independence ($\mathbb{E}(\varepsilon_i | x_i) = 0$), and Mean Independence in turn implies Weak Exogeneity via the Law of Iterated Expectations², the required condition for consistency is nested within the condition for unbiasedness, first stated in the Gauss-Markov Assumptions.

² $\mathbb{E}(x_i \varepsilon_i) = \mathbb{E}(\mathbb{E}(x_i \varepsilon_i | x_i)) = \mathbb{E}(x_i \mathbb{E}(\varepsilon_i | x_i)) = 0$ if $\mathbb{E}(\varepsilon_i | x_i) = 0$.

1.6.3 Asymptotic Normality of the least squares estimator

We are interested in understanding the behavior of $\sqrt{n}(\hat{\beta} - \beta)$. We can rewrite the sample counterpart of this object as follows

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \right)$$

Consider the set of assumptions we imposed for consistency, where naturally, using WLLN and Slutsky's Theorem, we have

$$\left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \xrightarrow{p} Q^{-1}$$

Now, we need to add two additional moment assumptions: $\mathbb{E}(\|x_i\|^4) < \infty$ and $\mathbb{E}(\varepsilon_i^4) < \infty$.³ We do not specify any structure on the errors yet. Under these assumptions, by Lindeberg-Lévy's CLT⁴, we have that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{d} \mathcal{N}(0, \Omega^*), \text{ where } \Omega^* = \mathbb{E}(\varepsilon_i^2 x_i x_i')$$

By CMT, we get that

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Q^{-1} \mathcal{N}(0, \Omega^*) = \mathcal{N}(0, Q^{-1} \Omega^* Q^{-1})$$

Now, suppose that we impose spherical errors. We can write the following, using LIE in the first equality

$$\mathbb{V}(x_i \varepsilon_i) = \mathbb{E}(\mathbb{V}(x_i \varepsilon_i | X)) = \mathbb{E}(x_i x_i' \mathbb{V}(\varepsilon_i | X)) = \sigma^2 \mathbb{E}(x_i x_i') = \sigma^2 Q$$

Therefore, we have that

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \sigma^2 Q^{-1} Q Q^{-1}) = \mathcal{N}(0, \sigma^2 Q^{-1})$$

The $\sqrt{n}$ normalization is used because it is the unique scaling factor that keeps the distribution of the estimator error "alive" for asymptotic analysis. The variance of the OLS estimator, $\mathbb{V}(\hat{\beta})$, is known to decrease at the rate of $\frac{1}{n}$ as the sample size n increases (e.g., in the simple case, $\mathbb{V}(\hat{\beta}) \approx \frac{\sigma^2}{n \cdot \mathbb{V}(X_i)}$). Because the variance collapses to zero at rate n , the standard deviation collapses at rate $\sqrt{n}$; consequently, multiplying the estimation error $(\hat{\beta} - \beta)$ by $\sqrt{n}$ exactly counteracts this collapse, yielding a distribution that converges to a non-degenerate Normal distribution.

1.7 Quick detour on intuition & collinearity

³These two assumptions are imposed to have a finite variance. Indeed, using Cauchy-Schwarz Inequality, it is easy to show that $\mathbb{E}(|\varepsilon_i x_i|)^2 \leq \sqrt{\mathbb{E}(\|x_i\|^4)} \sqrt{\mathbb{E}(\varepsilon_i^4)} < \infty$

⁴Note that Lindeberg-Lévy's CLT applies to the term $\frac{1}{\sqrt{n}} \sum x_i \varepsilon_i$ when the data is iid. We use the general notation Ω here, which is applicable even under heteroskedasticity (provided iid still holds). We will discuss later the Lindeberg-Feller condition for non-iid cases.

Remark 1.1: Some additional intuition on the variance

Consider the following expression for the variance

$$\mathbb{V}(\hat{\beta}) \approx \frac{\sigma^2}{n \cdot \mathbb{V}(X_i)}$$

This expression makes it clear *why* we want a lot of variation in our independent variables. Since $\mathbb{V}(\hat{\beta}) \propto 1/\mathbb{V}(X_i)$, the higher $\mathbb{V}(X_i)$, the lower the variance of the estimator.

This remark introduces the concept of multicollinearity, which is generally written as $\text{rank}(X) < k$, where k is the number of regressors. Note that this is saying that at least some columns of X are linearly dependent. Formally, this implies that $X'X$ is not invertible, and therefore the OLS estimates cannot be computed. A typical example is to consider a simple regression of wage on age, schooling and experience. Since experience is a combination of age and years of schooling, we have collinearity, and can't run OLS.

More formally, the uniqueness of the OLS solution and the invertibility of $X'X$ are directly tied to the rank of X . The rank of an $n \times K$ matrix X , denoted $\text{rank}(X)$, is the number of linearly independent columns (or rows) it contains. The condition for a unique OLS solution is that X must be full column rank, meaning $\text{rank}(X) = K$. If $\text{rank}(X) < K$, the columns of X are linearly dependent (perfect multicollinearity), which means the column space $\mathcal{R}(X)$ has a dimension less than K , and consequently, the $K \times K$ matrix $X'X$ is singular (non-invertible) according to Proposition 1.3. Without an invertible $X'X$, the OLS formula $\hat{\beta} = (X'X)^{-1}X'y$ cannot be computed, and the fitted vector $\hat{y}$ would be defined by a non-unique set of coefficients $\hat{\beta}$.

1.8 Function of Parameters

In most applications, a researcher is interested in a specific transformation of the coefficient vector β ⁵. For example, one may be interested in a single coefficient β_j , or a ratio β_j/β_l . More generally, we can write the parameter of interest θ as a function of the coefficients: $\theta = r(\beta)$ for some function $r : \mathbb{R}^k \rightarrow \mathbb{R}^q$ with $k \geq q$. The estimate of θ is therefore

$$\hat{\theta} = r(\hat{\beta})$$

Now, suppose that the conditions for consistency hold, $r(\cdot)$ is continuously differentiable in a neighborhood of (the true) β and $R := \frac{\partial r(\beta)'}{\partial \beta}$ has rank q . We now introduce the Delta method. In this context, it basically says that if the transformation is smooth enough, we can show $\hat{\theta}$ is asymptotically normal.

⁵Note that the next two parts are not directly part of the first year course at Bocconi, but are in my opinion very important to build towards hypothesis testing.

Theorem 1.10: Delta Method

Let $\mu \in \mathbb{R}^k$ and $g : \mathbb{R}^k \rightarrow \mathbb{R}^J$. If $\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} \xi$, where $g(u)$ is continuously differentiable in a neighborhood of μ , then as $n \rightarrow \infty$

$$\sqrt{n}(g(\hat{\mu}) - g(\mu)) \xrightarrow{d} \mathbf{G}'\xi$$

where $\mathbf{G}(u) = \frac{\partial g(u)}{\partial u'}$ and $\mathbf{G} = \mathbf{G}(\mu)$.

In particular, if $\xi \sim \mathcal{N}(\mathbf{0}, \mathbf{V})$ then as $n \rightarrow \infty$

$$\sqrt{n}(g(\hat{\mu}) - g(\mu)) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{G}\mathbf{V}\mathbf{G}')$$

In this context, we can write the following Taylor expansion

$$\hat{\theta} = r(\hat{\beta}) \approx r(\beta) + \left. \frac{\partial r(\beta)}{\partial \beta'} \right|_{\beta=\tilde{\beta}} (\hat{\beta} - \beta) \text{ where } \tilde{\beta} \text{ is on the line joining } \beta \text{ and } \hat{\beta}$$

Rewriting this, and normalizing by $\sqrt{n}$, we get

$$\sqrt{n}(r(\hat{\beta}) - r(\beta)) \approx \underbrace{\left. \frac{\partial r(\beta)}{\partial \beta'} \right|_{\beta=\tilde{\beta}}}_{\xrightarrow{p} \mathbf{R}} \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \mathbf{R}'\mathbf{V}\mathbf{R}) \text{ where } \mathbf{V} = \mathbf{Q}^{-1}\mathbf{\Omega}\mathbf{Q}^{-1}$$

Notice that the key step relies on the fact that $\hat{\beta} \xrightarrow{p} \beta$, which implies $\tilde{\beta} \xrightarrow{p} \beta$ and since the partial derivatives are continuous, the Jacobian converges in probability:

$$\left. \frac{\partial r(\beta)}{\partial \beta'} \right|_{\beta=\tilde{\beta}} \xrightarrow{p} \left. \frac{\partial r(\beta)}{\partial \beta'} \right|_{\beta=\beta} \equiv \mathbf{R}$$

Once that is derived, we can estimate the asymptotic variance of $\hat{\theta}$ by $\hat{R}'\hat{V}\hat{R}$ where we can show that a simple estimate of $\mathbf{V}$ is consistent, and $\hat{R} = \left. \frac{\partial r(\hat{\beta})}{\partial \beta'} \right|_{\beta=\hat{\beta}}$.

1.9 Asymptotic standard errors

If $\hat{\theta}$ is a scalar, the standard error of $\hat{\theta}$ is obtained as

$$s(\hat{\theta}) = \sqrt{\frac{1}{n} \hat{R}'\hat{V}\hat{R}}$$

A standard error is an estimator of the standard deviation of the sampling distribution of an estimator. Intuitively, it measures the precision of the estimate, quantifying the typical distance between your sample-based estimate $\hat{\theta}$ and the true population parameter θ . If $\hat{V}$ is an estimator of the variance-covariance matrix of $\hat{\beta}$, then standard errors are the square roots of the diagonal elements of the matrix:

$$s(\beta_j) = \sqrt{\frac{1}{n} \hat{V}_{jj}}$$

1.10 On the way to hypothesis testing: object construction

In this last subsection, I will construct the objects which will then be useful to proceed with hypothesis testing.

1.10.1 T-statistic

Again, suppose that $\hat{\theta}$ is a scalar. Based on the standard error $s(\hat{\theta})$, we can obtain the standardized object called t-statistic or t-ratio

$$T_n(\theta) = \frac{\hat{\theta} - \theta}{s(\hat{\theta})}$$

Now, by asymptotic distribution of $\hat{\theta}$, we can show that this statistic is *asymptotically pivotal*

$$T_n(\theta) = \frac{\hat{\theta} - \theta}{\sqrt{\frac{1}{n} \hat{R}' \hat{V} \hat{R}}} = \frac{\sqrt{n}(\hat{\theta} - \theta)}{\sqrt{\hat{R}' \hat{V} \hat{R}}} \xrightarrow{d} \frac{\mathcal{N}(0, R' V R)}{\sqrt{R' V R}} = \frac{\sqrt{R' V R}}{\sqrt{R' V R}} \mathcal{N}(0, 1) = \mathcal{N}(0, 1)$$

Asymptotically pivotal means that as $n \rightarrow \infty$, the statistic behaves predictably and does not depend on the unknown θ . This must not be true in finite samples, but this property simply says that the dependence on unknowns diminishes as n increases.

1.10.2 Confidence Intervals

One way to estimate the unknown parameter $\theta \in \mathbb{R}$ is to find a point estimator $\hat{\theta}$. If one is not interested in the precise value of θ , it may be reasonable to estimate θ by an interval $[L_n, U_n] \subset \mathbb{R}$ (called confidence interval) such that

$$\mathbb{P}(\theta \in [L_n, U_n]) \approx 1 - \alpha \text{ for some } \alpha \in (0, 1)$$

The goal is to set the coverage probability equal to a pre-specified target such 95%, with $\alpha = 0.05$. Since we rely upon asymptotic approximation, here we consider the interval $[L_n, U_n]$ such that

$$\mathbb{P}(\theta \in [L_n, U_n]) \rightarrow 1 - \alpha, \text{ called the } 100(1 - \alpha)\% \text{ asymptotic confidence interval}$$

One common way to construct the asymptotic confidence interval of θ is to base it upon the asymptotically pivotal object

$$T_n(\theta) = \frac{\hat{\theta} - \theta}{s(\hat{\theta})} \xrightarrow{d} \mathcal{N}(0, 1)$$

By the asymptotic distribution, for a desired confidence level $1 - \alpha$, we have

$$\mathbb{P}(-z_{\alpha/2} \leq T_n(\theta) \leq z_{\alpha/2}) \rightarrow 1 - \alpha$$

where $z_{\alpha/2}$ is the $100(1 - \alpha/2)$ percentile of $\mathcal{N}(0, 1)$. This is intuitive: the object we consider converges in distribution to a standard normal, so the probability of the statistic

falling between $-z_{\alpha/2}$ and $z_{\alpha/2}$ approaches $1 - \alpha$ as the sample size increases.

By isolating the true parameter θ , we get:

$$\mathbb{P}(\hat{\theta} - z_{\alpha/2}s(\hat{\theta}) \leq \theta \leq \hat{\theta} + z_{\alpha/2}s(\hat{\theta})) \rightarrow 1 - \alpha$$

Thus, the $100(1 - \alpha)\%$ asymptotic confidence interval of θ is:

$$[\hat{\theta} - z_{\alpha/2}s(\hat{\theta}), \quad \hat{\theta} + z_{\alpha/2}s(\hat{\theta})]$$

Notice that in this context (frequentist), we treat the confidence interval as a function of the data and hence random, while θ is fixed.

1.10.3 Wald Statistic

Now, consider the case where $\theta \in \mathbb{R}^q$. We want to find the asymptotically pivotal object. Instead of the t-ratio, consider the quadratic form

$$\sqrt{n}(\hat{\theta} - \theta)' A \sqrt{n}(\hat{\theta} - \theta)$$

If we set $A = \hat{V}_\theta^{-1} \equiv \hat{R}' \hat{V} \hat{R}$ we can derive the Wald statistic (asymptotically pivotal) using the CMT

$$W_n(\theta) = \sqrt{n}(\hat{\theta} - \theta)' \hat{V}^{-1} \sqrt{n}(\hat{\theta} - \theta) = \sqrt{n}(\hat{\theta} - \theta)' (\hat{R}' \hat{V} \hat{R})^{-1} \sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \chi_q^2$$

The confidence region for vector θ is given by $\{\theta : W_n(\theta) \leq \chi_{q,\alpha}^2\}$ where $\chi_{q,\alpha}^2$ is the $100(1 - \alpha)\%$ percentile of χ_q^2 .

We have now all the tools to move on to hypothesis testing. Yay!

2 Hypothesis Testing

2.1 Concepts

For parameter of interest $\theta = r(\beta)$, consider two-sided testing problem

$$\mathcal{H}_0 : \theta = \theta_0 \quad \text{against} \quad \mathcal{H}_1 : \theta \neq \theta_0$$

where θ_0 is a hypothetical value. $\mathcal{H}_0$ is called null hypothesis and $\mathcal{H}_1$ is called alternative hypothesis. In hypothesis testing, we assume that there is a true but unknown value of the parameter of interest θ , and this value either satisfies $\mathcal{H}_0$ or does not. The goal is therefore to assess whether or not $\mathcal{H}_0$ is true by asking if $\mathcal{H}_0$ is consistent with the observed data. The question is: Are the true coefficients zero? To answer this question the testing method asks the question: Are the observed estimates compatible with the hypothesis, in the sense that the deviation from the hypothesis can be reasonably explained by stochastic variation? Or are the observed estimates incompatible with the hypothesis, in the sense that that the

observed estimates would be highly unlikely if the hypothesis were true (Hansen, 2022)?

Let us consider a test in the form of

$$\begin{cases} \text{Accept } \mathcal{H}_0 & \text{if } T_n \leq c \\ \text{Reject } \mathcal{H}_0 & \text{if } T_n > c \end{cases}$$

where T_n is test the statistic and c the critical value.

In testing, we have two kinds of errors:

	Accept	Reject
$\mathcal{H}_0$ true	Correct	Type I
$\mathcal{H}_1$ true	Type II	Correct

Type I error probability can we written as

$$P(\text{Reject}|\mathcal{H}_0 \text{ true}) = P(T_n > c|\theta = \theta_0)$$

and represents false positives, e.g. an innocent person is convicted. Type II error probability can be written as

$$P(\text{Accept}|\mathcal{H}_1 \text{ true}) = P(T_n \leq c|\theta \neq \theta_0)$$

and represents false negative, e.g. a criminal is not convicted.

2.1.1 Optimality and The Neyman-Pearson Lemma

The primary goal in constructing a test is to manage the trade-off between the two error types. In classical testing, we choose to control the Type I error probability (the size of the test) at a fixed maximum level, α . We can choose c to satisfy

$$\mathbb{P}(\text{Type I Error}) \rightarrow \alpha$$

Given this fixed size, we then seek to find a test that minimizes the Type II error probability, which is equivalent to maximizing the power function, defined as $\pi(\theta) = \mathbb{P}(\text{Reject } \mathcal{H}_0|\mathcal{H}_1 \text{ true})$.

The **Neyman-Pearson (NP) Lemma** provides the theoretical foundation for this approach, stating that the most powerful test for a simple null hypothesis against a simple alternative is based on the **Likelihood Ratio (LR)**. This establishes the LR test as the theoretical benchmark for test optimality, and all other tests, such as the Wald and LM tests, are evaluated by their asymptotic equivalence to the LR test. More on the trinity of hypothesis testing later.

The UMP test can be written as follows $\varphi(x) = \mathbb{1}\{LR < k\}$ for some $k \in (0, 1)$ such that the probability of falsely rejecting the null is α . The likelihood ration can be written as

$$\Lambda = \frac{f_0(x)}{f_1(x)} = \frac{L(x_1, \dots, x_n, \theta_0)}{L(x_1, \dots, x_n, \theta_1)}$$

2.2 Linear hypotheses and joint significance

2.2.1 T-test

Based on the t-ratio constructed above, we can test $\mathcal{H}_0 : \theta = \theta_0$, for which it holds that $T_n(\theta) \xrightarrow{d} \mathcal{N}(0, 1)$. This implies that $\mathbb{P}(|T_n(\theta_0)| > z_{\alpha/2} | \mathcal{H}_0) \rightarrow \alpha$. Therefore, the asymptotic size α test is

$$\text{Accept } \mathcal{H}_0 \text{ if } |T_n(\theta_0)| \leq z_{\alpha/2} \quad \text{and} \quad \text{Reject } \mathcal{H}_0 \text{ if } |T_n(\theta_0)| > z_{\alpha/2}$$

Example 2.1: Single Linear Hypothesis

For a single linear hypothesis $\mathcal{H}_0 : \mathbf{c}'\beta = \gamma$, the test statistic is written as the T -test:

$$T = \frac{\mathbf{c}'\hat{\beta} - \gamma}{\text{SE}(\mathbf{c}'\hat{\beta})}$$

For the exact-sample test under the classical assumptions (spherical errors and normal errors $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$), the statistic follows a t -distribution:^a

$$T \sim t_{n-k} \quad \text{under } \mathcal{H}_0$$

The corresponding rejection rule for the exact-sample two-sided test is to Reject $\mathcal{H}_0$ if $|T| > t_{n-k, \alpha/2}$.

^aRecall that as the degrees of freedom increases, the t -distributions converge to the standard normal

In the exact-sample T -test, the test statistic follows the t -distribution with $n - k$ degrees of freedom, where n is the sample size and k is the total number of parameters estimated in the unrestricted model (including the intercept). The quantity $n - k$ represents the number of observations available to estimate the error variance after accounting for the k parameters that have been fit. It reflects the number of independent data points available for estimating the residual variation.

2.2.2 F-test

A joint linear hypothesis involves testing J restrictions on the parameter vector β simultaneously. We want to test $\mathcal{H}_0 : R\beta = c$ against $\mathcal{H}_1 : R\beta \neq c$, where R is a known $J \times K$ matrix of constants with full rank, where J is the number of restrictions, and c is a $J \times 1$ vector of known constants.

Example 2.2: Joint significance testing

Testing $\mathcal{H}_0 : \beta_2 = 1, \beta_3 = -1$ simultaneously for a model with $K = 4$ parameters:

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \varepsilon_i$$

This is subject to $\mathcal{H}_0 : \beta_2 = 1, \beta_3 = -1$, which implies $R\beta = c$:

$$R = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}$$

The test for joint hypotheses is based on verifying that the estimated restrictions $R\hat{\beta} - c$ are close to zero. This test ultimately uses the quadratic form we derived previously to obtain the following test statistic:

$$W = n(R\hat{\beta} - c)'[RVR']^{-1}(R\hat{\beta} - c) \xrightarrow{d} \chi_J^2$$

Notice that this can be done when σ^2 is known. If σ^2 is unknown, an alternative, exact-sample statistic is the F -test, which follows the $F_{J,n-k}$ distribution under $\mathcal{H}_0$ (under classical assumptions). As before, the F -distribution converges to a χ^2 distribution.

The most intuitive way to understand the F -test is by viewing it as a comparison between two models: the unrestricted model (OLS) and the restricted model (RLS) where the constraints $R\beta = c$ are imposed.

$$F = \frac{(RSS_R - RSS_{UR})/J}{RSS_{UR}/(n-k)} \sim F_{J,n-k} \quad \text{under } \mathcal{H}_0$$

The acceptance rule is to **Reject** $\mathcal{H}_0$ if $F \geq F_{(J,n-k),\alpha}$.

2.2.3 Restricted Least Squares Derivation

The RLS estimator $\tilde{\beta}$ is obtained by minimizing the sum of squared residuals subject to the linear constraints $R\beta = c$, using the Lagrangian method:

$$\mathcal{Q}(\beta, \lambda) = (y - X\beta)'(y - X\beta) - 2\lambda'(R\beta - c)$$

The first-order conditions (FOCs) are:

$$\left. \frac{\partial \mathcal{Q}}{\partial \beta} \right|_{\tilde{\beta}, \tilde{\lambda}} = 0 \implies -2X'y + 2(X'X)\tilde{\beta} - 2R'\tilde{\lambda} = 0$$

$$\left. \frac{\partial \mathcal{Q}}{\partial \lambda} \right|_{\tilde{\beta}, \tilde{\lambda}} = 0 \implies R\tilde{\beta} = c$$

Solving the system (see appendix) yields the RLS estimator $\tilde{\beta}$:

$$\tilde{\beta} = \hat{\beta} + (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}(c - R\hat{\beta})$$

If the null hypothesis $\mathcal{H}_0 : R\beta = c$ is true, the RLS estimator $\tilde{\beta}$ is BLUE and is more efficient than the unrestricted OLS estimator $\hat{\beta}$.

$$Var(\hat{\beta}) - Var(\tilde{\beta}) = \sigma^2(X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}R(X'X)^{-1} \geq 0$$

The implication is the an OLS with a restriction that reflects the true values is always more efficient than an unrestricted OLS.

Example 2.3: Significance of a Regression

A question of interest can be to know whether a regression as a whole is significant

$$y_i = \beta_1 + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \varepsilon_i$$

The test is a joint test of the hypothesis that all coefficients except the constant are zero

$$\mathcal{H}_0 : \beta_2 = \dots = \beta_k = 0$$

Notice that the number of restrictions here is $k - 1$, and the degrees of freedom of the unrestricted model $n - k$. We can define RSS_R as the restricted RSS, regress $y = \beta_1 + \varepsilon$ (under $\mathcal{H}_0$), and use the residuals to compute $\tilde{\varepsilon}_R' \tilde{\varepsilon}_R = TSS$. The F-test is therefore computed as

$$F = \frac{(RSS_R - RSS_{UR})/k - 1}{RSS_{UR}/(n - k)} = \frac{R^2/k - 1}{(1 - R^2)/n - k} \sim F_{k-1, n-k}$$

2.3 The Trinity of Testing

Before introducing the full Trinity of tests, we first need to establish the properties of the Maximum Likelihood Estimator (MLE), which forms the basis for the LR and LM statistics.

2.3.1 The ML Estimator

Let $f(y_i | \theta)$ be the probability density function (or probability mass function) of the data, where θ is the vector of parameters. As introduced above, the **Likelihood Function** for n independent observations is:

$$L(\theta) = \prod_{i=1}^n f(y_i | \theta)$$

The **Log-Likelihood Function** is:

$$\mathcal{L}(\theta) = \sum_{i=1}^n \ln f(y_i | \theta)$$

The **Maximum Likelihood Estimator (MLE)**, denoted $\hat{\theta}_{ML}$, is the value of θ that maximizes $\mathcal{L}(\theta)$. It is found by solving the first-order condition:

$$\left. \frac{\partial \mathcal{L}(\theta)}{\partial \theta} \right|_{\hat{\theta}_{ML}} = \mathbf{0}$$

Definition 2.1: Score

The vector of first derivatives of the log-likelihood function is called the **Score Function**:

$$\mathbf{S}(\theta) = \frac{\partial \mathcal{L}(\theta)}{\partial \theta}$$

Under regularity conditions, the expected value of the score function evaluated at the true parameter is zero: $\mathbb{E}[\mathbf{S}(\theta)] = \mathbf{0}$.

Definition 2.2: Hessian

The matrix of second derivatives of the log-likelihood function is the **Hessian Matrix**:

$$\mathbf{H}(\theta) = \frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta \partial \theta'}$$

Definition 2.3: Information Matrix

The **Information Matrix** is defined as the negative expected value of the Hessian:

$$\mathbf{I}(\theta) = -\mathbb{E}[\mathbf{H}(\theta)] = -\mathbb{E} \left[\frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta \partial \theta'} \right]$$

Under regularity conditions⁶, the MLE is consistent, asymptotically efficient, and asymptotically normally distributed:

$$\sqrt{n}(\hat{\theta}_{ML} - \theta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V})$$

The asymptotic covariance matrix $\mathbf{V}$ is the inverse of the Information Matrix:

$$\mathbf{V} = \mathbf{I}(\theta)^{-1}$$

Remark 2.1: The Fisher Information Identity

The Information Matrix can also be written as the expected outer product of the score function:

$$\mathbf{I}(\theta) = \mathbb{E}[\mathbf{S}(\theta)\mathbf{S}(\theta)']$$

In practice, we use the estimated (asymptotic) variance $\widehat{AVar}(\hat{\theta}_{ML})$. This can be obtained in two common ways by using different estimators for the Information Matrix $\mathbf{I}(\theta)$:

- Using the outer product of the scores (this estimates the Information Matrix using the empirical version of the Fisher Identity)

$$\widehat{AVar}(\hat{\theta}_{ML}) = \left[\sum_{i=1}^n \mathbf{S}_i(\hat{\theta}_{ML}) \mathbf{S}_i(\hat{\theta}_{ML})' \right]^{-1}$$

⁶More on this when we introduce extremum estimators.

- Using the negative of the Hessian (this estimates the Information Matrix using the observed (actual) Hessian from the maximum point):

$$\widehat{AVar}(\hat{\theta}_{ML}) = [-\mathbf{H}(\hat{\theta}_{ML})]^{-1}$$

With that, we can turn to the tests.

2.3.2 Wald Test

The Wald test measures the distance between the **unrestricted estimate** $\hat{\theta}$ and the value imposed by the null hypothesis. Consider $\mathcal{H}_0 : r(\theta) = r(\theta_0)$.

$$W = n \cdot (r(\hat{\theta}) - r(\theta_0))' [R(\hat{\theta})' \hat{V}(\theta) R(\hat{\theta})]^{-1} (r(\hat{\theta}) - r(\theta_0)) \xrightarrow{d} \chi_J^2$$

A problem with the Wald test is that it is not invariant to the formulation of the restrictions. For instance, if we test $\mathcal{H}_0 : \beta_1 \beta_2 = 1$ and $\mathcal{H}'_0 : \beta_1 = 1/\beta_2$, the Wald test provides different answers in finite sample, even if the two are asymptotically equivalent. Additionally, it has a tendency to over-reject.

To build additional intuition, consider the univariate case

$$W_{univ} = \frac{(\hat{\theta} - \theta_0)^2}{\widehat{Var}(\hat{\theta})} \sim_{\mathcal{H}_0} \chi_1^2$$

The numerator $(\hat{\theta} - \theta_0)^2$ measures the squared difference between our best estimate, $\hat{\theta}$, and the value hypothesized by the null, θ_0 . Higher deviation means stronger evidence against $\mathcal{H}_0$. The denominator $\widehat{Var}(\hat{\theta})$ measure precision. If we have a low variance it indicates a highly precise estimate (low $\widehat{Var}(\hat{\theta})$). A small deviation in the numerator is then magnified by the small denominator, resulting in a large W_{univ} . We are confident in $\hat{\theta}$, so we strongly reject $\mathcal{H}_0$. On the other hand, a flat likelihood function indicates an imprecise estimate (high $\widehat{Var}(\hat{\theta})$). The denominator is large, making it harder to reject $\mathcal{H}_0$. The data does not provide enough information to reliably distinguish $\hat{\theta}$ from θ_0 .

2.3.3 The Lagrange Multiplier or Score Test

The LM test measures how close the score function $S(\theta)$ is to zero when evaluated at the **restricted estimate** $\tilde{\theta}_R$. If $\mathcal{H}_0$ is true, $\tilde{\theta}_R$ should be close to the true parameter, and the score should be near zero. It relies only on the restricted estimation.

$$LM = \mathbf{S}(\tilde{\theta}_R)' \mathbf{I}(\tilde{\theta}_R)^{-1} \mathbf{S}(\tilde{\theta}_R) \xrightarrow{d} \chi_J^2$$

Example 2.4: LM

Consider the linear regression model under normality:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

We want to test the restriction $\mathcal{H}_0 : \beta_1 = 1$. The restricted model is:

$$y_i - x_{i1} = \beta_0 + \beta_2 x_{i2} + e_i$$

The MLE estimates for the restricted model are $\tilde{\beta}_0, \tilde{\beta}_2$ and $\tilde{\beta}_1 = 1$. Since the unrestricted likelihood whose gradient needs to be evaluated at the constraint estimates is given by:

$$\mathcal{L} = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2$$

Under the classical linear regression with normality, we want to verify that the Score (gradient) is close to zero at the restricted estimates $\tilde{\beta}$. The derivatives of the log-likelihood are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \beta_0} \Big|_{\tilde{\beta}, \tilde{\sigma}^2} &= \frac{1}{\tilde{\sigma}^2} \sum \tilde{e}_i \approx 0 \\ \frac{\partial \mathcal{L}}{\partial \beta_1} \Big|_{\tilde{\beta}, \tilde{\sigma}^2} &= \frac{1}{\tilde{\sigma}^2} \sum \tilde{e}_i x_{i1} \approx 0 \\ \frac{\partial \mathcal{L}}{\partial \beta_2} \Big|_{\tilde{\beta}, \tilde{\sigma}^2} &= \frac{1}{\tilde{\sigma}^2} \sum \tilde{e}_i x_{i2} \approx 0 \end{aligned}$$

The FOCs for $\beta_0, \beta_2, \sigma^2$ are satisfied (they form the constraint optimisation). The LM test focuses on the condition for the restricted parameter β_1 .

LM Test (Auxiliary Regression): The LM test is implemented by running an auxiliary regression: first, regress the restricted residuals $\tilde{e}_i$ on all regressors x_{i1} and x_{i2} . The LM statistic is then related to the R^2 of this auxiliary regression.

Intuition: If $\beta_1 \neq 1$ (i.e., $\mathcal{H}_0$ is false), the residuals from the restricted model $\tilde{e}_i$ will still display correlation with the excluded regressor x_{i1} . Hence, the covariance between $\tilde{e}_i$ and x_{i1} (which is the score component for β_1) will be high, resulting in a high R^2 and leading to rejection of $\mathcal{H}_0$.

2.3.4 The Likelihood Ratio Test

We introduce the following notation. Let $Q_n(\theta) := \mathcal{L}(\theta)$, and $Q(\theta) = \mathbb{E}[\ln f(Y_i, X_i | \theta)]$. The LR test compares the maximum log-likelihood under the **unrestricted model** ($\hat{\theta}$) to the maximum log-likelihood under the **restricted model** ($\tilde{\theta}$).

$$LR = 2 [Q_n(\hat{\theta}) - Q_n(\tilde{\theta})] \xrightarrow{d} \chi_J^2$$

Example 2.5: LR example

Suppose we want to test $\mathcal{H}_0 : \beta_1 = 1$. We need to run our MLE imposing $\beta_1 = 1$, and then run it without any restriction, obtaining $Q_n(\tilde{\beta})$ and $Q_n(\hat{\beta})$ respectively. We therefore obtain

$$LR = 2 (Q_n(\hat{\beta}) - Q_n(\tilde{\beta})) \xrightarrow{d} \chi_1^2$$

Figure 2.3.4 shows a graphical representation of the 3 tests.

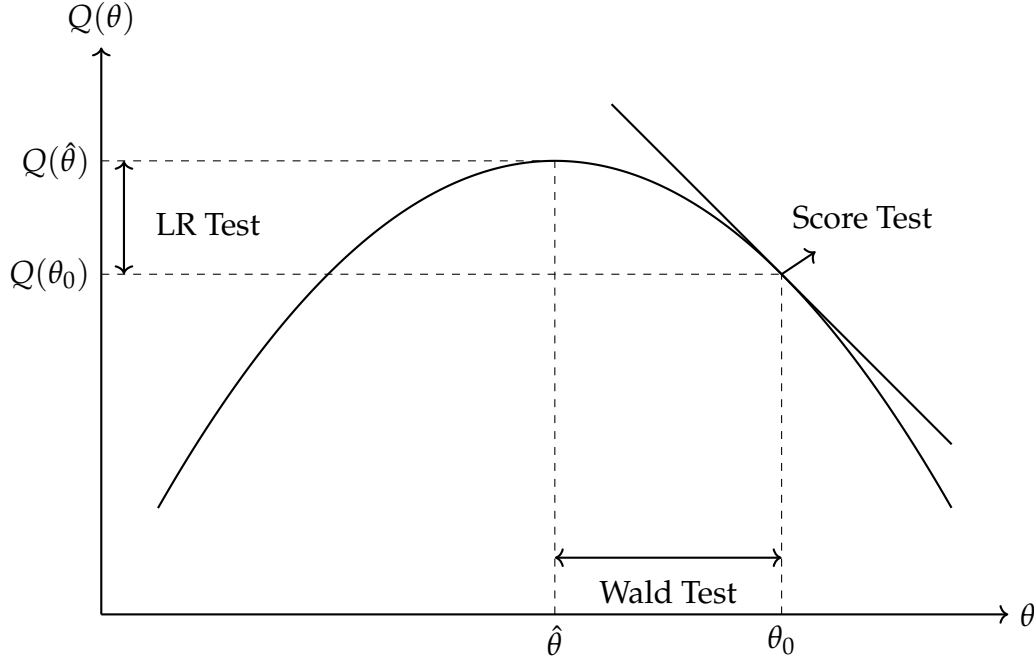


Figure 3: The trinity of hypothesis testing

3 Generalized Linear Regression

The whole point of this section is to relax assumption iv). That is, we do not assume spherical errors anymore. Indeed, we now consider a case where $\mathbb{V}(\varepsilon | X) = \Omega \neq \sigma^2 I_n$ ⁷, which we denote assumption iv)'. This can have different shapes. The most common one would be to think about **heteroskedasticity**, that is a case where

$$\mathbb{V}(\varepsilon_i) = \sigma_i^2 \implies \Omega = \begin{pmatrix} \sigma_1^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n^2 \end{pmatrix}$$

Another interesting example is one with **serial correlation**, for instance where ε_t follows an $AR(1)$ process:

⁷For consistency in the Generalized Least Squares (GLS) derivation, we define the full $n \times n$ covariance matrix as $\Omega \equiv \mathbb{V}(\varepsilon | X)$, which therefore absorbs the common scalar σ^2 from the initial assumption (iv).

$$\varepsilon_t = \rho \varepsilon_{t-1} + v_t, \text{ with } |\rho| < 1, v_t \sim iid(0, \sigma^2)$$

$$\Rightarrow \Omega = \frac{\sigma^2}{1 - \rho^2} \begin{pmatrix} 1 & \rho & \rho^2 & \dots & \rho^{T-1} \\ \rho & 1 & \rho & \dots & \rho^{T-2} \\ \rho^2 & \rho & 1 & \dots & \rho^{T-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{T-1} & \rho^{T-2} & \rho^{T-3} & \dots & 1 \end{pmatrix}$$

3.1 Finite-Sample Properties

Consider the Gauss-Markov assumptions i) to iii), with assumption iv)'. It is easy to show that the OLS estimator is still unbiased, since we rely on exogeneity assumption iii) to show unbiasedness. The conditional variance can be computed as follows

$$\begin{aligned} \mathbb{V}(\hat{\beta}|X) &= \mathbb{E} [(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])(\hat{\beta} - \mathbb{E}[\hat{\beta}|X])' | X] \\ &= \mathbb{E} \left[(\beta + (X'X)^{-1}X'\varepsilon - \beta)((\beta + (X'X)^{-1}X'\varepsilon - \beta))' | X \right] \\ &= \mathbb{E} \left[(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1} \right] = (X'X)^{-1}X'\Omega X(X'X)^{-1} \end{aligned}$$

OLS is therefore not BLUE anymore, since it is not efficient. We discuss later how to fix this.

3.2 Asymptotic Properties

First, for consistency, recall that we rely on a combination of a WLLN, Slutsky's Theorem, and assumption iii), we obtain that $\hat{\beta} \xrightarrow{p} \beta$. Since $\{x_i\varepsilon_i\}$ is no longer iid, we need to use a stronger version of the WLLN. Notably, we are lacking the **identically distributed** part. Based on Theorem 1.5, we have a suitable LLN that we can use for such cases, since we do not rely on any sort of identically distributed assumption. Notice that conditions of Theorem 1.5 are actually much stronger than what we need (see Appendix A.1).

Now, for asymptotic normality in presence of non-spherical errors, it is not possible to use Lindeberg-Levy's CLT (Theorem 1.6). Instead, we must use Lindeberg-Feller's CLT (Theorem 1.7). Using this CLT, provided that the Lindeberg condition holds, and following derivations done in the general case above, we get

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Q^{-1}\mathcal{N}(0, \Omega^*) = \mathcal{N}(0, Q^{-1}\Omega^*Q^{-1})$$

Below is an elegant example of when Lindeberg-Feller's CLT can be applied.

Consider

$$y_i = \beta x_i + u_i, \quad i = 1, \dots, n,$$

where $E(u_i^2) = \sigma^2$. Suppose the data are grouped into J unequal categories:

$$Y_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} y_i, \quad X_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i, \quad n = \sum_{j=1}^J n_j.$$

Estimate β by the no-intercept OLS regression of Y_j on X_j :

$$\hat{\beta} = \left(\sum_{j=1}^J X_j^2 \right)^{-1} \sum_{j=1}^J X_j Y_j.$$

We want to derive the limiting distribution of $\hat{\beta}$ and construct a test of the null hypothesis that $\beta = 0$ based on the grouped data.

First, let us write the model for a specific group $\mathcal{G}_j$:

$$Y_j = \beta X_j + U_j \text{ where } U_j = \frac{1}{n_j} \sum_{i \in \mathcal{G}_j} u_i$$

We need to put assumption on our model. A standard assumption is that $u_i \sim \mathcal{N}(0, \sigma^2)$ and is independent and identically distributed (iid). Based on this assumption, notice that for arbitrary group $\mathcal{G}_j$, we can compute the expectation

$$\mathbb{E}(U_j) = \mathbb{E} \left(\frac{1}{n_j} \sum_{i \in \mathcal{G}_j} u_i \right) = \frac{1}{n_j} \sum_{i \in \mathcal{G}_j} \mathbb{E}(u_i) = 0$$

and the variance

$$\begin{aligned} \mathbb{V}(U_j) &= \mathbb{V} \left(\frac{1}{n_j} \sum_{i \in \mathcal{G}_j} u_i \right) = \frac{1}{n_j^2} \sum_{i \in \mathcal{G}_j} \sum_{k \in \mathcal{G}_j} \text{Cov}(u_i, u_k) \\ &= \frac{1}{n_j^2} \left(\sum_{i \in \mathcal{G}_j} \mathbb{V}(u_i) + \sum_{i \in \mathcal{G}_j} \sum_{k \neq i, k \in \mathcal{G}_j} \text{Cov}(u_i, u_k) \right) = \frac{\sigma^2}{n_j} \equiv \sigma_j^2 \end{aligned}$$

The last equality comes from the fact that $\forall i \neq k, \text{Cov}(u_i, u_k) = 0$.

We can re-write $\hat{\beta}$ as follows:

$$\hat{\beta} = \beta + \left(\sum_{j=1}^J X_j^2 \right)^{-1} \sum_{j=1}^J X_j U_j$$

We are interested in the behavior of

$$\sqrt{J}(\hat{\beta} - \beta) = \sqrt{J} \left(\sum_{j=1}^J X_j^2 \right)^{-1} \sum_{j=1}^J X_j U_j = \frac{\frac{1}{\sqrt{J}} \sum_{j=1}^J X_j U_j}{\frac{1}{J} \sum_{j=1}^J X_j^2}$$

The key fact to notice here is that the variance varies for each group $\mathcal{G}_j$ (i.e. heteroskedasticity), which implies we can't apply Lindeberg-Levy's Central Limit Theorem, since this theorem only works for iid data⁸ Therefore, we need to use the Lindeberg-Feller CLT. I take the definition from a previous course, since we do not have the formal definition in this context.

We need to assume some **Lindeberg Condition**. This condition requires that the sum of the extreme-value variances becomes negligible compared to the total variance as $n \rightarrow \infty$.

In this context, we are interested in writing the Lindeberg condition to the sequence $Z_j := X_j U_j$, which is an independent but non-iid sequence of random variables. Notice that by LIE,

$$\mathbb{E}(Z_j) = \mathbb{E}[\mathbb{E}(X_j U_j | X_j)] = \mathbb{E}[X_j \mathbb{E}(U_j | X_j)] = 0$$

The variance can be computed as

$$\mathbb{V}(Z_j) = \mathbb{E}(X_j^2 U_j^2) = \mathbb{E}[X_j^2 \mathbb{E}(U_j^2 | X_j)] = \sigma_j^2 X_j^2 \equiv \tilde{\sigma}_j^2$$

We can now define

$$C_J := \sqrt{\sum_j \tilde{\sigma}_j^2}$$

We impose the following condition

$$\lim_{J \rightarrow \infty} \frac{\max_{j \in \{1, \dots, J\}} (X_j^2 / n_j)}{\sum_j (X_j^2 / n_j)} = 0$$

which is a sufficient condition for

$$\lim_{J \rightarrow \infty} \max_{j \in \{1, \dots, J\}} \frac{\tilde{\sigma}_j^2}{C_J^2} = 0$$

Now, assume the Lindeberg Condition holds, that is, assume, for any $\varepsilon > 0$:

$$\lim_{J \rightarrow \infty} \frac{1}{C_J^2} \sum_{j=1}^J \mathbb{E} \left[Z_j^2 \mathbb{1}_{\{|Z_j| \geq \varepsilon C_J\}} \right] = 0$$

We can finally apply Lindeberg-Feller's Theorem:

$$\frac{\sum_{j=1}^J Z_j}{\sqrt{\sum_j \tilde{\sigma}_j^2}} = \frac{\sum_{j=1}^J X_j U_j}{C_J} \xrightarrow{d} \mathcal{N}(0, 1)$$

⁸We cannot use the classical i.i.d. CLT because $\mathbb{V}(X_j U_j)$ varies with j through n_j . As J grows, the collection $\{X_j U_j\}_{j=1}^J$ forms a *triangular array* of independent but non-identically distributed random variables, so a Lindeberg-Feller CLT is required.

We can rewrite our object of interest as

$$\frac{1}{\sqrt{J}} \sum_{j=1}^J X_j U_j \frac{C_J}{\sqrt{J}} \cdot \frac{\sum_{j=1}^J X_j U_j}{C_J}$$

We analyze the limit of $\left(\frac{C_J}{\sqrt{J}}\right)^2$ since it represents the asymptotic variance of the numerator in the estimator:

$$\lim_{J \rightarrow \infty} \left(\frac{C_J}{\sqrt{J}}\right)^2 = \lim_{J \rightarrow \infty} \frac{C_J^2}{J} = \lim_{J \rightarrow \infty} \frac{1}{J} \sum_{j=1}^J \frac{\sigma^2 X_j^2}{n_j} := \sigma^2 \Omega$$

By Slutsky's Theorem, we have that

$$\frac{1}{\sqrt{J}} \sum_{j=1}^J X_j U_j \xrightarrow{d} \mathcal{N}(0, \sigma^2 \Omega)$$

Now, assume that

$$\frac{1}{J} \sum_{j=1}^J X_j^2 \xrightarrow{p} Q \implies \left(\frac{1}{J} \sum_{j=1}^J X_j^2\right)^{-1} \xrightarrow{p} Q^{-1} \text{ by Continuous Mapping Theorem}$$

We can therefore conclude that by Linderberg-Feller CLT and Slutsky's Theorem,

$$\sqrt{J}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \frac{\sigma^2 \Omega}{Q^2}\right)$$

Finally, we can perform the following T-test:

$$T = \frac{\hat{\beta}}{\sqrt{\sigma^2 \frac{\left(\sum_{j=1}^J \frac{X_j^2}{n_j}\right)}{\left(\sum_{j=1}^J X_j^2\right)^2}}}$$

Notice that under $H_0 : \beta = 0$, $T \xrightarrow{d} \mathcal{N}(0, 1)$

If σ^2 is unknown, we can use a standard method to estimate it.

3.3 Generalized Least Squares (GLS)

3.3.1 Theoretical GLS

The Generalized Least Squares (GLS) allows to adjust the linear regression model for it to satisfy the Gauss-Markov Conditions. The resulting OLS estimator equals the estimator that would minimize the generalized sum of squares

$$S(\beta) = (y - X\beta)' \Omega^{-1} (y - X\beta) \implies \hat{\beta}_{GLS} = (X' \Omega^{-1} X)^{-1} X' \Omega^{-1} y$$

The original model, $y = X\beta + \varepsilon$, has non-spherical errors, meaning $\mathbb{V}(\varepsilon | X) = \Omega$ (where Ω is an $n \times n$ matrix, and we are using your new, consistent notation). This violates the standard Gauss-Markov assumption (spherical errors), making OLS inefficient.

The solution is to premultiply the entire model by an invertible $n \times n$ matrix R such that:

$$R'R = \Omega^{-1}$$

The transformed model is:

$$\begin{aligned} Ry &= RX\beta + R\varepsilon \\ y^* &= X^*\beta + \varepsilon^* \end{aligned}$$

The errors in the new model, $\varepsilon^* = R\varepsilon$, now satisfy the spherical errors assumption, which is the whole point of the transformation:

$$\begin{aligned} \mathbb{V}(\varepsilon^* | X) &= \mathbb{E}(\varepsilon^* \varepsilon^{*'} | X) = \mathbb{E}(R\varepsilon \varepsilon' R' | X) \\ &= R\mathbb{E}(\varepsilon \varepsilon' | X)R' = R\Omega R' \end{aligned}$$

Since $R'R = \Omega^{-1}$, it follows that $\Omega = (R'R)^{-1} = R^{-1}(R')^{-1}$. Substituting this back:

$$R\Omega R' = R[R^{-1}(R')^{-1}]R' = (RR^{-1})[(R')^{-1}R'] = I_n I_n = I_n$$

The transformed error term, ε^* , now has a variance proportional to the identity matrix (I_n), meaning it is spherical. OLS applied to the transformed model (y^* regressed on X^*) is the Best Linear Unbiased Estimator (BLUE). In fact, if you apply OLS to the transformed model $(X^{*'}X^*)^{-1}X^{*'}y^*$, you will find it simplifies directly to the GLS formula:

$$\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$$

Theorem 3.1: Aitken Theorem

Under assumptions i) to iv), the GLS estimator $\hat{\beta}_{GLS}$ is efficient relative to all other unbiased estimators of β . This is a consequence of the transformation.

Assuming suitable regularity conditions⁹, the GLS estimator is consistent:

$$\hat{\beta}_{GLS} - \beta = \left(\frac{X'\Omega^{-1}X}{n} \right)^{-1} \frac{X'\Omega^{-1}\varepsilon}{n} \xrightarrow{p} 0$$

The asymptotic normality is given by:

$$\sqrt{n}(\hat{\beta}_{GLS} - \beta) \xrightarrow{d} \mathcal{N}\left(0, D_{\Omega^{-1}}^{-1}\right)$$

⁹On top of the usual exogeneity and moment conditions, we need to assume the $n \times n$ covariance matrix Ω must be positive definite, ensuring Ω^{-1} exists and is positive definite. This is necessary for the transformation $R'R = \Omega^{-1}$. Additionally, the limit matrix $D_{\Omega^{-1}} = \text{plim} \frac{X'\Omega^{-1}X}{n}$ must be finite and invertible (positive definite). This ensures the asymptotic variance is well-defined and finite.

where the $K \times K$ matrix $D_{\Omega^{-1}}$ is the probability limit of the normalized $X'\Omega^{-1}X$ term:

$$D_{\Omega^{-1}} = \text{plim} \frac{X'\Omega^{-1}X}{n} > 0$$

Note: The matrix $D_{\Omega^{-1}}$ here is analogous to the Q matrix in the spherical OLS case, representing the asymptotic precision of the transformed data. Unlike Ω^* in the OLS sandwich, $D_{\Omega^{-1}}$ requires no "sandwiching" because the GLS transformation has already ensured efficiency.

3.3.2 Feasible GLS

For all the results above, we need to assume Ω is known. The feasible version of GLS (FGLS) is a version where Ω is replaced by its estimator, $\hat{\Omega}$

$$\hat{\beta}_{FGLS} = (X'\hat{\Omega}^{-1}X)^{-1}X'\hat{\Omega}^{-1}y$$

Typically, $\hat{\beta}_{FGLS}$ is biased, but consistent and asymptotically normal and efficient. We can write the following two sufficient conditions for GLS and FGLS to be *asymptotically equivalent*

$$\text{plim} \frac{X'(\hat{\Omega}^{-1} - \Omega^{-1})X}{n} = 0 \text{ and } \text{plim} \frac{X'(\hat{\Omega}^{-1} - \Omega^{-1})\varepsilon}{\sqrt{n}} = 0$$

Condition 1 ensures the denominator (or the non-stochastic part, $\frac{X'\Omega^{-1}X}{n}$) of the FGLS estimator converges to the same limit as the GLS estimator. Condition 2 ensures the numerator (or the stochastic part, $\frac{X'\Omega^{-1}\varepsilon}{\sqrt{n}}$) of the FGLS estimator has the same asymptotic distribution as the GLS estimator. I prove the proposition below.

Proof. We want to show that

$$\text{plim} \frac{1}{N} X'(\hat{\Omega}^{-1} - \Omega^{-1})X = 0 \quad \text{and} \quad \text{plim} \frac{1}{\sqrt{N}} X'(\hat{\Omega}^{-1} - \Omega^{-1})u = 0.$$

$$\implies \sqrt{N}(\hat{\beta}_{GLS} - \beta) - \sqrt{N}(\hat{\beta}_{FGLS} - \beta) \xrightarrow{p} 0$$

Consider the following notation, which simplifies the algebra a lot:

$$Q_N := \frac{X'\Omega^{-1}X}{N} \quad \hat{Q}_N := \frac{X'\hat{\Omega}^{-1}X}{N} \quad W_N := \frac{X'\Omega^{-1}u}{\sqrt{N}} \quad \hat{W}_N := \frac{X'\hat{\Omega}^{-1}u}{\sqrt{N}}$$

Therefore, we want to show that

$$Q_N^{-1}W_N - \hat{Q}_N^{-1}\hat{W}_N \xrightarrow{p} 0$$

We can now add and subtract $\hat{Q}_N^{-1}W_N$, which yields:

$$(Q_N^{-1}W_N - \hat{Q}_N^{-1}W_N) + (\hat{Q}_N^{-1}W_N - \hat{Q}_N^{-1}\hat{W}_N)$$

We focus first on the first difference. By factoring and using the property of difference of inverse matrices ¹⁰, we can rewrite it as

$$(Q_N^{-1} - \hat{Q}_N^{-1})W_N = \left[Q_N^{-1}(\hat{Q}_N - Q_N)\hat{Q}_N^{-1} \right] W_N$$

Now, by our hypothesis, we know that $\hat{Q}_N - Q_N = o_p(1)$. Additionally, Assuming $\text{plim} \left(\frac{1}{N} X' \Omega^{-1} X \right) = Q$, a finite non-singular matrix, it follows by the Continuous Mapping Theorem that $\text{plim}(Q_N^{-1}) = Q^{-1}$, which implies $Q_N^{-1} = O_p(1)$. We also need to show that $\hat{Q}_N^{-1} = O_p(1)$. Notice that we know that $\text{plim}(\hat{Q}_N - Q_N) = 0$ and that $\text{plim} Q_N = Q$. Therefore, we can rewrite

$$\text{plim} \hat{Q}_N = \text{plim}(\hat{Q}_N - Q_N + Q_N) = \text{plim}(\hat{Q}_N - Q_N) + \text{plim}(Q_N) = Q$$

By the same argument as before (CMT), this implies $\hat{Q}_N^{-1} = O_p(1)$. Finally, note that $W_n = O_p(1)$ by central limit theorem. Notice that for any $Z_n = O_p(1)$, $A_n = o_p(1)$, $Z_n A_n = o_p(1)$, we have that the first difference is therefore

$$\left[Q_N^{-1}(\hat{Q}_N - Q_N)\hat{Q}_N^{-1} \right] W_n = O_p(1)o_p(1)O_p(1)O_p(1) = o_p(1)$$

We can move on to the second difference, which we rewrite as follows:

$$(\hat{Q}_N^{-1}W_N - \hat{Q}_N^{-1}\hat{W}_N) = \hat{Q}_N^{-1}(W_N - \hat{W}_N)$$

We have shown $\hat{Q}_N^{-1} = O_p(1)$, so we focus on the term $(W_N - \hat{W}_N)$, which we rewrite as

$$(W_N - \hat{W}_N) = \frac{1}{\sqrt{N}} X'(\Omega^{-1} - \hat{\Omega}^{-1})u$$

We are given that $\text{plim} \frac{1}{\sqrt{N}} X'(\hat{\Omega}^{-1} - \Omega^{-1})u = 0$. Notice that by continuity¹¹ of the plim operator, it follows that

$$\text{plim}(W_N - \hat{W}_N) = \text{plim} \left[-\frac{1}{\sqrt{N}} X'(\hat{\Omega}^{-1} - \Omega^{-1})u \right] = 0$$

Therefore, $(W_N - \hat{W}_N) = o_p(1)$, which implies $\hat{Q}_N^{-1}(W_N - \hat{W}_N) = O_p(1)o_p(1) = o_p(1)$

We have shown that

$$\begin{aligned} Q_N^{-1}W_N - \hat{Q}_N^{-1}\hat{W}_n &= (Q_N^{-1}W_N - \hat{Q}_N^{-1}W_N) + (\hat{Q}_N^{-1}W_N - \hat{Q}_N^{-1}\hat{W}_N)\hat{W}_N = o_p(1) + o_p(1) = o_p(1) \\ &\implies Q_N^{-1}W_N - \hat{Q}_N^{-1}\hat{W}_n \xrightarrow{p} 0 \end{aligned}$$

□

¹⁰For any two invertible matrices A and B of the same dimension,

$$A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}.$$

This identity follows from adding and subtracting $A^{-1}BB^{-1}$ and is valid whenever both inverses exist.

¹¹If $X_n \xrightarrow{p} X$, then for any constant a , $aX_n \xrightarrow{p} aX$. This follows from the continuity of convergence in probability (Continuous Mapping Theorem), and is often informally referred to as linearity of the probability limit.

Note that asymptotic efficiency of FGLS does not require the efficiency of the estimator of θ . Only consistency of $\hat{\theta}$ is required to achieve full efficiency for the FGLS estimator.

3.3.3 Application: Heteroskedasticity

This is a problem often encountered in cross-sectional and panel models. In these studies, we are interested in studying members of a *population*, say at a given point in time. We can think about firms, industries, geographical units (e.g. counties, states) or individual consumers. Members of such population may vary in many different characteristics (size, productivity, demographic composition,...), which can imply that individuals/firms are drawn from distributions with different variances: heteroskedasticity.

Let us define $\mathbb{V}(\varepsilon_i|X_i) = \sigma_i^2 := \sigma^2\omega_i^2$. The transformed model satisfying the GM conditions¹² is

$$\frac{y_i}{\omega_i} = \left(\frac{x_i}{\omega_i} \right)' \beta + \frac{\varepsilon_i}{\omega_i}$$

In this case, R is a diagonal matrix with $\frac{1}{\omega_i}$ on the diagonal, and Ω is a diagonal matrix of $\sigma^2\omega_i^2$. The weighted least squares estimator is therefore given by

$$\hat{\beta}_{WLS} = \left(\sum_i \frac{x_i x_i'}{\omega_i^2} \right)^{-1} \sum_i \frac{y_i x_i'}{\omega_i^2}$$

The idea is basically to reweight based on how noisy this observation is. If the observation has a high variance (and therefore a high ω_i^2), the denominator increases so we put less weight on this observation. We now introduce an example.

Example 3.1: Grouped-data Regression model

Rather than observing all individuals, we may observe only group averages (case in many very large datasets, or anonymized datasets etc).

The true data generating process satisfies all Gauss-Markov assumptions (including full independence):

$$y_{ij} = x'_{ij}\beta + \varepsilon_{ij} \quad j = 1, \dots, M_i, i = 1, \dots, n$$

where $\varepsilon_{ij} \sim iid(0, \sigma^2)$.

However, the estimable model is defined by the group means:

$$\bar{y}_i = \bar{x}'_i \beta + \bar{\varepsilon}_i$$

with $\bar{y}_i = \sum_{j=1}^{M_i} \frac{y_{ij}}{M_i}$, $\bar{x}_i = \sum_{j=1}^{M_i} \frac{x_{ij}}{M_i}$, and $\bar{\varepsilon}_i = \sum_{j=1}^{M_i} \frac{\varepsilon_{ij}}{M_i}$.

¹²It is easy to show that spherical errors hold in the transformed model: $Var(\varepsilon^* | X) = \mathbb{E}(\varepsilon^* \varepsilon^{*'} | X) = \frac{1}{\omega_i^2} \mathbb{E}(\varepsilon_i \varepsilon_i' | x_i) = \sigma^2$

The estimable model exhibits heteroskedasticity, as the variance of the average error is:

$$\mathbb{V}(\bar{\varepsilon}_i) = \mathbb{V}\left(\sum_{j=1}^{M_i} \frac{\varepsilon_{ij}}{M_i}\right) = \frac{\sigma^2}{M_i}$$

Therefore, the variance of the error term $\bar{\varepsilon}_i$ is inversely proportional to the group size M_i . Observations from larger groups (M_i large) are more precise (smaller variance) than observations from smaller groups.

GLS is an OLS on the transformed model, where the transformation uses the inverse square root of the variance, $\frac{1}{\sqrt{\mathbb{V}(\bar{\varepsilon}_i)}} = \frac{\sqrt{M_i}}{\sigma}$. Since σ is constant across all observations, the necessary weight ω_i is $\sqrt{M_i}$.

The transformed model is:

$$\sqrt{M_i}y_i = \sqrt{M_i}x_i'\beta + \sqrt{M_i}\bar{\varepsilon}_i$$

$$\text{or } y_i^* = x_i^{*'}\beta + \varepsilon_i^*$$

The transformed error $\varepsilon_i^* = \sqrt{M_i}\bar{\varepsilon}_i$ is now homoskedastic:

$$\mathbb{V}(\varepsilon_i^*) = \mathbb{V}(\sqrt{M_i}\bar{\varepsilon}_i) = M_i\mathbb{V}(\bar{\varepsilon}_i) = M_i\frac{\sigma^2}{M_i} = \sigma^2$$

Instead of using Weighted Least Squares (WLS), we can choose to use OLS, while only adjusting the standard errors. This approach does not require us to specify the functional form of the heteroskedasticity. The appropriate covariance matrix for the OLS estimator when heteroskedasticity is present is given by the sandwich formula, derived from the asymptotic variance:

$$\mathbb{V}(\hat{\beta} | X) = \left(\sum x_i x_i'\right)^{-1} \left(\sum \sigma_i^2 x_i x_i'\right) \left(\sum x_i x_i'\right)^{-1}$$

The estimator of this covariance matrix, known as the White (1980) estimator (or the Heteroskedasticity-Consistent Covariance Matrix Estimator, HCCME), replaces the unknown σ_i^2 with the squared OLS residuals $\hat{\varepsilon}_i^2$:

$$\widehat{\mathbb{V}(\hat{\beta} | X)} = \left(\sum x_i x_i'\right)^{-1} \left(\sum \hat{\varepsilon}_i^2 x_i x_i'\right) \left(\sum x_i x_i'\right)^{-1}$$

This estimator is consistent because it relies on the condition that the difference between the true normalized σ_i^2 and the estimated normalized $\hat{\varepsilon}_i^2$ vanishes asymptotically:

$$\text{plim } \frac{1}{n} \sum (\sigma_i^2 - \hat{\varepsilon}_i^2) x_i x_i' = 0$$

Standard errors calculated as the square root of the diagonal elements of this matrix are usually referred to as heteroskedasticity-consistent standard errors or simply White standard errors. In Stata, this is typically implemented using the 'vce(robust)' option.

Now, we introduce a theorem that provides necessary and sufficient conditions for equivalence between OLS and GLS.

Theorem 3.2: Kruskal's Theorem

Under assumptions i) to iii) and iv)', OLS is efficient if and only if

- The column space of X , denoted $\mathcal{R}(X)$ (i.e. the space spanned by the K columns of X), is spanned by K eigenvectors of the covariance matrix Ω
- The column space of the matrix product $X\Omega$ is that same as the column space of X , that is $\mathcal{R}(X\Omega) = \mathcal{R}(X)$

I find the second condition a bit more intuitive. $\mathcal{R}(X\Omega) = \mathcal{R}(X)$ means that applying the covariance structure Ω to the explanatory variables X does not "move" them out of their column space. The condition $\mathcal{R}(\Omega X) = \mathcal{R}X$ is the formal linear algebra requirement that ensures this proportionality holds, meaning the weighting matrix Ω^{-1} does not change the resulting projection (the coefficients $\hat{\beta}$).

3.3.4 Heteroskedasticity Tests

We introduce two tests for heteroskedasticity. First, the Goldfeld-Quandt Test (1965). We assume the heteroskedastic variance σ^2 is monotonically related to one observable variable, z_i . It is then a simple F test where we separate observations into two groups after reordering by z_i , omitting r central observations to improve power. Under $\mathcal{H}_0$ (homoskedasticity):

$$F = \frac{s_1^2}{s_2^2} \sim F_{m-k, m-k}$$

where $m = \frac{n-r}{2}$ and $s_j^2 = \hat{\varepsilon}'\hat{\varepsilon}/(m-k)$.

The second test is the White Test. It Does not specify anything about the form of heteroskedasticity. We regress the squared residuals $\hat{\varepsilon}^2$ on a constant and all p unique first moments, second moments, and cross-products of the original regressors. The test statistic is based on the R^2 from the auxiliary regression:

$$\text{White Test} = nR^2 \xrightarrow{p} \chi_p^2$$

It is also possible to apply the trinity of hypothesis testing to this context (see Limodio's slides).

4 Identification and Instrumental Variables

In this section, we relax the assumption of conditional mean independence, that is assumption iii). The asymptotic results derived in earlier section are all based on $\mathbb{E}(\varepsilon|X) = 0$.¹³

¹³A weaker condition, contemporaneous uncorrelatedness, which is implied by conditional mean independence, and written as $\mathbb{E}(x_i\varepsilon_i) = 0$ is sufficient to ensure the consistency of the OLS estimator.

However, in most cases, our regression has an endogeneity problem, that is $\mathbb{E}(x_i \varepsilon_i) \neq 0$. Using our asymptotic results from above, by Khinchine's WLLN, we obtain that

$$\frac{1}{n} \sum_{i=1}^n x_i x_i' \xrightarrow{p} \mathbb{E}(x_i x_i') \equiv Q \text{ and } \frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{p} \mathbb{E}(x_i \varepsilon_i) \neq 0$$

Therefore, by Slutsky's Theorem and CMT, we understand the OLS estimator is inconsistent

$$\beta \xrightarrow{p} \beta + Q^{-1} \mathbb{E}(x_i \varepsilon_i) \neq \beta$$

There are different causes of endogeneity, which I detail below.

4.1 Cause of Endogeneity

4.1.1 Measurement error

The true (unobservable) model starts with conditional mean independence $\mathbb{E}(\varepsilon_i | x_i^*) = 0$:

$$y_i = x_i^{*'} \beta + \varepsilon_i$$

We observe a measurement error u_i such that $x_i = x_i^* + u_i$, where $\mathbb{E}(u_i) = 0$. The key assumptions are that the measurement error u_i is independent of both the true regressor x_i^* and the equation error ε_i . These are strong assumptions, implying that the true value x_i^* reveals no information about the sign, size, or value of the measurement error u_i .

The regression model we estimate uses the observed regressor x_i and has a composite error term v_i :

$$y_i = x_i' \beta + v_i$$

where $v_i = \varepsilon_i - u_i' \beta$.

By construction, the observed regressor x_i and the new error v_i are correlated:

$$\mathbb{E}(x_i v_i) = \mathbb{E}[(x_i^* + u_i)(\varepsilon_i - u_i' \beta)] \neq 0$$

Specifically, $\mathbb{E}(x_i v_i) = \mathbb{E}(u_i v_i) = -\mathbb{E}(u_i u_i') \beta \neq 0$ because u_i and v_i are necessarily correlated as both are functions of the measurement error u_i .

Therefore, the OLS estimator $\hat{\beta}$ of equation is in general inconsistent. The regressor x_i becomes endogenous when measured with error, and the resulting parameter estimates will be biased, with the extent of this bias depending on the magnitude and pattern of the error.

Example 4.1: Special case: bivariate regression

The linear model is $y_i = \beta_0 + \beta_1 x_i^* + \varepsilon_i$ with conditional mean independence $\mathbb{E}(\varepsilon_i | \tilde{x}_i^*) = 0$. The true regressor $\tilde{x}_i^*$ is unobserved, and we instead observe the

error-ridden version $\tilde{x}_i = \tilde{x}_i^* + \tilde{u}_i$.

Suppose the measurement error $\tilde{u}_i$ is independent of the true regressor $\tilde{x}_i^*$ and the equation error ϵ_i , with $\mathbb{E}(\tilde{u}_i) = 0$ and variance $\mathbb{V}(\tilde{u}_i) = \sigma_{\tilde{u}}^2$.

Asymptotic Properties of $\hat{\beta}_1$

The probability limit of the slope estimator $\hat{\beta}_1$ is:

$$\text{plim } \hat{\beta}_1 = \frac{\text{plim } \frac{1}{n} \sum (\tilde{x}_i - \bar{\tilde{x}})(y_i - \bar{y})}{\text{plim } \frac{1}{n} \sum (\tilde{x}_i - \bar{\tilde{x}})^2} = \frac{\text{Cov}(\tilde{x}_i, y_i)}{\mathbb{V}(\tilde{x}_i)} = \beta_1 \frac{\mathbb{V}(\tilde{x}_i^*)}{\mathbb{V}(\tilde{x}_i^*) + \sigma_{\tilde{u}}^2}$$

Since the term $\frac{\mathbb{V}(\tilde{x}_i^*)}{\mathbb{V}(\tilde{x}_i^*) + \sigma_{\tilde{u}}^2}$ is less than 1, this is the typical case of "attenuation bias".

Therefore, $\hat{\beta}_1$ is inconsistent, and the amount of inconsistency is small when the "noise to signal ratio" ($\sigma_{\tilde{u}}^2 / \mathbb{V}(\tilde{x}_i^*)$) is small.

Asymptotic Properties of $\hat{\beta}_0$

The intercept estimator $\hat{\beta}_0$ is also inconsistent:

$$\text{plim } \hat{\beta}_0 = \text{plim } \bar{y} - \text{plim } \hat{\beta}_1 \text{plim } \bar{\tilde{x}} = \beta_0 + \beta_1 \text{plim } \bar{\tilde{x}}^* + \text{plim } \bar{\epsilon} - \text{plim}(\text{plim } \hat{\beta}_1 \text{plim } \bar{\tilde{x}})$$

$$\text{plim } \hat{\beta}_0 = \beta_0 + \beta_1 \mathbb{E}(\tilde{x}^*) \frac{\mathbb{V}(\tilde{x}^*)}{\mathbb{V}(\tilde{x}^*) + \sigma_{\tilde{u}}^2}$$

(The formula in the source is missing a β_1 term, assuming $\text{plim } \bar{\epsilon} = 0$ and $\text{plim } \bar{u} = 0$). Therefore, $\hat{\beta}_0$ is also inconsistent, unless $\mathbb{E}(\tilde{x}^*) = 0$ or $\beta_1 = 0$. The direction of inconsistency depends on the sign of β_1 and $\mathbb{E}(\tilde{x}^*)$.

In general, if only one regressor is measured with error, the estimator of its coefficient is asymptotically shrunk to zero. If more variables are measured with error, then very little can be said about the direction. Therefore, measurement error in more than one explanatory variable gives inconsistent OLS, not only for the parameters associated with the variables measured with error but for all parameters in general.

4.1.2 Simultaneity

Simultaneity arises when some of the independent variables are jointly determined with the dependent variable.

Consider the following linear market model which uses a structural form representation consisting of two equations:

- Demand equation: $q_t^d = \alpha p_t + X_t^d \beta + u_{1t}$
- Supply equation: $q_t^s = \gamma p_t + X_t^s \delta + u_{2t}$

Here, X_t^d and X_t^s are predetermined variables that capture exogenous shifts in supply and demand, satisfying conditional mean independence. These are structural equations derived from theory, each describing a particular aspect of the economy. The market equilibrium condition, $q_t^d = q_t^s = q_t$, ensures that price (p_t) and quantity (q_t) are jointly determined, making them both endogenous variables.

Our concern is the estimation of these structural form equations. To see the endogeneity problem, we rewrite the model in its reduced form by solving for p_t and q_t in terms of X_t^d , X_t^s , u_{1t} , and u_{2t} . The reduced form for price is:

$$p_t = \frac{1}{\alpha - \gamma} \left(X_t^s \delta - X_t^d \beta + u_{2t} - u_{1t} \right)$$

Clearly, p_t is correlated with both error terms u_{1t} and u_{2t} . Applying OLS to each structural form equation separately will therefore give inconsistent parameter estimates. OLS applied to the reduced form equations will provide consistent estimates of the reduced form parameters, but this is a nonlinear relation to the structural form parameters of interest.

4.1.3 Omitted Variable Bias

Consider the true model, which is called the long regression

$$y_i = x'_{1i}\beta_1 + x'_{2i}\beta_2 + u_i, \text{ with } \mathbb{E}(u_i|x_{1i}, x_{2i}) = 0$$

Suppose that we observe only x_{1i} and y_i , and therefore run the following short model, omitting x_{2i} :

$$y_i = x'_{1i}\beta_1 + \varepsilon_i, \text{ with } \varepsilon_i = x'_{2i}\beta_2 + u_i$$

Now, if we consider

$$\mathbb{E}(x_{1i}\varepsilon_i) = \mathbb{E}(x_{1i}(x'_{2i}\beta_2 + u_i)) = \beta_2 \mathbb{E}(x_{1i}x'_{2i}) \neq 0$$

unless $\beta_2 = 0$ or $\mathbb{E}(x_{1i}x'_{2i}) = 0$. This implies that the estimator is not consistent in most cases.

Now, consider the estimator of β_1 written in matrix form:

$$\hat{\beta}_1 = (X_1'X_1)^{-1}X_1'y = \beta_1 + (X_1'X_1)^{-1}X_1'\varepsilon = \beta_1 + (X_1'X_1)^{-1}X_1'X_2' + (X_1'X_1)^{-1}X_1'u$$

The last term vanishes by orthogonality condition, but the bias term $(X_1'X_1)^{-1}X_1'X_2'$ remains.

A well known example comes from Card, 2001a, where he shows that unobserved ability is an omitted variable when trying to evaluate the causal effect of education on earnings. Individuals with higher ability are likely to be more successful on the labor market by earning higher wages, and are likely to acquire more education. As such, unobserved ability affects both education ($\mathbb{E}(x_{1i}x'_{2i}) \neq 0$) and earnings ($\beta_2 \neq 0$), and the regressor, education, is correlated with the error term.

4.2 Instrumental Variables and Identification

4.2.1 Set-up

In this section, I introduce one of the solutions to this endogeneity problem: Instrumental Variables (IV).¹⁴ Consider the model

$$y_i = x_i' \beta + \varepsilon_i, \text{ where } \mathbb{E}(x_i \varepsilon_i) \neq 0$$

We first introduce z_i , which is an instrumental variable, for which we assume $\mathbb{E}(z_i \varepsilon_i) = 0$. Notice that z_i can overlap with x_i , that is any x_i that is exogenous is its own instrument. Note that we must have most instruments than endogenous regressors, that is $\dim(z_i) \geq \dim(x_i)$.

Now, consider what is called the *reduced form*¹⁵ relationship between x_i and z_i , found by linear projection of x_i on z_i :

$$x_i = z_i' \Pi + u_i$$

If we impose the moment condition $\mathbb{E}(z_i u_i') = 0$, we can recover Π :

$$\mathbb{E}(z_i u_i') = \mathbb{E}(z_i (x_i - z_i' \Pi)) = 0 \implies (\mathbb{E}(z_i z_i'))^{-1} \mathbb{E}(z_i x_i')$$

In matrix notation, the reduced form can be written as

$$X_{(n \times k)} = Z_{(n \times \ell)} \Pi_{(\ell \times k)} + U_{(n \times k)}$$

and we can consistently Π by OLS:

$$\hat{\Pi} = (Z'Z)^{-1} Z'X$$

If we plug this matrix reduced form equation into the original model, we get

$$y = X\beta + \varepsilon = (Z\Pi + U)\beta + \varepsilon = Z(\Pi\beta) + (U\beta + \varepsilon) \equiv Z\lambda + v$$

Note that λ is a valid projection coefficient since

$$\mathbb{E}(z_i v_i) = \mathbb{E}(z_i u_i') \beta + \mathbb{E}(z_i \varepsilon_i) = 0$$

Therefore, the OLS from y_i on z_i yields a consistent estimator of λ :

$$\hat{\lambda} = (Z'Z)^{-1} Z'y$$

We therefore have the following set of reduced form equations.

¹⁴I present things in a different order than Limodio's slides, but all the content should be there.

¹⁵Note that this is not directly what comes to mind when mentioning reduced form: this is more a first stage.

Definition 4.1: Reduced form equations

The reduced form equations in matrix notation are

1. $y = Z\lambda + v$
2. $X = Z\Pi + U$

Understand that so far we have just established:

1. The direct relationship between Z and y (what we think about when talking about reduced form usually): does the instrument have a direct impact on the outcome. For instance, if we instrument education by distance to school, the reduced form is the regression of income on distance to school directly.
2. The relation between X and Z , which is usually what we think about as first stage, and is therefore related to the relevance condition, which will be introduced shortly.

The question is now to understand if we can recover the original parameter of interest β from (λ, Π) : this is an **identification question**, that is a population question.

4.2.2 Identification

We say that the parameters β are **identified** if β can be recovered from (λ, Π) through the equation

$$\lambda_{(\ell \times 1)} = \Pi_{(\ell \times k)} \beta_{(k \times 1)}$$

It is easy to see that we basically want Π to be invertible. From Proposition 1.3, this is equivalent to Π being full column rank, that is having k linearly independent columns. Therefore, a sufficient condition for identification of β is

$$\text{rank} \Pi = k$$

To build intuition a bit more, recall that Π represents the "coefficients" of the impact of Z on X . Full rank means that there is non-zero covariance between at least k elements of Z and X . That is, at least k instruments should be correlated with the endogenous regressors X : this is a relevance condition. We must now distinguish between different cases.

First, consider the **just-identified** case, where $\ell = k$. The full rank condition guarantees invertibility, so β is identified as

$$\hat{\beta} = \hat{\Pi}^{-1} \hat{\lambda}$$

Indeed, plugging $\hat{\Pi}$ and $\hat{\lambda}$, we obtain

$$\begin{aligned} \hat{\Pi}^{-1} \hat{\lambda} &= \left[(Z'Z)^{-1} Z'X \right]^{-1} (Z'Z)^{-1} Z'y = \left[(Z'Z)^{-1} Z'X \right]^{-1} (Z'Z)^{-1} Z'(X\beta + \varepsilon) \\ &= \hat{\beta} + \left[(Z'Z)^{-1} Z'X \right]^{-1} (Z'Z)^{-1} Z'\varepsilon = \hat{\beta}_{IV} \end{aligned}$$

where the last equality comes by orthogonality of Z' and ε .

In the just-identified case ($\ell = k$), the IV estimator $\hat{\beta}_{IV} = \hat{\Pi}^{-1}\hat{\lambda}$ formalizes the intuition of "reduced form divided by first stage" that we usually think about when talking about IV. The numerator, $\hat{\lambda}$, is the estimated RF effect—the total impact of the instruments (Z) on the outcome (y). The denominator, $\hat{\Pi}$, is the estimated first-stage effect—the total impact of the instruments (Z) on the endogenous regressor (X). Since the instruments affect the outcome only through the endogenous regressor (the exclusion restriction), the structural coefficient β represents the causal effect of X on y . Therefore, dividing the total effect ($\hat{\lambda}$) by the mechanism's strength ($\hat{\Pi}$) effectively isolates the specific causal effect of interest, $\hat{\beta}_{IV}$, which is the change in y for a unit change in X induced only by the instrument. This calculation essentially normalizes the total impact by the degree of "relevance" provided by the instruments.

Now, consider the **overidentified** case, where $\ell > k$. We basically treat the reduced form relationship as having an error term e :

$$\lambda = \Pi\beta + e$$

and then minimize the sum of squared residuals $e'e$ with respect to β which yields a solution of the form

$$\Pi'\lambda = (\Pi'\Pi)\beta$$

Again, the full rank condition ensures that $(\Pi'\Pi)$ is invertible, which yields

$$\hat{\beta} = (\Pi'\Pi)^{-1}\Pi'\lambda$$

Finally, in the **underidentified** case ($\ell < k$), we do not have enough instrument to explain the endogenous regressors. Note that what really matters again is the rank: if we do not have full column rank in Π , some columns are linear combinations of each other (linear dependence) implying that we "lose" relevant information to instrument the endogenous variables.

4.3 Instrumental Variables and moments conditions

As we hinted at before, the instrumental variable estimator, $\hat{\beta}_{IV}$ is the method of moments estimator that solves, the following moment condition (validity)

$$\mathbb{E}(z_i\varepsilon_i) = \mathbb{E}(z_i(y_i - x_i'\beta)) = 0$$

The sample analog of this moment condition is

$$\frac{1}{n} \sum_{i=1}^n z_i(y_i - x_i'\beta) = 0$$

Assuming the rank condition (relevance), which implies that $\sum_{i=1}^n z_i x_i'$ is invertible, the IV estimator can be written as

$$\hat{\beta}_{IV} = \left(\sum_{i=1}^n z_i x_i' \right)^{-1} \sum_{i=1}^n z_i y_i \text{ or } \hat{\beta}_{IV} = (Z'X)^{-1}Z'y$$

Using the matrix form, we can take the expectation and observe that

$$\mathbb{E}(\hat{\beta}_{IV}|Z, X) = \beta + (Z'X)^{-1}Z'\mathbb{E}(\varepsilon|Z, X)$$

But notice that the main reason we use an IV is because $\mathbb{E}(\varepsilon|X) \neq 0$, which implies $\mathbb{E}(\varepsilon|Z, X) \neq 0$. Therefore, it is important to remember that **the IV estimator is generally biased in finite samples**.

Moving to the asymptotics, as per usual, we can re-write

$$\hat{\beta}_{IV} = \beta + \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)^{-1} \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i$$

We impose the following regularity conditions.

- $\{(x_i, z_i, \varepsilon_i)\}_{i=1}^n$ is iid
- Validity: $\mathbb{E}(z_i \varepsilon_i) = 0$
- Relevance: $\mathbb{E}(z_i x_i')$ is invertible
- Moment existence condition: $\mathbb{E}(z_i x_i') < \infty$ and $\mathbb{E}|z_i \varepsilon_i| < \infty$

Therefore, by Khinchine's WLLN, we obtain that

$$\frac{1}{n} \sum_{i=1}^n z_i x_i' \xrightarrow{p} \mathbb{E}(z_i x_i') \equiv Q_{IV} \text{ and } \frac{1}{n} \sum_{i=1}^n z_i \varepsilon_i \xrightarrow{p} \mathbb{E}(z_i \varepsilon_i) = 0$$

By Slutsky's Theorem and Continuous Mapping Theorem, we obtain

$$\hat{\beta}_{IV} \xrightarrow{p} \beta + \mathbb{E}(z_i x_i')^{-1} \mathbb{E}(z_i \varepsilon_i) = \beta + O_p(1) \cdot o_p(1) = \beta$$

This shows that under **validity** and **relevance** of the instrument, $\hat{\beta}_{IV}$ is consistent.

Moving to asymptotic normality, we impose two additional moment assumptions: $\mathbb{E}(\|z_i\|^4) < \infty$ and $\mathbb{E}(\varepsilon_i^4) < \infty$. Under these assumptions, by (Multivariate) Lindeberg-Lévy's CLT, we have that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i \varepsilon_i \xrightarrow{d} \mathcal{N}(0, \Omega_{IV}^*), \text{ where } \Omega_{IV}^* = \mathbb{E}(\varepsilon_i^2 z_i z_i')$$

By CMT, we get that

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Q_{IV}^{-1} \mathcal{N}(0, \Omega_{IV}^*) = \mathcal{N}(0, Q_{IV}^{-1} \Omega_{IV}^* Q_{IV}^{-1})$$

We now consider the estimation of the asymptotic variance for the IV estimator, $\hat{\beta}_{IV}$. A consistent estimator of the asymptotic variance is given by the sandwich formula:

$$\widehat{\text{Var}}(\hat{\beta}_{IV}) = \left(\frac{1}{n} \sum_{i=1}^n z_i x_i' \right)^{-1} \cdot \frac{1}{n} \sum_{i=1}^n (y_i - x_i' \hat{\beta}_{IV})^2 z_i z_i' \cdot \left(\frac{1}{n} \sum_{i=1}^n x_i z_i' \right)^{-1}$$

where the middle term estimates the Ω_{IV}^* matrix. The residuals used are the IV residuals, $\hat{\varepsilon}_{IV} = y - X\hat{\beta}_{IV}$.

If the errors are conditionally homoskedastic, meaning $\mathbb{E}(\varepsilon_i^2|z_i) = \sigma^2$ does not depend on z_i , then $\mathbb{E}(\varepsilon_i^2 z_i z_i') = \sigma^2 \mathbb{E}(z_i z_i')$. In this case, the asymptotic variance simplifies to:

$$\widehat{\text{Var}}(\hat{\beta}_{IV}) = s_{IV}^2 \left(\sum \frac{1}{n} z_i x_i' \right)^{-1} \cdot \frac{1}{n} \sum z_i z_i' \cdot \left(\sum \frac{1}{n} x_i z_i' \right)^{-1}$$

where s_{IV}^2 is a consistent estimator of the error variance σ^2 :

$$s_{IV}^2 = \frac{\hat{\varepsilon}_{IV}' \hat{\varepsilon}_{IV}}{n - k}$$

To show consistency of s_{IV}^2 , we substitute $y = X\beta + \varepsilon$ into the residuals $\hat{\varepsilon}_{IV}$:

$$\hat{\varepsilon}_{IV} = y - X\hat{\beta}_{IV} = y - X(Z'X)^{-1}Z'y = \left(I_n - X(Z'X)^{-1}Z' \right) \varepsilon$$

The consistency of s_{IV}^2 for σ^2 follows from the fact that $\text{plim} \frac{\varepsilon' \varepsilon}{n} = \sigma^2$ and the validity condition.

Example 4.2: Special case: $L = k = 2$

Consider the simple linear model $y_i = \alpha + \beta \tilde{x}_i + \varepsilon_i$, where the regressor $\tilde{x}_i$ is endogenous, $\mathbb{E}(\tilde{x}_i \varepsilon_i) \neq 0$. The vector of regressors is $x_i = (1, \tilde{x}_i)'$ (including the constant). The vector of instruments is $z_i = (1, \tilde{z}_i)'$, assuming the validity condition $\mathbb{E}(\tilde{z}_i \varepsilon_i) = 0$.

The relevance condition holds if the matrix $\mathbb{E}(z_i x_i')$ has full rank, which is equivalent to $\text{Cov}(\tilde{z}_i, \tilde{x}_i) \neq 0$:^a

$$\mathbb{E}(z_i x_i') = \begin{pmatrix} 1 & \mathbb{E}(\tilde{x}_i) \\ \mathbb{E}(\tilde{z}_i) & \mathbb{E}(\tilde{z}_i \tilde{x}_i) \end{pmatrix}$$

The IV estimator for the slope coefficient β and the intercept α are given by:

$$\hat{\beta}_{IV} = \frac{\widehat{\text{Cov}}(\tilde{z}_i, y_i)}{\widehat{\text{Cov}}(\tilde{z}_i, \tilde{x}_i)} = \frac{\sum (\tilde{z}_i - \bar{\tilde{z}})(y_i - \bar{y})}{\sum (\tilde{z}_i - \bar{\tilde{z}})(\tilde{x}_i - \bar{\tilde{x}})} \quad \text{and} \quad \hat{\alpha}_{IV} = \bar{y} - \hat{\beta}_{IV} \bar{\tilde{x}}$$

Under the assumption of homoskedasticity ($\mathbb{E}(\varepsilon_i^2|z_i) = \sigma^2$), the asymptotic variance of the slope estimator $\hat{\beta}_{IV}$ can be simplified and estimated as:

$$\mathbb{V}(\hat{\beta}_{IV}) = \sigma^2 \frac{\mathbb{V}(\tilde{x}_i)}{\text{Cov}(\tilde{z}_i, \tilde{x}_i)^2}$$

This can also be expressed using the squared correlation coefficient, $\text{Corr}(\tilde{z}_i, \tilde{x}_i)^2$:

$$\mathbb{V}(\hat{\beta}_{IV}) = \frac{\sigma^2}{\text{Corr}(\tilde{z}_i, \tilde{x}_i)^2 \mathbb{V}(\tilde{x}_i)}$$

For a given distribution of the regressor $\tilde{x}_i$, the variance $\mathbb{V}(\hat{\beta}_{IV})$ has the minimal possible value ($\sigma^2 / \mathbb{V}(\tilde{x}_i)$) when the correlation between $\tilde{x}_i$ and $\tilde{z}_i$ is unit. In general, the lower the correlation, the higher the variance of $\hat{\beta}_{IV}$.

1. Good instruments are not only uncorrelated with the regression error (which guarantees the consistency of the IV estimator) but also highly correlated with the explanatory variables (which gives the precision of the IV estimator).
2. The standard errors of IV regression are always higher than the OLS standard errors. We see that by the fact that as mentioned just above, $\mathbb{V}(\hat{\beta}_{IV})$ is minimized whenever $\text{Corr}(\tilde{z}_i, \tilde{x}_i) = 1$, that is whenever $\tilde{x}_i$ is an instrument for itself, the OLS case.

^aNote that here, $\text{Cov}(\tilde{z}_i, \tilde{x}_i) = \mathbb{E}(\tilde{z}_i \tilde{x}_i) - \mathbb{E}(\tilde{z}_i)\mathbb{E}(\tilde{x}_i) = \det(\mathbb{E}(z_i x_i'))$. Obviously, if $\det(\mathbb{E}(z_i x_i')) = 0$, the matrix is not invertible.

4.4 Two-Stage Least Squares (2SLS)

4.4.1 2SLS derivation

Let us now introduce the Two-Stage Least Squares (2SLS) estimator. We will get back to this when we cover GMM. For now, consider to overidentified case where, for any positive definite matrix $W_{\ell \times \ell}$, β can be expressed as

$$\beta = (\Pi' W \Pi)^{-1} \Pi' W \lambda$$

In the 2SLS case, we use

$$\hat{W} = \frac{1}{n} \sum_{i=1}^n z_i z_i' = \frac{1}{n} Z' Z$$

The 2SLS estimator is therefore

$$\tilde{\beta}_{2SLS} = (X' Z (Z' Z)^{-1} Z' X)^{-1} X' Z (Z' Z)^{-1} Z' y = (X' P_Z X)^{-1} X' P_Z y$$

The idea is that we are regressing X on Z first (first stage), to get the fitted values $\hat{X}$ and then regressing y on $\hat{X}$. This is what is called **Theil's interpretation**.

4.4.2 Asymptotic properties of 2SLS

The Two-Stage Least Squares (2SLS) estimator $\tilde{\beta}_{2SLS}$ is a generalization of the IV estimator used in the over-identified case ($\ell > k$). We analyze its consistency using asymptotic theory.

The 2SLS estimator can be written as:

$$\tilde{\beta}_{2SLS} = (X' P_Z X)^{-1} X' P_Z y$$

Substituting $y = X\beta + \varepsilon$ and manipulating the terms, we get:

$$\tilde{\beta}_{2SLS} = \beta + (X' P_Z X)^{-1} X' P_Z \varepsilon$$

Now, we replace the projection matrix $P_z = Z(Z'Z)^{-1}Z'$ and rewrite the expression using sample averages (multiplying and dividing by n):

$$\tilde{\beta}_{2SLS} = \beta + \left[\frac{1}{n} X'Z \left(\frac{1}{n} Z'Z \right)^{-1} \frac{1}{n} Z'X \right]^{-1} \frac{1}{n} X'Z \left(\frac{1}{n} Z'Z \right)^{-1} \frac{1}{n} Z'\varepsilon$$

We rely on the standard regularity conditions (i.i.d. observations, moment existence) and the two core IV assumptions: validity and relevance. By the WLLN (Khinchine's or similar), the sample averages converge in probability to their population expectations:

$$\begin{aligned} \text{plim } \frac{1}{n} X'Z &= \text{plim } \frac{1}{n} \sum x_i z_i' = \mathbb{E}(x_i z_i') \equiv Q_{xz} \\ \text{plim } \frac{1}{n} Z'X &= \text{plim } \frac{1}{n} \sum z_i x_i' = \mathbb{E}(z_i x_i') \equiv Q_{zx} \quad (\text{Note: } Q_{zx} = Q_{xz}') \\ \text{plim } \frac{1}{n} Z'Z &= \text{plim } \frac{1}{n} \sum z_i z_i' = \mathbb{E}(z_i z_i') \equiv Q_{zz} \\ \text{plim } \frac{1}{n} Z'\varepsilon &= \text{plim } \frac{1}{n} \sum z_i \varepsilon_i = \mathbb{E}(z_i \varepsilon_i) = 0 \quad (\text{Validity assumption}) \end{aligned}$$

Applying the WLLN and the Continuous Mapping Theorem (CMT) to the asymptotic expression for $\tilde{\beta}_{2SLS}$:

$$\text{plim } \tilde{\beta}_{2SLS} = \beta + \left[Q_{xz} Q_{zz}^{-1} Q_{zx} \right]^{-1} Q_{xz} Q_{zz}^{-1} \text{plim } \left(\frac{1}{n} Z'\varepsilon \right)$$

Substituting the validity condition $\text{plim}(\frac{1}{n} Z'\varepsilon) = 0$:

$$\text{plim } \tilde{\beta}_{2SLS} = \beta + \left[Q_{xz} Q_{zz}^{-1} Q_{zx} \right]^{-1} Q_{xz} Q_{zz}^{-1} \cdot 0$$

$$\text{plim } \tilde{\beta}_{2SLS} = \beta$$

This shows that the 2SLS estimator $\tilde{\beta}_{2SLS}$ is consistent under the standard validity and relevance conditions. The relevance condition ensures that the matrix $[Q_{xz} Q_{zz}^{-1} Q_{zx}]$ is invertible.

Having established the consistency of the 2SLS estimator $\tilde{\beta}_{2SLS}$, we now move to its asymptotic normality. We start with the expression for the difference between the 2SLS estimator and the true parameter vector, scaled by $\sqrt{n}$:

$$\sqrt{n}(\tilde{\beta}_{2SLS} - \beta) = \left[\frac{1}{n} X'Z \left(\frac{1}{n} Z'Z \right)^{-1} \frac{1}{n} Z'X \right]^{-1} \frac{1}{n} X'Z \left(\frac{1}{n} Z'Z \right)^{-1} \frac{1}{\sqrt{n}} Z'\varepsilon$$

Suppose the sample is i.i.d. as we have the valid moment conditions. By the Multivariate Lindeberg-Lévy Central Limit Theorem, the scaled sum of the moment condition terms converges in distribution to a normal distribution:

$$\frac{1}{\sqrt{n}} Z'\varepsilon = \frac{1}{\sqrt{n}} \sum z_i \varepsilon_i \xrightarrow{d} \mathcal{N}(0, \Omega_{zz})$$

where the asymptotic variance of the moment vector is $\Omega_{zz} = V(z_i \varepsilon_i) = \mathbb{E}(\varepsilon_i^2 z_i z_i')$.

We apply the Continuous Mapping Theorem (CMT) by taking the probability limit of the deterministic (non-stochastic) parts of the $\sqrt{n}(\tilde{\beta}_{2SLS} - \beta)$ expression, and substituting the CLT result for the stochastic part. Recall the probability limits:

$$\begin{aligned}\text{plim } \frac{1}{n} X'Z &= Q_{xz} \\ \text{plim } \frac{1}{n} Z'Z &= Q_{zz} \\ \text{plim } \frac{1}{n} Z'X &= Q_{zx}\end{aligned}$$

Applying these limits to the scaled expression:

$$\sqrt{n}(\tilde{\beta}_{2SLS} - \beta) \xrightarrow{d} \left[Q_{xz} Q_{zz}^{-1} Q_{zx} \right]^{-1} Q_{xz} Q_{zz}^{-1} \cdot \mathcal{N}(0, \Omega_{zz})$$

Therefore, the asymptotic distribution is:

$$\sqrt{n}(\tilde{\beta}_{2SLS} - \beta) \xrightarrow{d} \mathcal{N}(0, V_{2SLS})$$

with the asymptotic variance V_{2SLS} given by the generalized "sandwich" formula:

$$V_{2SLS} \equiv \left[Q_{xz} Q_{zz}^{-1} Q_{zx} \right]^{-1} Q_{xz} Q_{zz}^{-1} \Omega_{zz} Q_{zz}^{-1} Q_{zx} \left[Q_{xz} Q_{zz}^{-1} Q_{zx} \right]^{-1}$$

This is the standard, heteroskedasticity-robust formula for the asymptotic variance of the 2SLS estimator. Notice that the 2SLS estimator is inefficient compared to OLS. The variance is higher by design.

To use the derived asymptotic normality for inference (t-tests, confidence intervals), we need a consistent estimator for the asymptotic variance V_{2SLS} . We obtain such consistent estimator of the asymptotic variance V_{2SLS} by replacing all population moment matrices (Q_{xz} , Q_{zz} , and Ω_{zz}) with their consistent sample analogs.

The sample analog for the matrix $\Omega_{zz} = \mathbb{E}(\varepsilon_i^2 z_i z_i')$ is obtained by using the 2SLS residuals $\tilde{\varepsilon}_{i,2SLS} = y_i - x_i' \tilde{\beta}_{2SLS}$:

$$\frac{1}{n} \sum (y_i - x_i' \tilde{\beta}_{2SLS})^2 z_i z_i'$$

The consistent estimator $\hat{V}_{2SLS}$ is then the robust (Heteroskedasticity-Consistent) sandwich estimator, obtained by substituting the sample analogs into the general formula for V_{2SLS} .

If we assume conditional homoskedasticity of the errors with respect to the instruments, i.e., $\mathbb{E}(\varepsilon_i^2 | z_i) = \sigma^2$, the asymptotic variance simplifies. Under homoskedasticity, the middle matrix Ω_{zz} simplifies to:

$$\Omega_{zz} = \mathbb{E}(\varepsilon_i^2 z_i z_i') = \sigma^2 \mathbb{E}(z_i z_i') = \sigma^2 Q_{zz}$$

Substituting this into the V_{2SLS} formula and simplifying yields:

$$V_{2SLS} = \sigma^2 \left[Q_{xz} Q_{zz}^{-1} Q_{zx} \right]^{-1}$$

We replace the unknown population variance σ^2 with its consistent sample estimator, s^2 :

$$s^2 = \frac{(y - X\tilde{\beta}_{2SLS})'(y - X\tilde{\beta}_{2SLS})}{n - k} = \frac{\tilde{\varepsilon}'_{2SLS}\tilde{\varepsilon}_{2SLS}}{n - k}$$

The final homoskedastic estimator for the asymptotic variance is obtained by substituting s^2 and the sample moments for Q_{xz} , Q_{zz} , and Q_{zx} :

$$\hat{V}_{2SLS} = s^2 \left[\frac{1}{n} X'Z \left(\frac{1}{n} Z'Z \right)^{-1} \frac{1}{n} Z'X \right]^{-1}$$

4.5 Validity Tests and Weak Instruments

4.5.1 Testing for the Validity of Instruments: The J-Test

In the over-identified case ($\ell > k$), we have more instruments ($\mathbf{z}_i$, dimension ℓ) than endogenous regressors (x_i , dimension k). This provides $\ell - k$ extra moment conditions that can be used to test the validity of the instruments.

The core idea is that if the population moment conditions $\mathbb{E}(z_i \varepsilon_i) = 0$ are true (the null hypothesis $\mathcal{H}_0$), the sample analog $\frac{1}{n} \sum z_i (y_i - x_i' \tilde{\beta}_{2SLS})$ should be close to zero. This provides a basis for a model specification test of the null hypothesis ($\mathcal{H}_0$) against the alternative hypothesis ($\mathcal{H}_1$):

$$\mathcal{H}_0 : \mathbb{E}(z_i \varepsilon_i) = 0 \quad \text{versus} \quad \mathcal{H}_1 : \mathbb{E}(z_i \varepsilon_i) \neq 0$$

The test statistic is known as the J_n statistic (or Sargan's test in the homoskedastic case, or Hansen's J test in the heteroskedastic/GMM case):

$$J_n = n \left(\frac{1}{n} \sum z_i (y_i - x_i' \tilde{\beta}_{2SLS}) \right)' \left(s^2 \frac{1}{n} \sum z_i z_i' \right)^{-1} \left(\frac{1}{n} \sum z_i (y_i - x_i' \tilde{\beta}_{2SLS}) \right)$$

Under the null hypothesis $\mathcal{H}_0$, the J_n test statistic is asymptotically distributed as a chi-squared distribution with degrees of freedom equal to the number of overidentifying restrictions:

$$J_n \xrightarrow{d} \chi^2(\ell - k) \quad \text{under } \mathcal{H}_0$$

The J_n test is a test of the overidentifying restrictions. If we reject $\mathcal{H}_0$ (the p -value is small), it suggests that at least some of the moment conditions are invalid, implying that at least some of the instruments $\mathbf{z}_i$ are not truly exogenous. If the errors are conditionally heteroskedastic, the test must be constructed differently and should be based on an efficient GMM estimator, which will be discussed in the GMM topic.

4.5.2 Weak Instruments

Instruments are considered weak when they are only weakly correlated with the endogenous explanatory variables they instrument. They fail to provide sufficient variation to effectively isolate the exogenous component of the endogenous regressors. This can lead to biased and inconsistent parameters estimates, as well as inflated standard errors, making inference unreliable.

A way to test for weak instruments is to use a First-stage F-statistic. It aims at assessing the joint significance of the instruments in the first-stage regression. The initial rule of thumb, determined by Staiger and Stock, 1994 was an F-stat above 10 was enough to claim the instrument is not weak. New papers such as Lee et al., 2022 claims that it should actually be 100.

5 GMM and Extremum Estimation

In this section, we talk about the Generalized method of moments (GMM), and introduce extremum estimators.

5.1 Intro to GMM

Many estimation methods in econometrics are method-of-moments estimators, in which the k -dimensional parameter of interest θ_0 is assumed to satisfy an unconditional moment condition

$$\mathbb{E}(g(z_i, \theta_0)) = 0$$

for some J -dimensional vector of functions $g(z_i, \theta_0)$ of the observed data vector z_i and the parameter value $\theta \in \Theta$. If we assume that θ_0 is the unique solution of the population moment, and we are in a just-identified case ($J = k$). The method of moment estimator $\hat{\theta}$ is defined as a solution to the sample analogue of the population moment condition:

$$\frac{1}{n} \sum_{i=1}^n g(z_i, \hat{\theta}) = 0$$

The OLS estimation derived in section 1.2.2 is a perfect example of a method of moment estimator in this context, with the population moment condition being

$$\mathbb{E}(x_i(y_i - x_i'\beta)) = 0^{16}$$

Similarly, the IV estimator (denoting ζ_i the vector of instruments) is based on the moment condition

$$\mathbb{E}(\zeta_i(y_i - x_i'\beta)) = 0$$

The MLE estimator can also be written as a method-of-moments estimators

$$\mathbb{E}(s(z_i, \theta_0))$$

¹⁶Just to understand notation better, in this case we have $z_i \equiv (x_i', y_i')$ and $g(z_i, \beta) \equiv x_i(y_i - x_i'\beta)$

where $s(z_i, \theta_0)$ is the score function.

So far, we have dealt with cases where the number of equations is the same as the number of unknown parameters. Now, consider the overidentified where we have more moment conditions than parameters to estimate ($J > k$). For instance, we might have knowledge that the distribution of a scalar random variable z_i is not skewed, which allows us to have the moment $\mathbb{E}((z - \mathbb{E}(z))^3) = 0$. If we want to estimate $\theta_0 = \mathbb{E}(z)$, we have the following moment condition

$$\mathbb{E} \left[\left(\frac{z_i - \theta_0}{(z_i - \theta_0)^3} \right) \right] = 0$$

In general, whenever $J > k$, the system of equations

$$g_n(\theta) \equiv \frac{1}{n} \sum_{i=1}^n g(z_i, \theta) = 0$$

is overdetermined, meaning that there is no solution to this system. We provide the solution below.

Definition 5.1: Generalized Method of Moments Estimator

The Generalized Method of Moments (GMM) Estimator is defined as the estimator minimizing the weighted Euclidean norm, that is

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} g_n(\theta)' W_n g_n(\theta) \equiv \arg \min_{\theta \in \Theta} J_n(\theta)$$

where W_n is a symmetric positive matrix which converges in probability to a positive definite matrix W_0 .

The idea is that since we are in an over-identified case, we can't make each moment equal to zero in the sample. Instead, we try to minimize the sum of squares, that is we try to make all theoretical moments close to zero. The weight matrix just gives more importance to the moments that we theoretically believe are true.

Consider the basic linear model. We can write

$$g_n(\beta) = \frac{1}{n} \sum_{i=1}^n z_i(y_i - x_i' \beta) = \frac{1}{n} Z'(y - X\beta) \implies J_n(\beta) = \left[\frac{1}{n} Z'(y - X\beta) \right]' W_n \left[\frac{1}{n} Z'(y - X\beta) \right]$$

Taking the first-order condition with respect to β , we get¹⁷

$$\frac{\partial J_n(\beta)}{\partial \beta} = 0 \Leftrightarrow 2 \left[\frac{\partial g_n(\hat{\beta})}{\partial \beta} \right]' W_n g_n(\hat{\beta}) = 0 \Leftrightarrow \left[\frac{1}{n} Z'X \right]' W_n \left[\frac{1}{n} Z'(Y - X\beta) \right] = 0$$

¹⁷We use the following matrix multiplication rule:

$$\frac{\partial}{\partial x} \left[f(x)' B f(x) \right] = 2 \frac{\partial f(x)'}{\partial x} B f(x)$$

$$\Leftrightarrow \frac{1}{n}(X'Z)W_n\frac{1}{n}Z'Y = \frac{1}{n}(X'Z)W_n\left[\frac{1}{n}Z'X\hat{\beta}\right] \Leftrightarrow \hat{\beta} = [X'ZW_nZ'X]^{-1}X'ZW_nZ'y$$

Now, notice that if we impose $W_n = I_n$ and $Z = X$, we obtain that

$$\hat{\beta} = (X'XI_nX')^{-1}X'XI_nX'y = (X'X)^{-1}X'y = \hat{\beta}_{OLS}$$

If we impose $W_n = (Z'Z)^{-1}$, we obtain

$$\hat{\beta} = [X'Z(Z'Z)^{-1}Z'X]^{-1}X'Z(Z'Z)^{-1}Z'y = \hat{\beta}_{2SLS}$$

With these easy examples, we see that the different estimators are just version of GMM with a different weight matrix W_n . This leads us to our next concept, extremum estimators.

5.2 Extremum Estimators

The concept of extremum estimators is a class of estimator that is derived by finding extreme values - either a maximum of a minimum- of some objective function $Q_n(\theta)$:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} Q_n(\theta)$$

We discuss general asymptotic theory based on this class of estimators. Note that in the following asymptotic normality proof, we rely heavily on the fact that $\theta_0 \in \text{int } \Theta$. In Appendix A.4, we discuss the case where this is no longer the case: asymptotics at the boundary.

5.2.1 General Consistency

We start by introducing the general consistency theorem, which I prove (but this can be skipped).

Theorem 5.1: General Consistency Theorem (Newey and McFadden, 1994)

Suppose that

1. Θ is compact
2. The limit function Q_* is uniquely maximized at θ_0 (Identification)
3. Q_* is continuous in $\theta \in \Theta$
4. $\sup_{\theta \in \Theta} |Q_n(\theta) - Q_*(\theta)| \xrightarrow{p} 0$ for some $Q_* : \Theta \rightarrow \mathbb{R}$ (Uniform Convergence)

Then, we achieve consistency:

$$\hat{\theta} \xrightarrow{p} \theta_0$$

Proof. Pick any $\varepsilon > 0$. Since $\hat{\theta}$ maximizes $Q_n(\theta)$ by definition, we have that

$$Q_n(\hat{\theta}) > Q_n(\theta_0) - \frac{\varepsilon}{3}$$

By condition 4, for any $\theta \in \Theta$, we have that

$$|Q_n(\theta) - Q_*(\theta)| < \frac{\varepsilon}{3} \text{ with probability approaching 1 (w.p.a.1)}^{18}$$

Therefore, w.p.a.1, we have

$$|Q_n(\hat{\theta}) - Q_*(\hat{\theta})| < \frac{\varepsilon}{3} \implies Q_n(\hat{\theta}) - Q_*(\hat{\theta}) < \frac{\varepsilon}{3} \text{ since } (Q_n(\hat{\theta}) - Q_*(\hat{\theta})) > 0 \text{ and}$$

$$|Q_n(\theta_0) - Q_*(\theta_0)| < \frac{\varepsilon}{3} \implies Q_*(\theta_0) - Q_n(\theta_0) < \frac{\varepsilon}{3} \text{ since } (Q_n(\theta_0) - Q_*(\theta_0)) < 0$$

Combining these inequalities, w.p.a.1, we get

$$Q_*(\hat{\theta}) + \frac{\varepsilon}{3} > Q_n(\hat{\theta}) > Q_n(\theta_0) - \frac{\varepsilon}{3} > Q_*(\theta_0) - \frac{2\varepsilon}{3} \implies Q_*(\hat{\theta}) > Q_*(\theta_0) - \varepsilon \quad (*)$$

By definition of convergence in probability, we want $\mathbb{P}(\hat{\theta} \in \mathcal{N}) \rightarrow 1$ for any open neighborhood $\mathcal{N} \subset \Theta$ containing θ_0 . Pick such set. Since $\mathcal{N}$ is open, $\mathcal{N}^c$ is closed. Additionally, by condition 1, Θ is compact, which implies that $\Theta \cap \mathcal{N}^c$ is also compact. Since Q_* is continuous (by condition 3), Weierstrass theorem guarantees that there exists $\theta_* \in \Theta \cap \mathcal{N}^c$ such that

$$\sup_{\Theta \cap \mathcal{N}^c} Q_*(\theta) = Q_*(\theta_*)$$

Since Q_* is uniquely maximized at θ_0 by condition 4, we know that $Q_*(\theta_0) > Q_*(\theta_*)$. Set $\varepsilon' := Q_*(\theta_0) - Q_*(\theta_*) > 0$. Now, if we use equation (*) and set $\varepsilon = \varepsilon'$, we get

$$Q_*(\hat{\theta}) > Q_*(\theta_0) - \varepsilon' = Q_*(\theta_*) = \sup_{\Theta \cap \mathcal{N}^c} Q_*(\theta) \text{ w.p.a.1}$$

This implies that $\hat{\theta} \in \mathcal{N}$ with probability approaching 1, so we are done. $\square$

The proof is not easy, so I try to give some intuition below. The estimator $\hat{\theta}$ is chosen because it makes the sample objective function ($Q_n(\theta)$) as high as possible. Therefore, $Q_n(\hat{\theta})$ must be slightly greater than $Q_n(\theta_0)$.

The key step is establishing that the sample function $Q_n(\theta)$ is an excellent approximation of the true function $Q_*(\theta)$ everywhere (Uniform Convergence). This means that if we evaluate the true function Q_* at the sample peak $\hat{\theta}$, the result $Q_*(\hat{\theta})$ must be very close to the true peak value $Q_*(\theta_0)$. That's the meaning of the inequality $Q_*(\hat{\theta}) > Q_*(\theta_0) - \varepsilon$: the

¹⁸This holds because uniform convergence implies pointwise convergence. Additionally, w.p.a.1 is equivalent to convergence in probability. Recall that an event A_n holds with probability approaching 1 means $\lim_{n \rightarrow \infty} P(A_n) = 1$.

$\hat{\theta}$ chosen by the sample must give an outcome on the true function that is nearly optimal.

Finally, we use the Identification condition, which states that θ_0 is the unique point that maximizes Q_* . Because no other point $\theta \neq \theta_0$ can give a value of Q_* as high as $Q_*(\theta_0)$, the $\hat{\theta}$ that gives a nearly optimal value $Q_*(\hat{\theta})$ must itself be forced into a small neighborhood around θ_0 . If $\hat{\theta}$ were far away, the uniqueness condition would guarantee that $Q_*(\hat{\theta})$ would be significantly lower, which contradicts our finding from step 2.

In practice, most efforts are devoted to check condition 2 and 4. For condition 4, we typically need some kind of uniform law of large numbers.

Lemma 5.1: Uniform Law of Large Numbers (ULLN)

Suppose that

1. $\{z_i\}_{i=1}^n$ is i.i.d.
2. $g(z, \theta)$ is almost surely continuous at each $\theta \in \Theta$, and Θ is compact.
3. There is $d(z)$ such that $|g(z, \theta)| \leq d(z)$ for all $\theta \in \Theta$ and almost every z , and $E[d(z)] < \infty$.

Then, we have the uniform convergence condition:

$$\sup_{\theta \in \Theta} |\bar{g}(\theta) - E[g(z, \theta)]| \xrightarrow{p} 0$$

where $\bar{g}(\theta) = \frac{1}{n} \sum_{i=1}^n g(z_i, \theta)$ is the sample counterpart of the population moment $E[g(z, \theta)]$. Additionally, $E[g(z, \theta)]$ is continuous at each $\theta \in \Theta$.

Note that if we take $\frac{1}{n} \sum_{i=1}^n g(z_i)$, we are back in the usual WLLn case, where we need to assume $E(g(z)) < \infty$. Here, we have that $g(\cdot)$ is a function of θ , so we need to assume something stronger. Indeed, to guarantee uniform convergence, the moment condition needs to hold for the worst possible value of the function across the entire parameter space. The condition $E(\sup_{\theta \in \Theta} |g(z, \theta)|) < \infty$ ensures that the function $g(z, \theta)$ is well-behaved over the entire compact set Θ . This condition is known as the dominance condition.

5.2.2 General asymptotic normality

Suppose that we have achieved consistency. We now want to derive the asymptotic normal distribution in the form

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V)$$

We derive some the asymptotic distribution and then provide the formal theorem. Suppose that $Q_n(\theta)$ is continuously twice differentiable. We can look at the FOC for $\hat{\theta}$:

$$\frac{\partial Q_n(\hat{\theta})}{\partial \theta} = 0$$

We can write the Mean Value Theorem (MVT) around θ_0 as follows:

$$0 = \frac{\partial Q_n(\theta_0)}{\partial \theta} + \frac{\partial^2 Q_n(\tilde{\theta})}{\partial \theta \partial \theta'} (\hat{\theta} - \theta_0)$$

where $\tilde{\theta}$ is a point on the line joining θ_0 and $\hat{\theta}$. Assuming the inverse exists, we can therefore rewrite

$$\sqrt{n}(\hat{\theta} - \theta_0) = - \left(\frac{\partial^2 Q_n(\tilde{\theta})}{\partial \theta \partial \theta'} \right)^{-1} \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta}$$

We now move on to the theorem to understand what conditions are necessary and sufficient for asymptotic normality (I do not prove this theorem).

Theorem 5.2: General asymptotic normality theorem (Newey and McFadden, 1994)

Suppose that

- a. $\hat{\theta} \xrightarrow{p} \theta_0$ and $\theta_0 \in \text{int } \Theta$
- b. $Q_n(\theta)$ is continuously twice differentiable in a neighborhood $\mathcal{N}$ of θ_0
- c. $\sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} \xrightarrow{d} \mathcal{N}(0, \Sigma)$
- d. There exists $H(\theta)$ that is continuous at θ_0 such that

$$\sup_{\theta \in \mathcal{N}} \left| \frac{\partial^2 Q_n(\tilde{\theta})}{\partial \theta \partial \theta'} - H(\theta) \right| \xrightarrow{p} 0$$

and $H = H(\theta_0)$ non-singular.

Then, we obtain asymptotic normality

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, H^{-1} \Sigma H^{-1})$$

5.3 Linking the two concepts

As we have shown above, the GMM estimator is nothing but an extremum estimator where

$$Q_n(\theta) = -J_n(\theta) = -g_n(\theta)' \hat{W} g_n(\theta) \text{ and } \hat{\theta}_W = \arg \max_{\theta \in \Theta} Q_n(\theta)$$

Theorem 5.1 and 5.2 can be applied directly, but it would be nice to have some more primitive conditions. The following theorem provides the primitive conditions for consistency.

Theorem 5.3: Consistency of GMME

Suppose $\{z_i\}_{i=1}^n$ is iid and

- (i) Θ is compact

- (ii) $\hat{W} \xrightarrow{p} W_0$ and W_0 is symmetric and positive definite
- (iii) $g(z, \theta)$ is almost surely continuous at each $\theta \in \Theta$
- (iv) $\mathbb{E}(\sup_{\theta \in \Theta} |g(z, \theta)|) < \infty$
- (v) $W_0 \mathbb{E}(g(z, \theta)) = 0$ only if $\theta = \theta_0$

Proof. It is sufficient to check conditions 1-4 of Theorem 5.1. Condition 1 is guaranteed by i). We now check condition 4. In this context, we have that

$$Q_*(\theta) = -\mathbb{E}(g(z, \theta))' W \mathbb{E}(g(z, \theta))$$

We want to show that

$$\sup_{\theta \in \Theta} |Q_n(\theta) - Q_*(\theta)| = \sup_{\theta \in \Theta} |J_n(\theta) - J_*(\theta)| \xrightarrow{p} 0$$

We can rewrite the expression in the absolute value as

$$|Q_n(\theta) - Q_*(\theta)| = |g_n(\theta)' \hat{W}_n g_n(\theta) - \mathbb{E}[g(z, \theta)]' W_0 \mathbb{E}[g(z, \theta)]|$$

Let $\bar{g}(\theta) = g_n(\theta)$ and $\mu(\theta) = \mathbb{E}[g(z, \theta)]$. Also, let $\hat{W} = \hat{W}_n$ and $W = W_0$. Therefore, we have something of the form

$$|Q_n(\theta) - Q_*(\theta)| = |\bar{g}(\theta)' \hat{W} \bar{g}(\theta) - \mu(\theta)' W \mu(\theta)|$$

We use the identity $\bar{g}(\theta) = \mu(\theta) + [\bar{g}(\theta) - \mu(\theta)]$ and substitute it into the sample objective term:

$$|(\mu + (\bar{g} - \mu))' \hat{W} (\mu + (\bar{g} - \mu)) - \mu' W \mu|$$

Expanding the quadratic form:¹⁹

$$|\mu' \hat{W} \mu + \mu' \hat{W} (\bar{g} - \mu) + (\bar{g} - \mu)' \hat{W} \mu + (\bar{g} - \mu)' \hat{W} (\bar{g} - \mu) - \mu' W \mu|$$

We rearrange these terms to isolate the errors in the moments and the errors in the weighting matrix:

$$|\{(\bar{g} - \mu)' \hat{W} (\bar{g} - \mu)\} + \{\mu' (\hat{W} + \hat{W}') (\bar{g} - \mu)\} + \{\mu' (\hat{W} - W) \mu\}|$$

Using the Triangle Inequality²⁰, we separate the terms into the T_1, T_2, T_3 :

$$|Q_n(\theta) - Q_*(\theta)| \leq T_1(\theta) + T_2(\theta) + T_3(\theta)$$

where:

- $T_1(\theta) = |\{(\bar{g}(\theta) - \mu(\theta))' \hat{W} (\bar{g}(\theta) - \mu(\theta))\}|$

¹⁹ $(A + B)' W (A + B) = A' W A + A' W B + B' W A + B' W B$

²⁰ $|a + b + c| \leq |a| + |b| + |c|$

- $T_2(\theta) = |\mu(\theta)'(\hat{W} + \hat{W}')\{\bar{g}(\theta) - \mu(\theta)\}|$
- $T_3(\theta) = |\mu(\theta)'(\hat{W} - W)\mu(\theta)|$

Now, $\sup_{\theta \in \Theta} T_1(\theta) \xrightarrow{p} 0$ and $\sup_{\theta \in \Theta} T_2(\theta) \xrightarrow{p} 0$ by ULLN, which is guaranteed by iid assumption as well as i), iii) and iv). From ii), we obtain that $\sup_{\theta \in \Theta} T_3(\theta) \xrightarrow{p} 0$, and condition 4 is therefore verified. ULLN also implies that $\mathbb{E}[g(z, \theta)]$ is continuous in $\theta \in \Theta$, which in turns implies that $Q_*(\theta)$ is also continuous in $\theta \in \Theta$, so condition 3 is verified.

Next, we verify condition 2: the limit function Q_* is uniquely maximized at θ_0 . By the population moment condition, $\mathbb{E}[g(z, \theta_0)] = 0$, which implies $Q_*(\theta_0) = 0$. Since W_0 is positive definite, we have $Q_*(\theta) \leq 0$ for all $\theta \in \Theta$. Thus, it is sufficient to show that $Q_*(\theta) < 0$ for all $\theta \neq \theta_0$.

Since W_0 is symmetric and positive definite, there exists a matrix R such that $W_0 = R'R$. For any $\theta \neq \theta_0$, assumption (v) implies $W_0\mathbb{E}[g(z, \theta)] \neq 0$. This further implies that:

$$R\mathbb{E}[g(z, \theta)] \neq 0$$

Therefore, for any $\theta \neq \theta_0$:

$$\begin{aligned} Q_*(\theta) &= -\mathbb{E}[g(z, \theta)]'W_0\mathbb{E}[g(z, \theta)] \\ &= -\{R\mathbb{E}[g(z, \theta)]\}'\{R\mathbb{E}[g(z, \theta)]\} \\ &< 0 \end{aligned}$$

where the strict inequality follows from the fact that the inner product of a non-zero vector with itself is strictly positive. Thus, θ_0 is the unique maximizer of Q_* , and Condition 2 is verified. $\square$

This proof is hard, but allows us to see how setting primitive conditions yields the nice and general properties mentioned initially. We are now moving to the asymptotic normality counterpart. This is more difficult, but I really do find helpful to see how the asymptotic variance is derived.

Theorem 5.4: Asymptotic Normality of GMM

Suppose the assumptions for GMM consistency are satisfied. Additionally, assume:

- (i) $\theta_0 \in \text{int } \Theta$
- (ii) $g(z, \theta)$ is twice continuously differentiable in a neighborhood $\mathcal{N}$ of θ_0 with probability one
- (iii) $\mathbb{E}[|g(z, \theta_0)|^2] < \infty$, $\mathbb{E}\left[\left|\frac{\partial g(z, \theta_0)}{\partial \theta'}\right|\right] < \infty$, and $\mathbb{E}\left[\sup_{\theta \in \mathcal{N}} \left|\frac{\partial^2 g^{(j)}(z, \theta)}{\partial \theta \partial \theta'}\right|\right] < \infty$ for $j = 1, \dots, \dim g$
- (iv) $G'WG$ is nonsingular, where $G = \mathbb{E}\left[\frac{\partial g(z, \theta_0)}{\partial \theta'}\right]$

Then, the GMM estimator is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}\right)$$

where $\Omega = \mathbb{E}[g(z, \theta_0)g(z, \theta_0)']$ is the asymptotic variance of the moment conditions.

Proof. We provide a proof by construction, which is a bit different from what we have done in the last proof. We are not going to show why each primitive condition implies that 5.2 holds. Instead, we will construct the asymptotic distribution, highlighting where the assumptions come to play.

Consider $Q_n(\theta) = -\frac{1}{2} [g_n(\theta)' \hat{W} g_n(\theta)] \equiv f(g(\theta))$. Assuming i) and ii) allows us to write the MVT:

$$\sqrt{n}(\hat{\theta} - \theta_0) = - \left(\frac{\partial^2 Q_n(\tilde{\theta})}{\partial \theta \partial \theta'} \right)^{-1} \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta}$$

Let us take the first derivative:

$$\frac{\partial Q_n(\theta)}{\partial \theta} = - \left(\frac{\partial g_n(\theta)'}{\partial \theta} \right) \left(\frac{\hat{W} + \hat{W}'}{2} \right) g_n(\theta) = - \left(\frac{\partial g_n(\theta)}{\partial \theta'} \right)' \left(\frac{\hat{W} + \hat{W}'}{2} \right) g_n(\theta)$$

where the first equality comes from the Chain rule²¹ and the second inequality from the Transpose rule.²² Now, if we the second assumption of iii), that is $\mathbb{E}[|\frac{\partial g(z, \theta_0)}{\partial \theta'}|] < \infty$, we have that

$$\frac{\partial g_n(z, \theta_0)}{\partial \theta'} \xrightarrow{p} \mathbb{E} \left[\frac{\partial g(z, \theta_0)}{\partial \theta'} \right] \equiv G$$

Additionally, by consistency assumptions,

$$\frac{\hat{W} + \hat{W}'}{2} \xrightarrow{p} W \text{ by WLLN}$$

Now, we move to the variance, which can be written as follows by iid

$$\mathbb{V}(g_n(z_i, \theta_0)) - \mathbb{E}(g_n(z_i, \theta_0)g_n(z_i, \theta_0)') \equiv \Omega < \infty$$

It is finite by the first assumption of iii), $\mathbb{E}[|g(z, \theta_0)|^2] < \infty$ and Cauchy-Schwarz inequality. By Multivariate CLT, we have that

$$\sqrt{n}g_n(z_i, \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n g(z_i, \theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega)$$

²¹To take the derivative with respect to θ , we use the rule for a quadratic form $f(u) = u'Mu$ where u is a function of θ :

$$\frac{\partial f}{\partial \theta} = \frac{\partial u'}{\partial \theta} Mu + \frac{\partial u'}{\partial \theta} M' u = \frac{\partial u'}{\partial \theta} (M + M') u$$

²²Transpose rule:

$$\frac{\partial \bar{g}(\theta)'}{\partial \theta} = \left(\frac{\partial \bar{g}(\theta)}{\partial \theta'} \right)'$$

Therefore, by CMT, we have that

$$\sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} \xrightarrow{d} -G'W \cdot \mathcal{N}(0, \Omega) = \mathcal{N}(0, G'W\Omega WG) \equiv \mathcal{N}(0, \Sigma)$$

We can move to the denominator. Differentiating again with respect to θ' , we treat the score as a product of two functions of θ : the Jacobian term and the moment term. Using the product rule $(f \cdot g)' = f'g + fg'$, the j -th column of the Hessian (derivative with respect to θ_j) is expressed as:

$$\frac{\partial^2 Q_n(\theta)}{\partial \theta \partial \theta_j} = - \underbrace{\left(\frac{\partial^2 g_n(\theta)'}{\partial \theta \partial \theta_j} \right) \left(\frac{\hat{W} + \hat{W}'}{2} \right) g_n(\theta)}_{T_1(\theta)} - \underbrace{\left(\frac{\partial g_n(\theta)'}{\partial \theta} \right) \left(\frac{\hat{W} + \hat{W}'}{2} \right) \left(\frac{\partial g_n(\theta)}{\partial \theta_j} \right)}_{T_2(\theta)}$$

Now, assuming the last part of iii), that is $\mathbb{E} \left[\sup_{\theta \in \mathcal{N}} \left| \frac{\partial^2 g^{(j)}(z, \theta)}{\partial \theta \partial \theta'} \right| \right] < \infty$ for $j = 1, \dots, \dim g$, we have by ULLN that

$$T_1(\theta) = O_p(1) \cdot O_p(1) \cdot o_p(1) = o_p(1)$$

noting that $\mathbb{E}(g(z, \theta_0)) = 0$ by GMM assumption. We also have that, by ULLN,

$$T_2(\theta) \xrightarrow{p} G'WG \equiv H$$

which we assume to be non-singular in assumption iv).

Therefore, by CMT, we obtain the desired result

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, H^{-1}\Sigma H^{-1}) \text{ with } H^{-1}\Sigma H^{-1} = (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}$$

□

5.4 Optimal GMM

We can now introduce the optimal weighting matrix, that is the weighting matrix that minimizes the asymptotic variance.

Theorem 5.5: Optimal Weighting Matrix for GMM

The GMM asymptotic variance

$$V(W) = (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}$$

is minimized in the matrix sense (i.e., $V(W) - V(\Omega^{-1})$ is positive semi-definite) by choosing $W = \Omega^{-1}$. Under this choice, the variance collapses to:

$$\text{Avar}(\hat{\theta}) = (G'\Omega^{-1}G)^{-1}$$

Proof. We want to show that for any symmetric positive definite matrix W , $V(W) - V(\Omega^{-1})$ is a positive semi-definite matrix.

Let $A = (G'WG)^{-1}G'W$ and $B = G$. Note that by construction, $AB = I$. We are comparing:

$$V(W) = A\Omega A' \quad \text{and} \quad V(\Omega^{-1}) = (G'\Omega^{-1}G)^{-1}$$

Consider the following matrix, which is positive semi-definite by construction (as it is of the form XX'):

$$(A - (G'\Omega^{-1}G)^{-1}G'\Omega^{-1})\Omega(A - (G'\Omega^{-1}G)^{-1}G'\Omega^{-1})' \geq 0$$

Expanding this product:

$$\begin{aligned} & A\Omega A' - A\Omega\Omega^{-1}G(G'\Omega^{-1}G)^{-1} - (G'\Omega^{-1}G)^{-1}G'\Omega^{-1}\Omega A' \\ & + (G'\Omega^{-1}G)^{-1}G'\Omega^{-1}\Omega\Omega^{-1}G(G'\Omega^{-1}G)^{-1} \end{aligned}$$

Using the fact that $\Omega\Omega^{-1} = I$ and $AG = I$:

$$\begin{aligned} & = A\Omega A' - (AG)(G'\Omega^{-1}G)^{-1} - (G'\Omega^{-1}G)^{-1}(G'A') + (G'\Omega^{-1}G)^{-1}(G'\Omega^{-1}G)(G'\Omega^{-1}G)^{-1} \\ & = A\Omega A' - (G'\Omega^{-1}G)^{-1} - (G'\Omega^{-1}G)^{-1} + (G'\Omega^{-1}G)^{-1} \\ & = A\Omega A' - (G'\Omega^{-1}G)^{-1} \end{aligned}$$

Since we started with a quadratic form that is ≥ 0 , we have established that:

$$(G'WG)^{-1}G'W\Omega WG(G'WG)^{-1} \geq (G'\Omega^{-1}G)^{-1}$$

The equality holds when $A = (G'\Omega^{-1}G)^{-1}G'\Omega^{-1}$, which is satisfied when $W = \Omega^{-1}$. $\square$

The problem is of course that the optimal weight Ω is a function of θ_0 , and is therefore infeasible. This is why we introduce what is called the Two-step GMM.

Definition 5.2: Two Step GMM

The two step GMM is computed as follows.

1. Compute a 1st step GMME $\hat{\theta}_W$ using some W_n (e.g. $W_n = I_n$):

$$\hat{\theta}_W = \arg \min_{\theta \in \Theta} g_n(\theta)' W_n g_n(\theta)$$

2. Estimate $\hat{\Omega}$ as follows

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n g(z_i, \hat{\theta}_W) g(z_i, \hat{\theta}_W)'$$

3. Compute the 2-step GMME using the estimated optimal weight matrix $\hat{\Omega}^{-1}$

$$\hat{\theta}_{2step} = \arg \min_{\theta \in \Theta} g_n(\theta)' \hat{\Omega}^{-1} g_n(\theta)$$

Note that $\hat{\theta}_{2step}$ is more efficient than $\hat{\theta}_W$. Additionally, steps 2-3 can be repeated (which is called repeated or iterated GMM). While the 2-step GMM estimator is asymptotically efficient, the estimation of Ω depends on the initial (sub-optimal) $\hat{\theta}_W$. This suggests that repeating steps 2 and 3 iteratively—using the latest $\hat{\theta}$ to update $\hat{\Omega}$ —can improve finite sample performance.

The procedure is as follows. We continue updating $\hat{\Omega}(\hat{\theta}_{i-1})$ and re-estimating $\hat{\theta}_i$ until the difference $\|\hat{\theta}_i - \hat{\theta}_{i-1}\|$ falls below a small tolerance ϵ_θ (e.g., 10^{-6}). Note that all iterations beyond the second step share the same asymptotic distribution as the 2-step estimator. The primary gain is higher efficiency in finite samples.

5.5 Testing in GMM

5.5.1 Sargan-Hansen Test

One of the main tests in this context has to do with overidentified moment validity. The null/alternative hypotheses can be written as

$$\mathcal{H}_0 : \mathbb{E}(g(z_i, \theta)) = 0 \text{ for some } \theta \in \Theta \text{ v.s. } \mathcal{H}_1 : \mathbb{E}(g(z_i, \theta)) \neq 0 \text{ for all } \theta \in \Theta$$

Rejecting the null hypothesis here means that at least one moment condition does not hold in the data.

Definition 5.3: J-statistic

The J-statistic is written as follows

$$J_n = n \min_{\theta \in \Theta} g_n(\theta)' \hat{\Omega}^{-1} g_n(\theta) = n g_n(\hat{\theta})' \hat{\Omega}^{-1} g_n(\hat{\theta})$$

The intuition is that if the model is correct, $g_n(\hat{\theta})$ and J_n have to be close to zero. We basically need $g_n(\hat{\theta})' g_n(\hat{\theta})$ to go to zero faster than n goes to infinity. That is, we should have $g_n(\hat{\theta})' g_n(\hat{\theta}) = o(\frac{1}{n^\delta})$ for $\delta > 1$. This is a sufficient condition if we assume that $\hat{\Omega}^{-1} \xrightarrow{p} \Omega^{-1}$, meaning that $\hat{\Omega}^{-1} = O(1)$.

In other words, we know that $\sqrt{n}g_n(\hat{\theta})$ converges to a normal distribution. Since $\sqrt{n}g_n(\hat{\theta}) = O_p(1)$, it follows that:

$$g_n(\hat{\theta}) = O_p\left(\frac{1}{\sqrt{n}}\right)$$

Therefore, the quadratic form $g_n(\hat{\theta})' \hat{\Omega}^{-1} g_n(\hat{\theta})$ is $O_p(\frac{1}{n})$. When you multiply this by the n out front in the J-statistic formula, you get an $O_p(1)$ object, which allows it to converge to a χ^2 distribution rather than collapsing to zero or diverging to infinity. Therefore, under $\mathcal{H}_0$, we have that

$$J_n \xrightarrow{d} \chi^2_{(J-k)}$$

Note that J-test (or Sargan-Hansen test) is only applicable in overidentified models ($J > k$). A rejection suggests either that the moment conditions are invalid (instruments are not

exogenous) or that the functional form of the model is misspecified. It requires the use of the efficient (optimal) weighting matrix $\hat{\Omega}^{-1}$; if a sub-optimal W is used, the statistic does not follow a χ^2 distribution.

5.5.2 Hausman Test

The test compares two estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$, that behave differently under $\mathcal{H}_0$ and $\mathcal{H}_1$. $\hat{\theta}_1$ is consistent and efficient under $\mathcal{H}_0$, but inconsistent under $\mathcal{H}_1$. On the other hand, $\hat{\theta}_2$ is consistent under both $\mathcal{H}_0$ and $\mathcal{H}_1$, but is inefficient under $\mathcal{H}_0$. The idea is that if $\mathcal{H}_0$ is true, both estimators should converge to the same value, so the difference $(\hat{\theta}_2 - \hat{\theta}_1)$ should be close to zero. If it is far from zero, we reject $\mathcal{H}_0$ in favor of $\mathcal{H}_1$.

Example 5.1: Hausman Exogeneity Test: OLS vs. 2SLS

Consider a linear regression model $y_i = x_i' \beta_0 + \varepsilon_i$, where we suspect some regressors in x_i may be endogenous. We test the following hypotheses:

- $\mathcal{H}_0 : E[x_i \varepsilon_i] = 0$ (Exogeneity)
- $\mathcal{H}_1 : E[x_i \varepsilon_i] \neq 0$ (Endogeneity)

We compare two estimators with different properties:

- $\hat{\beta}_{OLS}$: Consistent and efficient under $\mathcal{H}_0$, but inconsistent under $\mathcal{H}_1$.
- $\hat{\beta}_{2SLS}$: Consistent under both $\mathcal{H}_0$ and $\mathcal{H}_1$, but inefficient under $\mathcal{H}_0$.

Let $\hat{q} = \hat{\beta}_{2SLS} - \hat{\beta}_{OLS}$. Under $\mathcal{H}_0$, $\hat{q} \xrightarrow{p} 0$ because both estimators are consistent. Under $\mathcal{H}_1$, $\hat{q} \xrightarrow{p} \text{plim } \hat{\beta}_{2SLS} - \text{plim } \hat{\beta}_{OLS} \neq 0$.

The Hausman test statistic H is defined as:

$$H = n(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})' \hat{V}_q^{-1} (\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) \xrightarrow{d} \chi^2(k)$$

where k is the dimension of β_0 . Crucially, because $\hat{\beta}_{OLS}$ is efficient under $\mathcal{H}_0$, Hausman showed that the variance of the difference simplifies to:

$$\hat{V}_q = \hat{V}_{2SLS} - \hat{V}_{OLS}$$

We reject exogeneity if $H > \chi^2_{1-\alpha}(k)$.

In practice, it is often difficult to know with certainty which instruments are truly valid. If both estimators use invalid instruments, the test loses its ability to distinguish between the models.

Additionally, the test relies on the difference in asymptotic variances $[\hat{V}(\hat{\beta}_{2SLS}) - \hat{V}(\hat{\beta}_{OLS})]$. If the additional instruments in the second-stage estimator do not actually improve the

asymptotic variance (i.e., they add no new information), this matrix difference may not be invertible, causing the test to fail.

Finally, if the "untrusted" instrument is invalid but happens to be uncorrelated with both the "trusted" instruments and the endogenous variables, $\hat{\beta}_1$ (the efficient estimator) might remain consistent even under $\mathcal{H}_1$. In this specific case, the test would fail to reject the null, despite the instrument being technically invalid.

6 Non-linear Regression

6.1 The Model

So far, we focused mostly on linear projection or regression model. We now consider the following nonlinear regression model

$$y_i = m(x_i, \theta_0) + e_i; \mathbb{E}(e_i | x_i) = 0$$

For example, we could have $m(x, \theta) = \exp(x'\theta)$, $m(x, \theta) = \theta_1 + \theta_2 x^{\theta_3}$, etc. One way to estimate θ_0 is to apply the least squares method where the Nonlinear least squares (NLLS) estimator is

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta))^2$$

where Θ is the parameter space.

6.2 Consistency

As in the GMM section, it is easy to write the NLLS estimator as an extremum estimator. Indeed, let

$$Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta))^2 \implies \hat{\theta} = \arg \max_{\theta \in \Theta} Q_n(\theta)$$

We can now write a set of primitive conditions that guarantee the General consistency Theorem (5.1).

Theorem 6.1: Consistency of NLLSE

Suppose the following conditions hold

- (i) Θ is compact
- (ii) $\{y_i, x_i'\}_{i=1}^n$ is iid and $\mathbb{E}(y|x) = m(x, \theta)$ almost surely only if $\theta = \theta_0$
- (iii) $m(x, \theta)$ is almost surely continuous at each $\theta \in \Theta$
- (iv) $\mathbb{E}(y^2) < \infty$ and $\mathbb{E}(\sup_{\theta \in \Theta} |m(x, \theta)|^2) < \infty$

Then, we achieve consistency:

$$\hat{\theta} \xrightarrow{p} \theta_0$$

Proof. As before, it is sufficient to check conditions 1-4 from Theorem 5.1. Condition 1 is guaranteed by (i). We now work towards proving uniform convergence (condition 4). In this case,

$$Q_*(\theta) = -\mathbb{E} \left((y - m(x, \theta))^2 \right)$$

so condition 4 amounts to checking

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta))^2 - \mathbb{E} \left((y - m(x, \theta))^2 \right) \right| \xrightarrow{p} 0$$

$\{y_i, x_i'\}_{i=1}^n$ iid ensures that the first condition for ULLN (Lemma 5.1) is satisfied. Condition (iii) (almost sure continuity) ensure that the second condition for ULLN is satisfied. Most work should be done for the third condition. We can rewrite

$$\begin{aligned} \mathbb{E}(\sup_{\theta \in \Theta} |y - m(x, \theta)|^2) &= \mathbb{E} \left(\sup_{\theta \in \Theta} |y^2 - 2m(x, \theta)y + m(x, \theta)^2| \right) \\ &\leq \mathbb{E}(y^2) + \mathbb{E}(\sup_{\theta \in \Theta} m(x, \theta)^2) + 2\mathbb{E}(\sup_{\theta \in \Theta} |m(x, \theta)y|) \end{aligned}$$

where used the triangle inequality. The first two terms are bounded by assumption (iv). The last term is bounded by Cauchy-Schwarz Inequality

$$2\mathbb{E}(\sup_{\theta \in \Theta} |m(x, \theta)y|) \leq 2 \left(\mathbb{E}(|\sup_{\theta \in \Theta}|^2) \right)^{\frac{1}{2}} \left(\mathbb{E}(y^2) \right)^{\frac{1}{2}} < \infty$$

Uniform convergence therefore follows from ULLN. Additionally, ULLN also guarantees of $Q_*(\theta)$, so condition 3 is also satisfied. We finish the proof by showing identification (condition 2). We want to show that $Q_*(\theta) = -\mathbb{E}((y - m(x, \theta))^2)$ is uniquely maximised at θ_0 . Recall that CEF $m(x) = \mathbb{E}(y|x)$ solves $\min_g \mathbb{E}((y - g(x))^2)$. Thus, by condition (i), condition 4 follows. $\square$

I explain the last part of the proof now. The idea is that we say that $m(x, \theta) = \mathbb{E}(y|x)$ only for $\theta = \theta_0$, and we prove that $m(x, \theta) = \mathbb{E}(y|x)$ uniquely minimizes the MSE. Hence, addition condition (i) ensure θ_0 is uniquely identified, as desired. Formally, let $m(x) = \mathbb{E}(y|x)$ and $a(x)$ any function such that $\mathbb{V}(a(x)) < \infty$. We can write

$$\begin{aligned} \mathbb{E}[(y - a(x))^2] &= \mathbb{E}[(y - m(x)) + (m(x) - a(x))]^2 \\ &= \mathbb{E}[(y - m(x))^2] + 2\mathbb{E}[(y - m(x))(m(x) - a(x))] + \mathbb{E}[(m(x) - a(x))^2]. \end{aligned}$$

Now, developing the terms and by LIE

$$\begin{aligned}
\mathbb{E}[(y - m(x))(m(x) - a(x))] &= \mathbb{E}[y m(x) - y a(x) - m(x)^2 + m(x)a(x)] \\
&= \mathbb{E}[y(m(x) - a(x)) - m(x)^2 + m(x)a(x)] \\
&= \mathbb{E}\left[\mathbb{E}[y(m(x) - a(x)) - m(x)^2 + m(x)a(x) \mid x]\right] \\
&= \mathbb{E}\left[(m(x) - a(x))\mathbb{E}[y \mid x] - m(x)^2 + m(x)a(x)\right] \\
&= \mathbb{E}[(m(x) - a(x))m(x) - m(x)^2 + m(x)a(x)] \\
&= \mathbb{E}[m(x)^2 - a(x)m(x) - m(x)^2 + m(x)a(x)] \\
&= 0.
\end{aligned}$$

Therefore,

$$\mathbb{E}[(y - a(x))^2] = \mathbb{E}[(y - m(x))^2] + \mathbb{E}[(m(x) - a(x))^2] \geq \mathbb{E}[(y - m(x))^2].$$

Now, if $a(x) \neq m(x)$, then

$$\mathbb{E}[(m(x) - a(x))^2] > 0,$$

so

$$\mathbb{E}[(y - a(x))^2] > \mathbb{E}[(y - m(x))^2] \quad \forall a(x) \neq m(x).$$

Hence the mean squared error has a unique minimum at

$$m(x) = \mathbb{E}[y \mid x]$$

6.3 Asymptotic Normality

We now move to the asymptotic normality theorem.

Theorem 6.2: Asymptotic Normality of NLLSE

Suppose the assumptions for NLLSE consistency are satisfied. Additionally, assume:

- (i) $\theta_0 \in \text{int } \Theta$
- (ii) $m(x, \theta)$ is twice continuously differentiable in a neighborhood $\mathcal{N}$ of θ_0 with probability one
- (iii) $\mathbb{E}[y^4] < \infty$, $\mathbb{E}[|\frac{\partial m(x, \theta_0)}{\partial \theta}|^4] < \infty$, and $\mathbb{E}\left[\sup_{\theta \in \mathcal{N}} \left|\frac{\partial^2 m(x, \theta)}{\partial \theta \partial \theta'}\right|^2\right] < \infty$
- (iv) $H \equiv \mathbb{E}\left[\frac{\partial m(x, \theta_0)}{\partial \theta} \frac{\partial m(x, \theta_0)}{\partial \theta'}\right]$ is nonsingular

Then, the NLLS estimator is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, H^{-1}\Sigma H^{-1}\right)$$

where $\Sigma \equiv \mathbb{E}\left[e^2 \frac{\partial m(x, \theta_0)}{\partial \theta} \frac{\partial m(x, \theta_0)}{\partial \theta'}\right]$ and $e \equiv y - m(x, \theta_0)$

Proof. As for the GMM proof, we provide a proof by construction highlighting where each primitive assumptions come to play. Consider $Q_n(\theta) := -\frac{1}{2n} \sum_{i=1}^n (y_i - m(x_i, \theta))^2$. Assuming i) and ii) allows us to write the MVT:

$$\sqrt{n}(\hat{\theta} - \theta_0) = - \left(\frac{\partial^2 Q_n(\tilde{\theta})}{\partial \theta \partial \theta'} \right)^{-1} \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta}$$

Let us focus on the first order term first. By Chain Rule, we have

$$\sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta_0)) \frac{\partial m(x_i, \theta_0)}{\partial \theta} \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n e_i \frac{\partial m(x_i, \theta_0)}{\partial \theta}$$

where $e_i \equiv m(x_i, \theta_0)$. Now, we can define the variance as follows

$$\Sigma := \mathbb{V} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n e_i \frac{\partial m(x_i, \theta_0)}{\partial \theta} \right)$$

Define

$$s_i := e_i \frac{\partial m(x_i, \theta_0)}{\partial \theta},$$

Then

$$\sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta} = \frac{1}{\sqrt{n}} \sum_{i=1}^n s_i.$$

By i.i.d. sampling,

$$\mathbb{V} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n s_i \right) = \frac{1}{n} \sum_{i=1}^n \mathbb{V}(s_i) = \mathbb{V}(s_1).$$

Now,

$$\mathbb{V}(s_1) = \mathbb{E}[s_1 s_1'] - \mathbb{E}[s_1] \mathbb{E}[s_1'].$$

Since $\mathbb{E}[e_1 | x_1] = 0$,

$$\mathbb{E}[s_1] = \mathbb{E} \left[e_1 \frac{\partial m(x_1, \theta_0)}{\partial \theta} \right] = \mathbb{E} \left[\frac{\partial m(x_1, \theta_0)}{\partial \theta} \mathbb{E}[e_1 | x_1] \right] = 0.$$

Therefore,

$$\Sigma = \mathbb{V} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n e_i \frac{\partial m(x_i, \theta_0)}{\partial \theta} \right) = \mathbb{E} \left[e_i^2 \frac{\partial m(x_i, \theta_0)}{\partial \theta} \frac{\partial m(x_i, \theta_0)}{\partial \theta'} \right]$$

To apply the Central Limit Theorem for iid data, we need $\Sigma < \infty$. By Hölder's Inequality, we have

$$\mathbb{E} \left[e_i^2 \frac{\partial m(x_i, \theta_0)}{\partial \theta} \frac{\partial m(x_i, \theta_0)}{\partial \theta'} \right] \leq \left[\mathbb{E}(e^4) \right]^{\frac{1}{4}} \left[\mathbb{E} \left(\left(\frac{\partial m(x_i, \theta_0)}{\partial \theta} \right)^4 \right) \right]^{\frac{1}{4}} < \infty$$

where the last inequality follows from the first two assumptions in (iii).
Now, for the second order term, we can write

$$\frac{\partial}{\partial \theta'} \left[\frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta_0)) \frac{\partial m(x_i, \theta_0)}{\partial \theta} \right] = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 m(x_i, \theta_0)}{\partial \theta \partial \theta'} (y_i - m(x_i, \theta_0)) - \frac{1}{n} \sum_{i=1}^n \left[\frac{\partial m(x_i, \theta_0)}{\partial \theta} \frac{\partial m(x_i, \theta_0)}{\partial \theta'} \right]$$

Addind the third assumption in (iii), i.e. $\mathbb{E} \left[\sup_{\theta \in \mathcal{N}} \left| \frac{\partial^2 m(x, \theta)}{\partial \theta \partial \theta'} \right|^2 \right] < \infty$, the first term vanishes asymptotically as $\mathbb{E}(e|x) = 0$. The second term converges to $H \equiv \mathbb{E} \left[\frac{\partial m(x, \theta_0)}{\partial \theta} \frac{\partial m(x, \theta_0)}{\partial \theta'} \right]$. Assuming (iv), that is non-singularity of H yields the result. $\square$

6.4 Numerical Computation of the NLLS Estimator

In practice, the NLLS estimator does not admit a closed form solution and must be computed numerically:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta))^2.$$

If $\dim(\theta)$ is small, one can perform a grid search over the parameter space. However, in higher dimensions, this quickly becomes infeasible. Instead, when the objective function

$$Q_n(\theta) := \frac{1}{n} \sum_{i=1}^n (y_i - m(x_i, \theta))^2$$

is sufficiently smooth, we can rely on gradient-based methods.

The estimator is characterized by the first order condition

$$\frac{\partial Q_n(\hat{\theta})}{\partial \theta} = 0.$$

Using a linear approximation around a current guess θ , we obtain

$$0 \approx \frac{\partial Q_n(\theta)}{\partial \theta} + \frac{\partial^2 Q_n(\theta)}{\partial \theta \partial \theta'} (\hat{\theta} - \theta),$$

which leads to the update rule

$$\theta^{(j+1)} = \theta^{(j)} - \left(\frac{\partial^2 Q_n(\theta^{(j)})}{\partial \theta \partial \theta'} \right)^{-1} \frac{\partial Q_n(\theta^{(j)})}{\partial \theta}.$$

This corresponds to a Newton-type method. The algorithm proceeds iteratively:

1. Choose an initial value $\theta^{(1)}$
2. Update using the rule above

3. Repeat until convergence, e.g. until $|\theta^{(j+1)} - \theta^{(j)}|$ is sufficiently small

More generally, we can consider updates of the form

$$\theta^{(j+1)} = \theta^{(j)} - \alpha^{(j)} D(\theta^{(j)}) \frac{\partial Q_n(\theta^{(j)})}{\partial \theta},$$

where $\alpha^{(j)} \in (0, 1)$ is a step size and $D(\theta^{(j)})$ is a direction matrix. Popular choices include quasi-Newton methods such as BFGS.

Finally, for non-smooth or highly irregular objective functions, derivative-free methods (e.g. simplex algorithms) can be used, often to obtain good starting values for gradient-based procedures.

7 Nonparametric Methods

7.1 Density estimation

Consider a sample $\{x_1, \dots, x_n\}$ of size n drawn from the distribution f . Our goal is to estimate this unknown distribution f without distributional assumptions, or nonparametrically. Fixing a point c , we could think of estimating $f(c)$, the value of the pdf at c , without imposing a parametric form. A first approach could be to use the empirical CDF

$$\hat{F}(x) \equiv \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \leq c\}$$

but this is not differentiable so we cannot use it to obtain the PDF. Indeed, $f(c)$ is the cdf $F(\cdot)$ at c :

$$f(c) = \frac{dF(x)}{dx} \Big|_{x=c} = \lim_{h \rightarrow 0} \frac{F(c + \frac{h}{2}) - F(c - \frac{h}{2})}{h} \approx \frac{\mathbb{P}(c - \frac{h}{2} \leq x_i \leq c + \frac{h}{2})}{h}, \text{ with small } h > 0$$

where we use the definition of a derivative. Now, we can estimate $\mathbb{P}(c - \frac{h}{2} \leq x_i \leq c + \frac{h}{2})$ as follows

$$\hat{Pr}(c - \frac{h}{2} \leq x_i \leq c + \frac{h}{2}) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{c - \frac{h}{2} \leq x_i \leq c + \frac{h}{2}\}$$

where we use the fact that for any event A_i , $\mathbb{P}(A_i) = \mathbb{E}(\mathbb{1}(A_i))$ and estimate this expectation in the sample. This estimated probability simply counts the number of time x_i falls between $c - \frac{h}{2}$ and $c + \frac{h}{2}$. This construction gives us the *histogram estimator*.

Definition 7.1: The Histogram Estimator

The histogram estimator is given by

$$\hat{f}(c) = \frac{1}{nh} \sum_{i=1}^n \mathbb{1}\{c - \frac{h}{2} \leq x_i \leq c + \frac{h}{2}\}$$

where h is called the bandwidth.

A histogram simply divides the sample space into bins B_ℓ of width h .

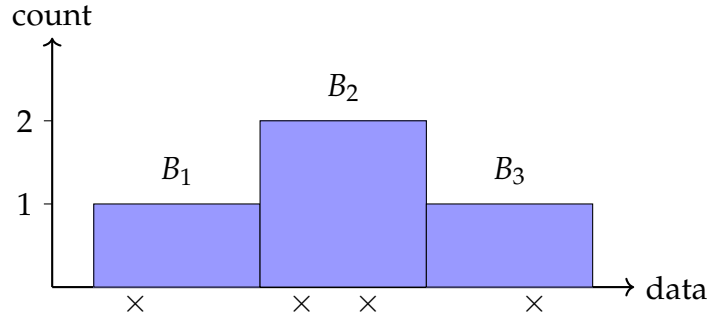


Figure 4: Illustration of a histogram estimator with bandwidth h .

Notice that the histogram depends on both the bin length and the bin origin, which adds unnecessary tuning parameters. Now, the histogram estimator can be written as

$$\hat{f}(c) = \frac{1}{nh} \sum_{i=1}^n \mathbb{1}\{c - \frac{h}{2} \leq x_i \leq c + \frac{h}{2}\} = \frac{1}{nh} \sum_{i=1}^n \mathbb{1}\{-\frac{1}{2} \leq \frac{x_i - c}{h} \leq \frac{1}{2}\} = \frac{1}{nh} \sum_{i=1}^n \mathbb{1}\{-\frac{1}{2} \leq u_i \leq \frac{1}{2}\}$$

where $u_i \equiv \frac{x_i - c}{h}$. Notice that $K(u) \equiv \mathbb{1}\{-\frac{1}{2} \leq u \leq \frac{1}{2}\}$ is the pdf of the $Uniform[-\frac{1}{2}, \frac{1}{2}]$. This specific kernel function defines a *parzen window*, which is very similar to a histogram but the hypercubes are centered around the inputs datapoints (=bin origin).

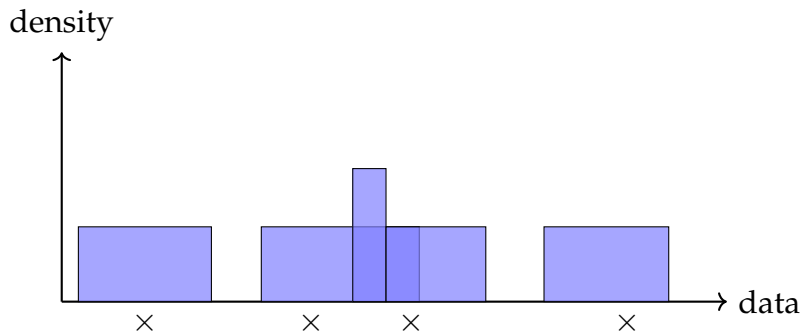


Figure 5: Parzen window illustration

Note that the windows is not perfect since it suffers from discontinuities due to the specific kernel function, and datapoints are weighted equally regardless of their distance to the estimated point x . This leads us to the kernel density estimator.

Definition 7.2: Kernel Density Estimator

Consider an iid sample $\{x_i\}_{i=1}^n$ with density function $f(x)$. Fix c . The kernel density estimator of $f(c)$ is defined as

$$\hat{f}(c) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - c}{h}\right)$$

where $K(\cdot)$ is called kernel function.

Below are some examples of Kernel functions.

Example 7.1: Common Kernel Functions

1. Uniform (Rectangular) kernel

$$K(u) = \frac{1}{2\sqrt{3}} \mathbb{1}\{-\sqrt{3} \leq u \leq \sqrt{3}\}.$$

2. Normal kernel

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right).$$

3. Epanechnikov kernel

$$K(u) = \frac{3}{4\sqrt{5}} \left(1 - \frac{u^2}{5}\right) \mathbb{1}\{-\sqrt{5} \leq u \leq \sqrt{5}\}.$$

4. Triangular kernel

$$K(u) = \frac{1}{\sqrt{6}} \left(1 - \frac{|u|}{\sqrt{6}}\right) \mathbb{1}\{-\sqrt{6} \leq u \leq \sqrt{6}\}.$$

7.2 Bias-Variance Trade-off and Asymptotics

This section discusses the bias-variance trade off that exists in this context between estimating something close to the truth (small bias) and with good precision (small variance). In this case, the larger the bandwidth h , the more observations are used and therefore the lower the variance. However, the larger h , the less well we fit the density locally. Deriving the asymptotes of both object is not trivial but helps understand what an optimal bandwidth would look like. We first put assumptions on our kernel function, and then provide expressions for the first moment, variance and provide a consistency result.

Assumption 7.1: Kernel Density Estimation

To analyze the properties of $\hat{f}(c)$, suppose:

1. $\{x_i\}_{i=1}^n$ is an i.i.d. sample with marginal pdf $f(\cdot)$.
2. c is an interior point of the support of x , and f is twice continuously differentiable in a neighborhood of c .
3. The kernel $K(\cdot)$ is a bounded pdf with bounded support $[-a, a]$, symmetric around zero, and satisfies

$$\int u^2 K(u) du = 1.$$

4. As $n \rightarrow \infty$, the bandwidth satisfies

$$h \rightarrow 0 \quad \text{and} \quad nh \rightarrow \infty.$$

Notice that symmetry of the kernel implies that

$$\int u K(u) du = 0$$

Additionally, the last assumption in 3. is a simplification. In practice, we only need the second moment to be bounded. Now that we have these assumptions, we can provide the main Theorem for this section.

Theorem 7.1: Bias, Variance and Consistency

Under conditions 1-4, we have

1. $\mathbb{E}(\hat{f}(c)) = f(c) + \frac{h^2}{2} f''(c) + o(h^2)$
2. $\mathbb{V}(\hat{f}(c)) = \frac{1}{nh} f(c) \int K(u)^2 du + o(\frac{1}{nh})$
3. $\hat{f}(c) \xrightarrow{p} f(c)$

These formulae characterize the effect of the bandwidth. If h is small, bias becomes smaller but variance becomes larger (and the other way around). For consistency, we need both $h \rightarrow 0$ and $nh \rightarrow \infty$

Proof. We start by deriving the mean of $\hat{f}(c)$. We can write

$$\mathbb{E}(\hat{f}(c)) = \frac{1}{h} \mathbb{E} \left[K \left(\frac{x_i - c}{h} \right) \right] = \frac{1}{h} \int K \left(\frac{x_i - c}{h} \right) f(x) dx$$

which follows by iid of $\{x_i\}_{i=1}^n$ and by definition of the expectation operator. Now, we proceed with a change of variables $u \equiv \frac{x-c}{h}$, which implies $f(x) = f(c + hu)$ and $du = \frac{1}{h} dx$. This allows us to rewrite

$$\mathbb{E}(\hat{f}(c)) = \int K(u) f(c + hu) du$$

Now, the Taylor expansion²³ of $f(c + hu)$ around c implies

$$\mathbb{E}(\hat{f}(c)) = \int K(u) \left\{ f(c) + f'(c)(hu) + \frac{1}{2}f''(c + hu^*)(hu)^2 \right\} du, \text{ for } u^* \in (0, u)$$

Now, we can develop the expression as follows

$$\begin{aligned} \mathbb{E}(\hat{f}(c)) &= f(c) \int K(u) du + hf'(c) \int K(u) u du + \frac{h^2}{2} \int K(u) f''(c + hu^*) u^2 du \\ &= f(c) + \frac{h^2}{2} \int K(u) f''(c + hu^*) u^2 du \\ &= f(c) + \frac{h^2}{2} f''(c) \int K(u) u^2 du + h^2 \int K(u) \{f''(c + hu^*) - f''(c)\} u^2 du \\ &= f(c) + \frac{h^2}{2} f''(c) + h^2 R_1, \text{ where } R_1 \equiv \frac{1}{2} \int K(u) \{f''(c + hu^*) - f''(c)\} u^2 du \end{aligned}$$

In the second equality, we have used the fact that densities integrate to one and symmetry of kernels. The penultimate equality simply adds and subtracts $f''(c) \int K(u) u^2 du$ and the last equality rearranges and uses the normalisation of the second moment from assumption 3.

It simply remains to show that $R_1 = o(1)$ as $h \rightarrow 0$. Pick any $\varepsilon > 0$ arbitrarily small. Since f'' is continuous in a neighborhood of c , there exists $\delta_\varepsilon > 0$ such that

$$|f''(c + v) - f''(c)| \leq \varepsilon, \forall v \in [-\delta_\varepsilon, \delta_\varepsilon]$$

For any $h \leq \frac{\delta_\varepsilon}{a}$, $|u| \leq a$ implies $|hu^*| \leq |hu| \leq |ha| \leq \delta_\varepsilon$, which implies that

$$|R_1| \leq \frac{1}{2} \int |f''(c + hu^*) - f''(c)| u^2 K(u) du \leq \varepsilon, \forall h \leq \frac{\delta_\varepsilon}{a}$$

so $R_1 = o(1)$ as $h \rightarrow 0$. We have therefore reached the fact that $\mathbb{E}(\hat{f}(c)) = f(c) + \frac{h^2}{2} f''(c) + o(h^2)$ where the bias is given by

$$B(\hat{f}(c)) \equiv \frac{h^2}{2} f''(c) + o(h^2)$$

We can now move to the variance. Using independence and the formula $\mathbb{V}(\sum_i Z_i) = \sum_i \mathbb{V}(Z_i)$ we get

$$\mathbb{V}(\hat{f}(c)) = \frac{1}{n^2 h^2} n \mathbb{V}\left(K\left(\frac{X_i - c}{h}\right)\right) = \frac{1}{n h^2} \left\{ \mathbb{E}\left[K\left(\frac{X_i - c}{h}\right)^2\right] - \left(\mathbb{E}\left[K\left(\frac{X_i - c}{h}\right)\right]\right)^2 \right\}.$$

We evaluate the two expectations by the change of variables $u = (x - c)/h$.

²³ $f(c + hu) \approx f(c) + f'(c)(c + hu - c) + \frac{1}{2}f''(c + hu^*)((c + hu - c)^2)$

First,

$$\mathbb{E}\left[K\left(\frac{X_i - c}{h}\right)^2\right] = \int K\left(\frac{x - c}{h}\right)^2 f(x) dx = h \int K(u)^2 f(c + hu) du.$$

Hence the first term equals

$$\frac{1}{nh^2} \mathbb{E}\left[K\left(\frac{X_i - c}{h}\right)^2\right] = \frac{1}{nh} \int K(u)^2 f(c + hu) du.$$

Second,

$$\mathbb{E}\left[K\left(\frac{X_i - c}{h}\right)\right] = \int K\left(\frac{x - c}{h}\right) f(x) dx = h \int K(u) f(c + hu) du,$$

so

$$\frac{1}{nh^2} \left(\mathbb{E}\left[K\left(\frac{X_i - c}{h}\right)\right]\right)^2 = \frac{1}{n} \left(\int K(u) f(c + hu) du\right)^2 = O\left(\frac{1}{n}\right).$$

The last equality follows because the integral is $O(1)$ (indeed $\int K(u) f(c + hu) du \rightarrow f(c)$ as $h \rightarrow 0$).

Combining the two pieces,

$$\mathbb{V}(\hat{f}(c)) = \frac{1}{nh} \int K(u)^2 f(c + hu) du + O\left(\frac{1}{n}\right).$$

To obtain the usual leading form, expand $f(c + hu) = f(c) + o(1)$ (continuity of f at c) and write

$$\int K(u)^2 f(c + hu) du = f(c) \int K(u)^2 du + R_2(h),$$

where

$$R_2(h) \equiv \int K(u)^2 \{f(c + hu) - f(c)\} du \xrightarrow{h \rightarrow 0} 0,$$

by continuity of f and bounded support/boundedness of K .

Therefore

$$\mathbb{V}(\hat{f}(c)) = \frac{1}{nh} f(c) \int K(u)^2 du + \frac{1}{nh} R_2(h) + O\left(\frac{1}{n}\right).$$

Under the bandwidth conditions $h \rightarrow 0$ and $nh \rightarrow \infty$ (Assumption 4), the remainder terms satisfy

$$\frac{1}{nh} R_2(h) = o\left(\frac{1}{nh}\right), \quad O\left(\frac{1}{n}\right) = o\left(\frac{1}{nh}\right),$$

so we conclude the standard asymptotic variance expansion

$$\boxed{\mathbb{V}(\hat{f}(c)) = \frac{1}{nh} f(c) \int K(u)^2 du + o\left(\frac{1}{nh}\right).}$$

Based on these formulae for mean and variance, condition 4 implies that

$$\mathbb{E}(\hat{f}(c)) \rightarrow f(c), \mathbb{V}(\hat{f}(c)) \rightarrow 0$$

Since convergence in mean square implies convergence in probability, we are done. $\square$

Now that we have done most of the heavy lifting, we can move to the optimal choice of h . One way to choose h is to perceive *optimality* in terms of mean squared error (MSE) of $\hat{f}(c)$. Recall the formula of MSE

$$MSE(c, h) = \mathbb{E} \left(\{\hat{f}(c) - f(c)\}^2 \right) = \mathbb{V}(\hat{f}(c)) + \mathbb{E} \left(\{\hat{f}(c) - f(c)\} \right)^2$$

Although we don't know the exact form of $MSE(c, h)$, we can compute the Approximate MSE (AMSE), using the asymptotic bias and variance

$$AMSE(c, h) = \frac{1}{4}h^4 f''(c)^2 + \frac{1}{nh} f(c) \int K(u)^2 du$$

Observe that this is a function of the sample size n , the bandwidth h and the kernel $K(u)$. As we hinted at before, the bias increases in h , while the variance decreases in h . Now, a global version of this measure is the approximate mean integrated squared error (AMISE), given by

$$\begin{aligned} AMISE(h) &= \int AMSE(x, h) dx = \frac{h^4}{4} \int_{c=x} (f''(x))^2 dx + \frac{1}{nh} \int K(u)^2 du \int f(x) dx \\ &= \frac{h^4}{4} \int_{c=x} (f''(x))^2 dx + \frac{1}{nh} \int K(u)^2 du \end{aligned}$$

AMISE is global: we want the global optimal bandwidth, when we don't want the bandwidth to depend on c . The optimal bandwidth are given by

$$h^*(c) = \arg \min_h AMSE(c, h) \text{ and } h^* = \arg \min_h AMSE(h)$$

In the first version, for each $c \in \mathbb{R}$, we have an $h(c)$ that minimizes the error. In the second case, each h generates a pdf that has different errors for each c , and there is one h that minimizes the sum of all these errors. Differentiating with respect to h gives the first-order condition for the AMISE-minimizing bandwidth h^* :

$$0 = \frac{\partial}{\partial h} AMISE(h) \approx \frac{\partial}{\partial h} \left(\frac{1}{4}h^4 A + \frac{1}{nh} B \right) = Ah^3 - \frac{B}{nh^2}$$

Equivalently,

$$Ah^5 = \frac{B}{n} \quad \implies \quad h^* = \left(\frac{B}{A} \right)^{1/5} n^{-1/5} = \left(\frac{\int K(u)^2 du}{\int f''(x)^2 dx} \right)^{1/5} n^{-1/5}$$

In particular, the optimal bandwidth satisfies the rate

$$h^* \propto n^{-1/5}$$

To pin down the number, we need to estimate the constant

$$\frac{\int K(u)^2 du}{\int f''(x)^2 dx}$$

$\int K(u)^2 du$ is a known constant, but $\int f''(x)^2 dx$ needs to be estimated. Note that the rate of convergence is slower than n^{-1} , which is the 'price to pay' for non-parametric estimation in general.

7.3 Non-parametric Regression/Function Approximation

What we did above is to approximate a density function, but we can also approximate $f(\cdot)$ in a different context, e.g.

$$Y = f(x) + \varepsilon$$

where x is scalar. This can be useful if we don't know the functional form of the true regression function and we have enough data. An idea could be to approximate the function by local averages.

Consider the nonparametric estimation of the conditional mean at a given point c

$$\mathbb{E}(y_i | x_i = c) = m(c)$$

If x_i is discrete, the conditional mean is estimated by the average of y_i 's with $x_i = c$:

$$\frac{\sum_{i=1}^n \mathbb{1}\{x_i = c\} y_i}{\sum_{i=1}^n \mathbb{1}\{x_i = c\}}$$

Similarly, if x_i is continuous, we can borrow data in a neighborhood of c and estimate by

$$\frac{\sum_{i=1}^n \mathbb{1}\{|x_i - c| \leq h\} y_i}{\sum_{i=1}^n \mathbb{1}\{|x_i - c| \leq h\}}$$

which is called a binned estimator and is represented in the figure below.

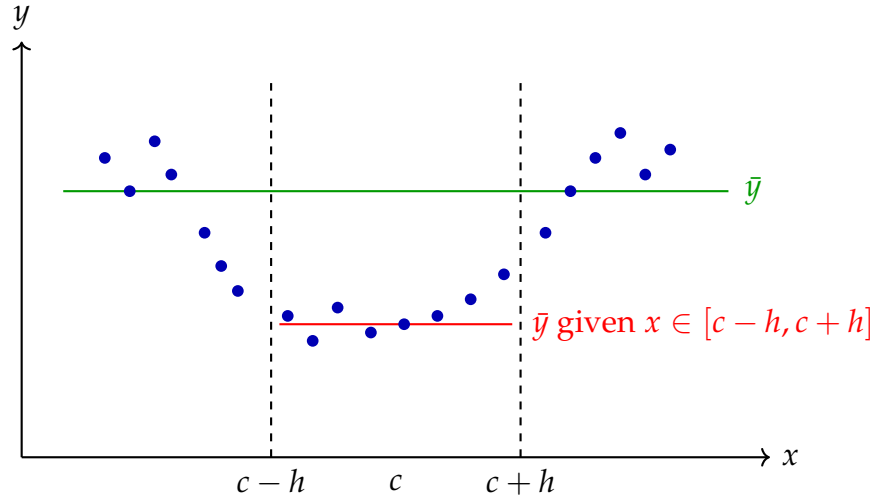


Figure 6: Global vs. local averaging: the green line is the overall mean $\bar{y}$, while the red segment shows the local average of observations with $x \in [c - h, c + h]$.

Similar to the density case, we can replace the indicator with a kernel function $K(\cdot)$ and obtain the kernel regression estimator.

Definition 7.3: Nadaraya-Watson Estimator

The Nadaraya-Watson Estimator is defined as

$$\hat{m}(c) := \frac{\sum_{i=1}^n K\left\{\frac{x_i - c}{h}\right\} y_i}{\sum_{i=1}^n K\left\{\frac{x_i - c}{h}\right\}}$$

Note that the NW estimator has different interpretations. First, it can be interpreted as a weighted average

$$\hat{m}(c) = \sum_{i=1}^n \omega_i(c) y_i \text{ where } \omega_i(c) \equiv \frac{K\left\{\frac{x_i - c}{h}\right\}}{\sum_{i=1}^n K\left\{\frac{x_i - c}{h}\right\}}$$

For instance, a triangular kernel puts more weight on observations close to c . Another interpretation is that $\hat{m}(c)$ is obtained as the least squares regression on constant using nearby observations of c :

$$\hat{m}(c) = \arg \min_{\alpha} \sum_{i=1}^n K\left(\frac{x_i - c}{h}\right) (y_i - \alpha)^2$$

The NW estimator essentially fits constants locally to describe the data. Note that this can perform poorly at boundary points or due to the asymmetry of the kernel function.

A natural extension is to allow for a local linear approximation of the regression function.

Definition 7.4: Local Linear Estimator

The local linear estimator $(\hat{\alpha}(c), \hat{\beta}(c))$ is defined as the solution to the weighted least squares problem

$$(\hat{\alpha}(c), \hat{\beta}(c)) = \arg \min_{\alpha, \beta} \sum_{i=1}^n K\left(\frac{x_i - c}{h}\right) (y_i - \alpha - \beta(x_i - c))^2.$$

The local linear estimator of the regression function is given by the estimated intercept:

$$\hat{m}(c) = \hat{\alpha}(c).$$

The slope coefficient $\hat{\beta}(c)$ provides an estimate of the derivative of the regression function at c :

$$\hat{\beta}(c) \approx m'(c) = \left. \frac{d}{dx} \mathbb{E}[Y | X = x] \right|_{x=c}$$

We can obviously extend this to nonparametric median, quantile regressions, etc.

7.4 Concluding Remarks

Nonparametric methods allow us to estimate unknown functions without imposing a finite-dimensional functional form. In this sense, *nonparametric* should be understood as

infinite-dimensional: estimating a function amounts to estimating a parameter at each point in its domain. Kernel-based estimators, such as density estimators, the Nadaraya–Watson regression, and local linear regression, approximate the target function locally by assigning more weight to observations close to the evaluation point. This flexibility comes at a cost: compared to parametric methods, nonparametric estimators converge at slower rates and require the choice of smoothing parameters, most notably the bandwidth, which governs the bias–variance trade-off.

More broadly, kernel methods are only one approach to nonparametric estimation. Alternative techniques, such as splines, series (sieve) estimators, or nearest-neighbor methods, also approximate functions using flexible, high-dimensional representations. Once we impose parametric structure—for instance by restricting the regression function to belong to a finite-dimensional family—the problem becomes parametric and standard estimation techniques apply. Nonparametric methods therefore provide a powerful and flexible starting point, allowing the data to inform the functional form before stronger modeling assumptions are introduced.

8 Limited Dependent Variables

Limited dependent variable (LDV) models arise when the outcome variable is discrete or otherwise constrained, making linear regression inappropriate. These models are typically estimated by Maximum Likelihood. Although the method has been introduced earlier, a full treatment of extremum estimators and their asymptotic properties is provided in Appendix A.3.2. Here, we focus on how likelihood-based methods are used in practice to estimate nonlinear models and interpret their parameters.

8.1 Binary dependent variable

We start this section by studying binary dependent variable (e.g. employed or not, college or not, etc.) where $y_i = 0$ or $y_i = 1$. We wish to explain the behavior of y_i by a vector of regressors x_i . The conditional mean gives

$$\mathbb{E}(y_i|x_i) = 1 \cdot \mathbb{P}(y_i = 1|x_i) + 0 \cdot \mathbb{P}(y_i = 0|x_i) = \mathbb{P}(y_i = 1|x_i)$$

so modeling the conditional mean is equivalent to modeling the conditional choice probability $\mathbb{P}(y_i = 1|x_i) \in [0, 1]$. Note that the linear model $\mathbb{E}(y_i|x_i) = x_i'\beta$ can take values outside $[0, 1]$. Thus, for choice probability, we can employ an index model

$$\mathbb{P}(y_i = 1|x_i) = F(x_i'\beta)$$

where $F(\cdot)$ is some known cdf (also called link function) where by construction $F(x_i'\beta) \in [0, 1]$.

Now that we defined the main objects, we can construct the so called **Latent variable model**. Consider a model where

$$y_i = \mathbb{1}\{y_i^* > 0\}; y_i^* = x_i'\beta + e_i \text{ where } e_i|x_i \sim F$$

where F is some known cdf. In this context, we observe y_i but not y_i^* . We could think of a case where we observe whether the individual is employed or not (y_i) but we don't observe the wage (y_i^*). If we assume the pdf of F is symmetric around zero, we can write

$$\mathbb{P}(y_i = 1 | x_i) = \mathbb{P}(y_i^* > 0 | x_i) = \mathbb{P}(e_i > -x_i'\beta | x_i) = 1 - F(-x_i'\beta) = F(x_i'\beta)$$

where the last case comes by symmetry.

To implement the model, we need to specify function forms of F . Common choices are Probit and Logit.

Definition 8.1: Probit and Logit CDFs

The Probit CDF corresponds to the standard normal CDF:

$$F(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{a^2}{2}} da$$

The Logit CDF corresponds to the standard logistic CDF:

$$F(z) = \frac{\exp(z)}{1 + \exp(z)}$$

Once we specify the distribution function F , we can apply maximum likelihood estimation. The index model implies that, for a given x_i ,

$$y_i = \begin{cases} 1 & \text{with probability } F(x_i'\beta), \\ 0 & \text{with probability } 1 - F(x_i'\beta) \end{cases}$$

The conditional probability mass function of $y_i | x_i$ is therefore

$$f(y_i | x_i, \beta) = F(x_i'\beta)^{y_i} [1 - F(x_i'\beta)]^{1-y_i}$$

The (conditional) log-likelihood function is

$$Q_n(\beta) = \frac{1}{n} \sum_{i=1}^n [y_i \log F(x_i'\beta) + (1 - y_i) \log (1 - F(x_i'\beta))]$$

For probit and logit models, $Q_n(\beta)$ is concave, so compactness of the parameter space is not required. By standard maximum likelihood theory, the estimator

$$\hat{\beta} = \arg \max_{\beta} Q_n(\beta)$$

is consistent and asymptotically normal:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, J^{-1}),$$

where

$$J = \mathbb{E} \left[\frac{f(x'_i \beta)^2}{F(x'_i \beta) (1 - F(x'_i \beta))} x_i x'_i \right]$$

Note that β does not have a simple interpretation as in linear regression models. The marginal effect of the k -th regressor x_{ik} is

$$\frac{\partial \mathbb{E}[y_i | x_i]}{\partial x_{ik}} = \frac{\partial \mathbb{P}(y_i = 1 | x_i)}{\partial x_{ik}} = f(x'_i \beta) \beta_k.$$

If $f(\cdot) > 0$, the sign of the marginal effect is the same as the sign of β_k . However, the magnitude of the marginal effect must be evaluated at specific values of x_i . Common choices include:

$$f(\bar{x}' \hat{\beta}) \hat{\beta}_k \quad \text{or} \quad \frac{1}{n} \sum_{i=1}^n f(x'_i \hat{\beta}) \hat{\beta}_k.$$

A common practice is to compute individual marginal effects

$$f(x'_i \hat{\beta}) \hat{\beta}_k$$

for each observation and report their sample average.

To estimate the index model, we need to specify F . However, economic theory is typically not informative enough to fully specify F . It is therefore desirable to treat F nonparametrically. Note that

$$\mathbb{E}[y_i | x'_i \beta = c] = \mathbb{P}(y_i = 1 | x'_i \beta = c) = F(c).$$

Thus, given β , the function $F(c)$ can be estimated by kernel regression of y_i on $x'_i \beta$:

$$\hat{F}(c) = \frac{\sum_{j=1}^n K\left(\frac{x'_j \beta - c}{h}\right) y_j}{\sum_{j=1}^n K\left(\frac{x'_j \beta - c}{h}\right)}.$$

For each $i = 1, \dots, n$, we can estimate $\hat{F}(x'_i \beta)$ via kernel regression. The semiparametric log-likelihood function is

$$Q_n^{SP}(\beta) = \frac{1}{n} \sum_{i=1}^n [y_i \log \hat{F}(x'_i \beta) + (1 - y_i) \log (1 - \hat{F}(x'_i \beta))].$$

The semiparametric MLE is defined as

$$\tilde{\beta} = \arg \max_{\beta \in B} Q_n^{SP}(\beta),$$

where B is a normalized parameter space (e.g. $\beta_1 = 1$).

This approach separates the estimation of the index $x'_i \beta$ from the functional form of F . Instead of imposing a parametric link (such as logit or probit), the conditional probability

$F(c)$ is estimated nonparametrically using kernel methods. As a result, the model is more flexible and robust to misspecification of the link function. However, this flexibility comes at a cost: estimation becomes computationally intensive and requires bandwidth selection, and identification of β is only up to scale, which necessitates a normalization such as $\beta_1 = 1$.

8.2 Multinomial dependent variable

In many applications, the dependent variable y_i can take more than two discrete values. These are referred to as multinomial dependent variables. A key distinction is whether the categories of y_i have a natural ordering.

Ordered data: the values of y_i have a meaningful ranking. For example,

$$y_i = \begin{cases} 0 & \text{low spending} \\ 1 & \text{medium spending} \\ 2 & \text{high spending} \end{cases} \quad \text{or} \quad y_i = \begin{cases} 0 & \text{dislike} \\ 1 & \text{neutral} \\ 2 & \text{like} \end{cases}$$

Unordered data: the values of y_i are simply labels with no intrinsic ordering. For example,

$$y_i = \begin{cases} 0 & \text{bus} \\ 1 & \text{train} \\ 2 & \text{car} \end{cases}$$

The appropriate econometric model depends on whether such an ordering exists.

8.2.1 Ordered Probit and Logit Models

Consider an ordered outcome $y_i \in \{0, 1, 2\}$ (the framework extends to more categories). We assume a latent variable model:

$$y_i^* = x_i' \beta + \varepsilon_i, \quad \varepsilon_i \mid x_i \sim F,$$

where F is a known cumulative distribution function (e.g. normal for probit, logistic for logit). The observed outcome is determined by threshold crossing:

$$y_i = \begin{cases} 0 & \text{if } y_i^* < \gamma_1, \\ 1 & \text{if } \gamma_1 \leq y_i^* < \gamma_2, \\ 2 & \text{if } \gamma_2 \leq y_i^*. \end{cases}$$

The parameters to estimate are

$$\theta = (\beta', \gamma_1, \gamma_2)'.$$

Using the latent variable representation, we can express the probabilities as:

$$P(y_i = 0 \mid x_i) = P(y_i^* < \gamma_1 \mid x_i) = F(\gamma_1 - x_i' \beta),$$

$$P(y_i = 2 \mid x_i) = P(y_i^* \geq \gamma_2 \mid x_i) = 1 - F(\gamma_2 - x_i' \beta) = F(x_i' \beta - \gamma_2),$$

$$P(y_i = 1 \mid x_i) = F(\gamma_2 - x_i' \beta) - F(\gamma_1 - x_i' \beta).$$

Let $\mathbb{1}\{y_i = j\}$ denote an indicator function. The log-likelihood is

$$\begin{aligned} Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n \Big[& \mathbb{1}\{y_i = 0\} \log F(\gamma_1 - x_i' \beta) + \mathbb{1}\{y_i = 1\} \log (F(\gamma_2 - x_i' \beta) - F(\gamma_1 - x_i' \beta)) \\ & + \mathbb{1}\{y_i = 2\} \log F(x_i' \beta - \gamma_2) \Big]. \end{aligned}$$

Standard maximum likelihood theory applies for estimation. Marginal effects measure how regressors affect outcome probabilities. For the k -th regressor:

$$\begin{aligned} \frac{\partial P(y_i = 0 \mid x_i)}{\partial x_{ik}} &= -\beta_k f(\gamma_1 - x_i' \beta), \\ \frac{\partial P(y_i = 2 \mid x_i)}{\partial x_{ik}} &= \beta_k f(\gamma_2 - x_i' \beta), \\ \frac{\partial P(y_i = 1 \mid x_i)}{\partial x_{ik}} &= \beta_k [f(\gamma_1 - x_i' \beta) - f(\gamma_2 - x_i' \beta)], \end{aligned}$$

where $f(\cdot)$ is the density corresponding to F .

The sign of β_k determines the direction of the effect on the extreme categories ($y = 0$ and $y = 2$), but the effect on the middle category is generally ambiguous.

The ordered probit/logit model assumes that outcomes arise from a continuous latent index y_i^* . The thresholds (γ_1, γ_2) partition this latent variable into ordered categories. Changes in x_i shift the distribution of y_i^* , redistributing probability mass across adjacent categories rather than affecting each category independently.

8.2.2 Unordered Multinomial Logit

When the outcome y_i is unordered, its values are simply labels and cannot be ranked. In this case, we directly model the choice probabilities. The multinomial logit model specifies

$$P(y_i = j \mid x_i) = \frac{\exp(x_{ij}' \beta^j)}{\sum_{h=0}^J \exp(x_{ih}' \beta^h)}, \quad j = 0, 1, \dots, J,$$

where x_{ij} are choice-specific regressors (which may include individual characteristics). To ensure identification, we impose a normalization such as $\beta^0 = 0$.

Now, note that the multinomial logit model can be derived from a random utility framework. Assume that the latent utility of individual i from choosing alternative j is

$$y_{ij}^* = x'_{ij}\beta^j + \varepsilon_{ij}, \quad j = 0, 1, \dots, J,$$

where $(\varepsilon_{i0}, \dots, \varepsilon_{iJ})$ are i.i.d. with type I extreme value distribution:

$$F(\varepsilon) = \exp(-\exp(-\varepsilon)).$$

The observed choice corresponds to the utility-maximizing alternative:

$$y_i = \arg \max_k y_{ik}^*.$$

Equivalently,

$$y_i = j \iff y_{ij}^* \geq y_{ik}^* \text{ for all } k \neq j.$$

Derivation of Logit Probabilities Using the latent utility model,

$$P(y_i = j \mid x_i) = P(y_{ij}^* \geq y_{ik}^* \forall k \neq j \mid x_i).$$

Substituting utilities,

$$= P(x'_{ij}\beta^j + \varepsilon_{ij} \geq x'_{ik}\beta^k + \varepsilon_{ik} \forall k \neq j).$$

Rewriting,

$$= P(\varepsilon_{ik} - \varepsilon_{ij} \leq x'_{ij}\beta^j - x'_{ik}\beta^k \forall k \neq j).$$

Under the type I extreme value assumption, this probability simplifies to the logit form:

$$P(y_i = j \mid x_i) = \frac{\exp(x'_{ij}\beta^j)}{\sum_{h=0}^J \exp(x'_{ih}\beta^h)}.$$

Now, the multinomial log-likelihood is

$$Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J \mathbb{1}\{y_i = j\} \left[x'_{ij}\beta^j - \log \left(\sum_{h=0}^J \exp(x'_{ih}\beta^h) \right) \right].$$

and standard maximum likelihood theory applies. Note that the multinomial logit model implies the *independence of irrelevant alternatives* (IIA): the odds ratio between any two alternatives does not depend on other available choices,

$$\frac{P(y_i = j \mid x_i)}{P(y_i = k \mid x_i)} = \frac{\exp(x'_{ij}\beta^j)}{\exp(x'_{ik}\beta^k)}.$$

This property can be restrictive in applications where alternatives are close substitutes. One could instead use a multinomial probit to avoid IIA property.

8.3 Censored vs. Truncated Dependent Variables

It is useful to distinguish between censoring and truncation, as they have different implications for estimation. A sample is *censored* if all observations are present in the data, but for some of them the dependent variable is only partially observed. In particular, values beyond a threshold are not observed exactly, but are instead recorded at the threshold. For example, if income is censored below a certain level c , we observe $y_i = c$ whenever the true income falls below c , without knowing the exact value.

A sample is *truncated* if some observations are systematically excluded from the dataset altogether. In this case, we do not observe any data (neither y_i nor x_i) for individuals whose outcome falls below (or above) a certain threshold. For instance, if incomes below c are truncated, individuals earning less than c do not appear in the sample at all.

The key distinction is therefore that censoring preserves all individuals but distorts part of the outcome variable, whereas truncation removes part of the population entirely, leading to selection in the observed sample.

8.3.1 Censored dependent variables - Tobit Models

Consider the following latent variable model

$$y_i^* = x_i' \beta + e_i, e_i | x_i \sim \mathcal{N}(0, \sigma^2)$$

However, y_i^* is unobservable. Instead, we observe

$$y_i = \max\{0, y_i^*\} = \mathbb{1}\{y_i^* > 0\}$$

Note that the threshold doesn't have to be zero, it could be any $c \in \mathbb{R}$. The figure below helps visualize.

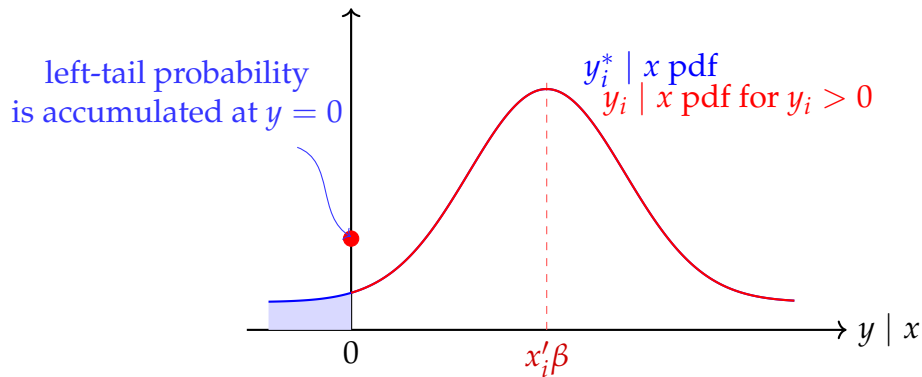


Figure 7: Left censoring in the Tobit model: probability mass below zero is concentrated at the censoring point, so the observed distribution still integrates to one. Note that for $y_i > 0$, the two curves overlap

In the censored case, the probability mass that would lie below the censoring point is not discarded; instead, it is concentrated at the boundary value. Hence, the observed distribution consists of a discrete mass at the censoring point and a continuous density elsewhere, and the total probability still sums to one.

Now, if y_i^* is observable, $\mathbb{E}(y_i^* | x_i) = x_i' \beta$ and OLS consistently estimates β . However, since $\max\{\cdot\}$ is convex, by Jensen's Inequality, we have

$$\mathbb{E}(y_i | x_i) = \mathbb{E}(\max\{0, y_i^*\} | x_i) \geq \max\{0, \mathbb{E}(y_i^* | x_i)\} = \max\{0, x_i' \beta\}$$

Therefore $\mathbb{E}(y_i | x_i) \neq x_i' \beta$ and OLS from y_i on x_i is inconsistent for β .

Now, define the standard normal pdf and cdf as

$$\phi(a) = \frac{1}{\sqrt{2\pi}} e^{-a^2/2},$$

$$\Phi(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-b^2/2} db.$$

Then, the pdf of a normal random variable $N(\mu, \sigma^2)$ can be written as

$$\frac{1}{\sigma} \phi\left(\frac{a - \mu}{\sigma}\right).$$

The conditional density of $y_i | x_i$ is a mixture of a point mass at zero and a continuous density for positive values:

$$f(y_i | x_i) = \begin{cases} \Phi\left(\frac{-x_i' \beta}{\sigma}\right) & \text{if } y_i = 0, \\ \frac{1}{\sigma} \phi\left(\frac{y_i - x_i' \beta}{\sigma}\right) & \text{if } y_i > 0. \end{cases}$$

The first part corresponds to the density below the threshold that accumulates into the mass point at $y_i = 0$:

$$\mathbb{P}(y_i^* \leq 0) = \mathbb{P}(x_i' \beta + e_i \leq 0) = \mathbb{P}(e_i \leq -x_i' \beta) = \mathbb{P}\left(\frac{e_i}{\sigma} \leq \frac{-x_i' \beta}{\sigma}\right) = \Phi\left(\frac{-x_i' \beta}{\sigma}\right)$$

The second term corresponds to uncensored observations (continuous density). The log-likelihood function is therefore

$$Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(y_i | x_i),$$

which can be written as

$$Q_n(\theta) = \frac{1}{n} \left[\sum_{i: y_i=0} \log \Phi\left(\frac{-x_i' \beta}{\sigma}\right) + \sum_{i: y_i>0} \log \left(\frac{1}{\sigma} \phi\left(\frac{y_i - x_i' \beta}{\sigma}\right) \right) \right].$$

Standard maximum likelihood theory applies.

8.3.2 Truncated dependent variables

In the truncated case, observations with $y_i^* \leq 0$ are not present in the sample at all. Therefore, the observed density is the latent density conditional on $y_i^* > 0$, which means that the remaining part of the distribution must be renormalized to integrate to one.

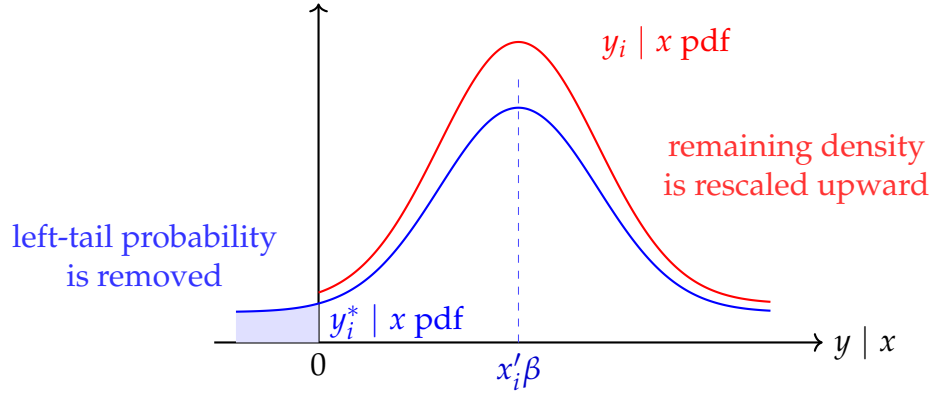


Figure 8: Truncated distribution: observations with $y_i^* \leq 0$ are not observed, so the density for $y_i > 0$ is renormalized

$$f(y_i | x_i, y_i^* > 0) = \frac{\frac{1}{\sigma} \phi\left(\frac{y_i - x_i' \beta}{\sigma}\right)}{\Phi\left(\frac{x_i' \beta}{\sigma}\right)}, \quad y_i > 0.$$

Example 8.1: Illustrative example

Suppose that 30% of the latent outcomes $y_i^* | x_i$ fall below zero and are therefore not observed. Then only 70% of the original distribution remains in the sample.

As a result, the observed density must be rescaled to integrate to one. Intuitively, each realization that appears once in the latent distribution now appears with relatively higher weight in the observed sample.

Formally, for $a > 0$,

$$f_{y_i | x_i}(a) = \frac{f_{y_i^* | x_i}(a)}{P(y_i^* > 0 | x_i)},$$

since y_i only takes positive values under truncation. The denominator ensures that the density integrates to one over the support $(0, \infty)$.

Generally, suppose that

$$y_i^* = x_i' \beta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1),$$

so that

$$y_i^* | x_i \sim \mathcal{N}(x_i' \beta, 1).$$

Assume that we only observe $y_i = y_i^*$ if $y_i^* \geq c$ (e.g. $c = 0$), and no observation otherwise.

The observed density is then the truncated normal density:

$$f(y_i | x_i, y_i^* \geq c) = \frac{\phi(y_i - x_i' \beta)}{P(y_i^* \geq c | x_i)}, \quad y_i \geq c.$$

Since

$$P(y_i^* \geq c | x_i) = 1 - \Phi(c - x_i' \beta),$$

we obtain

$$f(y_i | x_i, y_i^* \geq c) = \frac{\phi(y_i - x_i' \beta)}{1 - \Phi(c - x_i' \beta)}, \quad y_i \geq c.$$

The likelihood function is therefore

$$\mathcal{L}(\beta) = \prod_{i=1}^n \frac{\phi(y_i - x_i' \beta)}{1 - \Phi(c - x_i' \beta)},$$

and the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n [\log \phi(y_i - x_i' \beta) - \log (1 - \Phi(c - x_i' \beta))].$$

8.4 Sample selection models

This is the last in this section which builds toward the Heckman Selection Model. We could think about cases where wage data are observable only for people who actually enter the labor force, or the effect of social programs (outcomes are observable only for people who actually participated). In this case, data truncation may be caused by individual's decision, creating *selection*. As a simple example, consider the following latent variable model

$$y_i^* = x_{1i}' \beta + e_i \quad ; \quad d_i^* = x_{2i}' \gamma + \varepsilon_i$$

where (e_i, ε_i) is independent of $x_i \equiv (x_{1i}', x_{2i}')'$, $\varepsilon_i \sim N(0, 1)$ and $\mathbb{E}(e_i | \varepsilon_i) = \theta \varepsilon_i$.

Instead of (y_i^*, d_i^*) , we observe

$$y_i = \begin{cases} y_i^* & \text{if } d_i = 1, \\ \text{unobserved} & \text{if } d_i = 0, \end{cases}$$

and

$$d_i = \begin{cases} 1 & \text{if } d_i^* > 0, \\ 0 & \text{if } d_i^* \leq 0. \end{cases}$$

The covariates x_i are always observed. A typical example is one where y_i represents the wage while d_i indicator for labor force participation. Thus, wages are observed only for individuals who participate in the labor force.

Since y_i^* is observed only when $d_i = 1$, we can hope to estimate $\mathbb{E}(y_i^* | x_i, d_i = 1)$, that is the wage regression for people participating in the labor force. First, note that by independence

$$\mathbb{E}(y_i^* | x_i, \varepsilon_i) = x'_{1i}\beta + \mathbb{E}(e_i | x_i, \varepsilon_i) = x'_{1i}\beta + \mathbb{E}(e_i | \varepsilon_i) = x'_{1i}\beta + \theta\varepsilon_i$$

Then,

$$\mathbb{E}(y_i^* | x_i, d_i) = \mathbb{E}[\mathbb{E}(y_i^* | x_i, \varepsilon_i, d_i) | x_i, d_i] = \mathbb{E}[\mathbb{E}(y_i^* | x_i, \varepsilon_i) | x_i, d_i] = x'_{1i}\beta + \theta\mathbb{E}(\varepsilon_i | x_i, d_i)$$

If $\theta = 0$, OLS from y_i on x_{1i} (using only the selected sample) consistently estimates β . If $\theta \neq 0$, the model is

$$\mathbb{E}[y_i | x_i, d_i = 1] = x'_{1i}\beta + \theta h(x_i),$$

where

$$h(x_i) = \mathbb{E}[\varepsilon_i | x_i, d_i = 1].$$

If we knew $h(x_i)$, then OLS from y_i on x_{1i} and $h(x_i)$, using only the selected sample, would consistently estimate β and θ .

Now, using the formula for the Inverse Mills Ratio,

$$\mathbb{E}[Z | Z > c] = \frac{\phi(c)}{1 - \Phi(c)}, \quad Z \sim \mathcal{N}(0, 1),$$

we obtain

$$h(x_i) = \mathbb{E}[\varepsilon_i | x_i, \varepsilon_i > -x'_i\gamma] = \frac{\phi(x'_i\gamma)}{\Phi(x'_i\gamma)}.$$

Hence, the model used for estimation is

$$\mathbb{E}[y_i | x_i, d_i = 1] = x'_{1i}\beta + \theta \frac{\phi(x'_i\gamma)}{\Phi(x'_i\gamma)}.$$

This is the Heckman, 1979 selection model: the term $\frac{\phi(x'_i\gamma)}{\Phi(x'_i\gamma)}$ is the inverse Mills ratio, which corrects for selection bias.

One approach of estimation is nonlinear least squares using only the selected sample:

$$\min_{\beta, \theta, \gamma} \sum_{i=1}^n \left(y_i - x'_{1i}\beta - \theta \frac{\phi(x'_i\gamma)}{\Phi(x'_i\gamma)} \right)^2$$

Alternatively, we can use a two-step estimator:

1. Estimate γ by probit of d_i on x_i , obtaining $\hat{\gamma}$.
2. Estimate β and θ by OLS of y_i on x_{1i} and

$$\frac{\phi(x'_i\hat{\gamma})}{\Phi(x'_i\hat{\gamma})}$$

In the two-step procedure, the usual OLS standard errors are not valid, since they do not account for the estimation error in $\hat{\gamma}$.

For maximum likelihood, we further assume

$$\begin{pmatrix} e_i \\ \varepsilon_i \end{pmatrix} \bigg| x_i \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \theta \\ \theta & 1 \end{pmatrix} \right).$$

If $d_i = 0$, then y_i is unobserved and the likelihood contribution is

$$\mathbb{P}(d_i = 0 \mid x_i).$$

If $d_i = 1$, then y_i is observed and the likelihood contribution is

$$f(y_i \mid d_i = 1, x_i) \mathbb{P}(d_i = 1 \mid x_i).$$

Hence, the log-likelihood is

$$Q_n(\psi) = \sum_{i:d_i=0} \log \mathbb{P}(d_i = 0 \mid x_i) + \sum_{i:d_i=1} \log \{f(y_i \mid d_i = 1, x_i) \mathbb{P}(d_i = 1 \mid x_i)\},$$

where f is the conditional density of y_i given $d_i = 1$ and x_i .

The first term is

$$\mathbb{P}(d_i = 0 \mid x_i) = \Phi(-x_i' \gamma).$$

For the second term, note that

$$f(y_i \mid d_i = 1, x_i) \mathbb{P}(d_i = 1 \mid x_i) = f(y_i^* \mid x_i) \mathbb{P}(d_i = 1 \mid y_i^*, x_i).$$

Since

$$y_i^* = x_{1i}' \beta + e_i,$$

we have

$$f(y_i^* \mid x_i) = \frac{1}{\sigma} \phi \left(\frac{y_i^* - x_{1i}' \beta}{\sigma} \right).$$

By the properties of the bivariate normal distribution, we can write

$$\varepsilon_i = \frac{\theta}{\sigma^2} e_i + v_i,$$

where

$$v_i \sim \mathcal{N} \left(0, 1 - \frac{\theta^2}{\sigma^2} \right)$$

and v_i is independent of e_i .

Therefore,

$$\mathbb{P}(d_i = 1 \mid y_i^*, x_i) = \mathbb{P}(x_i' \gamma + \varepsilon_i > 0 \mid y_i^*, x_i).$$

Substituting for ε_i gives

$$\mathbb{P}(d_i = 1 \mid y_i^*, x_i) = \Phi \left(\frac{x_i' \gamma + \frac{\theta}{\sigma^2} (y_i^* - x_{1i}' \beta)}{\sqrt{1 - \frac{\theta^2}{\sigma^2}}} \right).$$

Combining the results, the log-likelihood is

$$Q_n(\psi) = \sum_{i:d_i=0} \log \Phi(-x_i' \gamma) + \sum_{i:d_i=1} \log \left[\frac{1}{\sigma} \phi \left(\frac{y_i - x_{1i}' \beta}{\sigma} \right) \Phi \left(\frac{x_i' \gamma + \frac{\theta}{\sigma^2} (y_i - x_{1i}' \beta)}{\sqrt{1 - \frac{\theta^2}{\sigma^2}}} \right) \right].$$

Standard maximum likelihood theory applies.

9 Panel Data

A panel data (pooled cross section and time series) regression differs from a regular time-series or cross-section regression in that it has a double subscript on all variables $\{y_{it}, x_{it}\}$ for $i = 1, \dots, n$ and $t = 1, \dots, T$. i denotes the cross-sectional dimension (individuals, households, firms, etc.) and t the time-series dimension (days, weeks, etc.). Different types of variables can be included in a panel. We can have variables that vary across individuals and time, such as wage, age, years of experience. Some variables are *time invariant* such as race. Finally, some variables vary only over time, but not across individuals, in general aggregate/macro factors (e.g. unemployment).

Another important bit of terminology is what we call complete or balanced panels, where individuals are observed over the entire sample periods. Unbalanced panels occur in case where some $\{it\}$ is missing. We focus on balanced panels. Typically, we will be concerned with situations where T is small and n is large: fixed T and $n \rightarrow \infty$.

9.1 Fixed Effects Models

Consider the following linear model

$$y_{it} = \alpha + x_{it}' \beta + u_{it}$$

with x_{it} a k dimensional vector of explanatory variables, not including a constant. For any t , we assume $x_{it} \sim i.i.d$ across individuals. An error component model (ECM) specifies the structure of the error. Let μ_i be an unobservable individual-specific effect, λ_t an unobservable time effect and v_{it} and idiosyncratic component, where we assume $v_{it} \sim iid(0, \sigma^2)$ on i and t . We call one-way ECM a model of the form

$$u_{it} = \mu_i + v_{it} \quad \text{or} \quad u_{it} = \lambda_t + v_{it}$$

We call a two-way ECM a model of the form

$$u_{it} = \mu_i + \lambda_t + v_{it}$$

If characteristics that have a direct effect on both dependent variable and explanatory variables are omitted, then explanatory variables are correlated with the regression error and, consequently, estimators of the coefficients will be inconsistent. A traditional response of econometrics to this problem has been instrumental variables models. However, it is often very difficult to find instruments. A major motivation for using panel data has been the ability to control for (possibly correlated) time-invariant heterogeneity without observing it.

9.1.1 General case

In fixed effects models, we treat μ_i as n fixed unknown parameters. Consider our one way ECM :

$$y_{it} = \alpha + \mu_i + x'_{it}\beta + v_{it}$$

The main assumption in this context is **strict exogeneity**

$$\mathbb{E}(v_{it}|x_{i1}, \dots, x_{iT}) = 0, \forall t$$

We can rewrite this model as

$$y_{it} = \alpha + \sum_{j=1}^n \mu_j d_{ij,t} + x'_{it}\beta + v_{it}$$

where $d_{ij,t} = \mathbb{1}\{i = j\}$. Note that if we have perfect collinearity with α (if we sum through all $d_{ij,t}$, we get 1). Therefore, we need to drop the constant and write

$$y_{it} = \sum_{j=1}^n \mu_j d_{ij,t} + x'_{it}\beta + v_{it}$$

To estimate the model efficiently, we stack the observations, dropping the subscript t . For a single individual i , the model is:

$$y_i = \iota_T \mu_i + X_i \beta + v_i,$$

where the components are:

$$\underbrace{y_i}_{(T \times 1)} = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{pmatrix}, \quad \underbrace{\iota_T}_{(T \times 1)} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \quad \underbrace{X_i}_{(T \times k)} = \begin{pmatrix} x'_{i1} \\ x'_{i2} \\ \vdots \\ x'_{iT} \end{pmatrix}, \quad \underbrace{v_i}_{(T \times 1)} = \begin{pmatrix} v_{i1} \\ v_{i2} \\ \vdots \\ v_{iT} \end{pmatrix}$$

Stacking all n individuals, we define the matrix equation:

$$y = D\mu + X\beta + v$$

With the following block definitions:

$$\underbrace{y}_{(nT \times 1)} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \underbrace{\mu}_{(n \times 1)} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix}, \quad \underbrace{v}_{(nT \times 1)} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}, \quad \underbrace{X}_{(nT \times k)} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

The matrix D is the matrix of individual dummies, defined using the Kronecker product:

$$\underbrace{D}_{(nT \times n)} = I_n \otimes \iota_T = \begin{pmatrix} \iota_T & 0 & 0 & \dots & 0 \\ 0 & \iota_T & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \iota_T \end{pmatrix}$$

Matrix D is a "selector." When you multiply $D\mu$, the first column of D picks out μ_1 and applies it to of person 1's observations (T of them), the second column picks μ_2 for person 2, and so on.

Example 9.1: Visualizing Panel Matrices ($n = 2, T = 2$)

Suppose we have two individuals, Alice ($i = 1$) and Bob ($i = 2$), observed for two years. Our model is $y_{it} = \mu_i + \beta x_{it} + v_{it}$.

For Alice ($i = 1$):

$$y_1 = \begin{pmatrix} y_{11} \\ y_{12} \end{pmatrix}, \quad X_1 = \begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix}, \quad \iota_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For Bob ($i = 2$):

$$y_2 = \begin{pmatrix} y_{21} \\ y_{22} \end{pmatrix}, \quad X_2 = \begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix}, \quad \iota_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The stacked outcome vector y and regressor matrix X (size $nT \times 1 = 4 \times 1$) are:

$$y = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \end{pmatrix}, \quad X = \begin{pmatrix} x_{11} \\ x_{12} \\ x_{21} \\ x_{22} \end{pmatrix}$$

$D = I_2 \otimes \iota_2$ creates the columns that assign μ_1 to Alice and μ_2 to Bob:

$$D = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

When we calculate $D\mu$, we get a vector where the first two rows are μ_1 and the last two are μ_2 . This allows each individual to have their own "fixed" intercept over time.

We estimate the stacked model $y = D\mu + X\beta + v$ via OLS. Under the assumption that $E[vv'] = \sigma_v^2 I_{nT}$, OLS is efficient. However, the following conditions must hold for identification:

- **Time Dimension:** $T \geq 2$. If $T = 1$, μ_i cannot be calculated as it is perfectly confounded with the error term.

- **Variation:** X must not contain time-invariant regressors, as these would be perfectly collinear with the columns of D .

Direct OLS estimation is often unattractive because it requires handling the $(nT \times n)$ matrix D and inverting a $(n+k) \times (n+k)$ matrix, which is computationally taxing for large n . We are typically only interested in β , making the estimation of all n values of μ unnecessary.

9.1.2 The Within Estimator (Demeaning)

A numerically efficient way to estimate β is to use the **Frisch-Waugh-Lovell (FWL) Theorem**. In our stacked model $y = D\mu + X\beta + v$, the matrix of individual dummies D plays the role of the variables we wish to partial out. Indeed, we write the partition as $[D : X]$.

We define the projection matrix M_D as:

$$M_D = I_{nT} - D(D'D)^{-1}D'$$

Recall $D = I_n \otimes \iota_T$. We derive $M_D = I_{nT} - D(D'D)^{-1}D'$ as follows:

1. $D'D = {}^{24}(I_n \otimes \iota_T)'(I_n \otimes \iota_T) = I_n \otimes \iota_T' \iota_T = TI_n$
2. $D(D'D)^{-1}D' = (I_n \otimes \iota_T)(\frac{1}{T}I_n)(I_n \otimes \iota_T') = I_n \otimes \frac{1}{T}J_T$ where $J_T = \iota_T \iota_T'$
3. $M_D = I_{nT} - I_n \otimes \frac{1}{T}J_T$

Now, look at the projection part $P_D = D(D'D)^{-1}D'$, which we have shown is equal to $I_n \otimes \frac{1}{T}J_T$, where J_T is a $T \times T$ matrix of all ones. When you multiply $\frac{1}{T}J_T$ by an individual's vector y_i , you get:

$$\frac{1}{T} \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix} \begin{pmatrix} y_{i1} \\ \vdots \\ y_{iT} \end{pmatrix} = \begin{pmatrix} \bar{y}_i \\ \vdots \\ \bar{y}_i \end{pmatrix}$$

This operation simply calculates the mean for that person and repeats it T times.

Now, apply the full $M_D = I_{nT} - P_D$ matrix to the vector y :

$$M_D y = (I_{nT} y) - (P_D y)$$

$$M_D y = \begin{pmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{nT} \end{pmatrix} - \begin{pmatrix} \bar{y}_1 \\ \bar{y}_1 \\ \vdots \\ \bar{y}_n \end{pmatrix} = \begin{pmatrix} y_{11} - \bar{y}_1 \\ y_{12} - \bar{y}_1 \\ \vdots \\ y_{nT} - \bar{y}_n \end{pmatrix}$$

Therefore, we get

$$M_D y = y - \bar{y}_i \quad \text{where} \quad \bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}$$

²⁴Using the properties of Kronecker products $(A \otimes B) \cdot (C \otimes D) = (AC \otimes BD)$

Thus, the FE estimator $\hat{\beta}_{FE} = (X'M_D X)^{-1} X'M_D y$ obtained by FWL is the OLS estimator of the regression of $(y_{it} - \bar{y}_i)$ on $(x_{it} - \bar{x}_i)$. $\hat{\beta}_{FE}$ is the estimator that we obtain if we first project out the group effects μ_i by measuring the individual observations in deviations from group means. This is known as **within variation**: we study how individuals varies compared to its own mean.

Intuition: The fixed effects estimator exploits variation *within* each individual over time, effectively comparing each person to themselves. By subtracting individual-specific means (demeaning), we remove all time-invariant characteristics—both observed and unobserved—that might confound our estimates. This transformation asks: when individual i experiences a change in x_{it} relative to their own average, how does y_{it} change relative to their own average? This "within" comparison controls for any fixed individual heterogeneity μ_i , whether it's ability, preferences, or other unchanging traits.

The estimator is unbiased under the strict exogeneity assumption:

$$\mathbb{E}(v_{it}|x_{i1}, \dots, x_{iT}) = 0 \quad \forall t$$

Substituting the model $y = D\mu + X\beta + v$ into the estimator formula:

$$\mathbb{E}(\hat{\beta}_W|X) = \mathbb{E} \left[(X'M_D X)^{-1} X'M_D (D\mu + X\beta + v) \right]$$

Since $M_D D = 0$, the μ term drops out:

$$\mathbb{E}(\hat{\beta}_W|X) = \beta + (X'M_D X)^{-1} X'M_D \mathbb{E}(v|X)$$

Under strict exogeneity, $\mathbb{E}(v|X) = 0$, so $\mathbb{E}(\hat{\beta}_W|X) = \beta$

The requirement for consistency is $E[(x_{it} - \bar{x}_i)(v_{it} - \bar{v}_i)] = 0$.

- **Small T** : Requires *Strict Exogeneity* (x_{it} uncorrelated with v_{is} for all s, t) because $\bar{v}_i$ contains errors from all periods.
- **Large T** : As $T \rightarrow \infty$, $\bar{v}_i$ and $\bar{x}_i$ converge to their expectations. In this case, *Contemporaneous Exogeneity* ($E[x_{it}v_{it}] = 0$) is sufficient for consistency.

This is why Fixed Effects is often considered "safer" in long panels (T is large) than in short panels where feedback loops can violate strict exogeneity.

Assuming the idiosyncratic errors are homoskedastic and serially uncorrelated ($V(v|X) = \sigma_v^2 I_{nT}$), the variance of the estimator is:

$$V(\hat{\beta}_W|X) = \sigma_v^2 (X'M_D X)^{-1}$$

σ_v^2 is consistently estimated as

$$\hat{\sigma}_v^2 = \frac{RSS}{nT - n - k}$$

where RSS is the residual sum of squares from the within-transformed model. Notice that we subtract $n + k$ because this is the number of unknown parameters (n fixed effects, k regressors).

While OLS is used for the within-transformed model, the disturbances $\tilde{v}_{it} = v_{it} - \bar{v}_i$ are correlated by construction. Usually, when disturbances are correlated, we use GLS to achieve efficiency. However, GLS is not an option here because the transformed idiosyncratic errors $\tilde{v}_{it} = v_{it} - \bar{v}_i$ necessarily sum to zero for each individual: $\sum_{t=1}^T \tilde{v}_{it} = 0$. This constraint implies that the T disturbances for each individual are linearly dependent, making the resulting $nT \times nT$ covariance matrix $\mathbb{E}[M_D v (M_D v)'] = \sigma_v^2 M_D$ singular. Consequently, the standard GLS estimator is not defined for the within-transformed model.

To account for this correlation and potential heteroskedasticity, we typically **cluster standard errors** at the individual (or treatment) level. Indeed, OLS standard errors assume that the residuals are i.i.d. However, in the FE model, even if the original v_{it} were i.i.d., the transformed residuals $(v_{it} - \bar{v}_i)$ are correlated because they all share the same $\bar{v}_i$ term. This correlation is strictly internal to the individual (the "cluster"). If we ignore this, our standard errors will be biased (usually downward), leading to over-rejection of the null hypothesis (t -statistics that are artificially high).

Formally,

$$\frac{RSS}{nT - k} \xrightarrow{p} \frac{T - 1}{T} \sigma_v^2$$

Since $T - 1 < T$, which is strictly smaller than the true variance. This is why we use

$$\hat{\sigma}_v^2 = \frac{RSS}{nT - n - k} = \frac{RSS}{nT - k} \cdot \frac{nT - k}{n(T - 1) - k} \xrightarrow{p} \frac{T - 1}{T} \sigma_v^2 \cdot \frac{T}{T - 1} = \sigma_v^2$$

Once we have obtained a consistent estimate $\hat{\beta}$ via the within-estimator, we can recover the estimated fixed effects $\hat{\mu}_i$. Using the FWL theorem, the vector of fixed effects is:

$$\hat{\mu} = (D'D)^{-1} D'(y - X\hat{\beta})$$

For each individual i , this simplifies to a very intuitive result:

$$\hat{\mu}_i = \bar{y}_i - \bar{x}_i' \hat{\beta}$$

The fixed effect μ_i is defined as the constant level of y for individual i that cannot be explained by the variation in x_{it} . Therefore, it is simply the difference between their average outcome and the predicted average outcome based on their characteristics.

While $\hat{\beta}$ is consistent as $n \rightarrow \infty$, remember that $\hat{\mu}_i$ is **not** consistent if T is fixed. We only have T observations to estimate each specific μ_i , so the estimation error in $\hat{\mu}_i$ does not vanish unless the number of time periods grows ($T \rightarrow \infty$).

9.1.3 Two-way ECM with FE

Consider our two way ECM :

$$y_{it} = \alpha + \mu_i + \lambda_t + x'_{it}\beta + v_{it}$$

Here again, we have a multicollinearity problem. Indeed, the sum of μ_i and the sum of λ_t both give 1. In general, we keep α and drop one individual and one time FE, say $\mu_n = 0$ and $\lambda_T = 0$.

The OLS estimation of this model is very similar to the one-way ECM. Omitting the derivations, we get

$$y_{it} - \bar{y}_i - \bar{y}_t + \bar{y} = (x_{it} - \bar{x}_i - \bar{x}_t + \bar{x})'\beta + v_{it} - \bar{v}_i - \bar{v}_t + \bar{v}$$

In practice, this is a version of the within-estimator in which we are also removing period-specific shocks across all individuals.

9.1.4 Drawbacks of the fixed-effect model

One of the main drawbacks of this model is that it fails to identify any components of β corresponding to regressors that are constant over time for a given individual (e.g. race, religion, etc.), since they are absorbed in the individual fixed effect. Indeed, they are perfectly collinear with the individual fixed effect D and are wiped out by the M_D transformation.

Coefficients of time-varying regressors are estimable, but these estimates may be very imprecise if most of the variation in a regressor is cross sectional rather than over time. We can diagnose this using the Stata command `xtsum`, which decomposes total variance into:

- **Between variation:** $\bar{x}_i - \bar{x}$, the variation across different individuals.
- **Within variation:** $x_{it} - \bar{x}_i$, the variation over time for the same individual.

If the within variation is small relative to the between variation, the FE estimator will have large standard errors because it relies solely on that thin "within" slice of data.

For example, in a short panel, education level is often time-invariant for most adults. Since M_D removes the cross-sectional differences in education levels, $\hat{\beta}_{educ}$ would be identified only by the few individuals who completed a degree during the sample period. This leads to a loss of power compared to a Random Effects (introduced below) or Pooled OLS model.

Additionally, prediction of the absolute conditional mean is not possible because the individual intercepts μ_i are treated as nuisance parameters and are not consistently estimated for fixed T . We can only predict *changes* in the dependent variable caused by **changes in time-varying regressors**.

9.2 Random Effects Models

Consider again the one way ECM:

$$y_{it} = \alpha + x'_{it}\beta + u_{it}$$

with x_{it} being i.i.d and $u_{it} = \mu_i + v_{it}$. In this case, individual effects μ_i are assumed to be random and are treated as drawings from the same distribution $\mu_i \sim i.i.d(0, \sigma_\mu^2)$. We further assume $v_{it} \sim i.i.d(0, \sigma_v^2)$. Additionally, we assume **strict exogeneity**

$$\mathbb{E}(v_{it}|x_{i1}, \dots, x_{iT}) = 0, \forall t$$

and assume $\mu_i \perp v_{it}$, $\mathbb{E}(\mu_i|x_{i1}, \dots, x_{iT}) = 0$. Note that here, there is no dummy variable trap, so we do not need to drop any constant term.

Denote the variance-covariance matrix $\Omega \equiv \mathbb{V}(u)$. The diagonal elements of Ω can be written as

$$\mathbb{V}(u_{it}) = \mathbb{V}(\mu_i + v_{it}) = \sigma_\mu^2 + \sigma_v^2 + 2\text{Cov}(\mu_i, v_{it}) = \sigma_\mu^2 + \sigma_v^2$$

For the covariance elements, we have

$$\text{Cov}(u_{it}, u_{is}) = \text{Cov}(\mu_i + v_{it}, \mu_i + v_{is}) = \mathbb{E}((\mu_i + v_{it})(\mu_i + v_{is})) = \sigma_\mu^2$$

This is true for any i across time periods within an individual. Now, $\text{Cov}(u_{it}, u_{is}) = 0$ for any $i \neq j$ since we assumed independence. Therefore, we can write $\Omega = \sigma_v^2 I_{nT} + \sigma_\mu^2 I_n \otimes J_T$

Therefore, the variance-covariance matrix of the composite errors, $\Omega = E[uu']$, is an $nT \times nT$ block-diagonal matrix. Each $T \times T$ block corresponds to an individual i and has a specific structure where the diagonal is $\sigma_\mu^2 + \sigma_v^2$ and all off-diagonals are σ_μ^2 .

$$\Omega = \begin{pmatrix} \begin{pmatrix} \sigma_\mu^2 + \sigma_v^2 & \cdots & \sigma_\mu^2 \\ \vdots & \ddots & \vdots \\ \sigma_\mu^2 & \cdots & \sigma_\mu^2 + \sigma_v^2 \end{pmatrix} & 0 & \cdots & 0 \\ 0 & \begin{pmatrix} \sigma_\mu^2 + \sigma_v^2 & \cdots & \sigma_\mu^2 \\ \vdots & \ddots & \vdots \\ \sigma_\mu^2 & \cdots & \sigma_\mu^2 + \sigma_v^2 \end{pmatrix} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \begin{pmatrix} \sigma_\mu^2 + \sigma_v^2 & \cdots & \sigma_\mu^2 \\ \vdots & \ddots & \vdots \\ \sigma_\mu^2 & \cdots & \sigma_\mu^2 + \sigma_v^2 \end{pmatrix} \end{pmatrix}$$

Rewriting the model as $y = X'\beta + u$, we can use a **pooled OLS** estimator:

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'y$$

OLS is unbiased and consistent, but not efficient because of the non-spherical errors. We can therefore use a **pooled GLS** estimator

$$\hat{\beta}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$$

which is consistent and BLUE.

Now, the **Between Estimator** discards all information regarding how individuals change over time and focuses exclusively on the variation *between* individuals. We apply the projection matrix $P_D = D(D'D)^{-1}D' = I_n \otimes \frac{1}{T}J_T$ to our model:

$$P_D y = P_D X \beta + P_D u$$

In scalar form, this is equivalent to running OLS on the individual time-averages:

$$\bar{y}_i = \alpha + \bar{x}_i' \beta + \bar{u}_i, \quad i = 1, \dots, n$$

where $\bar{u}_i = \mu_i + \bar{v}_i$. The between estimator is written as

$$\hat{\beta}_B = (X'P_DX)^{-1}X'P_D y$$

- **Consistency:** $\hat{\beta}_B$ is consistent if $E[\bar{u}_i | \bar{x}_i] = 0$, which is satisfied under the RE assumption $E[\mu_i | X] = 0$.
- **Efficiency:** It is generally **inefficient** compared to GLS because it collapses nT observations into only n averages, throwing away the "within" variation.

We can also rewrite our within estimator as hat

$$\beta_W = (X'M_DX)^{-1}X'M_D y$$

which is also unbiased and consistent but inefficient.

Now, recall P (which we called P_D) and M (which we called M_D) are symmetric, idempotent, and orthogonal projectors ($P \times M = 0$). Now, since $P = I_n \otimes \frac{1}{T}J_T$, we have $TP = I_n \otimes J_T$. Additionally, recall $M = I_{nT} - I_n \otimes \frac{1}{T}J_T$. It is easy to see that $M + P = I_{nT}$. Therefore, writing Ω as a function of these projectors, we get

$$\Omega = \sigma_v^2 M + (\sigma_v^2 + T\sigma_\mu^2)P$$

This is what we call a **spectral decomposition**: it expresses Ω as a weighted sum of its eigen-projections. Now, because P and M are orthogonal, any power of Ω is found by simply raising the scalar coefficients to that power. To find the inverse, we take the reciprocal of the coefficients:

$$\Omega^{-1} = \frac{1}{\sigma_v^2} M + \frac{1}{\sigma_v^2 + T\sigma_\mu^2} P$$

For GLS, we specifically need the "weighting" matrix $\Omega^{-1/2}$ to transform the data so that the errors become spherical. Applying the same logic, we take the square root of the coefficients:

$$\Omega^{-1/2} = \frac{1}{\sigma_v} M + \frac{1}{\sqrt{\sigma_v^2 + T\sigma_\mu^2}} P$$

By substituting $M = I_{nT} - P$ back into this equation, we arrive at the quasi-demeaning factor θ :

$$\Omega^{-1/2} = \frac{1}{\sigma_v} [I_{nT} - (1 - \sqrt{\theta})P] \text{ where } \theta = \frac{\sigma_v^2}{\sigma_v^2 + T\sigma_\mu^2}$$

θ is basically the ratio of the idiosyncratic variance to the total weighted variance. For each observation, the transformation is applied as follows:

$$y_{it} - (1 - \sqrt{\theta})\bar{y}_i = [x_{it} - (1 - \sqrt{\theta})\bar{x}_i]' \beta + r_{it}$$

where $\bar{y}_i$ and $\bar{x}_i$ are the individual-specific time averages. r_{it} is the transformed error term which is now spherical (homoskedastic and uncorrelated), allowing for efficient estimation. This is therefore consistent, BLUE and asymptotically normal.

Now, let's try to build some intuition. First, instead of subtracting the entire mean (as in Fixed Effects), we only subtract a portion of it, $(1 - \sqrt{\theta})$, which preserves the between-unit variation, and allows the model to estimate the effects of time-invariant variables and potentially increases efficiency.

Remark 9.1: Interpretation of θ

The parameter $\theta = \frac{\sigma_v^2}{\sigma_v^2 + T\sigma_\mu^2}$ determines the weight of the quasi-demeaning.

- If $\theta = 1$, we have **Pooled OLS**. This occurs when there is no individual-specific variance ($\sigma_\mu^2 = 0$).
- If $\theta = 0$, we have the **Within Estimator**. This occurs as the individual effects dominate ($\sigma_\mu^2 \rightarrow \infty$) or as the number of time periods $T \rightarrow \infty$.

Essentially, RE is a data-driven choice between OLS and FE based on the relative importance of within-unit and between-unit variance.

9.3 Fixed Effects vs. Random Effects

If both n and T are large, it makes no difference whether we treat individual specific effects μ_i as fixed or random because both the within estimator and the GLS estimator become the same. If T is small and n large, there are differences.

The advantage with FE is that there is no need to assume that individual effects are uncorrelated with covariates, but there is the issue of incidental parameters. For RE, a pro is

that the number of parameters is fixed, allowing for efficient estimation, but it relies on a strong assumption on uncorrelation of individual effects with covariates.

The choice can be viewed through two lenses:

- **Inference Type:** Use FE for *conditional inference* on a specific set of unique units (e.g., OECD countries). Use RE for *unconditional inference* where units are treated as homogeneous drawings from a larger population (e.g., a random sample of individuals).
- **The Consistency-Efficiency Trade-off:** RE uses GLS to provide efficient estimates but requires $\text{Cov}(\mu_i, x_{it}) = 0$. FE remains consistent regardless of this correlation but is less efficient as it discards between-unit variation.

Decision Rule: The **Hausman Test** is used to statistically determine if the RE assumption of zero correlation holds. If the test rejects, the RE estimates are biased, and FE must be used.

Chapter 2: Identification and Causal Inference

1 Introduction to Econometrics Reasoning, Statistical Learning and Micro-metrics

After the math-heavy treatment of econometrics in the first chapter, we move to the causal inference part. We rebuild some of the key-concepts, which might sound a bit redundant at first, but which I believe is important for understanding.

Econometrics studies data that record economic outcomes and the institutional environment that produced them. Applied microeconomists are often interested in *policy-relevant* questions: how do interventions (taxes, benefits, school reforms, class-size policies, etc.) affect outcomes such as earnings, labor supply, or learning? Modern microeconometrics focuses on across-agent heterogeneity and distributional consequences rather than representative-agent averages. Indeed, if the taste for goods and services varies across agents, policy actions must affect different agents differently. This heterogeneity can be with respect to *observable characteristics*, which is easily modeled, but also can also come from *unobservable characteristics*, which is what we are interested in.

1.1 Policy relevant parameters

Consider a simple model of human capital investment.

$$W_i = \alpha_0 + \alpha_1 S_i + \alpha_2 X_i + \alpha_3 \varepsilon_{2i} + \varepsilon_{1i} \quad (1.1)$$

$$S_i = \beta_0 + \beta_1 X_i + \varepsilon_{2i} \quad (1.2)$$

where S_i is a measure of investment, W_i is log wage, X_i are observable characteristics of individual i , and $\varepsilon_{1i}, \varepsilon_{2i}$ are unobservables. ε_{2i} allows individuals otherwise identical with respect to X_i to have different values of S_i , and can be thought of as endowed ability. ε_{1i} is purely random noise, and allows individuals with identical values of S_i, X_i and ε_{2i} to receive different wages. $\varepsilon_{1i}, \varepsilon_{2i}$ are *random variables*, and vary across individuals according to a probability distribution, which implies that W_i, S_i also are random variables conditional on X_i . The terms $\alpha_j, j \in \{1, 2, 3\}$ and $\beta_k, k \in \{0, 1\}$ are unknown **structural** parameters in the model. They are assumed to be deep and policy invariant. Indeed, S_i could change because of a policy, but α_1 would not.

Now, to generate W_i, S_i , considering X_i fixed, we need values of α_j and distributions for $\varepsilon_{1i}, \varepsilon_{2i}$. Such a fully specified data generating process (DGP) is called a structure, and equations in this structure (1.1, 1.2) are structural equations. A structural model puts restrictions on ε_{2i} to retrieve the parameters of interest. As economists, we are often interested in pinning down α_1 , i.e. the returns to schooling. In the linear model above, α_1 is the percent change in earnings resulting from one additional year spent at school, *ceteris paribus*. Note that the value of α_1 is the one generated by a **thought experiment**

in which individuals acquire more years of schooling independently of their observable and unobservable characteristics. In this sense, α_1 is the **causal effect** of education on earnings. Causality is defined by a counterfactual question, which is why it is important to think about what is compared to what. In this context, many cases could be considered: finishing college vs not finishing college; finishing college vs not finishing high-school, etc.

1.2 Identification

1.2.1 Understand Identification

Identification concerns whether a structural parameter can be uniquely determined from the distribution of the observable variables that generate the data. Formally, identification occurs if there exists a one-to-one mapping between the population distribution of the data and the parameter of interest. If multiple parameter values are consistent with the same observable distribution, the parameter is not identified.

It is crucial to distinguish identification from estimation. Identification is a population-level concept: it asks what could be learned if we had access to the entire population distribution (or, equivalently, an infinite sample). Estimation, by contrast, is a finite-sample problem: it concerns how to construct procedures that approximate the object of interest using observed data.

This distinction is closely related to the difference between an *estimand* and an *estimator*. The estimand is the population object we aim to learn about (for instance, a structural parameter such as α_1 or a causal parameter such as an average treatment effect). The estimator is a function of the sample data designed to approximate that estimand. Identification precedes estimation: if an estimand is not identified, no estimator—no matter how sophisticated—can consistently recover it, even with arbitrarily large samples.

1.2.2 Schooling example (continued)

We continue using our schooling example in this identification context. First, note that each structure implies a particular **probability distribution** for W_i and S_i given X_i . However, the same probability distribution for the observable random variables may be generated by different structures. Such structures are called observationally equivalent. Consider the linear wage-schooling model when we substitute for ε_{2i} in the wage equation:

$$W_i = \alpha_0 + \alpha_1 S_i + \alpha_2 X_i + \alpha_3 (S_i - \beta_0 - \beta_1 X_i) + \varepsilon_{1i}$$

$$W_i = (\alpha_0 - \beta_0 \alpha_3) + (\alpha_1 + \alpha_3) S_i + (\alpha_2 - \beta_1 \alpha_3) X_i + \varepsilon_{1i} \equiv \gamma_0 + \gamma_1 S_i + \gamma_2 X_i + \varepsilon_{1i}$$

Since the conditional expectation of W_i given S_i and X_i is linear in S_i and X_i , a regression of W_i on S_i and X_i identifies $\gamma_1 = \alpha_1 + \alpha_3$, not α_1 . In this context, no amount of statistical analysis can recover α_1 , because many different values of the structural parameters α generate the same reduced-form coefficients γ , and therefore the same observable distribution.

The question is therefore how can we establish a one-to-one correspondence between data and features of the structure (such as returns to schooling)? In the schooling case, as ε_{2i} is not observed, equation 1.2 is not restrictive enough to allow for identification of the β 's. If large values of ability ε_{2i} is associated with large values of school quality X_i , then data on X_i and S_i alone cannot distinguish the impact of X_i and ε_{2i} on S_i . This is why we need to impose **identifying restrictions**.

To achieve identification of the β 's, we must impose some restriction on the co-variation between ε_{2i} and X_i . Independence or weaker conditions like $\mathbb{E}(\varepsilon_{2i} \mid X_i = x) = 0$ are sufficient. The validity of this assumption may follow by *design*, for instance if school enrolment is determined by lotteries. It could also follow from economic arguments in the specific context. The study of correspondences between data and the structural feature of interest is called **identification strategy**.

“Identification logically comes first. Negative identification findings imply that statistical inference is fruitless: it makes no sense to try to use a finite sample to learn something that could not be learned even with an infinite sample. Positive identification findings imply that one should then examine the feasibility of statistical inference.”

— Manski (1995)

1.3 Statistical Reasoning

Statistical reasoning concerns the relationship between models and the observable distribution of the data. In contrast to identification, which asks what can be learned from the population distribution, statistical analysis focuses on how to summarize and approximate that distribution using sample data.

In practice, econometric methods often aim to estimate features of conditional distributions, such as the regression function

$$\mathbb{E}[W_i \mid S_i = s, X_i = x].$$

These objects are typically expressed as functionals of population moments. Estimation then proceeds by the *analogy principle*: population quantities (expectations, variances, regression functions) are replaced by their sample counterparts.

It is important to distinguish different notions of validity. **Statistical validity** concerns the precision of estimators and the reliability of inference in finite samples. **Internal validity** refers to whether the estimated relationship can be given a causal interpretation. **External validity** concerns the extent to which conclusions generalize beyond the population under study.

Modern microeconometrics often relies on flexible, data-driven tools—such as local regression and resampling methods—that impose minimal functional form assumptions. However, these statistical tools operate within an identification framework: they refine inference about parameters that must first be shown to be identified.

1.4 Policy Relevant Probability

1.4.1 The STAR Experiment

The Tennessee STAR (Student-Teacher Achievement Ratio) experiment was designed to evaluate the causal effect of class size on student performance. The policy question is whether reducing class size improves educational outcomes, given that smaller classes are costly. Beginning in 1985, a cohort of kindergarten students was randomly assigned to one of three types of classes:

- Regular-size classes (approximately 22-25 students),
- Regular-size classes with a full-time teacher's aide,
- Small classes (approximately 13-17 students).

Only schools with at least three classes per grade participated, and teachers were also randomly assigned to classes. Random assignment of students (and teachers) ensures that differences in outcomes across class types can be attributed to class size rather than to systematic differences in student characteristics. We use this example below to introduce potential outcomes for the first time.

1.4.2 Potential Outcomes

Let class size be described by a binary random variable D that takes value one for small classes and zero for regular classes. $\{Y(1), Y(0)\}$ are potential outcomes (scores) if enrolled in small or regular class, respectively (treatment is small class here). Note that potential outcomes are defined for the **same individual**. We can write the outcome revealed from the data as follows

$$Y = Y(0) + D(Y(1) - Y(0)) \quad (1.3)$$

$Y(1) - Y(0)$ is the class size effect, which is student specific and therefore unobservable, since we can only observe $Y(1)$ or $Y(0)$ simultaneously. The following moments can be retrieved from averages in the data:

Scores in small classes: $\mathbb{E}(Y \mid D = 1)$; Scores in regular classes: $\mathbb{E}(Y \mid D = 0)$

We can of course consider conditional versions of the above quantities (e.g. scores for students from low income families). No parametric form is imposed whatsoever. We can also estimate marginal distributions for $D = 0$ and $D = 1$: $F_Y(y \mid D = 0)$ and $F_Y(y \mid D = 1)$.

Since we cannot learn about unit-level effects, causal conclusions must rely on multiple units with different exposure to treatment. Potential outcomes should, in principle, be indexed by the entire vector of treatment assignments in the population, $D = (D_1, \dots, D_N)$. That is, the outcome of individual i could depend not only on her own treatment status D_i , but also on the treatment status of other individuals. In that case, we would write potential outcomes as $Y_i(D)$.

This allows for the possibility that causal effects vary with the overall pattern of treatment assignment (for instance, through peer effects or general equilibrium effects).

1.4.3 Objects of interest

In most empirical applications, however, we invoke an exclusion restriction—often referred to as the Stable Unit Treatment Value Assumption (SUTVA)—which assumes that an individual's potential outcomes depend only on her own treatment status. Under this assumption, we can write the potential outcomes simply as $Y_i(1)$ and $Y_i(0)$, without reference to the treatment status of other units.

We can now define our objects of interest. The first one is simply the Average Treatment Effect.

Definition 1.1: Average Treatment Effect (or gain)

The Average Treatment Effect (ATE) is defined as

$$\mathbb{E}[Y(1) - Y(0)].$$

The expectation is taken with respect to the population distribution of units. The ATE measures the average impact of the treatment across the population, computed over the distribution of individual-level treatment effects (based on each individual's potential outcomes).

Note that, by construction, the object $\mathbb{E}[Y(1) - Y(0)]$ is not directly observable, since we never observe both potential outcomes for the same individual. Linearity of expectation implies

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)].$$

However, the marginal expectations $\mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0)]$ are themselves not directly observable in general. What we observe instead are

$$\mathbb{E}[Y \mid D = 1] = \mathbb{E}[Y(1) \mid D = 1], \quad \mathbb{E}[Y \mid D = 0] = \mathbb{E}[Y(0) \mid D = 0],$$

where both equalities follow from SUTVA, which rules out interference (so that $Y_i(d)$ depends only on i 's own treatment) and imposes consistency (conditioning on $D = 1$ or $D = 0$ selects $Y(1)$ or $Y(0)$ as the realized outcome, respectively).

These quantities coincide with $\mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0)]$ only under appropriate identifying assumptions, such as random assignment. We will come back to this later.

We could also be interested in the distribution of gains.

Definition 1.2: Distribution of Gains

The distribution of gains is defined as the cumulative distribution function (CDF) of individual treatment effects:

$$F_{Y(1)-Y(0)}(t) = \mathbb{P}(Y(1) - Y(0) \leq t).$$

This object describes the full distribution of treatment effects in the population. For

each individual i , one can define the unit-level effect $Y_i(1) - Y_i(0)$, and the distribution of gains summarizes how these effects are distributed across individuals.

In general, the distribution of gains is not identified, because $Y(1)$ and $Y(0)$ are never jointly observed. Without additional assumptions on the joint distribution of potential outcomes, the distribution of their difference cannot be recovered from observable data.

We can now introduce quantiles formally.

Definition 1.3: Quantile

Let Z be a real-valued random variable with cumulative distribution function

$$F_Z(z) = \mathbb{P}(Z \leq z).$$

For $\tau \in (0, 1)$, the τ -quantile of Z is defined as

$$Q_Z(\tau) \equiv \inf \{z \in \mathbb{R} : F_Z(z) \geq \tau\}.$$

The quantile $Q_Z(\tau)$ represents the smallest value z such that at least a fraction τ of the population has $Z \leq z$. In other words, it is the value below which a proportion τ of the distribution lies.

Proposition 1.1: Identification of QTE

Assume $\{Y_d : d \in \mathcal{D}\} \perp\!\!\!\perp D$, where $Y_d = Y(d)$ for each $d \in \mathcal{D}$. Then the quantile treatment effect

$$QTE_\tau(d, d') \equiv Q_{Y_d}(\tau) - Q_{Y_{d'}}(\tau),$$

is identified for every $\tau \in (0, 1)$

Proof. Let $\tau \in (0, 1)$ be the percentile in the distribution. For any $d \in \mathcal{D}$,

$$Q_{Y_d}(\tau) \equiv \inf\{y \in \mathbb{R} : F_{Y_d}(y) \geq \tau\}$$

provides what is the grade that corresponds to a specific τ . The quantile treatment effects

$$QTE_\tau(d, d') \equiv Q_{Y_d}(\tau) - Q_{Y_{d'}}(\tau)$$

therefore provides the difference in quantiles for $d, d' \in \mathcal{D}$.

Now, we can work towards the identification question. First, notice that the potential outcome CDF for any $d \in \mathcal{D}$ is written as

$$F_{Y_d}(y) = \mathbb{P}(Y_d \leq y)$$

By the independence assumption $\{Y_d\} \perp\!\!\!\perp D$ and since $Y = Y_D$, we have

$$F_{Y_d}(y) = \mathbb{P}(Y_d \leq y) = \mathbb{P}(Y_d \leq y | D = d) \stackrel{\perp\!\!\!\perp}{=} \mathbb{P}(Y \leq y | D = d)$$

provided $\mathbb{P}(D = d) > 0$.

The potential outcomes CDF can therefore be computed using observable data (Y, D) only, which allows to compute the quantiles for any $d, d' \in \mathcal{D}$ and recover the quantile treatment effects. Identification comes from independence and the fact that $QTE_\tau(d, d')$ depends only on the marginal distributions F_{Y_d} and $F_{Y_{d'}}$. $\square$

Proposition 1.2: Identification of QoTE

Assume $\{Y_d : d \in \mathcal{D}\} \perp\!\!\!\perp D$, where $Y_d = Y(d)$ for each $d \in \mathcal{D}$. The quantile of treatment effects

$$QoTE_\tau(d, d') \equiv Q_{\Delta_{d,d'}}(\tau), \quad \Delta_{d,d'} \equiv Y_d - Y_{d'},$$

is not identified in general.

Proof. We have that for any $d, d' \in \mathcal{D}$, the treatment effect is defined as

$$\Delta_{d,d'} \equiv Y_d - Y_{d'}$$

The CDF of this object can be written as

$$F_{\Delta_{d,d'}}(t) = \mathbb{P}(\Delta_{d,d'} \leq t) = \mathbb{P}(Y_d - Y_{d'} \leq t)$$

While the independence assumption identifies the marginal distributions F_{Y_d} and $F_{Y_{d'}}$, the distribution of the difference $F_{Y_d - Y_{d'}}$ depends on the joint distribution of $(Y_d, Y_{d'})$, which we can never compute since we never observe both potential outcomes for the same individual.

To see this concretely, consider the following example. Let $d = 1, d' = 0$ and suppose $Y_0, Y_1 \sim \text{Bernoulli}(1/2)$ (so the marginals are identical across two data-generating processes). Consider two joint laws with the same marginals:

- *Perfect positive dependence:* $Y_1 = Y_0$ a.s. Then $\Delta_{1,0} = 0$ a.s., so $Q_{\Delta_{1,0}}(\tau) = 0$ for all τ .
- *Perfect negative dependence:* $Y_1 = 1 - Y_0$ a.s. Then $\Delta_{1,0} \in \{-1, 1\}$ with $\Pr(\Delta_{1,0} = 1) = \Pr(Y_0 = 0) = 1/2$, so for example $Q_{\Delta_{1,0}}(0.75) = 1$.

Both joint laws produce identical observable marginals F_{Y_0} and F_{Y_1} , but yield different values of $QoTE_{0.75}(1, 0)$. Therefore $QoTE_\tau(d, d')$ is not identified in general. $\square$

We have just shown that the quantile treatment effect $QTE_\tau(d, d')$ and the quantile of treatment effects $QoTE_\tau(d, d')$ capture fundamentally different objects. The $QTE_\tau(d, d')$ compares the τ -quantiles of the two marginal distributions Y_d and $Y_{d'}$, and therefore measures how the τ -th percentile of outcomes shifts across treatment states. It depends only on the marginal distributions F_{Y_d} and $F_{Y_{d'}}$. In contrast, the $QoTE_\tau(d, d')$ is the τ -quantile of the individual-level treatment effect $\Delta_{d,d'} = Y_d - Y_{d'}$. It therefore summarizes the distribution of treatment effects across individuals and depends on the joint distribution of $(Y_d, Y_{d'})$. While independence identifies the marginals and thus the $QTE_\tau(d, d')$, it does not identify the joint dependence structure, so $QoTE_\tau(d, d')$ is generally not identified.

1.4.4 Relevant Variability

Potential outcomes $\{Y(1), Y(0)\}$ should be viewed as **realizations** of random variables (or attributes/traits) that remain fixed for each student irrespective of the value of D . The random variables yielding these attributes need not be identically distributed across students, for example because they may vary with demographics. If class size is randomly assigned, we have

$$\mathbb{P}(D = 1 | Y(1) = y_1, Y(0) = y_0) = \mathbb{P}(D = 1)$$

Now, if N is the size of students in the STAR cohort, the distribution of $Y(1) - Y(0)$ across students is centered at

$$\frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$$

which is the (population) average effect of enrolling in a small class.

Example 1.1: The Perfect Doctor and Selection Based on Gains

Consider $N = 8$ patients with potential outcomes under standard surgery ($Y(0)$) and new surgery ($Y(1)$). The table below lists the assignment indicator D (treatment chosen by the “perfect doctor”) together with $(Y(0), Y(1))$ for each patient. The perfect doctor chooses $D = 1$ whenever $Y_i(0) < Y_i(1)$.

D	$Y(0)$	$Y(1)$
1	13	14
0	6	0
0	4	1
0	5	2
0	6	3
0	6	1
1	8	10
1	8	9

True averages. The population (true) averages of the potential outcomes are

$$\bar{Y}(0) = \frac{1}{8} \sum_{i=1}^8 Y_i(0) = 7, \quad \bar{Y}(1) = \frac{1}{8} \sum_{i=1}^8 Y_i(1) = 5,$$

so the true average treatment effect (ATE) is

$$\text{ATE} = \bar{Y}(1) - \bar{Y}(0) = 5 - 7 = -2.$$

Observed outcomes and naive comparison. The observed outcome is $Y_i = Y_i(0) + D_i(Y_i(1) - Y_i(0))$. Given the assignment D in the table, the observed sample means by group are

$$\bar{Y} |_{D=0} = \frac{6 + 4 + 5 + 6 + 6}{5} = \frac{27}{5} = 5.4, \quad \bar{Y} |_{D=1} = \frac{14 + 10 + 9}{3} = \frac{33}{3} = 11.$$

A naive comparison of group means would therefore suggest a treatment effect of $11 - 5.4 = 5.6$, which is the opposite sign and very different in magnitude from the true ATE: -2 .

This example illustrates the importance of the assignment mechanism: when assignment is endogenous (here chosen by a “perfect doctor” who selects treatment for units expected to benefit), naive comparisons of observed means do not recover causal effects.

2 Inference and Randomized Experiments

In this section, we begin by discussing selection bias formally, decomposing the naive difference in means into a causal effect and a bias term. We then show how this decomposition can be expressed equivalently using potential outcomes notation and linear regression. Next, we introduce a taxonomy of research designs using Directed Acyclic Graphs (DAGs) to clarify identification assumptions. We then turn to randomized experiments, distinguishing between sampling-based and design-based uncertainty. Finally, we study inference in experiments through permutation (randomization) inference and Neyman’s approach to variance estimation and confidence intervals.

2.1 Selection Bias, Randomization and Regression

Recall that $Y = Y(0) + D(Y(1) - Y(0))$. The naive comparison is given by comparing the difference between the observed outcome conditional on treatment and control

$$\mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0) = \mathbb{E}(Y(1) \mid D = 1) - \mathbb{E}(Y(0) \mid D = 0),$$

This does not necessarily identify the causal effect of interest. Indeed, by adding and subtracting the counterfactual $\mathbb{E}(Y(0) \mid D = 1)$, we obtain the following decomposition (using linearity)

$$\mathbb{E}(Y(1) - Y(0) \mid D = 1) + \left[\mathbb{E}(Y(0) \mid D = 1) - \mathbb{E}(Y(0) \mid D = 0) \right].$$

The first term, is the average treatment effect on the treated (ATT). It can be written as

$$\mathbb{E}(Y(1) - Y(0) \mid D = 1) = \mathbb{E}(Y(1) \mid D = 1) - \mathbb{E}(Y(0) \mid D = 1),$$

where the first term is observed and the second is the counterfactual mean outcome for treated units under control. The counterfactual can be estimated if the selection bias is equal to zero. All methods we could think about in applied microeconometrics boil down to the imputation of this counterfactual given the assumptions that make the bias equal zero.

The second term is the selection bias. It compares score in a world where all classes are large (no treatment) of kids assigned to treatment (small classes) compared to kids in the

control group. If the selection bias is positive, we say there is positive selection, meaning that on average, kids that get the small class treatment have better scores than kids that don't get access to treatment. Random assignment solves the selection problem by design in this context.

Indeed, randomization balances the distribution of subjects across treatment groups. Claiming that $D = 1$ and $D = 0$ would have had the same counterfactual scores $Y(0)$ amounts to stating the following **orthogonality condition**

$$F_{Y(0)}(y|D = 0) = F_{Y(0)}(y|D = 1)$$

which implies $\mathbb{E}(y_0|D = 0) = \mathbb{E}(y_0|D = 1)$ and therefore no selection bias. Note that this condition can be stated simply as $Y(0) \perp\!\!\!\perp D$.

We can now show that the potential outcome notation is equivalent to a regression analysis.

Proposition 2.1: Potential Outcomes and Linear Regression

Suppose treatment effects are homogeneous, so that $Y(1) - Y(0) = \beta$ for all units. Then the potential outcomes model

$$Y = Y(0) + D(Y(1) - Y(0))$$

can be written as the linear regression model

$$Y = \alpha + \beta D + U,$$

where $\alpha = \mathbb{E}(Y(0))$ and $U = Y(0) - \mathbb{E}(Y(0))$.

Proof. First, assume that we are in the case of homogeneous returns for simplicity. Again, recall that $Y = Y(0) + D(Y(1) - Y(0))$. Adding and subtracting $\mathbb{E}(Y(0))$, we can write

$$Y = \mathbb{E}(Y(0)) + D(Y(1) - Y(0)) + Y(0) - \mathbb{E}(Y(0)) \equiv \alpha + \beta D + U$$

Now, consider the feasible regression of Y on D

$$\mathbb{E}(Y|D = d) = \alpha + \beta d + \mathbb{E}(U|D = d)$$

In our binary case, we can evaluate this at $D = 1$ and $D = 0$, which yields

$$\mathbb{E}(Y|D = 1) = \alpha + \beta + \mathbb{E}(U|D = 1) \text{ and } \mathbb{E}(Y|D = 0) = \alpha + \mathbb{E}(U|D = 0)$$

Under homogeneous treatment effects, we have that $Y(1) - Y(0)$ is constant, and therefore $Y(1) - Y(0) = \mathbb{E}(Y(1) - Y(0))$. Taking the difference between the two expressions above using this, we get

$$\mathbb{E}(Y|D = 1) - \mathbb{E}(Y|D = 0) = \beta + \mathbb{E}(U|D = 1) - \mathbb{E}(U|D = 0)$$

Now, recall that $U \equiv Y(0) - \mathbb{E}(Y(0))$. Hence, for $d \in \{0, 1\}$,

$$\mathbb{E}(U \mid D = d) = \mathbb{E}(Y(0) - \mathbb{E}(Y(0)) \mid D = d) = \mathbb{E}(Y(0) \mid D = d) - \mathbb{E}(Y(0)),$$

since $\mathbb{E}(Y(0))$ is a constant. Therefore,

$$\begin{aligned} \mathbb{E}(U \mid D = 1) - \mathbb{E}(U \mid D = 0) &= [\mathbb{E}(Y(0) \mid D = 1) - \mathbb{E}(Y(0))] - [\mathbb{E}(Y(0) \mid D = 0) - \mathbb{E}(Y(0))] \\ &= \mathbb{E}(Y(0) \mid D = 1) - \mathbb{E}(Y(0) \mid D = 0). \end{aligned}$$

Substituting back yields

$$\mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0) = \beta + [\mathbb{E}(Y(0) \mid D = 1) - \mathbb{E}(Y(0) \mid D = 0)].$$

□

This proof allows us to see that random assignment to classes is equivalent to stating that the error U is uncorrelated/mean independent with treatment D , or that treatment is exogenous.

2.2 Taxonomy of Research Designs

To begin with, consider the general OVB example through the following example where we regress Y on treatment D and observables X :

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \beta_3 U + \varepsilon$$

where U is unobservable and ε is a random error component (harmless).

Proposition 2.2: Omitted Variable Bias (OVB)

Suppose the true (long) model is

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \beta_3 U + \varepsilon,$$

where ε satisfies $\mathbb{E}(\varepsilon \mid D, X, U) = 0$.

Consider instead the short regression of Y on D only. Then the population coefficient on D in the short regression is

$$\beta_{\text{short}} = \frac{\text{Cov}(D, Y)}{\text{V}(D)} = \beta_1 + \beta_2 \frac{\text{Cov}(D, X)}{\text{V}(D)} + \beta_3 \frac{\text{Cov}(D, U)}{\text{V}(D)}.$$

In particular, the short regression coefficient equals the true causal effect β_1 plus bias terms arising from the correlation between D and the omitted variables.

Proof. By definition of the population regression coefficient,

$$\beta_{\text{short}} = \frac{\text{Cov}(D, Y)}{\mathbb{V}(D)}$$

Substituting the true model,

$$\text{Cov}(D, Y) = \text{Cov}(D, \beta_1 D + \beta_2 X + \beta_3 U + \varepsilon).$$

By linearity of covariance and using $\text{Cov}(D, \varepsilon) = 0$, we obtain

$$\text{Cov}(D, Y) = \beta_1 \mathbb{V}(D) + \beta_2 \text{Cov}(D, X) + \beta_3 \text{Cov}(D, U).$$

Dividing by $\mathbb{V}(D)$ yields the result. □

Example 2.1: OVB sign with parental income

Suppose X is parental income and the true model is

$$Y = \beta_0 + \beta_1 D + \beta_2 X + \beta_3 U + \varepsilon,$$

where Y is a test score and $D = 1$ denotes assignment to a small class.

If wealthier parents are more likely to enroll their children in schools with small classes, then treatment is positively associated with income:

$$\text{Cov}(D, X) > 0.$$

If parental income has a positive causal effect on test scores, $\beta_2 > 0$, then the OVB term

$$\beta_2 \frac{\text{Cov}(D, X)}{\mathbb{V}(D)}$$

is positive. Consequently the short regression coefficient

$$\frac{\text{Cov}(D, Y)}{\mathbb{V}(D)} = {}^a \beta_1 + \beta_2 \frac{\text{Cov}(D, X)}{\mathbb{V}(D)} + \beta_3 \frac{\text{Cov}(D, U)}{\mathbb{V}(D)}$$

is upward biased relative to β_1 (holding the U -term fixed).

For the binary treatment case one useful identity simplifies intuition: when $D \in \{0, 1\}$,

$$\frac{\text{Cov}(D, X)}{\mathbb{V}(D)} = \mathbb{E}[X \mid D = 1] - \mathbb{E}[X \mid D = 0],$$

so the bias term from X equals $\beta_2 (\mathbb{E}[X \mid D = 1] - \mathbb{E}[X \mid D = 0])$. Thus, if treated units have higher average parental income and $\beta_2 > 0$, the omitted-income bias is positive and the naive comparison of means overstates the causal effect of small class assignment.

^aTo see this, let $p = \Pr(D = 1)$. Since $D \in \{0, 1\}$,

$$\text{Cov}(D, X) = \mathbb{E}[DX] - \mathbb{E}[D]\mathbb{E}[X].$$

Note that $\mathbb{E}[DX] = p \mathbb{E}[X | D = 1]$ and

$$\mathbb{E}[X] = p \mathbb{E}[X | D = 1] + (1 - p) \mathbb{E}[X | D = 0].$$

Substituting yields

$$\text{Cov}(D, X) = p(1 - p)(\mathbb{E}[X | D = 1] - \mathbb{E}[X | D = 0]).$$

Because $\mathbb{V}(D) = p(1 - p)$ for a binary treatment,

$$\frac{\text{Cov}(D, X)}{\mathbb{V}(D)} = \mathbb{E}[X | D = 1] - \mathbb{E}[X | D = 0].$$

We can now introduce a nice visualisation object for causal inference: Directed Acyclic Graphs (DAGs), which are directed graphs with no directed cycles. In the following example, we omit ε because it is an innocuous variable. Dashed lines between nodes involve unobserved variables. We start with a general case.

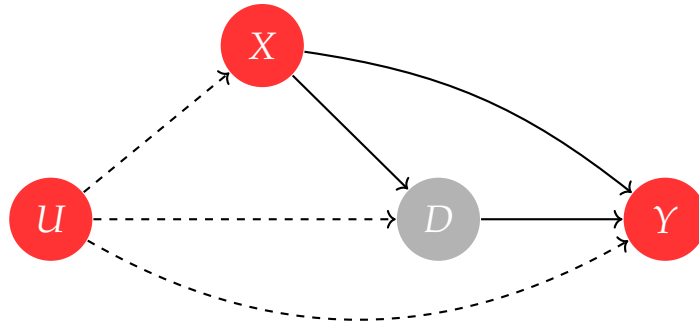


Figure 9: General case with backdoor paths

Figure 9 presents a general case where the paths $D \leftarrow U \rightarrow Y$ and $D \leftarrow X \rightarrow Y$ are called **back-door paths**. The idea is that when we regress Y on D , we use variation in D to explain variation in Y . Since D is related to X, U , variation in D may reflect variation in X, U . The observed association between D and Y may reflect these back-door paths rather than a causal effect of D . The path $D \leftarrow X \rightarrow Y$ can be blocked since X is observable, but U is a confounder since it is not. Controlling for X effectively implies that we are holding X constant (it's like placing your hand on the X node).

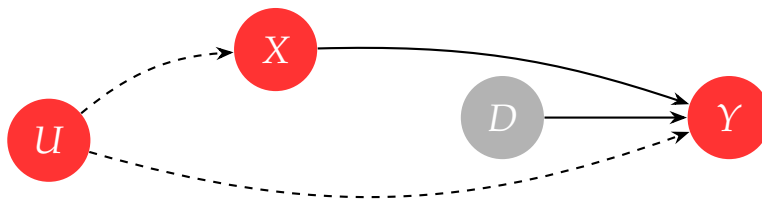


Figure 10: Random Assignment

Figure 10 presents a DAG in the case of random assignment (RA) of treatment. Under RA,

there are no arrows pointing into D , implying there are no back-door paths from D to Y . In particular, D is independent of X and U . As a result, observables and unobservables play no role for identification, since treatment variation is orthogonal to all pre-treatment characteristics.

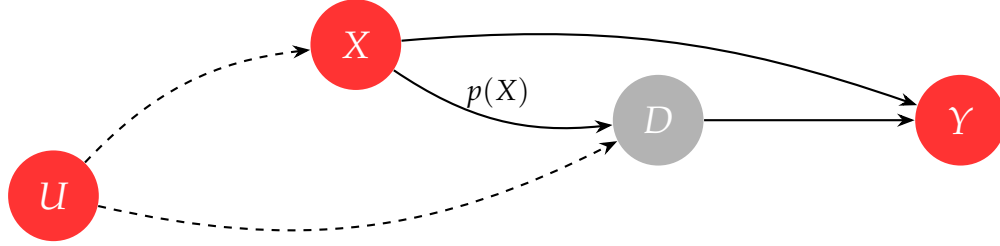


Figure 11: Selection on Observables

Figure 11 presents a case where we use conditioning to shut back-doors. First, note that this relies on a strong assumption: **selection on observables**. This assumption says that once we control for observables, there are no more confounders (all confounding operates through X). Indeed, on the DAG, note there is no edge linking U and Y , and all back-door paths are blocked once conditioning on X . In other words, $D \leftarrow X \rightarrow Y$ is the only back-door path, which is blocked once we control for X . Note that an important assumption in this context is the common support assumption, which can be written as follows

$$p(X) \equiv \mathbb{P}(D = 1|X) \in (0, 1)$$

where $p(X)$ is the propensity score. This implies that holding X fixed, there is still variation in D . We discuss selection on observables more later.

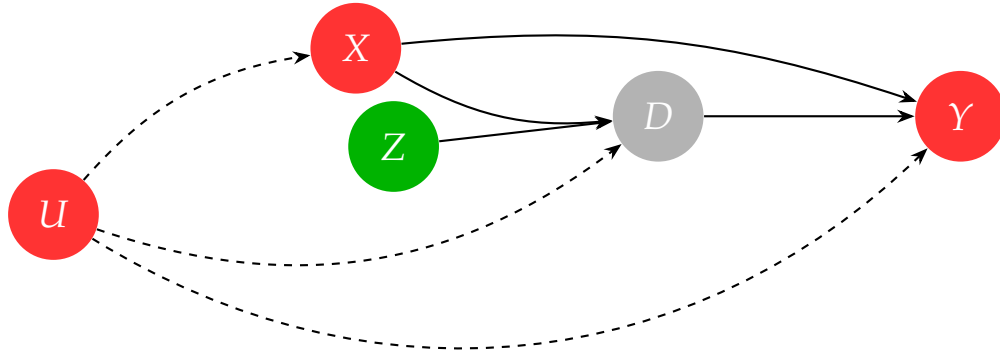


Figure 12: Instrumental Variable

Figure 12 illustrates the instrumental variable (IV) design. The instrument Z affects treatment D (relevance), but has no direct effect on the outcome Y (exclusion restriction). Moreover, there are no unobserved confounders affecting both Z and either D or Y (exogeneity). Under these conditions, variation in D induced by Z can be used to identify a causal effect of D on Y , even in the presence of unobserved confounding between D and Y .

2.3 Randomized Experiments

2.3.1 Taxonomy of Randomized Experiments

The assignment mechanism describes how each of N units in the population is assigned to treatment, and can be characterized by:

$$\mathbb{P}(\mathbf{D} \mid \mathbf{Y}(0), \mathbf{Y}(1), \mathbf{X}),$$

where $\mathbf{Y}(0)$, $\mathbf{Y}(1)$, $\mathbf{D}$ and $\mathbf{X}$ are the N -component vectors of potential outcomes, treatment status and covariates, with i -th element equal to $Y_i(0)$, $Y_i(1)$, D_i and X_i , respectively.

In randomized experiments, this probability is known and independent of potential outcomes. In a full randomization of $N_1 \equiv \sum_{i=1}^N D_i$ units, we have:

$$\mathbb{P}(\mathbf{D} \mid \mathbf{Y}(0), \mathbf{Y}(1), \mathbf{X}) = \binom{N}{N_1}^{-1},$$

where $N_0 \equiv \sum_{i=1}^N (1 - D_i)$, $N_0 = N - N_1$. This probability is based on all the ways to assign N_1 individuals in the treatment group, out of the N individuals. The realized assignment of N_1 units is one of the $\binom{N}{N_1}$ possible assignments that could have occurred.

Stratification on blocks $\mathcal{B}_j$ defined from covariates ensures variability in treatment assignment within each stratum, for $j = 1, \dots, J$.

In paired randomized experiments, the units included in the randomization (e.g., villages) are matched into pairs, and one unit in each pair is randomly assigned to treatment.

In a stratified randomization of $N_1(j) \equiv \sum_{i \in \mathcal{B}_j} D_i$ units, we have:

$$\mathbb{P}(\mathbf{D} \mid \mathbf{Y}(0), \mathbf{Y}(1), \mathbf{X}) = \prod_{j=1}^J \binom{N(j)}{N_1(j)}^{-1},$$

where $N_0(j) \equiv \sum_{i \in \mathcal{B}_j} (1 - D_i)$, $N(j) \equiv N_0(j) + N_1(j)$. One can think of different lotteries across strata, where the unit-level probability of treatment is

$$e(X_i) \equiv \frac{N_1(j)}{N(j)}.$$

Experiments cannot eliminate the problem of missing counterfactuals, but they ensure that missingness occurs at random. Randomization is valuable because it yields **exchangeability** ($Y(0), Y(1) \perp\!\!\!\perp D$) or **conditional exchangeability** ($Y(0), Y(1) \perp\!\!\!\perp D \mid X \in \mathcal{B}_j$). We say that groups $D = 1$ and $D = 0$ are exchangeable if their outcomes would coincide under identical exposure histories. Recall that causality is **design-based**: experiments require manipulation D_i to assess its relationship with Y_i . A research design defines the counterfactual and the parameter of interest.

2.3.2 Rethinking the Thought Experiment

We now discuss two very important concepts: Sampling-Based (SB) and Design-Based (DB) uncertainty. Let $R_i \in \{0, 1\}$ denote if i was sampled and $D_i \in \{0, 1\}$ if unit i was treated. Let n the total sample size from a population of size N . Size by treatment status is defined similarly (e.g. n_1 and N_1). Note that we can write $n_1 = \sum_{i=1}^N R_i D_i$ and $n_0 = \sum_{i=1}^N R_i (1 - D_i)$. There are two distinct sources of randomness, coming from sampling (via R_i) and treatment assignment (D_i). The difference-in-means estimator is

$$\hat{\tau} \equiv \frac{1}{n_1} \sum_{i=1}^{n_1} Y_i - \frac{1}{n_0} \sum_{i=1}^{n_0} Y_i = \frac{1}{n_1} \sum_{i=1}^N R_i D_i Y_i - \frac{1}{n_0} \sum_{i=1}^N R_i (1 - D_i) Y_i.$$

Multiplying and dividing by N , this can be written as

$$\hat{\tau} = \frac{1}{N} \sum_{i=1}^N \left(\frac{N R_i D_i}{n_1} \right) Y_i - \frac{1}{N} \sum_{i=1}^N \left(\frac{N R_i (1 - D_i)}{n_0} \right) Y_i.$$

This representation makes explicit that the estimator is a re-weighted population average. This is consistent with the general statement that any treatment comparison of weighted averages. In this context, $\omega_1 \equiv \frac{N R_i D_i}{n_1}$, $\omega_0 \equiv \frac{N R_i (1 - D_i)}{n_0}$. Imputation methods build down to understanding what the weights are.

Randomness enters through R_i (sampling variation from selection of units in the sample) and D_i (variation induced by the particular treatment allocation). Often, we think about the first source of uncertainty, since this is what impacts power computations. To do inference on the whole population, we need to take both sources of uncertainty into account. Taking $n \rightarrow \infty$ only solves the problem related to Sampling-Based uncertainty.

Now, suppose we are trying to evaluate the impact of a specific policy using a RCT. The population has size N , but we can only work with a sample of size n . After randomly selection this sample, we administer a baseline survey to all n units. Next, we randomly assign n_1 of these units to the intervention:

$$\mathbb{P}(R_i D_i = 1) = \mathbb{P}(D_i = 1 \mid R_i = 1) \mathbb{P}(R_i = 1) = \frac{n_1}{N} \frac{n}{N} = \frac{n_1}{N}$$

Similarly,

$$\mathbb{P}(R_i (1 - D_i) = 1) = \frac{n_0}{N}$$

The comparison of outcome averages can be written as

$$\hat{\tau} \equiv \frac{1}{N} \sum_{i=1}^N \left[\frac{R_i D_i}{\mathbb{P}(R_i D_i = 1)} - \frac{R_i (1 - D_i)}{\mathbb{P}(R_i (1 - D_i) = 1)} \right] Y_i$$

We also need to define the specific causal parameters we identify through this comparison. When conditioning on the realized sample (i.e. treating R_i as fixed), the only remaining

source of uncertainty is the treatment assignment D_i . This is what we call a design-based view. In this context, we study $\mathbb{E}(\hat{\tau} \mid Y(0), Y(1), R)$ and the estimand is the average effect for the realized sample, or S-ATE

$$\tau_S = \frac{1}{n} \sum_{i=1}^N R_i (Y_{1i} - Y_{0i}).$$

In a sampling-based view, the sample is randomly drawn, and the treatment is randomly assigned within the sample. Both R_i and D_i are random and hence source of uncertainty. Therefore, we study $\mathbb{E}(\hat{\tau})$, averaging over sampling and assignment randomness. The population average treatment effect (P-ATE) is defined as

$$\tau_P = \frac{1}{N} \sum_{i=1}^N (Y_{1i} - Y_{0i}).$$

As a summary, remember that S-ATE answers an internal validity question: given the people I have in my experiment, what is the average effect? The P-ATE tackles external validity: for the population these people represent, what is the average effect?

Importantly, the estimator of the treatment effect is the same under both perspectives: in practice, we compute the difference in means (or the coefficient on D in a regression of Y on D). What differs is the variance calculation, since the source of uncertainty depends on whether we condition on the realized sample (design-based view) or average over sampling as well (sampling-based view). Standard softwares (e.g. Stata) do not explicitly distinguish between these two perspectives; in randomized experiments, conventional heteroskedasticity-robust standard errors coincide with, or provide a conservative upper bound for, the design-based variance derived by Neyman (see below).

2.3.3 Permutation Inference

Assume that all the population is considered so that uncertainty follows from the assignment mechanism. Indeed, permutation inference deals with cases where we have population level data, and therefore tackles **design-based uncertainty**. Other methods like bootstrap can be used to solve for **sampling uncertainty**. The aim is to derive the randomization distribution of a test-statistic, i.e. the distribution induced by the random assignment of treatment given fixed potential outcomes under a sharp null hypothesis. For instance, we could consider the following sharp hypothesis: $\mathcal{H}_0 : Y_{1i} = Y_{0i}, \forall i$. Permutation inference allows us to obtain the exact p-value. The test statistic can be defined as

$$T \equiv \frac{1}{N_1} \sum_{i=1}^{N_1} Y_i - \frac{1}{N_0} \sum_{i=1}^{N_0} Y_i$$

D	Permutation	Y_0	Y_1	Exchangeability	
	D^*			Y_0^{null}	Y_1^{null}
0	0	6	–	6	6
0	1	4	–	4	4
0	1	5	–	5	5
1	1	–	2	2	2
1	0	–	10	10	10
1	0	–	9	9	9
Observed Averages		5	7	8.4	3.7

Table 1: Permutation and Exchangeability under a Sharp Null

Consider the example in Table 1, which exemplifies one of the possible permutations. The exact distribution under the null is derived by calculating values of the statistic for each possible combination of randomized assignment. In the example, $N = 6$ and $N_1 = 3$, so there are $\binom{6}{3} = 20$ different assignment vectors. Fisher’s p-value is the proportion of permutations where the test statistic is at least as extreme as the observed statistic ($T \geq T_{obs}$).

We now provide a concrete example. The following table displays the potential outcomes $Y_i(D_i)$ as well as the treatment status D_i and the observed outcome Y_i for four individuals ($i \in \{1, 2, 3, 4\}$). Assume a completely randomized design with exactly one treated and three control units (all assignments equally likely).

Unit i	$Y_i(0)$	$Y_i(1)$	D_i	Y_i
1	1	1	0	1
2	4	4	0	4
3	10	10	1	10
4	5	5	0	5

Table 2: Design: one treated and three controls; sharp null $Y_i(1) = Y_i(0)$.

Suppose we are interested in the average treatment effect $\mathbb{E}[Y_i(1) - Y_i(0)]$, and we assume $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$. We can rewrite the observed outcome Y_i as follows

$$Y_i = Y_i(0) + D_i(Y_i(1) - Y_i(0))$$

As usual, we only observe one of the two potential outcomes at the same time. For each unit i , the data is missing because we only observe $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$. The missing value is $Y_i(1)$ if $D_i = 0$, and $Y_i(0)$ if $D_i = 1$. Now, notice that for all i ,

$$\mathbb{E}(Y_i | D = 1) = \mathbb{E}(Y_i(1) | D = 1) \stackrel{\perp\!\!\!\perp}{=} \mathbb{E}(Y_i(1))$$

$$\mathbb{E}(Y_i | D = 0) = \mathbb{E}(Y_i(0) | D = 0) \stackrel{\perp\!\!\!\perp}{=} \mathbb{E}(Y_i(0))$$

Therefore, by independence and linearity of the expectation operator, we can compute

$$\mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}(Y_i(1)) - \mathbb{E}(Y_i(0)) = \mathbb{E}(Y_i|D = 1) - \mathbb{E}(Y_i|D = 0)$$

We can observe the right hand side of the equation and therefore identify the ATE.

Now, we consider the sharp null hypothesis of no treatment effect: $\mathcal{H}_0 : Y_i(0) = Y_i(1)$. We use the standard Fisher statistic:

$$T = |\bar{Y}_{treat} - \bar{Y}_{control}|$$

In the data, where the 3rd unit was treated, we have, $\bar{Y}_{treat} = 10$, $\bar{Y}_{control} = \frac{10}{3}$ and therefore $T_{obs} = |10 - 3.33| = 6.67$. The first potential permutation is the case where unit 1 is treated instead of unit 3. In this context, we have that $T = |1 - 6.33| = 5.33$. The second permutation would have unit 2 being treated, and hence $T = |1 - 6.33| = 1.33$. The last permutation has unit 4 treated, and therefore $T = 0$.

Fisher's p-value is the proportion of permutations where the test statistic is at least as extreme as the observed statistic ($T \geq T_{obs}$):

$$p\text{-value} = \frac{\#\{T \geq 6.67\}}{4} = \frac{1}{4}$$

Now suppose the individual units can be grouped in two villages: $i \in \{1, 2\}$ belong to village A and $i \in \{3, 4\}$ belong to village B. Suppose the treatment is participation in a workshop and that treated individuals share information learned at the workshop with other individuals in the same village.

Formally, let $\mathcal{G}_j$ correspond to village j (here $j = \{1, 2\}$). We can formalize this idea of perfect spillover in the village as follows. Let $\mathbf{D}_j = (D_{j,1}, D_{j,2})$ be the treatment vector for village j . We have a within-village violation of SUTVA

$$Y_i = Y_i(D_{j,1}, D_{j,2})$$

The idea is that in a village where at least one individual i is treated, all other individuals $-i$ are control, but are contaminated by the treatment of i and are therefore not a pure control, **making the identification of the ATE impossible**.

We could recover a village level ATE if we impose some additional conditions. First, we must assume that there is at least one pure control group, that is:

$$\exists \mathcal{G}_j \text{ s.t. } \forall i \in \mathcal{G}_j, D_i = 0$$

Additionally, we must assume that there is no spillover between villages (an individual in village j is not sharing information from the workshop to any other individual from villages $-j$). This is basically a SUTVA at the village level assumption.

The randomization inference from before is not valid anymore since the control groups are contaminated in each permutation. The SUTVA violation means that potential outcomes are not "fixed" at the individual level and depend on the treatment assignment of others in the village. We could introduce randomization inference at the village level under the additional assumptions just mentioned, reducing the number of permutations from 4 to 2. The test statistic would be the difference between the average outcomes of the treated village and the control village. Note that with only two villages, the p-value will be at least 0.5, meaning we would lack some statistical power to reject the null.

2.3.4 Neyman's Approach

Permutation inference relies on a sharp null hypothesis and derives the exact distribution of a test statistic from the random assignment mechanism. In many applications, however, we are not interested in testing a sharp null such as $Y_i(1) = Y_i(0)$ for all i , but rather in conducting inference on the population average treatment effect

$$\tau \equiv \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0)).$$

Unlike a sharp null, this hypothesis does not allow us to fill in missing potential outcomes. As a result, exact permutation inference is no longer available. Instead, following Neyman (1923), we treat potential outcomes as fixed and derive the expectation and variance of the difference-in-means estimator under the random assignment mechanism. In this framework, uncertainty arises from treatment assignment, while potential outcomes are regarded as fixed attributes of the population.

Proposition 2.3: Unbiasedness of Difference-in-Means under Randomization

Consider a finite population of size N with fixed potential outcomes $\{Y_i(0), Y_i(1)\}_{i=1}^N$. Suppose exactly n_1 units are assigned to treatment by complete randomization (so $n_0 \equiv N - n_1$). Define the difference-in-means estimator

$$\hat{\tau} \equiv \bar{Y}_1^{obs} - \bar{Y}_0^{obs} = \frac{1}{n_1} \sum_{i=1}^N D_i Y_i(1) - \frac{1}{n_0} \sum_{i=1}^N (1 - D_i) Y_i(0),$$

and the population average treatment effect

$$\tau \equiv \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0)).$$

Then, taking expectation over the random assignment mechanism,

$$\mathbb{E}[\hat{\tau} \mid \mathbf{Y}(0), \mathbf{Y}(1)] = \tau.$$

^aNote that $\frac{1}{n_d} \sum_{i=1}^N D_i Y_i(d) = \frac{1}{n_d} \sum_{i=1}^{n_d} Y_i(d), \forall d \in \{0, 1\}$

Proof. Under complete randomization, each unit has the same probability of being assigned to treatment:

$$\mathbb{E}[D_i] = \mathbb{P}(D_i = 1) = \frac{n_1}{N}.$$

Therefore,

$$\mathbb{E}\left[\frac{1}{n_1} \sum_{i=1}^N D_i Y_i(1) \mid \mathbf{Y}(0), \mathbf{Y}(1)\right] = \frac{1}{n_1} \sum_{i=1}^N Y_i(1) \mathbb{E}[D_i] = \frac{1}{N} \sum_{i=1}^N Y_i(1).$$

Similarly, since $\mathbb{E}[1 - D_i] = n_0/N$,

$$\mathbb{E}\left[\frac{1}{n_0} \sum_{i=1}^N (1 - D_i) Y_i(0) \mid \mathbf{Y}(0), \mathbf{Y}(1)\right] = \frac{1}{N} \sum_{i=1}^N Y_i(0).$$

Subtracting the two expressions yields

$$\mathbb{E}[\hat{\tau} \mid \mathbf{Y}(0), \mathbf{Y}(1)] = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0)) = \tau.$$

□

This result implies that under complete randomization, the difference-in-means estimator recovers the true population average treatment effect in expectation, with unbiasedness arising solely from the random assignment mechanism. We now derive its variance, taking expectations over the assignment mechanism while treating potential outcomes as fixed.

Definition 2.1: Neyman's Variance Formula (1923)

Under complete randomization,

$$\mathbb{V}[\hat{\tau} \mid \mathbf{Y}(0), \mathbf{Y}(1)] = \frac{S_0^2}{n_0} + \frac{S_1^2}{n_1} - \frac{S_{01}^2}{N},$$

where

$$S_\omega^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i(\omega) - \bar{Y}(\omega))^2, \quad \omega \in \{0, 1\},$$

and

$$S_{01}^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i(1) - Y_i(0) - \tau)^2.$$

Several features of this variance formula are important. First, the variance depends on the population variances of the potential outcomes (S_0^2 and S_1^2). These reflect heterogeneity in outcomes within treatment and control groups. Indeed, this is the variance coming from randomness in the treated/control group mean. If outcomes under treatment $Y(1)$ are very dispersed across units (large S_1^2), then which units end up in treatment matters more.

Obviously, more treated units n_1 implies a lower variance.

The third term, S_{01}^2 , represents the variance of the individual treatment effects. Its negative sign reflects the fact that we are estimating the average effect for a fixed group of N units. In this finite population setting, the more the individual effects vary, the more the uncertainty about the 'missing' counterfactuals is already captured within the observed variance of the treatment and control groups, leading to a smaller overall variance for the population average than if we were generalizing to an infinite super-population.

When considering the asymptotic case where $N \rightarrow \infty$, we hold the sample sizes n_1 and n_0 fixed. As the total population N grows relative to the sample, the finite population correction term vanishes. This simplifies the variance to the standard "infinite population" approximation

$$\mathbb{V}[\hat{\tau} \mid \mathbf{Y}(0), \mathbf{Y}(1)] \approx \frac{S_0^2}{n_0} + \frac{S_1^2}{n_1}.$$

Recall that uncertainty may arise from both sampling (R_i) and treatment assignment (D_i). In the general framework, the sample analogues of the population variances of potential outcomes are given by the group-level empirical variances:

$$s_0^2 = \frac{1}{n_0 - 1} \sum_{i=1}^N R_i(1 - D_i) \left[Y_i - \left(\frac{1}{n_0} \sum_{i=1}^N R_i(1 - D_i) Y_i \right) \right]^2,$$

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^N R_i D_i \left[Y_i - \left(\frac{1}{n_1} \sum_{i=1}^N R_i D_i Y_i \right) \right]^2,$$

Since the term S_{01}^2 cannot be estimated (because both potential outcomes are never jointly observed), a conservative estimator of the variance is given by

$$\widehat{\mathbb{V}}(\hat{\tau}) = \frac{s_0^2}{n_0} + \frac{s_1^2}{n_1}.$$

If $S_{01}^2 > 0$ (heterogeneous effects), $\widehat{\mathbb{V}}(\hat{\tau})$ will be larger than the true variance. This makes the t -statistics smaller and your confidence intervals wider, which is conservative because it makes it harder to claim a significant result unless the effect is very strong.

3 Causality in Designs with Longitudinal Variation

Definition 3.1: Longitudinal Variation

Longitudinal variation refers to variation in outcomes and/or treatment status for the same statistical units observed repeatedly over time. Formally, for units $i = 1, \dots, N$ and periods $t = 0, \dots, T$, longitudinal data consist of observations $\{Y_{it}, D_{it}\}$, where identification exploits within-unit changes over time, cross-sectional differences across units, or a combination of both.

3.1 Intro to DiD - Back to New Jersey

This section introduces the concept of Difference in Differences (DiD) using the seminal paper by Card and Krueger, 1993. The simple question the authors asked was *how do employers in a low-wage labor market respond to an increase in the minimum wage?* They use the fact that in April 1992, New Jersey (NJ) raised the state minimum from \$4.25 to \$5.05 while Pennsylvania (PA) stayed at \$4.25 throughout this period. The general idea is to control for temporal changes of the potential no-treatment outcomes by assuming that a no-treatment group is subject to the changes the treatment group would have experienced. Therefore, they compare individuals in NJ fast food restaurants (treated group) to individuals in fast food restaurants of eastern PA (control group).

Let $(Y_{ist}(0), Y_{ist}(1))$ potential employment at restaurant i , state s and period t if there is a high (NJ, $s = 1$) or low (PA, $s = 0$) minimum wage regime. The indicator for having a high minimum wage is D_{st} . If $t = 0$ denotes pre-policy periods and $t = 1$ post-policy periods, we have $D_{s0} = 0$ for both states and D_{s1} only for NJ. Note that D_{st} is an interaction between a post-policy dummy and a NJ dummy. In other words, treatment varies only along the interaction of state and time: only NJ in the post-policy period is exposed to the higher minimum wage. This interaction term is precisely the source of identifying variation in the DiD design.

The DiD setup imposes an **additive structure** for $Y_{ist}(0)$ in the no-treatment state

$$Y_{ist}(0) = \eta_s + \beta_t + \varepsilon_{ist}$$

where η_s is a state fixed effect, β_t a time fixed effect and ε_{ist} a restaurant-specific idiosyncratic shock. This implies that in the absence of a minimum wage, change in employment is determined by the sum of a time-invariant state effect and a year effect. Therefore, it imposes a **common (parallel) trends** assumption: absent the policy change, average employment in NJ and PA would have evolved in parallel over time, so that any differential post-policy change can be attributed to the treatment. This model is a Two-Way-Fixed-Effects (TWFE) model.

We now continue with our example with the simplest version of DiD one can think about, exemplified by Table 3.

$\mathbb{E}(Y_{ist})$	Before (February)	After (November)
Policy on (New Jersey)	$\eta_1 + \beta_0$	$\eta_1 + \beta_1 + \tau$
Policy off (Pennsylvania)	$\eta_0 + \beta_0$	$\eta_0 + \beta_1$

Table 3: 2×2 Difference-in-Differences (DiD) Design

First, recall that in this context, the observed outcome is

$$Y_{ist} = Y_{ist}(0) + D_{st} (Y_{ist}(1) - Y_{ist}(0))$$

The key identifying assumption in this content is that employment trends $\beta_1 - \beta_0$ would have been the same in both states without the policy. Now, understand that taking the difference after and before for NJ, and then subtract the difference after and before for PA identifies the causal effect τ :

$$\left(\underbrace{\eta_1 + \beta_1 + \tau}_{\text{NJ, After}} - \underbrace{(\eta_1 + \beta_0)}_{\text{NJ, Before}} \right) - \left(\underbrace{\eta_0 + \beta_1}_{\text{PA, After}} - \underbrace{(\eta_0 + \beta_0)}_{\text{PA, Before}} \right) = \tau$$

Proposition 3.1: Equivalence of DiD and TWFE in the 2×2 Design

Consider a two-group ($s \in \{0, 1\}$), two-period ($t \in \{0, 1\}$) setting with treatment indicator $D_{st} = NJ_s \times P_t$. Assume the untreated potential outcome satisfies

$$Y_{ist}(0) = \eta_s + \beta_t + \varepsilon_{ist},$$

and define the regression model

$$Y_{ist} = \alpha + \beta_1 P_t + \eta_1 NJ_s + \tau D_{st} + U_{ist}.$$

where P_t an indicator for post-policy period ($t = 1$), NJ_s a NJ indicator, and $D_{st} \equiv NJ_s \times P_t$. Then the population OLS coefficient τ equals the difference-in-differences estimand:

$$\begin{aligned} \tau = & [\mathbb{E}(Y_{ist} \mid P_t = 1, NJ_s = 1) - \mathbb{E}(Y_{ist} \mid P_t = 0, NJ_s = 1)] \\ & - [\mathbb{E}(Y_{ist} \mid P_t = 1, NJ_s = 0) - \mathbb{E}(Y_{ist} \mid P_t = 0, NJ_s = 0)]. \end{aligned}$$

Proof. From the regression specification,

$$Y_{ist} = \alpha + \beta_1 P_t + \eta_1 NJ_s + \tau D_{st} + U_{ist}.$$

Taking conditional expectations yields the four cell means:

$$\begin{aligned} \mathbb{E}(Y_{ist} \mid P_t = 0, NJ_s = 0) &= \alpha, \\ \mathbb{E}(Y_{ist} \mid P_t = 1, NJ_s = 0) &= \alpha + \beta_1, \\ \mathbb{E}(Y_{ist} \mid P_t = 0, NJ_s = 1) &= \alpha + \eta_1, \\ \mathbb{E}(Y_{ist} \mid P_t = 1, NJ_s = 1) &= \alpha + \beta_1 + \eta_1 + \tau. \end{aligned}$$

Taking the difference across periods within New Jersey gives

$$(\alpha + \beta_1 + \eta_1 + \tau) - (\alpha + \eta_1) = \beta_1 + \tau.$$

Taking the same difference within Pennsylvania gives

$$(\alpha + \beta_1) - \alpha = \beta_1.$$

Subtracting the two differences eliminates the common time effect:

$$(\beta_1 + \tau) - \beta_1 = \tau.$$

Hence the TWFE coefficient on D_{st} equals the difference-in-differences estimand. $\square$

Remark 3.1: Why OLS equals the four cell means

In the 2×2 design, the regression includes an intercept, a time dummy, a group dummy, and their interaction. These four regressors generate exactly four predicted values — one for each (P_t, NJ_s) cell.

Because the regressors do not vary within a cell, the predicted value is constant inside each cell. OLS therefore chooses the coefficients so that the predicted value in each cell equals that cell's average outcome. Hence, the regression is simply fitting the four cell means.

As a result, the coefficient on the interaction term equals the difference between the two before-after differences, i.e. the difference-in-differences estimand.

Proposition 3.2

Let $s \in \{0, 1\}$ index states ($NJ = 1$, $PA = 0$) and $t \in \{0, 1\}$ index time (before = 0, after = 1). Define the population DiD contrast

$$\Delta^{DiD} \equiv \left(\mathbb{E}[Y_{is1} | s = 1] - \mathbb{E}[Y_{is0} | s = 1] \right) - \left(\mathbb{E}[Y_{is1} | s = 0] - \mathbb{E}[Y_{is0} | s = 0] \right).$$

Then

$$\Delta^{DiD} = \underbrace{\mathbb{E}[Y_{i11}(1) - Y_{i11}(0)]}_{\text{causal effect on treated in } t = 1} + \underbrace{\left(\mathbb{E}[Y_{i11}(0) - Y_{i10}(0)] - \mathbb{E}[Y_{i01}(0) - Y_{i00}(0)] \right)}_{\text{difference in counterfactual trends (selection bias)}}.$$

In particular, under parallel trends, $\mathbb{E}[Y_{i11}(0) - Y_{i10}(0)] = \mathbb{E}[Y_{i01}(0) - Y_{i00}(0)]$, the second term is zero and Δ^{DiD} identifies the causal effect.

Proof. Using $Y_{ist} = Y_{ist}(0) + D_{st}(Y_{ist}(1) - Y_{ist}(0))$ and that only $(s, t) = (1, 1)$ is treated, we have

$$\mathbb{E}[Y_{i11}] = \mathbb{E}[Y_{i11}(1)], \quad \mathbb{E}[Y_{i10}] = \mathbb{E}[Y_{i10}(0)], \quad \mathbb{E}[Y_{i01}] = \mathbb{E}[Y_{i01}(0)], \quad \mathbb{E}[Y_{i00}] = \mathbb{E}[Y_{i00}(0)].$$

Therefore,

$$\begin{aligned} \Delta^{DiD} &= \left(\mathbb{E}[Y_{i11}(1)] - \mathbb{E}[Y_{i10}(0)] \right) - \left(\mathbb{E}[Y_{i01}(0)] - \mathbb{E}[Y_{i00}(0)] \right) \\ &= \left(\mathbb{E}[Y_{i11}(1)] - \mathbb{E}[Y_{i11}(0)] \right) + \left(\mathbb{E}[Y_{i11}(0)] - \mathbb{E}[Y_{i10}(0)] \right) - \left(\mathbb{E}[Y_{i01}(0)] - \mathbb{E}[Y_{i00}(0)] \right), \end{aligned}$$

which is the stated decomposition, where we added and subtracted the counterfactual $\mathbb{E}[Y_{i11}(0)]$ in the last step. $\square$

This decomposition makes clear that we are again facing a missing counterfactual problem. In particular, $\mathbb{E}[Y_{i11}(0)]$ is unobserved and represents what would have happened in New Jersey after the policy in the absence of treatment. Moreover, the decomposition shows that the selection bias term is not specific to this application, but arises generically in causal inference: the DiD estimand equals the treatment effect plus the difference in counterfactual trends across groups. Imposing the parallel trends assumption makes the last term vanish, thereby allowing identification of the treatment effect. Importantly, parallel trends is an assumption about **potential outcomes**. It is hence fundamentally untestable, since it concerns counterfactual quantities that are never observed. In practice, researchers examine pre-treatment trends to assess whether the groups evolve similarly before treatment. While such tests can help build credibility, they do not verify the assumption itself. Ultimately, the plausibility of parallel trends must be justified by institutional and economic reasoning specific to the context under study.

Note that in practice, the parallel trends restriction may be more plausible after conditioning on covariates X_{is0} :

$$\mathbb{E}[Y_{i11}(0) - Y_{i10}(0) \mid X_{is0} = x] = \mathbb{E}[Y_{i01}(0) - Y_{i00}(0) \mid X_{is0} = x]$$

Note that covariates must be **pre-determined** and not affected by the intervention; otherwise, the estimated causal parameter τ misses the portion of effect mediated by X_{is0} . The issue is avoided by conditioning exclusively on baseline pre-policy covariates, hence the subscript $t = 0$.

The equivalence between DiD and TWFE in the simple two-groups, two-periods design has led researchers to apply TWFE regressions in increasingly complex settings—multiple periods, multiple groups, staggered adoption, fuzziness, and treatments that switch on and off. However, once we move beyond the canonical 2×2 case, this equivalence breaks down and the interpretation of the TWFE coefficient becomes more subtle. In the next section, we examine how treatment variation over time and across groups affects identification and what causal parameter is actually being estimated in these richer designs.

3.2 Data Structures and Research Designs

3.2.1 Data Structures

We consider the causal effects of a binary treatment $D_{it} \in \{0, 1\}$ on a scalar outcome Y_{it} , where $i = 1, \dots, N$ represents units and $t = 0, \dots, T$ represents time periods. The data can be organized into the following matrices:

$$\mathbf{Y} = \begin{pmatrix} Y_{10} & Y_{11} & \dots & Y_{1T} \\ Y_{20} & Y_{21} & \dots & Y_{2T} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{N0} & Y_{N1} & \dots & Y_{NT} \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & 1 & \dots & 0 \\ 0 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \dots & 0 \end{pmatrix}$$

This framework allows us to discuss a taxonomy of ways in which treatment varies across units and over time. For the current discussion, we assume no dynamic treatment effects,

meaning outcomes depend only on the contemporaneous treatment status. The current outcome is affected only by the current value of the treatment, not by the historical treatment path. Under this assumption, we have two potential outcomes for each unit and time period, $Y_{it}(0)$ and $Y_{it}(1)$, with their corresponding matrices $\mathbf{Y}(0)$ and $\mathbf{Y}(1)$.

Potential outcomes are also indexed to the timing of treatment. We must establish a baseline comparison: is the effect measured with respect to the outcome if the unit is never-treated, always-treated, or not-yet-treated? The causal effect for a specific unit at a specific time is defined as $\tau_{it} \equiv Y_{it} - Y_{it}(0)$. In practice, we often look for a meaningful weighted average of these individual effects across treated units:

$$\sum_{i,t:D_{it}=1} \omega_{it} \tau_{it}$$

where weights ω_{it} are convex. Estimates of causal effects for treated units at time t typically take the form of an Average Treatment Effect on the Treated (ATT):

$$\sum_{i,t} \left(\frac{D_{it}}{\sum_{i,t} D_{it}} \right) (Y_{it} - Y_{it}(0))$$

Because we never observe $Y_{it}(0)$ when $D_{it} = 1$, the missing counterfactuals must be imputed using methods such as regression-based or matching estimators.

The following matrices illustrate the patterns of observed ($\checkmark$) versus missing (?) counterfactual outcomes for $Y(0)$ under different treatment regimes. The highlighted cells represent the missing potential outcomes that must be imputed.

$$\mathbf{Y}(0)_{general} = \begin{pmatrix} ? & ? & \dots & \checkmark & \checkmark \\ \checkmark & ? & \dots & ? & \checkmark \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \checkmark & ? & \dots & \checkmark & ? \\ \checkmark & \checkmark & \dots & ? & ? \end{pmatrix} \quad \mathbf{Y}(0)_{block} = \begin{pmatrix} \checkmark & \checkmark & \checkmark & \dots & \checkmark \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \checkmark & \checkmark & \checkmark & \dots & \checkmark \\ \checkmark & \checkmark & ? & \dots & ? \\ \checkmark & \checkmark & ? & \dots & ? \end{pmatrix}$$

$$\mathbf{Y}(0)_{staggered} = \begin{pmatrix} \checkmark & \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \checkmark & \dots & \checkmark & ? \\ \checkmark & \checkmark & \dots & ? & ? \\ \vdots & \vdots & \dots & \vdots & \vdots \\ \checkmark & ? & \dots & ? & ? \end{pmatrix}$$

3.2.2 Research Designs

Lags as Controls This approach is typically used in settings with large N and small T , where some units are assigned to treatment in the last period. The structure of the missing

potential outcomes for $Y(0)$ looks like this:

$$\mathbf{Y}(0)_{N \gg T} = \begin{pmatrix} \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \dots & \checkmark & ? \\ \checkmark & \dots & \checkmark & ? \\ \vdots & \vdots & \vdots & \vdots \\ \checkmark & \dots & \checkmark & ? \end{pmatrix}$$

Matching-type methods are used here to adjust for lagged outcomes. For instance, the imputation stems from a horizontal regression involving T parameters:

$$\operatorname{argmin} \sum_{i:D_{iT}=0} \left(Y_{iT} - \sum_{t=0}^{T-1} \beta_t Y_{it} \right)^2$$

The counterfactual outcome for a treated unit i is then computed as:

$$\hat{Y}_{iT}(0) = \sum_{t=0}^{T-1} \hat{\beta}_t Y_{it}$$

This method essentially assumes that treatment and control units share a common correlation pattern between the outcome at time T and the outcome in previous periods. This is considered a stronger assumption than parallel trends because we are matching on the specific pre-policy values of the outcome. Lagged values must not be manipulated by units in expectation of treatment (no anticipation). While it works better with small T , regularization techniques (e.g., Lasso, elastic net) are frequently used in empirical work to handle larger sets of lags.

Synthetic Control This method is typically used in settings where only one unit (usually the last one) is treated, but the time dimension T is large, allowing for multiple pre-treatment periods (T_{pre}) and post-treatment periods. The missing potential outcome matrix for $Y(0)$ reflects this specific structure:

$$\mathbf{Y}(0)_{N \ll T} = \begin{pmatrix} \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \dots & \checkmark \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \checkmark & \checkmark & \checkmark & \checkmark & \checkmark & \dots & \checkmark \\ \checkmark & \checkmark & \checkmark & ? & ? & \dots & ? \end{pmatrix}$$

Synthetic control is still a matching adjustment, but it utilizes a vertical regression (N parameters and T_{pre} observations):

$$\operatorname{argmin} \sum_{t:D_{Nt}=0} \left(Y_{Nt} - \gamma_0 - \sum_{i=1}^{N-1} \gamma_i Y_{it} \right)^2$$

The original implementation of this idea imposes specific constraints: $\gamma_0 = 0$, $\gamma_i \geq 0$, and $\sum_{i=1}^{N-1} \gamma_i = 1$. While these restrictions provide interpretability, they can make it difficult to obtain good predictions for treated units with extreme outcomes. The counterfactual outcome for unit i in period t when $D_{it} = 1$ is computed as:

$$\hat{Y}_{Nt}(0) = \hat{\gamma}_0 + \sum_{i=1}^{N-1} \hat{\gamma}_i Y_{it}$$

It assumes that the correlation pattern across treatment and control units is stable over time. The framework can be extended to handle more than one treatment unit. It works better with a large T_{pre} to ensure the synthetic control unit closely tracks the treated unit's pre-treatment path. More on this later on!

3.3 Back to DiD: On The Way to Staggered Adoption

3.3.1 How DiD works

TWFE/DiD can be used when we are in a block assignment situation where a block of units are treated from a certain time onwards or are never treated and T can grow large.

$$\mathbf{Y}(0)_{block} = \begin{pmatrix} \checkmark & \checkmark & \checkmark & \dots & \checkmark \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \checkmark & \checkmark & \checkmark & \dots & \checkmark \\ \checkmark & \checkmark & ? & \dots & ? \\ \checkmark & \checkmark & ? & \dots & ? \end{pmatrix}$$

We can run the TWFE, which corresponds to

$$Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}, \quad (3.1)$$

As explained before, the TWFE imposes parallel trends on the untreated potential outcomes since

$$Y_{it}(0) = \alpha_i + \beta_t + \varepsilon_{it}$$

Y_{it}	$t = 0$	$t = 1$	$t = 2$
Unit 1 (treated)	$\alpha_1 + \beta_0$	$\alpha_1 + \beta_1 + \tau_{11}$	
Unit 2 (control)	$\alpha_2 + \beta_0$	$\alpha_2 + \beta_1$	
Unit 3			

Table 4: TWFE with Common Treatment Timing

Table 4 presents a table similar to Table 3, which is the canonical 2×2 DiD specification. We will extend this table below once we get to staggered adoption. As before,

$$(Y_{11} - Y_{10}) - (Y_{21} - Y_{20}) = \tau_{11}.$$

We now get into how DiD actually works. In a balanced panel of $N \times (T + 1)$ observations, we include dummies for each unit i (α_i) and each time period t (β_t):

$$Y_{it} = \sum_{i=1}^N \alpha_i A_{it} + \sum_{t=1}^T \beta_t B_{it} + \tau D_{it} + e_{it}$$

In this form, A_{it} is a dummy variable that equals 1 if the observation is for unit i and 0 otherwise. B_{it} is a dummy variable that equals 1 if the observation is for time t and 0 otherwise. α_i and β_t are the unit and time fixed effects (the intercepts we want to "partial out").

According to the **Frisch-Waugh-Lovell (FWL) Theorem**, the estimate of τ can be obtained by regressing both Y_{it} and D_{it} on the fixed effects and then regressing the residuals on each other. Formally, (i) regress D_{it} on all unit and time dummies to get residuals $\tilde{D}_{it}$, (ii) regress Y_{it} on all unit and time dummies to get residuals $\tilde{Y}_{it}$, (iii) regress $\tilde{Y}_{it}$ on $\tilde{D}_{it}$ to get $\hat{\tau}$. In (i), we are projecting D_{it} onto the space spanned by $N + T$ dummies. In a balanced panel, the projections of D_{it} onto the unit and time dummy spaces are exactly the unit-level means $\bar{D}_{i.}$ and time-level means $\bar{D}_{.t}$:

$$\bar{D}_{i.} = \frac{1}{T+1} \sum_{t=0}^T D_{it}^{25}, \quad \bar{D}_{.t} = \frac{1}{N} \sum_{i=1}^N D_{it}$$

Because $\bar{D}_{i.}$ and $\bar{D}_{.t}$ capture the entirety of the variation in D_{it} that the unit and time fixed effects could possibly explain, they are "sufficient statistics". Including them as linear controls partials out the exact same variation as including $N + T$ dummies. Consequently, the high-dimensional regression is numerically equivalent to a model with only two controls:

$$Y_{it} = \vartheta_0 + \vartheta_1 \bar{D}_{i.} + \vartheta_2 \bar{D}_{.t} + \tau D_{it} + v_{it} \quad (3.2)$$

To summarize, fixed effects remove from D_{it} precisely the variation explained by its unit and time averages. Once these averages are controlled for, the remaining variation in D_{it} identifies τ . Note that what we have derived so far is true generally, and does not rely on timing of treatment being common.

²⁵Consider regressing D_{it} on unit fixed effects only: $D_{it} = \alpha_i + u_{it}$. Within each unit i , the regressor α_i is constant over time. OLS therefore chooses α_i to minimize

$$\sum_{t=0}^T (D_{it} - \alpha_i)^2.$$

Taking the first-order condition with respect to α_i gives

$$\sum_{t=0}^T (D_{it} - \alpha_i) = 0 \implies \alpha_i = \frac{1}{T+1} \sum_{t=0}^T D_{it} = \bar{D}_{i.}.$$

Thus, projecting D_{it} onto the space spanned by unit dummies replaces D_{it} by its unit-level mean $\bar{D}_{i.}$.

Now, this framework collapses into the classic TWFE DiD when the timing of treatment is common across all treated units. In this case, $D_{it} = D_i \cdot P_t$, where D_i is a "treated group" dummy and P_t is a "post-treatment" dummy. Under common timing, the sufficient statistics $\bar{D}_{i\cdot}$ and $\bar{D}_{\cdot t}$ become perfectly collinear with the group and time indicators respectively:

$$\bar{D}_{i\cdot} = D_i \underbrace{\left(\frac{1}{T+1} \sum_{t=0}^T P_t \right)}_{\text{constant}} \propto D_i, \quad \bar{D}_{\cdot t} = P_t \underbrace{\left(\frac{1}{N} \sum_{i=1}^N D_i \right)}_{\text{constant}} \propto P_t$$

Equation 3.2 is hence algebraically equivalent to the standard DiD regression:

$$Y_{it} = \vartheta_0 + \tilde{\vartheta}_1 D_i + \tilde{\vartheta}_2 P_t + \tau D_{it} + v_{it}$$

Remark 3.2: Intuition

The FWL theorem implies that τ is identified solely by the variation in D_{it} that **cannot** be explained by the unit's average treatment intensity ($\bar{D}_{i\cdot}$) or the global treatment intensity at that specific time ($\bar{D}_{\cdot t}$). In the common timing case, this residual variation is exactly the interaction term $D_i \cdot P_t$.

3.3.2 Why DiD works

In the case of common treatment timing, the OLS estimator for τ is algebraically equivalent to the difference between the average change in the treated group and the average change in the control group:

$$\tau = \underbrace{\frac{1}{N_1} \sum_{i=1}^{N_1} \Delta \bar{Y}_i}_{\text{treated}} - \underbrace{\frac{1}{N - N_1} \sum_{j=1}^{N - N_1} \Delta \bar{Y}_j}_{\text{controls}} \quad (3.3)$$

where N_1 is the number of treated units and $N - N_1$ is the number of control units. The individual-level change $\Delta \bar{Y}_i$ is defined as the difference between the post-treatment average and the pre-treatment average for unit i :

$$\Delta \bar{Y}_i = \underbrace{\frac{1}{T_1} \sum_{t:D_{it}=1} Y_{it}}_{\bar{Y}_i^{post}} - \underbrace{\frac{1}{T+1-T_1} \sum_{t:D_{it}=0} Y_{it}}_{\bar{Y}_i^{pre}} = \bar{Y}_i^{post} - \bar{Y}_i^{pre}$$

where T_1 is the number of post treated periods and $T+1-T_1$ the number of pre treatment units. The unit fixed effects α_i cancel out when we take the difference $\bar{Y}_i^{post} - \bar{Y}_i^{pre}$. The time fixed effects β_t cancel out when we subtract the control group's average change from the treated group's average change.

Now, by defining the individual causal effect as $\tau_{it} = Y_{it}(1) - Y_{it}(0)$, we can show that the TWFE estimator τ identifies a specific weighted average of these effects.

Using the potential-outcome decomposition, we can rewrite Y_{it} as

$$Y_{it} = Y_{it}(0) + D_{it}\tau_{it}, \quad \tau_{it} \equiv Y_{it}(1) - Y_{it}(0),$$

We now analyze treated and control units separately.

Treated units. For i such that $D_i = 1$, we have $D_{it} = 1$ in post periods and $D_{it} = 0$ in pre periods. Hence

$$\begin{aligned} \bar{Y}_i^{post} &= \frac{1}{T_1} \sum_{t:D_{it}=1} (Y_{it}(0) + \tau_{it}) = \frac{1}{T_1} \sum_{t:D_{it}=1} Y_{it}(0) + \frac{1}{T_1} \sum_{t:D_{it}=1} \tau_{it} \equiv \bar{Y}_i^{post}(0) + \frac{1}{T_1} \sum_{t:D_{it}=1} \tau_{it}, \\ \bar{Y}_i^{pre} &= \frac{1}{T+1-T_1} \sum_{t:D_{it}=0} Y_{it}(0) \equiv \bar{Y}_i^{pre} \end{aligned}$$

Therefore,

$$\Delta \bar{Y}_i = (\bar{Y}_i^{post}(0) - \bar{Y}_i^{pre}(0)) + \frac{1}{T_1} \sum_{t:D_{it}=1} \tau_{it} \quad (3.4)$$

Control units. For j such that $D_j = 0$, we have $Y_{jt} = Y_{jt}(0)$ for all t , so

$$\Delta \bar{Y}_j = \bar{Y}_j^{post}(0) - \bar{Y}_j^{pre}(0) \quad (3.5)$$

Substituting (3.4) and (3.5) into (3.3) gives

$$\tau = \underbrace{\left[\frac{1}{N_1} \sum_{i:D_i=1} (\bar{Y}_i^{post}(0) - \bar{Y}_i^{pre}(0)) - \frac{1}{N_0} \sum_{j:D_j=0} (\bar{Y}_j^{post}(0) - \bar{Y}_j^{pre}(0)) \right]}_{\text{difference in counterfactual trends}} + \frac{1}{N_1} \sum_{i:D_i=1} \frac{1}{T_1} \sum_{t:D_{it}=1} \tau_{it}$$

Under the parallel trends assumption for untreated potential outcomes, the bracketed term equals zero. Hence,

$$\tau = \frac{1}{N_1} \sum_{i:D_i=1} \frac{1}{T_1} \sum_{t:D_{it}=1} \tau_{it} = \sum_{i,t:D_{it}=1} \frac{1}{N_1 T_1} \tau_{it}.$$

Thus, in the common-timing (block) design, TWFE estimates an equally weighted average of treatment effects across treated unit-time observations. All weights are positive, equal to $\frac{1}{N_1 T_1}$, and sum to one (convex weights).

3.3.3 Why TWFE does not work for staggered adoption

Table 5 presents a case of staggered adoption, similar to the following matrix, where not all units are treated at the same calendar time t :

$$\mathbf{Y}(0)_{\text{staggered}} = \begin{pmatrix} \checkmark & \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \checkmark & \dots & \checkmark & ? \\ \checkmark & \checkmark & \dots & ? & ? \\ \vdots & \vdots & \dots & \vdots & \vdots \\ \checkmark & ? & \dots & ? & ? \end{pmatrix}$$

Y_{it}	$t = 0$	$t = 1$	$t = 2$
Unit 1 (treated at $t = 1$)	$\alpha_1 + \beta_0$	$\alpha_1 + \beta_1 + \tau_{11}$	$\alpha_1 + \beta_2 + \tau_{12}$
Unit 2 (treated at $t = 2$)	$\alpha_2 + \beta_0$	$\alpha_2 + \beta_1$	$\alpha_2 + \beta_2 + \tau_{22}$
Unit 3 (control)	$\alpha_3 + \beta_0$	$\alpha_3 + \beta_1$	$\alpha_3 + \beta_2$

Table 5: TWFE with Staggered Treatment Timing

With staggered assignment, TWFE estimates time effects by leveraging contrasts between units with no change in treatment. For example, it compares unit 1 and 2 at times 1 and 2 since the first unit did not have a change in treatment status at period 2. Using data from the table, it is easy to see that the change in outcome for unit 1 is written as

$$Y_{12} - Y_{11} = (\beta_2 - \beta_1) + (\tau_{12} - \tau_{11})$$

Similarly, for unit 2,

$$Y_{22} - Y_{21} = \tau_{22} + (\beta_2 - \beta_1)$$

The difference in differences estimator is therefore computed as the difference between these two quantities, which yields

$$\tau_{22} - (\tau_{12} - \tau_{11}) = \tau_{22} \iff \tau_{12} = \tau_{11}$$

Thus, TWFE compares late adopters to already-treated units, and hence implicitly uses already-treated units as controls for later-treated units. The resulting contrast identifies τ_{22} only if treatment effects for early adopters do not change over time (i.e., $\tau_{12} = \tau_{11}$). This is a strong homogeneity assumption. When treatment effects are heterogeneous or dynamic, TWFE combines different effects with potentially non-convex weights, generating so-called **forbidden comparisons**.

3.4 Staggered Adoption and Causal Interpretation

3.4.1 Event Study Design

Definition 3.2: Event-study

An **event study** is a staggered adoption design where units are treated at different times and there may be never-treated units.

Consider a setting with N units observed over $T + 1$ time periods (balanced panel). For each $i \in \{1, \dots, N\}$ and $t \in \{0, \dots, T\}$, we observe the outcome Y_{it} and treatment status $D_{it} \in \{0, 1\}$:

$$D_{it} = \begin{cases} 1 & \text{if unit } i \text{ is treated in period } t, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 3.3: Absorbing Treatment

There is an event date E_i when D_{it} switches from zero to one and remains one forever. This imposes a *monotonicity restriction*: the treatment path is a non-decreasing sequence of zeros followed by ones, so that

$$D_{is} \leq D_{it} \quad \text{if } s \leq t.$$

Treatment cohorts $\{0, \dots, T, +\infty\}$ are indexed to the time treatment started:

$$E_i = \min\{t : D_{it} = 1\}.$$

For never-treated units, $E_i = +\infty$. We model explicitly treatment dynamics by including leads and lags in the estimating equation.

Define relative event time (time relative to treatment) and treatment as follows

$$K_{it} = t - E_i, \quad D_{it} = \mathbb{1}(K_{it} \geq 0).$$

The event-study specification is

$$Y_{it} = \alpha_i + \beta_t + \sum_{h=-a}^{-2} \mu_h \mathbb{1}(K_{it} = h) + \sum_{h=0}^{b-1} \mu_h \mathbb{1}(K_{it} = h) + \mu_b \mathbb{1}(K_{it} \geq b) + \varepsilon_{it}. \quad (3.6)$$

The first summation captures **pre-trends** (leads), while the second captures post-treatment **effects** (lags). For treated units, $K_{it} \in [-T, T]$, if treatment occurs in period T , then $K_{i0} = -T$. If treatment occurs in period 0, then $K_{iT} = T$. Never-treated units are infinitely far from the event: $K_{it} = -\infty$.

The first lead, $\mathbb{1}(K_{it} = -1)$, is excluded as a normalization and the coefficients on the remaining leads are used for testing. Some relative periods must be excluded to avoid multicollinearity, since cohort effects (α_i), time effects (β_t), and relative time indicators are related.²⁶

$$\sum_{h=-T}^T \mathbb{1}(K_{it} = h) = 1.$$

The core of an **event-study plot** is to display the estimated coefficients

$$\{\hat{\mu}_h\}_{h=-a}^{b-1}.$$

Applied work often shows estimated effects at horizons outside the time window over which treatment is expected to affect the outcome. Now, out of all possible relative periods:

$$H^{\text{incl}} = \{[-a, -2] \cup [0, T]\} \quad \text{are included,}$$

$$H^{\text{excl}} = \{[-T, -a) \cup \{-1\}\} \quad \text{are excluded.}$$

Finally, a static specification is obtained by pooling all post-treatment periods:

$$Y_{it} = \alpha_i + \beta_t + \mu_{\text{static}} D_{it} + \varepsilon_{it}.$$

²⁶See the underidentification result in Borusyak et al., 2024

3.4.2 Causal Interpretation

First, potential outcomes are indexed to treatment cohorts $Y_{it}(E_i)$. The observed outcomes is

$$Y_{it} = Y_{it}(\infty) + \sum_{e=0}^T \mathbb{1}(E_i = e) [Y_{it}(e) - Y_{it}(\infty)]$$

Note that we are implicitly imposing SUTVA. Indeed, $Y_{it} = Y_{it}(E_i)$ assumes (i) no interference across units, meaning that unit i 's outcome depends only on its own treatment path and not on the treatment timing of other units, and (ii) a well-defined treatment version (absorbing treatment with a unique adoption date). If spillovers or general equilibrium effects were present, potential outcomes would need to be indexed by the entire vector of treatment timings, and the decomposition above would no longer hold. We now define the two main assumptions in this model.

Assumption 3.1: Parallel Trends

Consider this conditions when treatment timing is randomized. Parallel trends between any two periods t and s can be written as

$$\mathbb{E} [Y_{is}(\infty) - Y_{it}(\infty) | E_i = e] = \mathbb{E} [Y_{is}(\infty) - Y_{it}(\infty)]$$

This version of the parallel trends assumptions says that the difference in outcome between period s and t is independent of starting time (or starting time is as good as random). This condition is the cohort-based version of the usual parallel trends assumption: in the absence of treatment, units that are treated earlier or later would have followed the same outcome trends over time, so differences in treatment timing are orthogonal to outcome dynamics. I repeat myself again: parallel trends is an assumption on **potential outcomes** and therefore cannot be tested for!

Assumption 3.2: No anticipations effects

For units treated at time e , treatment has no effect before adoption:

$$\mathbb{E} [Y_{ie+\ell}(e) - Y_{ie+\ell}(\infty) | E_i = e] = 0, \quad \ell < 0$$

This assumption states that outcomes prior to treatment are unaffected by future treatment, so treated units behave like never-treated units before the treatment date. Note that if we observe pre-trends, we can't distinguish between a violation of parallel trends and no anticipation.

We can now move to the definition of an important object, the Cohort-ATT.

Definition 3.4: CATT

The cohort e ATT ℓ periods from initial treatment is

$$CATT_{e\ell} = \mathbb{E} [Y_{ie+\ell}(e) - Y_{ie+\ell}(\infty) | E_i = e], \quad \ell \geq 0$$

3.4.3 Bias (forbidden comparisons)

Theorem 3.1: TWFE Event-Study Decomposition (Sun–Abraham)

Under SUTVA, parallel trends, and no anticipation, the population event-study coefficient μ_h from the TWFE regression admits the decomposition

$$\mu_h = \sum_{e \neq \infty} \omega_{eh} CATT_{eh} + \sum_{\substack{\ell \geq 0 \\ \ell \in \mathcal{H}^{inc} \setminus \{h\}}} \sum_{e \neq \infty} \omega_{e\ell} CATT_{e\ell}.$$

The first term is a weighted average of cohort-specific treatment effects at relative time h , while the second term reflects contamination from treatment effects at other horizons due to comparisons with already-treated units (forbidden comparisons).

To prove this theorem, we need to introduce the following proposition (proposition 1 from Sun and Abraham, 2021).

Proposition 3.3: Regression decomposition (simplified notation)

Under the event-study specification with unit and time fixed effects, the population coefficient on relative period h , denoted μ_h , can be written (purely algebraically) as a linear combination of cohort and relative-period cell differences:

$$\begin{aligned} \mu_h = & \sum_{e \neq \infty} \omega_{eh} \left(\mathbb{E}[Y_{i,e+h}(e) - Y_{i0}(\infty) \mid E_i = e] - \mathbb{E}[Y_{i,e+h}(\infty) - Y_{i0}(\infty)] \right) \\ & + \sum_{\ell \in \mathcal{H}^{incl} \setminus \{h\}} \sum_{e \neq \infty} \omega_{e\ell} \left(\mathbb{E}[Y_{i,e+\ell}(e) - Y_{i0}(\infty) \mid E_i = e] - \mathbb{E}[Y_{i,e+\ell}(\infty) - Y_{i0}(\infty)] \right) \\ & + \sum_{\ell \in \mathcal{H}^{excl}} \sum_{e \neq \infty} \omega_{e\ell} \left(\mathbb{E}[Y_{i,e+\ell}(e) - Y_{i0}(\infty) \mid E_i = e] - \mathbb{E}[Y_{i,e+\ell}(\infty) - Y_{i0}(\infty)] \right) \\ & + \sum_t \omega_{\infty t} \left(\mathbb{E}[Y_{it}(e) - Y_{i0}(\infty) \mid E_i = \infty] - \mathbb{E}[Y_{it}(\infty) - Y_{i0}(\infty)] \right). \end{aligned}$$

The weights $\omega_{..}$ are determined solely by the residualized relative-period indicators (they depend on the design and not on outcomes). This identity simply expresses μ_h as a linear combination of differences in trends across cohort $\times$ relative-period cells (contemporaneous h , other included relative periods, excluded relative periods, and never-treated).

Proof. I only provide an outline of the proof. The proof begins by applying the **Frisch–Waugh–Lovell (FWL)** theorem. This reduces the multi-variable TWFE regression to a bivariate relationship. The population coefficient μ_h is expressed as the covariance between

the residualized outcome $\tilde{Y}_{it}$ and the residualized treatment indicator $\tilde{W}_{it}^h$:

$$\mu_h = \frac{\mathbb{E} [\sum_t \tilde{W}_{it}^h Y_{it}]}{\mathbb{E} [\sum_t (\tilde{W}_{it}^h)^2]}.$$

This establishes that μ_h is a weighted sum of outcomes, where the weights ω_{it} are determined by the design (the timing of treatment) and the residuals from projecting the treatment indicator onto all other fixed effects and covariates.

The observed outcome Y_{it} is replaced by its potential outcome representation $Y_{it}(e)$. To ensure the algebra reflects differences in trends rather than levels, a baseline term—the never-treated outcome at a reference period $Y_{i0}(\infty)$ —is subtracted. Because the TWFE model includes unit and time fixed effects, these constants sum to zero across the weights, leaving the OLS coefficient μ_h unchanged but centered:

$$Y_{it} \rightarrow (Y_{it}(e) - Y_{i0}(\infty)).$$

The proof then "re-bins" the data. Instead of summing over units i and years t , it partitions the observations into adoption cohorts (e) and relative periods ($\ell = t - e$).

Finally, the proof groups these cohort-relative cells into four distinct contributions. By collecting the weights $\omega_{e\ell}$ for each cell, the identity reveals that μ_h is a linear combination of:

- **Contemporaneous effects** ($\ell = h$): The actual effect we intend to measure.
- **Other included periods** ($\ell \in \mathcal{H}^{\text{incl}} \setminus \{h\}$): Contamination from other estimated coefficients.
- **Excluded periods** ($\ell \in \mathcal{H}^{\text{excl}}$): Contamination from the omitted categories.
- **Never-treated trends** ($e = \infty$): Differences in the underlying paths of the control group.

□

Now that we have this decomposition, we can prove Theorem 3.4.3.

Proof. Let us focus on the first term in the decomposition

$$\sum_{e \neq \infty} \omega_{eh} \left(\underbrace{\mathbb{E}[Y_{i,e+h}(e) - Y_{i0}(\infty) \mid E_i = e]}_{\text{Actual Term A}} - \underbrace{\mathbb{E}[Y_{i,e+h}(\infty) - Y_{i0}(\infty)]}_{\text{Actual Term B}} \right)$$

For term A, we can add/subtract the counterfactual $Y_{i,e+h}(\infty)$

$$\mathbb{E}[Y_{i,e+h}(e) - Y_{i,e+h}(\infty) + Y_{i,e+h}(\infty) - Y_{i0}(\infty) \mid E_i = e]$$

Rearranging gives

$$\underbrace{\mathbb{E}[Y_{i,e+h}(e) - Y_{i,e+h}(\infty) \mid E_i = e]}_{CATT_{eh}} + \underbrace{\mathbb{E}[Y_{i,e+h}(\infty) - Y_{i0}(\infty) \mid E_i = e]}_{\text{Cohort-specific trend}}$$

Plugging this back in the full expression yields

$$\mu_h \propto \sum_{e \neq \infty} \omega_{eh} \left(CATT_{eh} + \underbrace{\mathbb{E}[Y_{i,e+h}(\infty) - Y_{i0}(\infty) \mid E_i = e]}_{\text{Trend for cohort } e} - \underbrace{\mathbb{E}[Y_{i,e+h}(\infty) - Y_{i0}(\infty)]}_{\text{Average trend}} \right)$$

Parallel trend implies

$$\mathbb{E}[Y_{it}(\infty) - Y_{i0}(\infty) \mid E_i = e] = \mathbb{E}[Y_{it}(\infty) - Y_{i0}(\infty)]$$

so

$$\mu_h \propto \sum_{e \neq \infty} \omega_{eh} CATT_{eh}$$

This same logic applies to the other relative-period summations ($\ell \in \mathcal{H}^{incl}$ and $\ell \in \mathcal{H}^{excl}$), so we get that under parallel trends, the population coefficient μ_h is expressed as a linear combination of $CATT$ s across cohorts and relative periods:

$$\mu_h = \sum_{e \neq \infty} \omega_{eh} CATT_{eh} + \sum_{\ell \in \mathcal{H}^{incl} \setminus \{h\}} \sum_{e \neq \infty} \omega_{e\ell} CATT_{e\ell} + \sum_{\ell \in \mathcal{H}^{excl}} \sum_{e \neq \infty} \omega_{e\ell} CATT_{e\ell}$$

Now, no anticipation effects implies that $CATT_{e\ell} = 0, \ell < 0$, which yields the desired result

$$\mu_h = \sum_{e \neq \infty} \omega_{eh} CATT_{eh} + \sum_{\substack{\ell \geq 0 \\ \ell \in \mathcal{H}^{incl} \setminus \{h\}}} \sum_{e \neq \infty} \omega_{e\ell} CATT_{e\ell}.$$

□

Note additionally that for a pre-treatment coefficient ($h < 0$), the parameter μ_h still depends on post-treatment $CATT$ s:

$$\mu_h = \sum_{\{\ell \geq 0\} \in \mathcal{H}^{incl}} \sum_{e \neq \infty} \omega_{e\ell} CATT_{e\ell}$$

This means a non-zero "pre-trend" test result might actually be "contamination" from future treatment effect heterogeneity rather than a genuine failure of parallel trends.

In the Sun and Abraham, 2021 framework, the weights $\omega_{e\ell}$ are purely mechanical and determined by the regression design, specifically by regressing the cohort-relative period indicator $\mathbb{1}(K_{it} = \ell) \mathbb{1}(E_i = e)$ on the TWFE specification.

The weights are not restricted to the $[0, 1]$ interval and satisfy the following summation properties:

- **Contemporaneous Period:** The weights associated with the relative period of interest h sum to one across all treated cohorts: $\sum_{e \neq \infty} \omega_{eh} = 1$.
- **Other Included Periods:** For any other period ℓ included in the regression ($\ell \in \mathcal{H}^{\text{incl}} \setminus \{h\}$), the weights sum to zero: $\sum_{e \neq \infty} \omega_{e\ell} = 0$.
- **Excluded Periods:** The weights for all periods excluded from the regression sum to negative one: $\sum_{\ell \in \mathcal{H}^{\text{excl}}} \sum_{e \neq \infty} \omega_{e\ell} = -1$.

The fundamental issue with the TWFE event-study estimator is revealed by the weighting properties derived in the decomposition. For any included relative period ℓ other than the one of interest h , the weights sum to zero across cohorts:

$$\sum_{e \neq \infty} \omega_{e\ell} = 0, \quad \text{for } \ell \in \mathcal{H}^{\text{incl}} \setminus \{h\}.$$

This property is a mechanical consequence of the OLS residualization process.

If treatment effects are constant across cohorts ($CATT_{e\ell} = ATT_{\ell}$), the contamination term becomes $ATT_{\ell} \sum_e \omega_{e\ell}$. Since the weights sum to zero, the entire term vanishes, and μ_h correctly identifies the treatment effect. If treatment effects vary by cohort ($CATT_{e\ell} \neq CATT_{e'\ell}$), the sum $\sum_e \omega_{e\ell} CATT_{e\ell}$ generally does **not** equal zero. Because the weights $\omega_{e\ell}$ can be negative, the OLS estimator "subtracts" the effects of early-treated cohorts from the effects of late-treated cohorts.

Remark 3.3: Applied Perspective: Why Should You Care?

The decomposition above has a concrete empirical implication. When you run a standard TWFE event-study in a staggered setting, the coefficients you report are *not* simply weighted averages of cohort-specific effects at each horizon. They are contaminated by treatment effects at *other* horizons — and the contaminating weights can be negative.

This has two practical consequences. First, even if every cohort has a positive treatment effect at every horizon, your estimated $\hat{\mu}_h$ can be *negative* — not because the treatment is harmful, but purely because of how OLS reweights already-treated cohorts. Second, and more subtly, a statistically significant pre-trend coefficient ($h < 0$) need not indicate a violation of parallel trends at all: it may simply reflect contamination from heterogeneous post-treatment dynamics bleeding back into the pre-period.

In short, with staggered adoption and heterogeneous effects, the standard TWFE event-study is neither a clean test of parallel trends nor a clean estimate of dynamic treatment effects. Both the identification test and the object of interest are simultaneously corrupted.

3.4.4 Solution 1: Sun and Abraham, 2021

The general idea of the recent literature is to draw inference by *reverse engineering* a well-defined causal estimand from disaggregated causal parameters. Instead of relying on

the implicit and often non-convex weights generated by TWFE, the target parameter is constructed explicitly as a convex aggregation of cohort-specific treatment effects:

$$\sum_{e \neq \infty} \lambda_{eh} CATT_{eh}, \quad \text{with} \quad \sum_{e \neq \infty} \lambda_{eh} = 1.$$

The weights λ_{eh} are chosen transparently so that the overall estimand is a proper weighted average of economically meaningful cohort-level effects.

The implementation proceeds in three steps. First, one estimates all $CATT_{eh}$ directly, using carefully chosen pre-periods and a clean control group (typically not-yet-treated or never-treated units). Second, estimation of each cohort-specific effect is based on a plain difference-in-differences comparison, thereby preserving a clear causal interpretation under parallel trends and no anticipation. Third, the cohort-specific estimates are aggregated using weighted averages, where the weights are typically proportional to each cohort's sample share.

The objective of this procedure is to eliminate the non-convex and sometimes negative weighting that arises mechanically in TWFE event-study regressions. Conceptually, this approach mirrors what the matching literature emphasized in the early 2000s: define the causal parameter of interest explicitly, estimate disaggregated building blocks, and then aggregate them using transparent and economically interpretable weights.

$$\mathbf{Y}(0)_{\text{staggered}} = \begin{pmatrix} \checkmark & \checkmark & \dots & \checkmark & \checkmark \\ \checkmark & \checkmark & \dots & \checkmark & ? \\ \checkmark & \checkmark & \dots & ? & ? \\ \vdots & \vdots & \dots & \vdots & \vdots \\ \checkmark & ? & \dots & ? & ? \end{pmatrix}$$

Cohort $E_i = \infty$ has checkmarks ($\checkmark$) in every column. This means they remain in the control state for the entire duration of the study. Cohort $E_i = T - 1$ has checkmarks in almost all columns, but they switch to a question mark (?) in the second-to-last column. That specific column is time $t = T - 1$. Cohort $E_i = T - 2$ switches from a checkmark to a question mark even earlier, in the third-to-last column ($t = T - 2$). The goal is to impute the missing counterfactuals (?).

Now, what is the information used to estimate the $CATT_{e1}$'s?

To estimate the effect for $T - 1$, we only look at the history of units that haven't been treated yet (or will never be), ensuring the control group isn't currently experiencing its own dynamic treatment effects. To understand what would have happened to that yellow cell if treatment hadn't occurred, Sun and Abraham, 2021 use the never-treated units ($E_i = \infty$) across all time (so the top row with all but one columns) as well as the pre-treatment history of the cohort itself (the pink checkmarks in the second row). Note that the pre-treatment history of the treated cohort is not used as a control group per se;

rather, it is used to difference out cohort-specific levels and trends, while the actual control group consists of never-treated units (or in the last treated cohort).

$$\mathbf{Y}(0)_{\text{staggered}} = \begin{pmatrix} \checkmark & \checkmark & \cdots & \checkmark & \checkmark & \checkmark & \checkmark \\ \checkmark & \checkmark & \cdots & \checkmark & \checkmark & ? & ? \\ \checkmark & \checkmark & \cdots & \checkmark & ? & ? & ? \\ \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{array}{l} E_i = \infty \\ E_i = T - 1 \text{ (treated)} \\ E_i = T - 2 \\ \vdots \end{array}$$

Now, to estimate the effect for $T - 2$, we use the never-treated units ($E_i = \infty$) across all time (so the top row with all but one columns) as well as the pre-treatment history of the cohort itself (the pink checkmarks in the third row). Note that this is because Sun and Abraham, 2021 advocate for a stricter control group than a standard "not-yet-treated" approach. For cohort $E_i = T - 2$, later observations of the never-treated units are not used because each $CATT_{e\ell}$ is identified from differences at the same calendar time $t = e + \ell$. Periods occurring after this comparison window do not enter the DiD moment condition and are therefore mechanically excluded.

$$\mathbf{Y}(0)_{\text{staggered}} = \begin{pmatrix} \checkmark & \checkmark & \cdots & \checkmark & \checkmark & \checkmark & \checkmark \\ \checkmark & \checkmark & \cdots & \checkmark & \checkmark & ? & ? \\ \checkmark & \checkmark & \cdots & \checkmark & ? & ? & ? \\ \vdots & \vdots & \cdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \begin{array}{l} E_i = \infty \\ E_i = T - 1 \\ E_i = T - 2 \text{ (treated)} \\ \vdots \end{array}$$

In practice, the cohort–relative-time effects $CATT_{eh}$ are estimated using a fully interacted event-study specification that is saturated in relative time K_{it} and cohort E_i , while omitting indicators for control units:

$$Y_{it} = \alpha_i + \beta_t + \sum_{h \neq -1} \sum_{e \in \mathcal{C}} \tau_{eh} \mathbb{1}(K_{it} = h) \mathbb{1}(E_i = e) + u_{it}.$$

Here, $\mathcal{C}$ denotes the set of treated cohorts excluding the last-treated cohort, $\mathcal{C} = \{e : e \neq \max(E_i)\}$, which is omitted from the interacted specification and used as the control group. The last-treated cohort is omitted from the interacted dummies and serves as the control cohort: because it remains untreated when earlier cohorts are treated, it provides valid counterfactual outcomes against which the $CATT_{e,\ell}$ are identified.

3.4.5 Solution (∞): the very dense DiD literature

I try to synthesise the modern DiD literature below. Roth et al., 2023 propose an amazing practitioners guide to DiD. I discuss Borusyak et al., 2024 in details in Appendix B.

Estimator	Main Contribution	How It Is Built	Control Group	When to Use / Caveats
Callaway and Sant'Anna, 2021 R: did / csdid Stata: csdid	Estimates cohort-ATTs (group-time average treatment effects) $ATT(g, t)$ directly, then aggregates with user-specified weights. Avoids negative weighting. Easily extended to conditional PT via DR.	Step 1. Estimate $ATT(g, t)$ for each cohort g and post-treatment period t using a 2×2 DiD. Step 2. Aggregate to event-study or overall ATT using λ_g weights (e.g. cohort-share).	Two options: (a) never-treated units only ($G_i = \infty$); (b) not-yet-treated units ($G_i > t$ at each t). Option (b) requires PT only from $g_{\min} - 1$ (weaker assumption, wider comparison group). Option (a) requires global PT.	Best default. Use when you want maximum transparency on control group and minimal PT assumptions. Not the most efficient under homoskedastic errors. Needs enough never/not-yet-treated units.
Borusyak et al., 2024 R/Stata: did2s, didimputation	Efficiency: proven efficient under homoskedastic errors. Estimates $\hat{Y}_{it}(0)$ via a TWFE fit on clean (untreated) observations, then directly imputes counterfactuals for treated units.	Step 1. Fit TWFE using <i>only untreated</i> obs: $\hat{\alpha}_i, \hat{\beta}_t$ from $Y_{it} = \alpha_i + \beta_t + \varepsilon_{it}$ on $\{D_{it} = 0\}$. Step 2. Impute $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\beta}_t$ for all treated cells. Step 3. Compute $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$ and average with chosen weights.	Never- and not-yet-treated units are used jointly in step 1 to estimate the fixed effects. Uses <i>all</i> pre-treatment periods (vs. only $g - 1$), so imposes PT over a longer horizon.	Prefer when outcome is not too serially correlated and you are confident in long-run PT. Can have larger bias than CS if PT holds only approximately over shorter windows.

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Table (continued)

Estimator	Main Contribution	How It Is Built	Control Group	When to Use / Caveats
De Chaise-martin and d'Haultfoeuille 2020 R/Stata: <code>DIDmultipt</code>	Non-absorbing treatment: allows D_{it} to switch on <i>and off</i> over time. Identifies the ATT in the period of switch-on (or switch-off) under a “no carryover” assumption. Also handles multi-valued and continuous D_{it} .	Computes $\hat{\delta}$ by comparing units whose treatment <i>changed</i> between $t - 1$ and t to units that did <i>not</i> change, weighting by the size of the change: $\hat{\delta} \approx \sum_{i,t} \mathbb{1}[\Delta D_{it} \neq 0]$ Formally a Wald-DiD extending the standard 2×2 DiD.	Units whose treatment did not change between $t - 1$ and t (“stayers”). For switch-ons the switchers are compared to stable-untreated stayers; for switch-offs, to stable-treated stayers.	Use when treatment is genuinely reversible or multi-valued. Requires no-carryover : $Y_{it}(d)$ depends only on current D_{it} , not on history. Restrictive if treatment has persistent effects. In absorbing-treatment settings, equivalent to ATT_0^w in CS with specific weights.

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Table (continued)

Estimator	Main Contribution	How It Is Built	Control Group	When to Use / Caveats
Sun and Abraham, 2021 Stata: <code>eventstudyint</code> R: <code>fixest</code> (<code>sunab</code>)	Clean event-study coefficients: derives TWFE contamination analytically (interaction-weighted estimator). Replaces contaminated $\hat{\mu}_h$ with a proper weighted average of $CATT_{el}$. Tight connection to TWFE framework: easy to implement.	Runs a fully saturated regression interacting relative-time indicators with cohort dummies: Then aggregates $\hat{\delta}_{eh}$ using cohort-share weights λ_{eh} to recover $ATT_h = \sum_e \lambda_{eh} CATT_{eh}$.	Never-treated (if they exist) <i>or</i> last-treated cohort (the cohort with the latest adoption date $\max_i E_i$, which is untreated when earlier cohorts are treated). Notably, it does <i>not</i> use not-yet-treated units beyond the last cohort.	Prefer when you want clean event-study plots and a close mapping to TWFE. Slightly stronger PT assumption than CS (not-yet-treated) version. Last-treated as control is only valid if <i>it</i> has no treatment effect yet.

Note. All estimators require: (i) *parallel trends* on untreated potential outcomes, and (ii) *no anticipation*. PT can be imposed unconditionally or conditional on pre-treatment covariates X_i (all four estimators support conditioning). Packages listed are the main implementations; `did2s` provides a unified interface to CS, BJS, and SA simultaneously.

3.5 Recent Advances

Honest Approach to Parallel Trends Violations

The approaches above *assume* PT holds. In practice, PT is untestable (it concerns unobserved potential outcomes). Pre-trend tests—plotting $\hat{\mu}_h$ for $h < 0$ —are intuitive but suffer from at least three pathologies: **(1) low power** (a linear violation that passes the pre-test will also generate a spurious post-treatment effect with the same probability); **(2) pre-test bias** (conditioning on passing a pre-test selects a biased sample from the DGP); **(3) contamination** (under SA's decomposition, even clean pre-trend coefficients may reflect heterogeneous post-treatment effects bleeding backwards).

Rambachan and Roth, 2023 formalise PT violations by restricting how much the post-treatment trend can deviate from the pre-treatment trend. Indeed, instead of testing $\delta = 0$,

they partially identify the treatment effects under a range of restrictions on the possible violations of parallel trends. This answers the following question: "What is the set of treatment effects consistent with the observed pre-trends, assuming the post-treatment violations aren't too different from the pre-treatment ones?"

Define δ_s as the violation of PT in period s . They impose restrictions such as:

$$|\delta_{\text{post}}| \leq \bar{M} \cdot \max_{s < 0} |\delta_s|$$

which just says that the magnitude of the post-treatment violation of parallel trends can be no larger than a constant M times the maximal pre-treatment violation. This yields **robust confidence sets** that remain valid under the chosen class of violations.

Section 4.6 of Roth et al., 2023 provides a great summary of the best practices for robustness. One key object they suggest to use is what they call the *breakdown value* $\bar{M}^*$. This represents how big would the post-treatment violation of parallel trends have to be relative to the largest pre-treatment violation to invalidate a particular conclusion. They "encourage researchers to routinely report the results of the sensitivity analyses alongside the event study plots". Obviously, one should also discuss possible violations of parallel trend in the specific context in hand. Specifically, it is useful to provide a discussion of the types of time-varying factors that may have differentially affected the outcome for the treated group.

Continuous and Multi-Valued Treatment

Standard DiD assumes $D_{it} \in \{0, 1\}$. Two main extensions exist:

- De Chaisemartin and d'Haultfoeuille, 2018 introduce *fuzzy DiD*: all groups are treated in both periods, but the *intensity* of treatment changes in one group. Generalises the Wald-DiD to the continuous case. Requires an analogue of PT and a rank condition (monotone first-stage), analogous to IV. Estimates a LATE-type parameter—the effect of a unit increase in treatment for compliers.
- Callaway et al., 2024 extend the CS framework to doses (multi-valued or continuous D_{it}), defining cohort-dose-ATTs and aggregating them. Requires a generalisation of PT to hold across dose levels.

Both approaches emphasise that a significant pre-trend in the continuous case is harder to assess and that the parallel-trends assumption must be plausible across the *entire support* of the treatment distribution.

3.6 Synthetic Control

4 Identification Under Conditional Exogeneity

4.1 Conditional Exogeneity

We start by defining two important assumptions in this world. The first one is conditional independence.

Assumption 4.1: Conditional Independence (CI)

The data satisfies Conditional Independence (CI) if

$$(Y(0), Y(1)) \perp\!\!\!\perp D | X = x$$

We also call this assumption selection on observables. Once we control for X , we are in a world akin to randomization. The average treatment effect for the subpopulation with $X = x$ is given by

$$\tau[x] = \mathbb{E}(Y(1) - Y(0) | X = x)$$

An additional assumption is common support/overlap.

Assumption 4.2: Common Support (CS)

The data satisfies Common Support (CS) if

$$0 < e(x) \equiv \mathbb{E}(D | X = x) < 1$$

where $e(x)$ the propensity score.

This basically says that every subject/group has a non-zero probability of receiving any of the treatments. Holding X fixed, there is still variation in D . CI and CS together imply **Strong Ignorability (SI)**.

The ATT, $\mathbb{E}(Y(1) - Y(0) | D = 1)$ can be constructed in two steps under SI. First, the conditional effect $\tau[x]$ is identified from the data

$$\tau[x] = \mathbb{E}(Y(1) | X = x) - \mathbb{E}(Y(0) | X = x) = \mathbb{E}(Y | D = 1, X = x) - \mathbb{E}(Y | D = 0, X = x)$$

The last equality comes from CI, which implies that within a group with $X = x$, treatment is as good as random, so treated and untreated units have the same distribution of potential outcomes. Once we condition on X , selection bias disappears, so observed outcomes within treatment groups recover the corresponding potential outcomes. Assuming X is discrete, we can now rewrite our ATT as follows

$$\begin{aligned} \mathbb{E}(Y(1) - Y(0) | D = 1) &= \mathbb{E}[\mathbb{E}(Y(1) - Y(0) | D = 1, X) | D = 1] = \mathbb{E}(\tau[x] | D = 1) \\ &= \sum_x \tau[x] \mathbb{P}(X = x | D = 1) = \sum_x \tau[x] \frac{\mathbb{P}(D = 1 | x) \mathbb{P}(x)}{\sum_{x'} \mathbb{P}(D = 1 | x') \mathbb{P}(x')} = \sum_x \tau[x] \frac{e(x) \mathbb{P}(X = x)}{\sum_{x'} e(x') \mathbb{P}(X = x')} \equiv \sum_x \tau[x] \omega(x) \end{aligned}$$

where the first equality comes from conditional LIE²⁷, the second equality uses CI (as shown above) and the penultimate from Bayes Theorem. Note that CS allows to identify $\tau[x]$ from the data. The last step shows that the ATT is a weighted average of group-specific treatment effects, with weights

$$\omega(x) = \frac{e(x)\mathbb{P}(X = x)}{\sum_{x'} e(x')\mathbb{P}(X = x')} = \mathbb{P}(X = x | D = 1)$$

The higher $e(x)$, the higher the probability that an individual with $X = x$ is treated, hence the higher the weight. We basically assign more weight to TEs for groups that have more treated units.

Interestingly, the regression counterpart of this idea does not generally recover the same quantity.

Proposition 4.1: OLS with Discrete Controls

Suppose $D \in \{0, 1\}$ and X is discrete. Assume

1. Conditional Independence: $(Y(0), Y(1)) \perp D \mid X$,
2. Common Support: $0 < e(x) < 1$ for all x with $\mathbb{P}(X = x) > 0$,

where $e(x) = \mathbb{P}(D = 1 \mid X = x)$.

Consider the population linear projection

$$Y = \beta_0 + \beta_1 D + \sum_x \beta_{2x} \mathbb{1}(X = x) + U.$$

Then the coefficient on D satisfies

$$\beta_1 = \sum_x \tau(x) \omega(x), \quad \omega(x) = \frac{e(x)(1 - e(x)) \mathbb{P}(X = x)}{\sum_{x'} e(x')(1 - e(x')) \mathbb{P}(X = x')}.$$

Proof. By FWL, we have that

$$\beta_1 = \frac{\mathbb{E}(\tilde{D}\tilde{Y})}{\mathbb{E}(\tilde{D}^2)} \text{ where } \tilde{Y} \equiv Y - \mathbb{E}(Y|X), \tilde{D} \equiv D - \mathbb{E}(D|X)$$

Now, we can write the observed outcome as

$$Y = Y(0) + D(Y(1) - Y(0)).$$

Now, let

$$\eta \equiv (Y(1) - Y(0)) - \tau(X), \quad \text{so that } \mathbb{E}[\eta \mid X] = 0.$$

²⁷For any r.v. X, Y, Z , $\mathbb{E}(X|Z) = \mathbb{E}(\mathbb{E}(X|Z, Y)|Z)$

Add and subtract $\tau(X)$ inside the treatment effect:

$$Y(1) - Y(0) = \tau(X) + \underbrace{\left((Y(1) - Y(0)) - \tau(X) \right)}_{\eta},$$

Substituting into the outcome equation gives

$$Y = Y(0) + D\tau(X) + D\eta.$$

Taking conditional expectations given X and using $\mathbb{E}[D | X] = e(X)$ and $\mathbb{E}[\eta | X] = 0$,

$$\mathbb{E}[Y | X] = \mathbb{E}[Y(0) | X] + e(X)\tau(X).$$

Hence the X -residualized outcome is

$$\tilde{Y} \equiv Y - \mathbb{E}[Y | X] = (Y(0) - \mathbb{E}[Y(0) | X]) + \tau(X)(D - e(X)) + D\eta.$$

Now, noting that with discrete X , $\mathbb{E}(D|X) = e(x)$ so $\tilde{D} = D - e(x)$, we have

$$\mathbb{E}[\tilde{D}\tilde{Y}] = \mathbb{E}[(D - e(X))(Y(0) - \mathbb{E}[Y(0) | X])] + \mathbb{E}[\tau(X)(D - e(X))^2] + \mathbb{E}[(D - e(X))D\eta].$$

The first term is zero because $\mathbb{E}[D - e(X) | X] = 0$.²⁸ The third term is zero because

$$\mathbb{E}[(D - e(X))D\eta | X] = \mathbb{E}[(D - e(X))D | X] \mathbb{E}[\eta | X] = 0.$$

Hence,

$$\mathbb{E}[\tilde{D}\tilde{Y}] = \mathbb{E}[\tau(X)(D - e(X))^2].$$

The denominator is therefore the variance of the treatment D that is unexplained by X :

$$\mathbb{E}[\tilde{D}^2] = \mathbb{E}[(D - e(X))^2] = \mathbb{E}[\mathbb{V}(D | X)].$$

Note that we can find the last equality as follows

$$\begin{aligned} \mathbb{V}(D | X) &= \mathbb{E}[(D - \mathbb{E}(D|X))^2 | X] = \mathbb{E}[(D - e(X))^2 | X] \\ \Leftrightarrow \mathbb{E}(\mathbb{V}(D | X)) &= \mathbb{E}\left(\mathbb{E}[(D - e(X))^2 | X]\right) = \mathbb{E}[(D - e(X))^2] \end{aligned}$$

Therefore,

$$\beta_1 = \frac{\mathbb{E}[\tau(X)\mathbb{V}(D | X)]}{\mathbb{E}[\mathbb{V}(D | X)]}.$$

²⁸By LIE, $\mathbb{E}[(D - e(X))(Y(0) - \mathbb{E}[Y(0) | X])] = \mathbb{E}[\mathbb{E}[(D - e(X))(Y(0) - \mathbb{E}[Y(0) | X]) | X]]$. Conditioning on X , the term $(Y(0) - \mathbb{E}[Y(0) | X])$ is measurable with respect to X and factors out, giving $\mathbb{E}[\mathbb{E}[D - e(X) | X] \cdot (Y(0) - \mathbb{E}[Y(0) | X])] = 0$, since $\mathbb{E}[D - e(X) | X] = e(X) - e(X) = 0$ by definition of the propensity score.

Since $D \in \{0, 1\}$, $\mathbb{V}(D \mid X = x) = e(x)(1 - e(x))$. By the Law of Iterated Expectations:

$$E[\tilde{D}^2] = \sum_x e(x)[1 - e(x)]P[X = x]$$

Therefore, for discrete X ,

$$\beta_1 = \sum_x \tau(x) \frac{e(x)(1 - e(x))\mathbb{P}(X = x)}{\sum_{x'} e(x')(1 - e(x'))\mathbb{P}(X = x')}.$$

□

Now, recall that selection bias arises when two groups are not representation of the same population: it is a composition problem. The key assumption in this section is that this composition problem can be solved by controlling for observables X . Now, the propensity score can be described as a **balancing score**:

$$X \perp\!\!\!\perp D \mid e(x)$$

Proposition 4.2: Propensity Score as a Balancing Score

Let $e(X) \equiv \mathbb{P}(D = 1 \mid X)$. Then $X \perp\!\!\!\perp D \mid e(X)$; i.e., conditional on the propensity score, the distribution of X is the same in the treated and untreated groups.

Proof. Fix any covariate value x and any score value p such that $\mathbb{P}(X = x, e(X) = p) > 0$. Note that $e(X)$ is a function of X , so on the event $\{X = x\}$ we have $e(X) = e(x)$. Hence $\mathbb{P}(D = 1 \mid X = x, e(X) = p) = \mathbb{P}(D = 1 \mid X = x) = e(x)$. Moreover, if $\mathbb{P}(X = x \mid e(X) = p) > 0$, then necessarily $p = e(x)$, so the previous display equals p . Therefore, for all such (x, p) ,

$$\mathbb{P}(D = 1 \mid X = x, e(X) = p) = p = \mathbb{P}(D = 1 \mid e(X) = p).$$

This is exactly $D \perp\!\!\!\perp X \mid e(X)$. □

All biases due to observable covariates can be eliminated by conditioning on the propensity score. Indeed, if units are homogenous with respect to $e(x)$, they share the same distribution of X . Additionally, the propensity score provides **dimensionality reduction**: prevents from conditioning on all covariates. The propensity score can be thought of as a sufficient statistic.

4.2 Assessing Common Support (Overlap)

Lack of overlap in covariate distributions makes conventional estimators (e.g. regression) highly sensitive to specification choices and may yield imprecise or unstable estimates. Overlap is most effectively assessed using the propensity score $e(X)$.

- Overlap in each covariate's marginal distribution is not sufficient.
- Plot the distribution of $e(X)$ separately for $D = 1$ and $D = 0$.

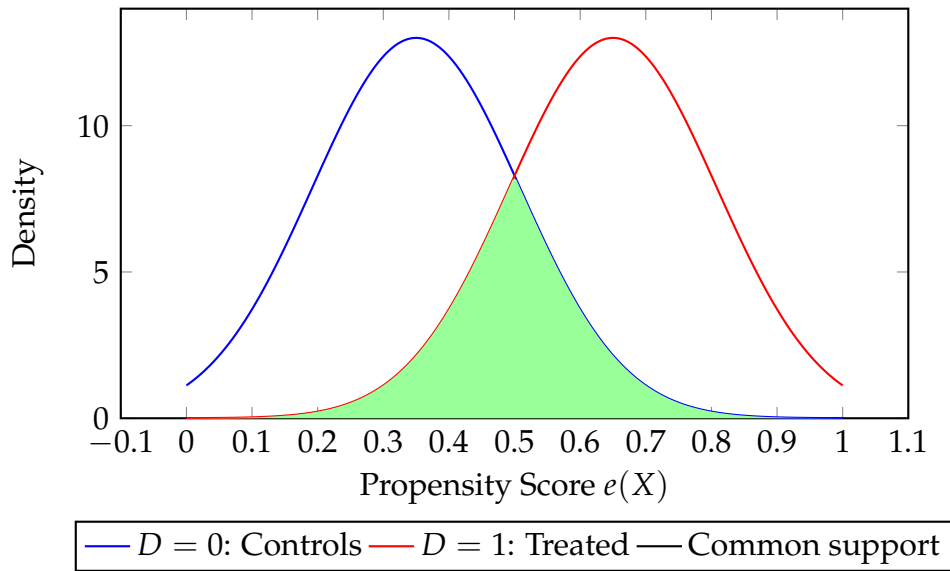
- Compare odds ratios across groups:

$$\frac{e(X)}{1 - e(X)} \propto \frac{f(X | D = 1)}{f(X | D = 0)}.$$

Estimating $e(X)$ is essentially a prediction problem. One can select a parsimonious model from a potentially large set of pre-treatment covariates (e.g. LASSO):

1. Use LASSO to select covariates predicting Y .
2. Use LASSO to select covariates predicting D .
3. Take the union of both sets.

Graphical assessment of overlap.



The shaded region represents the range of propensity scores where both treated and control units are observed. Lack of overlap (i.e. regions where one density is close to zero) implies extrapolation and weak identification.

4.3 The Analogy Principle and Estimable Representations

Consider the ATT as before

$$\tau \equiv \mathbb{E}(Y(1) - Y(0) | D = 1) = \mathbb{E}(Y(1) | D = 1) - \mathbb{E}(Y(0) | D = 1)$$

As always, we are interesting in the counterfactual term, which can be written as follows by conditional LIE

$$\mathbb{E}(Y(0) | D = 1) = \mathbb{E}[\mathbb{E}(Y(0) | D = 1, X) | D = 1]$$

Estimation can be obtained via the analogy principle, considering sample analogs of the above quantities, which defines alternative classes of estimators for τ .

Now, understand that Strong Ignorability implies identification of the conditional outcome means

$$\mathbb{E}(Y|D = 1, X = x) = \mathbb{E}(Y(1)|D = 1, X = x) = \mathbb{E}(Y(1)|X = x) \equiv \mu_1(x)$$

$$\mathbb{E}(Y|D = 0, X = x) = \mathbb{E}(Y(0)|D = 0, X = x) = \mathbb{E}(Y(0)|X = x) \equiv \mu_0(x)$$

where the first step just follows from consistency and the second step from SI. Note that the same holds if Y is replaced by any function $g(Y)$. For instance, marginal distribution are obtained from $g(y) = \mathbb{1}(Y \leq y)$.

Now that we have established this, we can show that τ can be rewritten as

$$\tau = \mathbb{E}(Y|D = 1) - \mathbb{E}[\mathbb{E}(Y(0)|\mathbf{D=0}, X) | D = 1] = \mathbb{E}(Y|D = 1) - \mathbb{E}[\mu_0(x)|D = 1]$$

The key step is the first equality, which uses CI (or selection on observables). Indeed, CI implies that $\mathbb{E}(Y(0)|D = 1, X) = \mathbb{E}(Y(0)|D = 0, X) = \mathbb{E}(Y(0)|X)$. Once we condition on X , treatment assignment is independent of $Y(0)$. consistency is then used to express $\mu(x)$ in terms of observed outcomes among controls. The empirical analog of this expression computed from a sample of n_1 units is given by

$$\frac{1}{n_1} \sum_{i=1}^{n_1} (y_i - \mu_0(x_i))$$

where $\mu_0(x_i)$ is the counterfactual outcome and y_i is observed for treated group. The treatment effect is therefore estimated as

$$\frac{1}{n_1} \sum_{i=1}^{n_1} (y_i - \hat{\mu}_0(x_i))$$

Everything boils down to how to estimate this counterfactual, as always. Note that under CI, one can also write

$$\mathbb{E}(Y|D = 1, e(x)) = \mathbb{E}(Y(1)|D = 1, e(x)) = \mathbb{E}(Y(1)|e(x)) \equiv \mu_1(e(x))$$

$$\mathbb{E}(Y|D = 0, e(x)) = \mathbb{E}(Y(0)|D = 0, e(x)) = \mathbb{E}(Y(0)|e(x)) \equiv \mu_0(e(x))$$

Similar calculations yield the following alternative estimator for the TE

$$\frac{1}{n_1} \sum_{i=1}^{n_1} (y_i - \hat{\mu}_0(e(x_i)))$$

To sum up, the identification result reduces estimation of the ATT to

$$\tau = \mathbb{E}[Y - \mu_0(X) | D = 1],$$

so that the central statistical problem is how to estimate the counterfactual regression function $\mu_0(X)$ (or $\mu_0(e(X))$).

Several challenges arise:

- Avoiding strong functional form assumptions on conditional expectations;
- Ensuring robustness to small implementation choices;
- Dealing with high-dimensional covariates (curse of dimensionality).

The estimators discussed below differ in how they address these trade-offs.

4.3.1 Regression Estimators

Assuming scalar X for simplicity, a direct approach consists of estimating the conditional mean function

$$\mu_0(x) = \mathbb{E}[Y \mid D = 0, X = x]$$

using observations in the control group only. For instance, under a linear specification,

$$y = \alpha_0 + \beta_0 x_i + u \implies \hat{\mu}_0(x) = \hat{\alpha}_0 + \hat{\beta}_0 x_i,$$

we obtain $\hat{\mu}_0(x)$ by OLS on the subsample with $D = 0$. The ATT estimator becomes

$$\begin{aligned} \hat{\tau} &= \frac{1}{n_1} \sum_{i:D_i=1} (y_i - \hat{\mu}_0(x_i)) = \frac{1}{n_1} \sum_{i:D_i=1} y_i - \frac{1}{n_1} \sum_{i:D_i=1} \hat{\mu}_0(x_i) \\ &= \bar{y}_1 - \frac{1}{n_1} \sum_{i:D_i=1} (\hat{\alpha}_0 + \hat{\beta}_0 x_i) = \bar{y}_1 - \frac{1}{n_1} \sum_{i:D_i=1} (\bar{y}_0 - \hat{\beta}_0 \bar{x}_0 + \hat{\beta}_0 x_i) = \bar{y}_1 - \bar{y}_0 - \hat{\beta}_0 (\bar{x}_1 - \bar{x}_0) \end{aligned}$$

where we used $\hat{\alpha}_0 = \bar{y}_0 - \hat{\beta}_0 \bar{x}_0$. This just shows that the raw difference in mean outcomes is regression-adjusted for differences in mean covariates X across groups. This regression-adjusted estimator may rely heavily on extrapolation when treated and control covariate distributions differ substantially, is sensitive to lack of common support, and—while handling dimensionality parametrically—can suffer from bias if the conditional expectation function is misspecified. Other options can involve a polynomial specification to model the propensity score or a local linear (kernel) regression.

4.3.2 Weighting Estimators

An alternative way to address dimensionality arises from rewriting the unknown counterfactual as follows

$$\mathbb{E}(Y(0) \mid D = 1) = \mathbb{E}(\mathbb{E}(Y(0) \mid X) \mid D = 1) = \int_{\mathbb{R}} \mu_0(x) dF(x \mid D = 1)$$

where in the first equality, we use LIE and CI (to drop D inside), and the second equality is by definition of conditional expectation. Now, if we multiply and divide²⁹ by $dF(x \mid D = 0)$, we can rewrite

$$\mathbb{E}(Y(0) \mid D = 1) = \int_{\mathbb{R}} \mu_0(x) \frac{dF(x \mid D = 1)}{dF(x \mid D = 0)} dF(x \mid D = 0)$$

²⁹Formally, this step is a change of measure. Let $F_1(\cdot) = F(\cdot \mid D = 1)$ and $F_0(\cdot) = F(\cdot \mid D = 0)$. Under Common Support, F_1 is absolutely continuous with respect to F_0 (denoted $F_1 \ll F_0$), meaning that $F_0(A) = 0$ implies $F_1(A) = 0$ for any measurable set A . By the Radon–Nikodym theorem, there exists a derivative

Note that this step requires CS so that $dF(x|D = 0) \neq 0$. Now, Bayes' Theorem gives

$$f(x|D = 1) = \frac{\mathbb{P}(D = 1|X = x)f(x)}{\mathbb{P}(D = 1)}; f(x|D = 0) = \frac{\mathbb{P}(D = 0|X = x)f(x)}{\mathbb{P}(D = 0)}$$

Taking the ratio of the two gives convex non-negative weights $p(x)$:

$$p(x) \equiv \frac{e(x)}{1 - e(x)} \frac{\mathbb{P}(D = 0)}{\mathbb{P}(D = 1)}$$

To conclude, we can rewrite

$$\begin{aligned} \mathbb{E}(Y(0)|D = 1) &= \int_{\mathbb{R}} \mathbb{E}(Y(0)|X)p(x)f(x|D = 0)dx = \int_{\mathbb{R}} \mathbb{E}(Y(0)p(x)|X)f(x|D = 0)dx \\ &= \mathbb{E}(p(x)Y|D = 0) \end{aligned}$$

where the last object is identified and is equivalent to the weighted average of the outcome for the control group.

The empirical counterpart of this expression based on a sample of n_0 control units is given by

$$\frac{1}{n_0} \sum_{j=1}^{n_0} p(x_j)y_j$$

The key idea is to represent the unknown counterfactual using the control group, by assigning importance weights to its units. The weights are functions of the estimated propensity score. This yields the following estimator of the treatment effect

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^{n_1} y_i - \frac{1}{n_0} \sum_{j=1}^{n_0} \hat{p}(x_j)y_j \text{ where } \hat{p}(x) = \frac{\hat{e}(x)}{1 - \hat{e}(x)} \frac{n_0}{n_1}$$

Note that we used n_0, n_1 as the estimates for $\mathbb{P}(D = 0), \mathbb{P}(D = 1)$, respectively. The key challenge here is to estimate the propensity score correctly.

Note that for computation, it is convenient (e.g. when calculating bootstrap standard errors) to rewrite the estimator as follows. Rewrite both sums over the full sample using $D_i \in \{0, 1\}$:

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n D_i y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - D_i) \hat{p}(x_i) y_i.$$

$\frac{dF_1}{dF_0}(x)$ such that for any integrable function g ,

$$\int g(x) dF_1(x) = \int g(x) \frac{dF_1}{dF_0}(x) dF_0(x).$$

When densities exist, this derivative reduces to the ratio $\frac{f(x|D=1)}{f(x|D=0)}$, which yields the propensity-score weights.

Substituting $\hat{p}(x_i) = \frac{\hat{e}(x_i)}{1-\hat{e}(x_i)} \frac{n_0}{n_1}$ gives

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n D_i y_i - \frac{1}{n_1} \sum_{i=1}^n (1 - D_i) \frac{\hat{e}(x_i)}{1 - \hat{e}(x_i)} y_i = \frac{1}{n_1} \sum_{i=1}^n \left[D_i - (1 - D_i) \frac{\hat{e}(x_i)}{1 - \hat{e}(x_i)} \right] y_i.$$

Finally, using $n_1 = n (n_1/n)$,

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \underbrace{\left[\frac{D_i}{n_1/n} - \frac{(1 - D_i)}{n_1/n} \frac{\hat{e}(x_i)}{1 - \hat{e}(x_i)} \right]}_{\lambda_i \text{ (weights)}} y_i.$$

Thus, IPW rebalances the control group so that its covariate distribution matches that of the treated group:

$$dF(x \mid D = 1) = p(x) dF(x \mid D = 0).$$

In practice, the distribution of weights should be inspected carefully. Large weights arise when $\hat{e}(x)$ is close to 0 or 1, reflecting weak overlap. Weights are typically normalized to sum to one in finite samples.

4.3.3 Matching Estimators

Matching estimators approximate the counterfactual outcome for each treated unit by pairing it with one (or a small number of) observationally similar control units. The key idea is to estimate $\mu_0(x_i)$ locally rather than modeling it globally.

Let ξ_{ij} denote a distance measure between treated unit i ($D_i = 1$) and control unit j ($D_j = 0$). A control unit j is the *nearest neighbor* to treated unit i if

$$\xi_{ij} = \min\{\xi_{i1}, \xi_{i2}, \dots, \xi_{in_0}\}.$$

Common choices of distance include:

- **Euclidean distance:**

$$\xi_{ij} = \|x_i - x_j\|;$$

- **Mahalanobis distance** (with Σ_X the covariance matrix of X):

$$\xi_{ij} = (x_i - x_j)' \Sigma_X^{-1} (x_i - x_j);$$

- **Propensity score distance:**

$$\xi_{ij} = |e(x_i) - e(x_j)|.$$

For simplicity, consider one-to-one matching with replacement. Let $\ell(i)$ denote the control unit closest to treated unit i . Define the estimated counterfactual outcome for unit i as

$$\hat{y}_{i0} \equiv y_{\ell(i)}.$$

This can be written as a weighted average over controls:

$$\hat{y}_{i0} = \sum_{j=1}^{n_0} \mathbb{1}\{j = \ell(i)\} y_j,$$

The treatment effect for treated unit i is then estimated as

$$\hat{\tau}_i = y_i - \hat{y}_{i0}.$$

Averaging over treated units yields the matching estimator of the ATT:

$$\hat{\tau} = \frac{1}{n_1} \sum_{i:D_i=1} (y_i - \hat{y}_{i0}).$$

Now, if we define

$$K_j \equiv \sum_{i=1}^{n_1} \mathbb{1}\{j = \ell(i)\},$$

that is, the number of times control unit j is used as a match. Then, the matching estimator of the ATT becomes

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^{n_1} y_i - \frac{1}{n_1} \sum_{j=1}^{n_0} K_j y_j.$$

Equivalently, writing the estimator over the full sample,

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \lambda_i y_i,$$

where the weights are given by

$$\lambda_i = \frac{D_i}{n_1/n} - \frac{(1 - D_i)K_i}{n_1/n}$$

Indeed,

$$\hat{\tau} = \frac{1}{n_1} \sum_{i:D_i=1} y_i - \frac{1}{n_1} \sum_{i:D_i=0} K_i y_i = \frac{1}{n_1} \sum_{i=1}^n [D_i - (1 - D_i)K_i] y_i.$$

Since $n_1 = n(n_1/n)$,

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - (1 - D_i)K_i}{n_1/n}}_{\lambda_i} y_i.$$

Matching estimators can be interpreted as local nonparametric approximations of $\mu_0(x)$, where the number of neighbors plays a role analogous to the bandwidth in kernel regression.

4.3.4 Concluding Remarks

All estimators considered can be written in a unified form as

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n \lambda_i y_i,$$

where the weights λ_i depend only on the treatment indicator D_i and covariates X_i , but not on the outcomes themselves. Under independence across units, their variance can be expressed as

$$\mathbb{V}(\hat{\tau} \mid X, D) = \frac{1}{n} \sum_{i=1}^n \lambda_i^2 \mathbb{V}(y_i \mid X_i, D_i),$$

making clear that precision is directly governed by the dispersion of the weights. Different estimators correspond to different constructions of these weights and therefore involve distinct bias–variance trade-offs. Weighting estimators may become unstable when the propensity score approaches zero or one, reflecting weak overlap. Regression-based approaches can suffer from extrapolation bias when treated and control covariate distributions differ substantially. Matching estimators avoid global functional form assumptions but may introduce non-trivial bias if the quality of matches is poor. Hybrid procedures, such as bias-corrected matching or doubly robust estimators, aim to mitigate these risks by combining outcome modeling and reweighting, though they introduce additional modeling and tuning choices. Ultimately, all methods operationalize the same identification result, differing only in how they approximate the counterfactual distribution of untreated outcomes for the treated group.

5 Instrumental Variation

In Section 4 (of Chapter 1), we introduced instrumental variables in the classical linear model, focusing on identification, consistency and asymptotic properties of the IV and 2SLS estimators under homogeneous treatment effects.

In this section, we revisit IV from a potential outcomes perspective. Rather than asking whether β is identified in a linear model, we ask a deeper question:

What causal parameter does IV estimate when treatment effects are heterogeneous?

This shift in perspective leads to the concepts of compliance types, Local Average Treatment Effects (LATE), and Marginal Treatment Effects (MTE), which clarify the interpretation and limitations of IV estimators.

5.1 The Traditional View of Instrumental Variables

We first provide a quick additional word on the traditional approach to instrumental variables, which is developed in a linear framework with *constant returns to treatment*. To fix ideas, consider the relationship between schooling S and wages W :

$$\begin{aligned} W &= \alpha_0 + \alpha_1 S + \alpha_2 Z + \alpha_3 \varepsilon_2 + \varepsilon_1, \\ S &= \beta_0 + \beta_1 Z + \varepsilon_2, \end{aligned}$$

where Z is an observable characteristic, ε_1 is a wage shock ("luck") and ε_2 represents ability and affects both schooling and wages. Because ε_2 enters both equations, schooling S is endogenous.

Now, suppose we impose:

$$\mathbb{E}[\varepsilon_1 | Z] = 0 \quad \text{and} \quad \mathbb{E}[\varepsilon_2 | Z] = 0.$$

Then Z is as good as randomly assigned and uncorrelated with unobservables in the wage equation. Substituting the schooling equation into the wage equation yields the *reduced form*:

$$W = (\alpha_0 + \alpha_1 \beta_0) + (\alpha_2 + \alpha_1 \beta_1) Z + (\alpha_1 + \alpha_3) \varepsilon_2 + \varepsilon_1.$$

Taking expectations conditional on Z :

$$\begin{aligned} \mathbb{E}[W | Z = z] &= (\alpha_0 + \alpha_1 \beta_0) + (\alpha_2 + \alpha_1 \beta_1) z, \\ \mathbb{E}[S | Z = z] &= \beta_0 + \beta_1 z. \end{aligned}$$

These are reduced-form regressions. The first equation is what we think about when we mention reduced form (direct impact of instrument on outcome variable) while the second equation is the first stage.

If we impose the exclusion restriction ($\alpha_2 = 0$), then Z affects wages only through schooling. From the reduced-form regressions we obtain:

$$\beta_1 = \frac{\text{Cov}(S, Z)}{\text{V}(Z)} \quad (\text{first stage}),$$

and

$$\alpha_1 \beta_1 = \frac{\text{Cov}(W, Z)}{\text{V}(Z)} \quad (\text{reduced form}).$$

Dividing the reduced form coefficient by the first-stage coefficient yields:

$$\alpha_1 = \frac{\alpha_1 \beta_1}{\beta_1} = \frac{\text{Cov}(W, Z)}{\text{Cov}(S, Z)},$$

provided that $\text{Cov}(S, Z) \neq 0$ (relevance). This makes explicit the standard interpretation of IV as

$$\text{IV} = \frac{\text{Reduced Form}}{\text{First Stage}}$$

Equivalently, for two distinct values z_1 and z_2 of the instrument, we can write the Wald Ratio:

$$\alpha_1 = \frac{\mathbb{E}[W \mid Z = z_2] - \mathbb{E}[W \mid Z = z_1]}{\mathbb{E}[S \mid Z = z_2] - \mathbb{E}[S \mid Z = z_1]},$$

as long as the denominator is non-zero. This ratio interpretation highlights the logic of IV:

The causal effect of schooling is obtained by dividing the total effect of the instrument on wages (reduced form) by the effect of the instrument on schooling (first stage).

In practice, the IV estimator is implemented via Two-Stage Least Squares:

1. Regress S on Z to obtain fitted values $\hat{S}$.
2. Regress W on $\hat{S}$.

Under the exclusion and relevance conditions, this procedure consistently estimates α_1 . The traditional view therefore delivers a clean interpretation: *IV identifies a single constant causal parameter*. In the next section, we relax the assumption of constant returns and ask what IV identifies when treatment effects are heterogeneous.

5.2 Heterogeneous Returns

5.2.1 General approach

The key question of this section is: **what do IV regressions estimate if gains are not constant?**

To fix ideas, consider the case of a binary causal variables $D \in \{0, 1\}$ (e.g. participation in a training for unemployed individuals) and a binary instrument $Z \in \{0, 1\}$ (e.g. randomized invitation to training program). Potential outcomes are now defined with respect to the 2×2 scenarios that result from the cross tabulation of D and Z : $Y_i(D_i, Z_i)$. The observed outcome Y_i coincides with the potential outcome realized for the observed combination ($D_i = d, Z_i = z$). Individual's i effect resulting from different values of the instrument is $Y_i(D_i, 1) - Y_i(D_i, 0)$. The same quantity computed with respect to the causal variable is $Y_i(1, Z_i) - Y_i(0, Z_i)$. We also need to adjust notation to express the idea that Z has an effect on D : $D_i(Z_i)$ are potential states for the causal variable. $D_i(0)$ represents the potential participation if I do not receive the letter while $D_i(1)$ is the potential participation if I receive the letter.

First, we need to assume that Z is (as good as) randomly assigned

$$\{Y_i(1, 1), Y_i(0, 1), Y_i(1, 0), Y_i(0, 0), D_i(0), D_i(1)\} \perp\!\!\!\perp Z_i$$

Now, by random assignment³⁰, we can write the **first-stage** equation as

$$\mathbb{E}(D_i | Z_i = 1) - \mathbb{E}(D_i | Z_i = 0) = \mathbb{E}(D_i(1) - D_i(0))$$

³⁰Note that we always assume some form of consistency/SUTVA: $\mathbb{E}(D_i | Z_i = z) = \mathbb{E}(D_i(z) | Z_i = z)$. SUTVA ensures that the observed treatment is exactly the potential treatment corresponding to the realized instrument value.

implying that the first stage regression from 2SLS captures the average causal effect of Z (receiving the letter) on D (participation). Similarly, one can show

$$\mathbb{E}(Y_i|Z_i = 1) - \mathbb{E}(Y_i|Z_i = 0) = \mathbb{E}(Y_i(1) - Y_i(0))$$

implying that the reduced form regression captures the average effect of Z on Y , often called the intent to treat (which could be an interesting parameter of interest by itself).

Now, in this context, the **exclusion restriction** can be stated as follows

$$Y_i(0,1) = Y_i(0,0) \text{ and } Y_i(1,1) = Y_i(1,0)$$

The first part of the assumption means that the potential outcome is the same whether the person received the letter or not given they did not go. The second part of the assumption says that if the individual was going anyway, the letter has no effect on the outcome. This is akin to the usual interpretation we have: the instrument affects the outcome only through the treatment. Note that in the context, the assumption reduces the dimensionality of potential outcomes from 4 to 2. Together with random assignment,

$$\{Y_i(0), Y_i(1), D_i(0), D_i(1)\} \perp\!\!\!\perp Z_i,$$

it follows that

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp Z_i \mid D_i(0), D_i(1).$$

This formalizes the idea that the effect on Z on the outcome is solely the result of the first stage effect on D .

Now, depending on the values taken by $\{D(1), D(0)\}$, the following four groups are defined in the population:

- Always-Takers ($\mathcal{A}$): $D(1) = 1, D(0) = 1$ (always get the treatment regardless of assignment outcome)
- Never-Takers ($\mathcal{N}$): $D(1) = 0, D(0) = 0$ (never get the treatment regardless of assignment outcome)
- Compliers ($\mathcal{C}$): $D(1) = 1, D(0) = 0$ (always comply with assignment outcome)
- Defiers ($\mathcal{D}$): $D(1) = 0, D(0) = 1$ (never comply with assignment outcome)

Proposition 5.1: Reduced form decomposition

Let $(Y_i(1), Y_i(0))$ denote the potential outcomes under treatment and no treatment, respectively, and let $D_i(1), D_i(0)$ denote the potential treatment states under $Z_i = 1$ and $Z_i = 0$. Assume exclusion and random assignment among the relevant subpopulation:

$$\{Y_i(1), Y_i(0), D_i(1), D_i(0)\} \perp\!\!\!\perp Z_i,$$

together with consistency (SUTVA). Then the reduced form difference satisfies

$$\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0] = \mathbb{E}[Y_i(1) - Y_i(0) | \mathcal{C}] \pi_{\mathcal{C}} - \mathbb{E}[Y_i(1) - Y_i(0) | \mathcal{D}] \pi_{\mathcal{D}},$$

where $\pi_{\mathcal{T}} \equiv \mathbb{P}(\mathcal{T})$ for $\mathcal{T} \in \{\mathcal{A}, \mathcal{N}, \mathcal{C}, \mathcal{D}\}$, and $\mathcal{C}, \mathcal{D}$ denote compliers and defiers.

Proof. Let $(Y_i(1), Y_i(0))$ be the potential outcomes in case of treatment and no treatment, respectively. We assume exclusion, that is

$$\{Y_i(1), Y_i(0), D_i(1), D_i(0)\} \perp Z_i.$$

Adding random assignment allows to write

$$\{Y_i(1), Y_i(0)\} \perp Z_i | D_i(1), D_i(0).$$

We can write the following decomposition by LIE³¹:

$$\begin{aligned} \mathbb{E}[Y_i | Z_i] &= \mathbb{E}[Y_i | \mathcal{C}, Z_i] \mathbb{P}(\mathcal{C} | Z_i) \\ &\quad + \mathbb{E}[Y_i | \mathcal{A}, Z_i] \mathbb{P}(\mathcal{A} | Z_i) \\ &\quad + \mathbb{E}[Y_i | \mathcal{N}, Z_i] \mathbb{P}(\mathcal{N} | Z_i) \\ &\quad + \mathbb{E}[Y_i | \mathcal{D}, Z_i] \mathbb{P}(\mathcal{D} | Z_i). \end{aligned}$$

Now, for any $\mathcal{T} \in \{\mathcal{A}, \mathcal{N}, \mathcal{C}, \mathcal{D}\}$, since Z_i is as good as random, we have that

$$\{D_i(1), D_i(0)\} \perp Z_i.$$

Now, understand that the compliance type is fully determined by the pair $(D_i(1), D_i(0))$: $\mathcal{T} = f(D_i(1), D_i(0))$. Therefore, it inherits the independence from random assignment, and we can write

$$\mathbb{P}(\mathcal{T} | Z_i) = \mathbb{P}(\mathcal{T}) \equiv \pi_{\mathcal{T}}.$$

Hence, the mixture weights do not depend on Z_i . Moreover, by consistency $Y_i = Y_i(D_i(Z_i))$, and by random assignment and exclusion the mixture components do not depend on Z_i , so we can write³²

$$\begin{aligned} \mathbb{E}[Y_i | Z_i] &= \mathbb{E}[Y_i(Z_i) | \mathcal{C}] \pi_{\mathcal{C}} \\ &\quad + \mathbb{E}[Y_i(1) | \mathcal{A}] \pi_{\mathcal{A}} \\ &\quad + \mathbb{E}[Y_i(0) | \mathcal{N}] \pi_{\mathcal{N}} \\ &\quad + \mathbb{E}[Y_i(1 - Z_i) | \mathcal{D}] \pi_{\mathcal{D}}. \end{aligned}$$

³¹Let $\mathcal{T} \in \{\mathcal{A}, \mathcal{N}, \mathcal{C}, \mathcal{D}\}$ denote the compliance type. Since these types form a partition of the population, LIE gives

$$\mathbb{E}[Y_i | Z_i] = \sum_{t \in \{\mathcal{A}, \mathcal{N}, \mathcal{C}, \mathcal{D}\}} \mathbb{E}[Y_i | Z_i, \mathcal{T} = t] \mathbb{P}(\mathcal{T} = t | Z_i).$$

³²With a slight abuse of notation, $Y_i(Z_i)$ denotes $Y_i(D_i(Z_i))$. Under exclusion, outcomes depend only on the treatment received, so $Y_i(D_i, Z_i) = Y_i(D_i)$ and the instrument affects Y_i only through $D_i(Z_i)$.

It follows that

$$\begin{aligned}\mathbb{E}[Y_i | Z_i = 1] &= \mathbb{E}[Y_i(1) | \mathcal{C}] \pi_{\mathcal{C}} \\ &\quad + \mathbb{E}[Y_i(1) | \mathcal{A}] \pi_{\mathcal{A}} \\ &\quad + \mathbb{E}[Y_i(0) | \mathcal{N}] \pi_{\mathcal{N}} \\ &\quad + \mathbb{E}[Y_i(0) | \mathcal{D}] \pi_{\mathcal{D}},\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[Y_i | Z_i = 0] &= \mathbb{E}[Y_i(0) | \mathcal{C}] \pi_{\mathcal{C}} \\ &\quad + \mathbb{E}[Y_i(1) | \mathcal{A}] \pi_{\mathcal{A}} \\ &\quad + \mathbb{E}[Y_i(0) | \mathcal{N}] \pi_{\mathcal{N}} \\ &\quad + \mathbb{E}[Y_i(1) | \mathcal{D}] \pi_{\mathcal{D}}.\end{aligned}$$

This allows us to recover the intention to treat:

$$\begin{aligned}\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0] &= \mathbb{E}[Y_i(1) - Y_i(0) | \mathcal{C}] \pi_{\mathcal{C}} \\ &\quad - \mathbb{E}[Y_i(1) - Y_i(0) | \mathcal{D}] \pi_{\mathcal{D}}.\end{aligned}$$

□

Proposition 5.2: First stage and Wald Ratio

Under the same assumptions, the first-stage difference satisfies

$$\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0] = \pi_{\mathcal{C}} - \pi_{\mathcal{D}}.$$

Consequently, the Wald ratio equals

$$\begin{aligned}\mathcal{W} &= \frac{\mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0]} \\ &= \frac{\mathbb{E}[Y_i(1) - Y_i(0) | \mathcal{C}] \pi_{\mathcal{C}} - \mathbb{E}[Y_i(1) - Y_i(0) | \mathcal{D}] \pi_{\mathcal{D}}}{\pi_{\mathcal{C}} - \pi_{\mathcal{D}}}\end{aligned}$$

In general, this object is not equal to the average treatment effect. In particular, it depends on both compliers and defiers, and may even have the wrong sign.

Proof. Let $D_i(1), D_i(0)$ denote the potential treatment status if $Z_i = 1$ or $Z_i = 0$, respectively.

Recall that Z_i is as good as random:

$$Z_i \perp (D_i(0), D_i(1)).$$

We assume SUTVA, so that $D_i = D_i(Z_i)$ and there is no spillovers across units. Now, by independence and LIE, we can write

$$\mathbb{E}[D_i | Z_i] = \mathbb{E}[D_i(Z_i)] = \mathbb{E}[D_i(Z_i) | \mathcal{C}] \mathbb{P}(\mathcal{C}) + \mathbb{E}[D_i(Z_i) | \mathcal{A}] \mathbb{P}(\mathcal{A}) + \mathbb{E}[D_i(Z_i) | \mathcal{N}] \mathbb{P}(\mathcal{N})$$

$$+\mathbb{E}[D_i(Z_i) \mid \mathcal{D}] \mathbb{P}(\mathcal{D})$$

By definition of the types, we can rewrite

$$\mathbb{E}[D_i \mid Z_i = 1] = \pi_{\mathcal{C}} + \pi_{\mathcal{A}}, \quad \mathbb{E}[D_i \mid Z_i = 0] = \pi_{\mathcal{A}} + \pi_{\mathcal{D}},$$

where $\pi_{\mathcal{T}} \equiv \mathbb{P}(\mathcal{T})$ for $\mathcal{T} \in \{\mathcal{A}, \mathcal{N}, \mathcal{C}, \mathcal{D}\}$.

Therefore,

$$\gamma_1 = \pi_{\mathcal{C}} - \pi_{\mathcal{D}}.$$

We can therefore recover the Wald ratio by dividing the intention to treat by the first stage:

$$\mathcal{W} \equiv \frac{\mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{C}] \pi_{\mathcal{C}} - \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{D}] \pi_{\mathcal{D}}}{\pi_{\mathcal{C}} - \pi_{\mathcal{D}}}.$$

□

5.2.2 LATE

We have seen that under the assumptions of random assignment of the instrument, exclusion restriction and relevance, the IV estimator does not recover the ATE. Hence, we need to impose an additional assumption.

Proposition 5.3: LATE under monotonicity

Suppose, in addition to exclusion, random assignment, and consistency, that monotonicity holds:

$$D_i(1) \geq D_i(0) \quad \text{for all } i,$$

so that there are no defiers ($\pi_{\mathcal{D}} = 0$). Then the Wald ratio identifies the Local Average Treatment Effect (LATE):

$$\mathcal{W} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{C}].$$

Proof. Under monotonicity, $\pi_{\mathcal{D}} = 0$. From Proposition 2,

$$\mathcal{W} = \frac{\mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{C}] \pi_{\mathcal{C}} - \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{D}] \pi_{\mathcal{D}}}{\pi_{\mathcal{C}} - \pi_{\mathcal{D}}}.$$

Substituting $\pi_{\mathcal{D}} = 0$ yields

$$\mathcal{W} = \frac{\mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{C}] \pi_{\mathcal{C}}}{\pi_{\mathcal{C}}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathcal{C}],$$

provided that $\pi_{\mathcal{C}} \neq 0$ (relevance). □

This result shows that instrumental variables do not generally recover the average treatment effect. Instead, under monotonicity, IV identifies the average causal effect for the subpopulation of compliers, that is, those individuals whose treatment status is affected by the instrument. The parameter therefore depends on the instrument used and on the behavioral margin it shifts.

5.2.3 Rethinking Monotonicity

The replacement of homogeneity by monotonicity triggered a renaissance of IV methods in the 1990s. Monotonicity requires that the instrument moves treatment decisions in the same direction for all individuals:

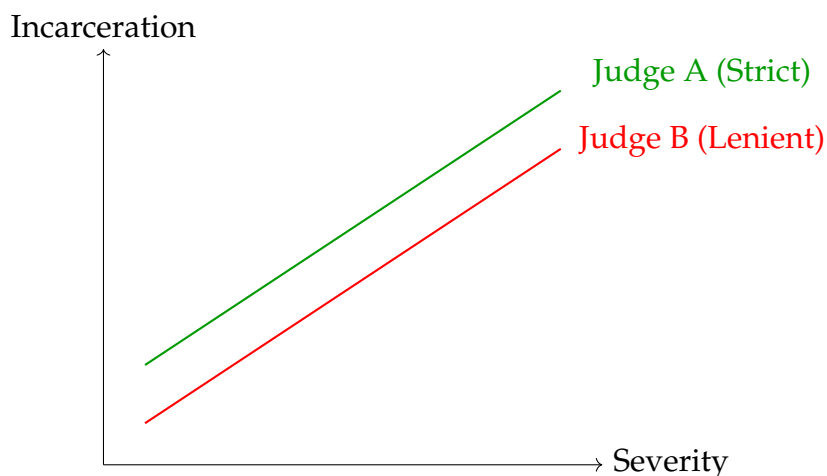
$$D_i(1) \geq D_i(0) \quad \text{for all } i.$$

In words, if a “lenient” judge incarcerates a defendant, a “strict” judge would also incarcerate that defendant.

To understand the restriction imposed by monotonicity, consider the example of randomly assigned judges with different sentencing rates, where Z_i indicates assignment to a strict ($Z_i = 1$) or lenient ($Z_i = 0$) judge, and D_i denotes incarceration.

Panel A: Monotonicity Holds

In this case, the strict judge always imposes a (weakly) more severe sentence than the lenient judge for every level of crime severity.



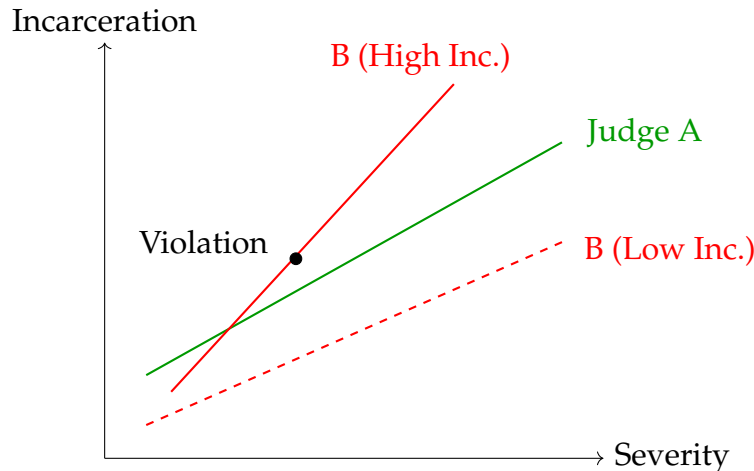
Here, the strict judge’s incarceration rule lies everywhere above that of the lenient judge. Hence,

$$D_i(1) \geq D_i(0)$$

for all defendants, and monotonicity holds.

Panel B: Violation of Monotonicity

Now suppose judges differ along multiple dimensions. Judge A is strict based only on crime severity, while Judge B is generally lenient but particularly harsh toward high-income defendants.



In this case, the sentencing rules cross. For some defendants,

$$D_i(1) < D_i(0),$$

so defiers exist and monotonicity fails.

Monotonicity therefore imposes a strong behavioral restriction: the instrument must shift treatment decisions in a uniform direction across all individuals. When agents differ along multiple unobserved dimensions, such uniformity may be implausible.

5.2.4 Is LATE Relevant for Policy Counterfactuals?

The Local Average Treatment Effect (LATE) identifies the average causal effect for the subpopulation of *compliers*, that is, individuals whose treatment status is affected by the instrument. This raises an important question: **is LATE the parameter of interest for policy analysis?**

First, LATE is inherently *instrument-specific*. The group of compliers depends on the particular instrument used. An individual who is a complier with respect to one instrument (e.g. distance to college) may not be a complier with respect to another instrument (e.g. tuition subsidies). Hence, IV does not identify a universal treatment effect, but rather the effect at a specific behavioral margin.

Second, the margin affected by the instrument need not coincide with the margin relevant for a policy counterfactual. The individuals whose behavior changes under the instrument may differ from those whose behavior would change under a new policy reform. If treatment effects are heterogeneous, the average effect for these two groups may differ substantially. In such cases, IV estimates do not necessarily recover the parameter relevant for the policy of interest.

Third, when the instrument represents a past policy change, IV identifies the causal effect for those individuals whose treatment choice responded to that historical change. Using this estimate to evaluate a *new* counterfactual policy (e.g. expanding or contracting eligibility) generally requires extrapolation beyond the identified margin.

Therefore, LATE is best interpreted as identifying *what is learnable from the available variation*. It provides credible causal effects for a well-defined subpopulation, but translating these effects into policy predictions requires additional assumptions about how treatment effects vary across individuals.

5.2.5 Characterizing Compliers: A Practical Guide

Compliers are not directly observable: we do not know which individuals satisfy $D_i(1) = 1$ and $D_i(0) = 0$. However, under random assignment, exclusion, and monotonicity, we can characterize their observable features.

The key observation is that although compliance types are latent, their average characteristics are identified.

Let X_i denote a generic observable characteristic and let $g(\cdot)$ be any measurable function (e.g. $g(X_i) = X_i$, $g(X_i) = \mathbb{1}\{X_i = 1\}$, or higher-order moments). Under the same assumptions as for LATE and assuming no defiers, we have:

$$\mathbb{E}[g(X_i) | \mathcal{C}] = \frac{\mathbb{E}[g(X_i)D_i | Z_i = 1] - \mathbb{E}[g(X_i)D_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0]}.$$

Proof. Start from the decomposition (under random assignment, exclusion, SUTVA, and monotonicity):

$$\mathbb{E}[g(X_i)D_i | Z_i = z] = \mathbb{E}[g(X_i)D_i(z) | \mathcal{C}]\pi_{\mathcal{C}} + \mathbb{E}[g(X_i) | \mathcal{A}]\pi_{\mathcal{A}}.$$

Evaluating at the two instrument values:

$$\mathbb{E}[g(X_i)D_i | Z_i = 1] = \mathbb{E}[g(X_i) | \mathcal{C}]\pi_{\mathcal{C}} + \mathbb{E}[g(X_i) | \mathcal{A}]\pi_{\mathcal{A}},$$

$$\mathbb{E}[g(X_i)D_i | Z_i = 0] = \mathbb{E}[g(X_i) | \mathcal{A}]\pi_{\mathcal{A}}.$$

Taking differences eliminates always-takers:

$$\mathbb{E}[g(X_i)D_i | Z_i = 1] - \mathbb{E}[g(X_i)D_i | Z_i = 0] = \mathbb{E}[g(X_i) | \mathcal{C}]\pi_{\mathcal{C}}.$$

From the first stage under monotonicity:

$$\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0] = \pi_{\mathcal{C}}.$$

Dividing the two expressions yields the result

$$\mathbb{E}[g(X_i) | \mathcal{C}] = \frac{\mathbb{E}[g(X_i)D_i | Z_i = 1] - \mathbb{E}[g(X_i)D_i | Z_i = 0]}{\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0]}.$$

□

Thus, the average characteristic of compliers is identified by a Wald-type ratio: the difference in $g(X_i)D_i$ across instrument states divided by the first stage.

The idea is that only compliers change treatment status when Z_i changes. Always-takers and never-takers contribute equally to both $Z_i = 1$ and $Z_i = 0$ groups. Taking differences therefore isolates the contribution of compliers. In practice, characterizing compliers is straightforward:

1. Select a characteristic X_i (e.g. age, income, gender).

2. Compute

$$\mathbb{E}[g(X_i)D_i \mid Z_i = 1] \quad \text{and} \quad \mathbb{E}[g(X_i)D_i \mid Z_i = 0].$$

3. Divide their difference by the first-stage estimate

$$\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0].$$

4. Repeat for each observable characteristic.

Equivalently, this can be implemented via a 2SLS regression of $g(X_i)D_i$ on D_i using Z_i as an instrument. Similarly, setting $g(Y_i) = \mathbb{1}\{Y_i \leq y\}$ or $g(Y_i) = Y_i$ allows one to recover $\mathbb{E}[Y_i(1) \mid \mathcal{C}]$ and $\mathbb{E}[Y_i(0) \mid \mathcal{C}]$ separately.

A typical empirical application presents:

- **Table 1: Sample averages.** Means of X_i for the full sample and by instrument status.
- **Table 2: Complier characteristics.** Estimated $\mathbb{E}[X_i \mid \mathcal{C}]$ for each observable X_i , alongside the overall sample mean for comparison.
- **Table 3 (optional): Outcome comparisons.** Estimated $\mathbb{E}[Y_i(1) \mid \mathcal{C}]$ and $\mathbb{E}[Y_i(0) \mid \mathcal{C}]$, which allow comparison of compliers to always-takers and never-takers in terms of outcomes.

Characterizing compliers is often described as a “low-hanging fruit”: once LATE is identified, their observable features can be recovered using simple sample moments. While we cannot identify individual compliers, we can describe their average characteristics and compare them to the overall population or to other compliance types.

5.3 Beyond Average Effects

5.3.1 Local Treatment Effects on Distributions

The proof establishing identification of LATE extends to any functional of potential outcomes. Let $g\{\cdot\}$ be a measurable, well-behaved function. Under random assignment, exclusion, SUTVA, and monotonicity, we can write:

$$\begin{aligned}\mathbb{E}[g(Y_i) \mid Z_i = 1] &= \mathbb{E}[g(Y_i(1)) \mid \mathcal{C}] \pi_{\mathcal{C}} + \mathbb{E}[g(Y_i(1)) \mid \mathcal{A}] \pi_{\mathcal{A}} + \mathbb{E}[g(Y_i(0)) \mid \mathcal{N}] \pi_{\mathcal{N}}, \\ \mathbb{E}[g(Y_i) \mid Z_i = 0] &= \mathbb{E}[g(Y_i(0)) \mid \mathcal{C}] \pi_{\mathcal{C}} + \mathbb{E}[g(Y_i(1)) \mid \mathcal{A}] \pi_{\mathcal{A}} + \mathbb{E}[g(Y_i(0)) \mid \mathcal{N}] \pi_{\mathcal{N}}.\end{aligned}$$

Taking differences eliminates always-takers and never-takers:

$$\mathbb{E}[g(Y_i) \mid Z_i = 1] - \mathbb{E}[g(Y_i) \mid Z_i = 0] = \mathbb{E}[g(Y_i(1)) - g(Y_i(0)) \mid \mathcal{C}] \pi_{\mathcal{C}}.$$

Dividing by the first stage $\pi_{\mathcal{C}}$ yields:

$$\frac{\mathbb{E}[g(Y_i) \mid Z_i = 1] - \mathbb{E}[g(Y_i) \mid Z_i = 0]}{\pi_{\mathcal{C}}} = \mathbb{E}[g(Y_i(1)) - g(Y_i(0)) \mid \mathcal{C}].$$

Thus, LATE identifies not only mean treatment effects but any functional of the potential outcome distributions for compliers. In particular, if

$$g(Y_i) = \mathbb{1}\{Y_i \leq y\},$$

the LATE estimand identifies the distance between cumulative potential outcome distributions:

$$\mathbb{E}[g(Y_i(1)) - g(Y_i(0)) \mid \mathcal{C}] = F_{Y(1)}(y \mid \mathcal{C}) - F_{Y(0)}(y \mid \mathcal{C}).$$

5.3.2 Recovering Distributional Effects: An Operational Approach

The two mixture equations

$$\begin{aligned}\mathbb{E}[g(Y_i) \mid Z_i = 1] &= \mathbb{E}[g(Y_i(1)) \mid \mathcal{C}] \pi_{\mathcal{C}} + \mathbb{E}[g(Y_i(1)) \mid \mathcal{A}] \pi_{\mathcal{A}} + \mathbb{E}[g(Y_i(0)) \mid \mathcal{N}] \pi_{\mathcal{N}}, \\ \mathbb{E}[g(Y_i) \mid Z_i = 0] &= \mathbb{E}[g(Y_i(0)) \mid \mathcal{C}] \pi_{\mathcal{C}} + \mathbb{E}[g(Y_i(1)) \mid \mathcal{A}] \pi_{\mathcal{A}} + \mathbb{E}[g(Y_i(0)) \mid \mathcal{N}] \pi_{\mathcal{N}},\end{aligned}$$

contain two unknown objects, $\mathbb{E}[g(Y_i(1)) \mid \mathcal{C}]$ and $\mathbb{E}[g(Y_i(0)) \mid \mathcal{C}]$. While identification follows from these equations, direct estimation is inconvenient. An alternative representation yields a convenient operational solution.

Multiplying by D_i and $1 - D_i$ separately gives:

$$\begin{aligned}\mathbb{E}[g(Y_i) D_i \mid Z_i = 1] &= \mathbb{E}[g(Y_i(1)) \mid \mathcal{C}] \pi_{\mathcal{C}} + \mathbb{E}[g(Y_i(1)) \mid \mathcal{A}] \pi_{\mathcal{A}}, \\ \mathbb{E}[g(Y_i) D_i \mid Z_i = 0] &= \mathbb{E}[g(Y_i(1)) \mid \mathcal{A}] \pi_{\mathcal{A}},\end{aligned}$$

$$\begin{aligned}\mathbb{E}[g(Y_i)(1 - D_i) \mid Z_i = 1] &= \mathbb{E}[g(Y_i(0)) \mid \mathcal{N}] \pi_{\mathcal{N}}, \\ \mathbb{E}[g(Y_i)(1 - D_i) \mid Z_i = 0] &= \mathbb{E}[g(Y_i(0)) \mid \mathcal{C}] \pi_{\mathcal{C}} + \mathbb{E}[g(Y_i(0)) \mid \mathcal{N}] \pi_{\mathcal{N}}.\end{aligned}$$

Taking differences eliminates always-takers and never-takers:

$$\mathbb{E}[g(Y_i(1)) \mid \mathcal{C}] = \frac{\mathbb{E}[g(Y_i) D_i \mid Z_i = 1] - \mathbb{E}[g(Y_i) D_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]},$$

$$\mathbb{E}[g(Y_i(0)) \mid \mathcal{C}] = \frac{\mathbb{E}[g(Y_i)(1 - D_i) \mid Z_i = 1] - \mathbb{E}[g(Y_i)(1 - D_i) \mid Z_i = 0]}{\mathbb{E}[(1 - D_i) \mid Z_i = 1] - \mathbb{E}[(1 - D_i) \mid Z_i = 0]}.$$

Setting

$$g(Y_i) = \mathbb{1}\{Y_i \leq y\},$$

we obtain the complier potential outcome distributions:

$$F_{Y(1)}(y \mid \mathcal{C}) = \frac{\mathbb{E}[\mathbb{1}\{Y_i \leq y\}D_i \mid Z_i = 1] - \mathbb{E}[\mathbb{1}\{Y_i \leq y\}D_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]},$$

$$F_{Y(0)}(y \mid \mathcal{C}) = \frac{\mathbb{E}[\mathbb{1}\{Y_i \leq y\}(1 - D_i) \mid Z_i = 1] - \mathbb{E}[\mathbb{1}\{Y_i \leq y\}(1 - D_i) \mid Z_i = 0]}{\mathbb{E}[(1 - D_i) \mid Z_i = 1] - \mathbb{E}[(1 - D_i) \mid Z_i = 0]}.$$

Quantile treatment effects for compliers are obtained by solving:

$$\tau = F_{Y(1)}(q_{1\tau} \mid \mathcal{C}), \quad \tau = F_{Y(0)}(q_{0\tau} \mid \mathcal{C}),$$

and computing

$$QTE_\tau = q_{1\tau} - q_{0\tau}.$$

Replacing the indicator function by a kernel $K_h(Y_i - y)$ yields estimators of the density of $Y_i(1)$ and $Y_i(0)$ for compliers:

$$K_h(Y_i - y)D_i \quad \text{or} \quad K_h(Y_i - y)(1 - D_i),$$

where $K_h(\cdot)$ is a kernel function with bandwidth h .

The representation above shows that once LATE is identified, distributional effects follow mechanically from the same logic. In particular, the procedure does not rely on additional structural assumptions beyond those required for LATE. As emphasized by Abadie (2002), this provides a simple operational tool: by replacing the outcome with transformed objects such as $\mathbb{1}\{Y_i \leq y\}$ or kernel functions, one can recover entire potential outcome distributions for compliers. The method is therefore largely mechanical — it exploits the same IV structure repeatedly — yet it allows researchers to estimate rich objects such as quantile treatment effects and density functions without imposing parametric restrictions.

5.4 Latent Index Models

We start by considering a Generalized Roy Model (GRM) in the context of a study on returns to education. Let $Y(0), Y(1)$ be potential earnings from dropping out after high school and from attending college, respectively. The school choice decision (drop out after high school vs college) is D . We can write the potential outcome equations for $j \in \{0, 1\}$ as

$$Y(j) = \mu_j(x) + U_j$$

where X is a set of observables, $\mu_j(x) \equiv \mathbb{E}(Y(j) \mid X = x)$ and U_j is unobservable or latent. Note that by definition of $\mu_j(x)$, we normalize $\mathbb{E}(U_j \mid X = x) = 0$. We are not assuming independence between U_j and X , but we are just defining U_j as the residual after projecting

on X .

Choices are made by agents under imperfect information about costs and outcomes. Let $\mathcal{I}$ the information set at the time of decision. We can defined the expected benefit at the time of decision as

$$I_D \equiv \mathbb{E}(Y(1) - Y(0) - C \mid \mathcal{I})$$

where C represents a cost (e.g. tuition fees). In this context, we have that $D = \mathbb{1}\{I_D > 0\}$ (the individual goes to college if, given their information set, the expected benefit of attending college is positive).

Now, consider $X \subset \mathbf{Z} \equiv [X, Z]$ where $\mathbf{Z}$ may consist of multiple excluded instruments (e.g. shift in tuition fees). We assume that Z is as good as random conditional on X and can write the first stage as

$$C = \mu_C(\mathbf{Z}) + U_C$$

where $\mu_C(\mathbf{Z}) \equiv \mathbb{E}(C \mid \mathbf{Z} = \mathbf{z})$. Note that the way we defined the potential outcome and the first stage implies an exclusion restriction, since Z affects costs through $\mu_C(\mathbf{Z})$, but does not enter the earnings equations $Y(j)$. We can think of this as partialling out X and using variation in Z that is orthogonal to unobservables.

We can now plug the definition of the potential outcomes and the first stage into the expected benefit expression:

$$I_D = \mathbb{E}(\mu_1(x) + U_1 - \mu_0(x) - U_0 - \mu_C(\mathbf{Z}) - U_C \mid \mathcal{I}) = \mu_D(\mathbf{Z}) - V$$

where $\mu_D(\mathbf{Z}) \equiv \mathbb{E}(\mu_1(x) - \mu_0(x) - \mu_C(\mathbf{Z}) \mid \mathcal{I})$ represents observable heterogeneity and $V \equiv -\mathbb{E}(U_1 - U_0 - U_C \mid \mathcal{I})$ unobserved heterogeneity. With these definitions, we can rewrite

$$D = \mathbb{1}\{\mu_D(\mathbf{Z}) - V > 0\} = \mathbb{1}\{\mu_D(\mathbf{Z}) > V\}$$

First, $\mu_D(\mathbf{Z})$ depends on observables X and instruments Z (through costs). For instance, if tuition decreases due to a change in Z , then $\mu_D(\mathbf{Z})$ increases. It can therefore be interpreted as the part of the college gain that is predictable using observed variables. Second, V collects all unobserved determinants of the perceived net benefit of college — such as unobserved ability, tastes for education, or private information about future earnings. The decision rule

$$D = \mathbb{1}\{\mu_D(\mathbf{Z}) > V\}$$

means that an individual attends college if their observable net return exceeds their unobserved resistance (or “distaste”) for college. Thus, treatment occurs when a latent index crosses an unobserved threshold, which is why this framework is referred to as a *latent index model*.

We can now define the propensity score with respect to $\mathbf{Z}$ as follows

$$\mathbb{P}(\mathbf{Z}) \equiv \mathbb{P}(\mu_D(\mathbf{Z}) > V) = F_V(\mu_D(\mathbf{Z}))$$

where $F_V(\cdot)$ the CDF of random variable V , that we assume to have a continuous distribution. Applying the CDF on both sides of the decision rule function gives

$$D = \mathbb{1}\{F_V(\mu_D(\mathbf{Z})) > F_V(V)\} = \mathbb{1}\{\mathbb{P}(\mathbf{Z}) > U_D\}$$

where U_D is a random variable uniformly distributed over $[0, 1]$.³³

Note that to obtain the representation

$$D = \mathbb{1}\{\mathbb{P}(\mathbf{Z}) > U_D\},$$

we had to impose the following assumptions:

1. **Continuity.** The unobserved heterogeneity term V is continuously distributed with cumulative distribution function $F_V(\cdot)$. This ensures that $U_D \equiv F_V(V) \sim U[0, 1]$ by the probability integral transform.
2. **Exogeneity of the instrument.** Conditional on X , the instrument Z is independent of unobserved heterogeneity:

$$Z \perp (U_0, U_1, U_C) \mid X,$$

which implies $Z \perp V \mid X$. Therefore, variation in Z shifts the decision rule only through the observable component $\mu_D(\mathbf{Z})$.

3. **Exclusion restriction.** The instrument Z affects the schooling decision through costs (via $\mu_C(\mathbf{Z})$), but does not enter the potential earnings equations $Y(j)$.

The propensity score represents the probability of attending college for individuals with characteristics $\mathbf{Z}$. In the latent index representation, the propensity score acts as a threshold that is compared to the uniform unobserved heterogeneity U_D . Thus, changes in Z shift the propensity score and move individuals across the participation margin in a monotone way. We reached a key result which follows from Vytlacil, 2002: equivalence between LATE and the Latent Index Model.

Proposition 5.4: Equivalence between LATE and the Latent Index Model

Suppose the instrument Z satisfies:

1. **Independence:**

$$Z \perp (Y(0), Y(1), D(0), D(1)) \mid X.$$

2. **Monotonicity:**

$$D(1) \geq D(0) \quad \text{almost surely.}$$

³³We used the probability integral transform: suppose that a random variable W has a continuous distribution for which the CDF is F_W . Then, the random variable $\mathcal{X}$ defined as $\mathcal{X} := F_W(W)$ has a standard uniform distribution.

Then there exists a scalar unobservable $U_D \sim U[0, 1]$ such that the treatment decision can be written as a latent index threshold crossing rule:

$$D = \mathbb{1}\{\mathbb{P}(Z, X) > U_D\},$$

where $\mathbb{P}(Z, X) = \mathbb{P}(D = 1 \mid Z, X)$ is the propensity score.

Conversely, any treatment assignment generated by a latent index model of the form

$$D = \mathbb{1}\{\mathbb{P}(Z, X) > U_D\},$$

with U_D scalar and independent of Z conditional on X , satisfies independence and monotonicity.

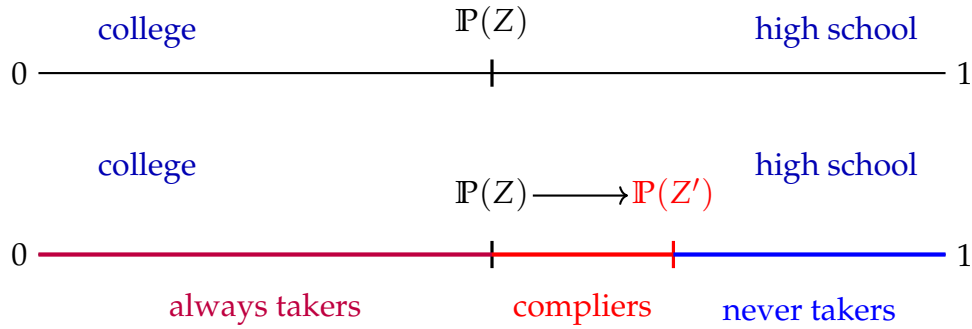


Figure 13: Latent index representation and movement of the threshold.

Figure 13 shows a case where a reform takes place (e.g. randomized reduction in tuition fees). We can see that the interval between $\mathbb{P}(Z)$ and $\mathbb{P}(Z')$ represents the compliers. The group on the left of $\mathbb{P}(Z)$ is the group of Always Takers (were already going to college before the change in tuition), while the group on the right of $\mathbb{P}(Z')$ is the Never Takers (still not going to college even with the reduction). The absence of defiers follows mechanically from the threshold crossing structure. Since treatment is assigned whenever $\mathbb{P}(Z) > U_D$, a higher value of the instrument can only increase (never decrease) the probability of treatment for any individual. This monotonicity is a direct consequence of additive separability and the assumption that selection heterogeneity is one-dimensional.

The non-trivial direction of the equivalence result shows that independence and monotonicity alone are sufficient to guarantee the existence of a latent index representation.³⁴ Intuitively, monotonicity implies that as the instrument changes, individuals can only switch treatment status in one direction (from 0 to 1), which means they can be ordered according to a single dimension of resistance to treatment. This ordering allows one to define a scalar latent variable that ranks individuals by their propensity to participate.

³⁴The proof relies on measure theoretic arguments and is therefore omitted here

After an appropriate normalization, treatment can be written as a threshold-crossing rule of the form $D = \mathbb{1}\{\mathbb{P}(Z) > U_D\}$. Thus, even without starting from a structural Roy model, the LATE assumptions force selection to operate through a single-index latent variable.

5.5 Local Instrumental Variables

In this subsection, we introduce the idea of marginal treatment effect. We start again from the result of the GRM, where the decision rule is given by $D = \mathbb{1}\{\mathbb{P}(Z) > U_D\}$ where $U_D \sim U[0, 1]$. We can rewrite the outcome decomposition as

$$Y = Y(0) + \mathbb{1}\{\mathbb{P}(Z) > U_D\}(Y(1) - Y(0))$$

To understand how the instrument shifts outcomes through changes in selection, we study the conditional expectation³⁵ $\mathbb{E}[Y \mid \mathbb{P}(Z) = p]$, i.e., the average outcome for populations facing a given value of the propensity score. Plugging the outcome decomposition yields

$$\begin{aligned} \mathbb{E}[Y \mid \mathbb{P}(Z) = p] &= \mathbb{E}[Y(0) + \mathbb{1}\{\mathbb{P}(Z) > U_D\}(Y(1) - Y(0)) \mid \mathbb{P}(Z) = p] \\ &= \mathbb{E}[Y(0) + \mathbb{1}\{p > U_D\}(Y(1) - Y(0))] = \mathbb{E}[Y(0)] + \mathbb{E}[\mathbb{1}\{p > U_D\}(Y_1 - Y_0)] \equiv \mathbb{E}[Y(0)] + S(p) \end{aligned}$$

$S(p)$ as the total expected gain for the portion of the population that opts into treatment ($U_D < p$). By the Law of Iterated Expectations, we can condition on $U_D = u$ and integrate over its density $f(u)$:

$$S(p) = \int_0^1 \mathbb{E}[\mathbb{1}\{p > U_D\}(Y(1) - Y(0)) \mid U_D = u] f(u) du$$

Since $f(u) = 1$ for a standard uniform distribution, and the indicator $\mathbb{1}\{p > U_D\}$ is inside the expectation, the integral simplifies to:

$$S(p) = \int_0^p \mathbb{E}[Y(1) - Y(0) \mid U_D = u] du$$

Now, by Fundamental Theorem of Calculus³⁶, we can recover the marginal treatment effect at a given p

$$\frac{\partial \mathbb{E}[Y \mid \mathbb{P}(Z) = p]}{\partial p} = \frac{\partial}{\partial p} \int_0^p \mathbb{E}[Y(1) - Y(0) \mid U_D = u] du = \mathbb{E}[Y(1) - Y(0) \mid U_D = p]$$

Making the conditioning on X explicit, we have our definition of the marginal treatment effect (MTE).

³⁵The conditioning on X is implicit for the derivation.

³⁶We apply the Fundamental Theorem of Calculus: if $F(p) = \int_0^p g(u) du$ and $g(\cdot)$ is continuous (or sufficiently regular), then $\frac{d}{dp} F(p) = g(p)$. Here, $g(u) = \mathbb{E}[Y(1) - Y(0) \mid U_D = u]$, so differentiating the integral with respect to its upper limit evaluates the integrand at $u = p$.

Definition 5.1: Marginal Treatment Effect

The Marginal Treatment Effect at $U_D = u$ and $X = x$ is given by

$$MTE(u, x) = \mathbb{E}[Y(1) - Y(0) | U_D = u, X = x]$$

This is the return for the agent on a margin of indifference between participation (college) and non-participation (high school). Since $U_D \in [0, 1]$, the MTE describes how returns vary across quantiles of the unobserved component U_D . Each quantile identifies who is induced to go to college by a marginal change in $\mathbb{P}(Z)$.

Now, we can link LATE and LIV. Let $p_2 > p_1$. Since $D = \mathbb{1}\{p > U_D\}$, we have

$$\mathbb{1}\{p_1 > U_D\} = 1 \Rightarrow \mathbb{1}\{p_2 > U_D\} = 1,$$

and

$$\mathbb{1}\{p_2 > U_D\} = 0 \Rightarrow \mathbb{1}\{p_1 > U_D\} = 0.$$

Hence, treatment changes only for individuals with $p_1 \leq U_D \leq p_2$.

Recall that

$$S(p) = \mathbb{E}[(Y(1) - Y(0))\mathbb{1}\{U_D \leq p\}].$$

Then

$$S(p_2) - S(p_1) = \mathbb{E}[(Y(1) - Y(0))\mathbb{1}\{p_1 \leq U_D \leq p_2\}].$$

Since $U_D \sim U[0, 1]$, this can be written as

$$S(p_2) - S(p_1) = \mathbb{E}[Y(1) - Y(0) | p_1 \leq U_D \leq p_2] (p_2 - p_1) = \int_{p_1}^{p_2} MTE(\eta) d\eta.$$

Therefore,

$$LATE(p_1, p_2) = \frac{S(p_2) - S(p_1)}{p_2 - p_1} = \frac{\int_{p_1}^{p_2} MTE(\eta) d\eta}{p_2 - p_1}.$$

Thus, *LATE is the average of the MTE over the interval $[p_1, p_2]$.*

Finally, Local IV is obtained as the limit of LATE over shrinking intervals:

$$\lim_{p_2 \rightarrow p_1} LATE(p_1, p_2) = \lim_{p_2 \rightarrow p_1} \frac{S(p_2) - S(p_1)}{p_2 - p_1} = MTE(p_1).$$

It follows that the Marginal Treatment Effect provides a unifying representation of traditional treatment effect parameters. In particular, integrating the MTE over the full support of U_D yields the average treatment effect,

$$ATE = \int_0^1 MTE(\eta) d\eta,$$

which corresponds to averaging treatment effects across all individuals in the population. More generally, many policy-relevant parameters can be expressed as weighted averages of the MTE:

$$\Delta_j(x) = \int_0^1 \text{MTE}(u, x) h_j(u, x) du,$$

where the weights $h_j(u, x)$ are parameter-specific. Thus, different treatment parameters correspond to placing different weights on individuals located at different points of the resistance distribution.

5.6 Shift-Share IV

I base my discussion of SSIV mostly on the two following theoretical papers Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020. The first one discusses exogenous shocks, while the second one exogenous shares. Another useful resource is Borusyak et al., 2025 which provides a checklist for both cases. I use [these slides](#) from Peter Hull as an additional reference. Finally, I use some of the references used by Xavier Jaravel in my EC475 course at LSE.

5.6.1 SSIV Basics

Shift-share, or Bartik, instruments are widely used when treatment exposure varies across observational units because different units are differentially exposed to common shocks. Let observational units be indexed by $\ell = 1, \dots, L$ and shock categories by $n = 1, \dots, N$. A shift-share instrument takes the form

$$z_\ell = \sum_{n=1}^N s_{\ell n} g_n,$$

where $s_{\ell n}$ measures unit ℓ 's exposure share to shock n , and g_n is a shock common to all units exposed to category n .

The typical IV regression is

$$y_\ell = \beta x_\ell + w'_\ell \gamma + \varepsilon_\ell,$$

where the endogenous treatment x_ℓ is instrumented by z_ℓ . The first stage is

$$x_\ell = \pi z_\ell + w'_\ell \delta + v_\ell.$$

The canonical example is based on Bartik, 1991, where y_ℓ is the wage growth and x_ℓ the employment growth in location ℓ . β captures the inverse local labor supply elasticity. The residual ε_ℓ would then contain all labor supply shocks, such as those arising from demographic, human capital, or migration changes, that are not asymptotically correlated with the control vector w_ℓ . In this context, g_n may for instance denote new government subsidies to the output of different industries n and $s_{\ell n}$ may be location ℓ 's lagged shares of industry employment.

All IV design must satisfy some exogeneity condition $\mathbb{E}(z_i \varepsilon_i) = 0$. The challenge with SSIV is that there are two sources of variation, shocks and shares. To argue convincingly for SSIV exogeneity, one needs to explain which properties of the shocks and shares matters. One view treats the shocks g_n as quasi-random, allowing the shares $s_{\ell n}$ to be endogenous. Another view treats the shares as quasi-random or exogenous, allowing the shocks to be non-random. These two interpretations imply different identifying assumptions, different diagnostics, and different approaches to inference, which I discuss below.

5.6.2 Many Exogenous Shocks

The idea is to use exogenous shocks that are as-if randomly assigned and tailored to the treatment of interest (i.e. satisfy exclusion restriction affecting outcome only through treatment). For example, imagine a lottery that randomly assigns a subsidy level g_n to each industry n . In the above labor supply setting, local employment growth x_i can be instrumented for by a weighted average of the subsidies, using initial local employment shares as weights. Subsidies can be viewed as only affecting wages by shifting labor demand and do not have direct effects on labor supply.

The key insight is that “shift-based identification stems from a simple observation: a share-weighted average of random shifts is itself as-good-as-random. This is true even if the shares are econometrically endogenous, in the sense that units with different shares may have systematically different unobservables” (Borusyak et al., 2025). Mathematically, this means that we can have $\text{Cov}(z_\ell, \varepsilon_\ell) = 0$ even if $\text{Cov}(s_{\ell n}, \varepsilon_\ell) \neq 0$. The bottom line is that if you have a valid instrument at the shock level, you can translate it to the unit level, instrumenting $\sum_n s_{\ell n} x_{\ell n}$ with $\sum_n s_{\ell n} g_n$.

The following equivalence formalizes this idea. Although the regression is run at the unit level, the SSIV estimator can be written as a weighted IV regression at the shock level.

Proposition 5.5: Shock-level equivalence

Consider the shift-share instrument

$$z_\ell = \sum_{n=1}^N s_{\ell n} g_n.$$

Let $y_\ell^\perp$ and $x_\ell^\perp$ denote residuals from projecting y_ℓ and x_ℓ on the controls. Define

$$s_n = \frac{1}{L} \sum_{\ell=1}^L s_{\ell n}, \quad \bar{v}_n^\perp = \frac{\sum_{\ell=1}^L s_{\ell n} v_\ell^\perp}{\sum_{\ell=1}^L s_{\ell n}}.$$

Then the SSIV coefficient can be written as

$$\hat{\beta} = \frac{\sum_n s_n g_n \bar{y}_n^\perp}{\sum_n s_n g_n \bar{x}_n^\perp}.$$

Equivalently, $\hat{\beta}$ is the coefficient from an s_n -weighted shock-level IV regression of $\bar{y}_n^\perp$ on $\bar{x}_n^\perp$ using g_n as the instrument.

Proof. By substituting $z_\ell = \sum_n s_{\ell n} g_n$ into the residualized IV estimand (which can be written that way by FWL),

$$\hat{\beta} = \frac{\sum_\ell z_\ell y_\ell^\perp}{\sum_\ell z_\ell x_\ell^\perp} = \frac{\sum_\ell \sum_n s_{\ell n} g_n y_\ell^\perp}{\sum_\ell \sum_n s_{\ell n} g_n x_\ell^\perp}.$$

Changing the order of summation gives

$$\hat{\beta} = \frac{\sum_n g_n \sum_\ell s_{\ell n} y_\ell^\perp}{\sum_n g_n \sum_\ell s_{\ell n} x_\ell^\perp}.$$

Using

$$\sum_\ell s_{\ell n} v_\ell^\perp = \left(\sum_\ell s_{\ell n} \right) \bar{v}_n^\perp = L s_n \bar{v}_n^\perp,$$

we obtain

$$\hat{\beta} = \frac{\sum_n s_n g_n \bar{y}_n^\perp}{\sum_n s_n g_n \bar{x}_n^\perp}.$$

This is the sample IV estimand from the shock-level regression weighted by s_n . □

While the math is fairly standard, I find the intuition behind this equivalence somehow convoluted, and try to make it clearer below. Consider the example above where g_n is a subsidy to industry n . Different regions specialize differently in each specific industries, making the exposure shares $s_{n\ell}$ heterogenous. The subsidy for the specific lottery n is a single number (one realization from a lottery), which affects the L regions differently. This implies that one shock is used for L different regions. Running the regression at the shock level, while equivalent mathematically, uses this single shock only once. Instead of asking "region by region, how does exposure-weighted subsidy correlate with the outcome?", we ask "industry by industry, does a bigger subsidy g_n line up with better outcomes in the places that specialize in n ?"

The key takeaway from this proposition is therefore that the "location-level" IV procedure is equivalent to a "industry-level" IV regression with model

$$\bar{y}_n^\perp = \alpha + \bar{x}_n^\perp \beta + \bar{\varepsilon}_n$$

instrumented by g_n with weights s_n . Before turning to identification, it's important to understand what composes the error term in this shock level regression. $\bar{\varepsilon}_n$ is an exposure-weighted average,

$$\bar{\varepsilon}_n = \frac{\sum_{\ell=1}^L s_{\ell n} \varepsilon_\ell}{\sum_{\ell=1}^L s_{\ell n}}$$

so region ℓ enters $\bar{\varepsilon}_n$ with weight $s_{n\ell}$. A region with high share of n gets a big weight in the error term. This residual can therefore be thought of as the average residual of observations with a high share of n .

We can now discuss consistency of $\hat{\beta}_{SSIV}$. The moment condition can be written as

$$\mathbb{E} \left[\frac{1}{L} \sum_{\ell=1}^L z_{\ell} \varepsilon_{\ell} \right] = 0$$

Note that this differs from the usual IV moment condition since we cannot assume iid sampling given the fact that different regions share the same shock.

Assumption 5.1: Quasi-Random Shock Assignment

$$\mathbb{E}(g_n | \bar{\varepsilon}_n, s) = \mu \text{ for all } n$$

Proposition 5.6: Moment condition

Under complete shares, the SSIV moment condition is satisfied under Assumption 5.1.

Proof. Rewriting the moment condition gives the following

$$\mathbb{E} \left[\frac{1}{L} \sum_{\ell=1}^L z_{\ell} \varepsilon_{\ell} \right] = \mathbb{E} \left[\frac{1}{L} \sum_n g_n \sum_{\ell} s_{\ell n} \varepsilon_{\ell} \right]$$

Now, by definition,

$$s_n = \frac{1}{L} \sum_{\ell} s_{\ell n} \text{ and } \bar{\varepsilon}_n = \frac{\sum_{\ell=1}^L s_{\ell n} \varepsilon_{\ell}}{\sum_{\ell=1}^L s_{\ell n}}$$

Rearranging and combining the two gives

$$\sum_{\ell} s_{\ell n} \varepsilon_{\ell} = L s_n \bar{\varepsilon}_n$$

Plugging this into the moment condition and using LIE gives

$$\begin{aligned} \mathbb{E} \left[\frac{1}{L} \sum_n g_n \sum_{\ell} s_{\ell n} \varepsilon_{\ell} \right] &= \mathbb{E} \left[\sum_n g_n s_n \bar{\varepsilon}_n \right] = \mathbb{E} \left[\mathbb{E} \left(\sum_n g_n s_n \bar{\varepsilon}_n | \bar{\varepsilon}, s \right) \right] = \mathbb{E} \left[\sum_n s_n \mathbb{E}(g_n | \bar{\varepsilon}, s) \bar{\varepsilon}_n \right] \\ &= \mu \mathbb{E} \left[\sum_n s_n \bar{\varepsilon}_n \right] \end{aligned}$$

where the last step uses Assumption 5.1. Now, note if we assume complete shares, meaning that shares sum to one across industries, we can write

$$\mu \mathbb{E} \left[\sum_n s_n \bar{\varepsilon}_n \right] = \mu \mathbb{E} \left[\frac{1}{L} \sum_n \sum_{\ell} s_{\ell n} \varepsilon_{\ell} \right] = \mu \mathbb{E} \left[\frac{1}{L} \sum_{\ell} \varepsilon_{\ell} \right] = 0$$

where the last step just assume the error term to be mean zero, which concludes the proof. $\square$

Intuitively, Borusyak et al., 2022 put it nicely. The proposition shows that the shift-share instrument is valid (in the sense that the IV moment condition holds) when the underlying shocks are as-good-as-randomly assigned: each g_n has the same mean μ regardless of the realizations of the relevant unobservables $\bar{\epsilon}$ (and average exposures s). The conditioning on s matters: exposure is allowed to be endogenous, and only the shock must be clean.

In the labour supply example, this assumption means that import tariffs should not have been chosen strategically based on labour supply trends, or in a way that is correlated with such trends. Crucially, it is not a problem that regions specializing in different industries are on different trends — that heterogeneity is absorbed by the shares. The threat is a correlation between the tariff and the residual unobservable $\bar{\epsilon}_n$. For example, if manufacturing-heavy regions are on worse labour supply trends, Assumption 5.1 requires for tariffs not to be systematically larger for manufacturing on that account: the expected tariff must be the same for high- and low- $\bar{\epsilon}_n$ industries.

Assumption 5.1 delivers the right moment condition, but for the shock-level average to actually converge we need a second ingredient: enough effectively independent shocks. Assumption A2 supplies this.

Assumption 5.2: Many uncorrelated shocks

- $\mathbb{E}[\sum_n s_n^2] \rightarrow 0$: the expected Herfindahl index of average shock exposure converges to zero. No single industry dominates the exposure structure, which implies the number of shocks $N \rightarrow \infty$.
- $\text{Cov}(g_n, g_{n'} \mid \bar{\epsilon}, s) = 0$ for all $n' \neq n$: shocks are mutually uncorrelated given the unobservables.

Together these imply a shock-level law of large numbers,

$$\sum_n s_n g_n \bar{\epsilon}_n \xrightarrow{p} 0,$$

so the endogenous part of the estimator washes out as the number of shocks grows. Intuitively, 5.1 makes each term mean-zero and 5.2 ensures there are many weakly-dependent terms for the averaging to work: neither the shocks nor their exposures may be concentrated in a few industries. Both conditions, while novel for SSIV, would be standard for a shock-level IV regression run with weights s_n and instrument g_n . These two assumptions are the baseline assumptions for the quasi-experimental framework: given a standard relevance condition and the additional regularity conditions, they are sufficient for SSIV consistency.

Having established consistency, I turn to implementation. Borusyak et al., 2025 give a seven-step checklist for the exogenous-shifts approach. I state each step briefly and then expand on the two that most affect estimates in practice: the incomplete-share control (Step 3) and inference (Step 7).

1. **Motivate with a shift-level idealized experiment.** Specify which unobserved confounders bias OLS, then describe a hypothetical experiment (e.g. subsidies assigned at random across industries) that would generate treatment variation unrelated to those confounders.
2. **Bridge the gap between observed and ideal shifts.** Real shifts are rarely randomized. Argue quasi-random assignment conditional on controls: shift-level confounders q_n enter as shift-share aggregates $\sum_n s_{\ell n} q_n$, and unit-level confounders enter directly.
3. **Include the incomplete-share control.** When shares do not sum to one, control for $S_\ell = \sum_n s_{\ell n}$.
4. **Lag shares to the start of the natural experiment.** Measure shares before the shifts are realized, so that shifts cannot affect the shares; lagging further than necessary only weakens the instrument.
5. **Report descriptive statistics for shifts, not just units.** Treat the N shocks as the effective sample: report the mean and s.d. of g_n , and the effective number of shocks $1 / \sum_n s_n^2$ (the inverse Herfindahl of importance weights s_n).
6. **Balance-test the shifts, not just the instrument.** Regress g_n on shift-level proxies for confounders (e.g. industry skill or immigrant intensity) to check that shocks are not systematically larger for industries on different trends.
7. **Produce estimates with exposure-robust standard errors and check sensitivity.** Conventional SEs are invalid; use the shock-level equivalent regression.

As promised, I now discuss step 3 and 7. First, the *incomplete shares* problem. When shares sum to one ($S_\ell \equiv \sum_n s_{\ell n} = 1$), the instrument $z_\ell = \sum_n s_{\ell n} g_n$ is a weighted *average* of the shocks: an average of lottery draws is itself a lottery outcome, centered at μ regardless of the region. When shares are incomplete and S_ℓ varies across regions, z_ℓ becomes a weighted *sum*, which scales with total exposure. With positively-centered shocks, a region with high S_ℓ mechanically receives a larger z_ℓ — not from luckier draws but from having more shares to add up. Since S_ℓ is itself endogenous (a region's total exposure to the shock-generating sector correlates with its unobservables), z_ℓ inherits a correlation with ε_ℓ and the instrument is invalid. Controlling for S_ℓ solves this: among regions with the same total exposure, z_ℓ varies only through *which* shocks were drawn, and the remaining variation is again as-good-as-random.

Now, valid inference requires more care than a standard IV regression. The problem is the same non-iid structure that motivated the shock-level view throughout: regions specializing in the same industries share common shocks, so their instruments z_ℓ are mechanically correlated, and their errors ε_ℓ may be correlated too through exposure to the same unobserved industry shocks. Conventional standard errors, which treat the L regions as independent observations, ignore this dependence and are invalid.

Borusyak et al., 2022 give a clean fix that reuses the shock-level equivalence. Rather than devising a new variance formula, one simply runs the equivalent shift-level IV regression: the outcome and treatment are residualized on the unit-level controls and collapsed to the shock level as the exposure-weighted averages $\bar{y}_n^\perp$ and $\bar{x}_n^\perp$, the shocks g_n serve directly as the instrument, shock-level controls $\bar{q}_n$ enter as controls, and the regression is weighted by average shares s_n . This regression produces *identical* $\hat{\beta}$ to the region-level procedure (by the equivalence proposition), but its standard errors are now valid because estimation happens at the level at which the shocks are actually assigned — so the dependence across regions sharing a shock is automatically accounted for. The `ssaggregate` package (Stata and R) automates the transformation, and the same regression yields exposure-robust first-stage F -statistics for judging instrument strength.

5.6.3 Exogenous Shares

Identification in a shift-share design requires that *either* the shocks *or* the shares be exogenous. The framework above develops the first approach. The alternative, developed by Goldsmith-Pinkham et al., 2020, instead treats the shocks as given and potentially endogenous objects, and requires the exposure shares $s_{\ell n}$ to be as-good-as-randomly assigned conditional on controls. Identification can then be understood as pooling many difference-in-differences designs that compare units with differing exposure to the same shock, with the shares (rather than the shifts) supplying the quasi-experimental variation. This approach is valuable when a credible shock-level experiment is unavailable but a small number of shares can be argued to be exogenous.

For example, in the immigration setting of Card, 2001b, the shocks g_n are national inflows of immigrants from each origin country n , and the shares $s_{\ell n}$ are the fractions of earlier immigrants from country n already settled in city ℓ . The shares path treats these historical settlement shares as as-good-as-randomly assigned with respect to current local labour-market shocks: because they were determined by migration decisions decades earlier, a city's past exposure to a given origin country is plausibly unrelated to its present-day wage or employment unobservables, even if national inflows themselves are endogenous.

I refer to Goldsmith-Pinkham et al., 2020 and Borusyak et al., 2025 for details and do not develop it further.

6 Treatment Effects with Jumps and Kinks

6.1 Basics

The idea of the RD Design is based on a triplet: a score (or running variable), a cutoff and a treatment. By design, treatment depends on the running variable and a known cutoff. Units are treated if their running variable is greater than the cutoff. Note that this is **selection on one observable**, the running variable.

We can now introduce notation. X_i is the running variable and c is a known cutoff. $D_i \in \{0, 1\}$ is the treatment status. $\mathbb{E}(Y_i(j)|X_i = x)$ is discontinuous at $X_i = c$ and the jump is sharp if $D_i = \mathbb{1}(X_i \geq c)$. As always, our potential outcomes notation and decomposition holds

$$Y_i = Y_i(0) + D_i (Y_i(1) - Y_i(0))$$

We focus on $\mu_j \equiv \mathbb{E}(Y_i(j)|X_i = x), j \in \{0, 1\}$, but could work with other parts of potential outcome distributions as well.

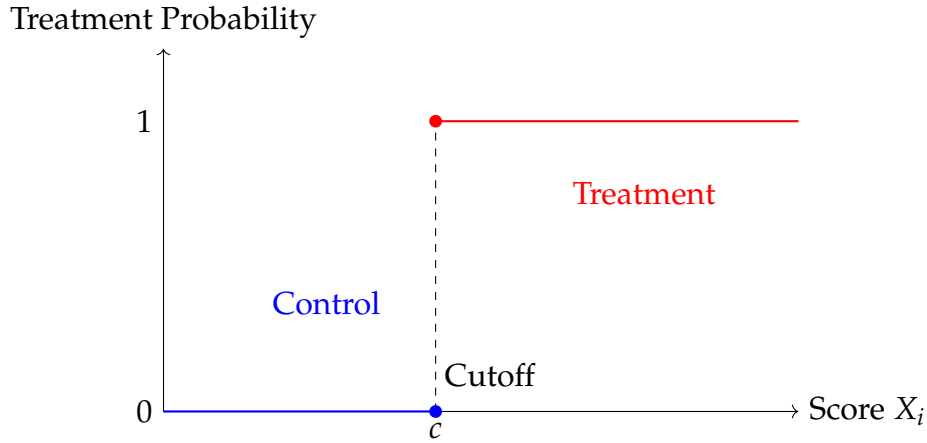


Figure 14: Sharp RDD: Treatment assignment rule

All relevant variables but D_i must have sufficiently smooth profiles with respect to X_i . Discontinuity designs rely on the continuity of key variables W_i at the cutoff. In particular, the identifying assumption is that conditional expectations are continuous in the running variable X_i at c . Formally, we assume:

$$\lim_{x \uparrow c} \mathbb{E}[W_i | X_i = x] = \mathbb{E}[W_i | X_i = c^-] \equiv \mu_0^-,$$

$$\lim_{x \downarrow c} \mathbb{E}[W_i | X_i = x] = \mathbb{E}[W_i | X_i = c^+] \equiv \mu_1^+.$$

Thus, any discontinuity in the conditional expectation at c can be attributed to the treatment assignment rule rather than to a structural break in the underlying relationship between W_i and X_i .

Now, because D_i is a deterministic function of X_i , we can write

$$\mathbb{E}(Y_i|X_i = x) = \mathbb{E}(Y_i(0)|X_i = x) + \mathbb{1}(x \geq c)\mathbb{E}(Y_i(1) - Y_i(0)|X_i = x)$$

Then, it is easy to see that

$$\mathbb{E}(Y_i|X_i = c^+) = \mathbb{E}(Y_i(0)|X_i = c^+) + \mathbb{E}(Y_i(1) - Y_i(0)|X_i = c^+)$$

$$\mathbb{E}(Y_i|X_i = c^-) = \mathbb{E}(Y_i(0)|X_i = c^-)$$

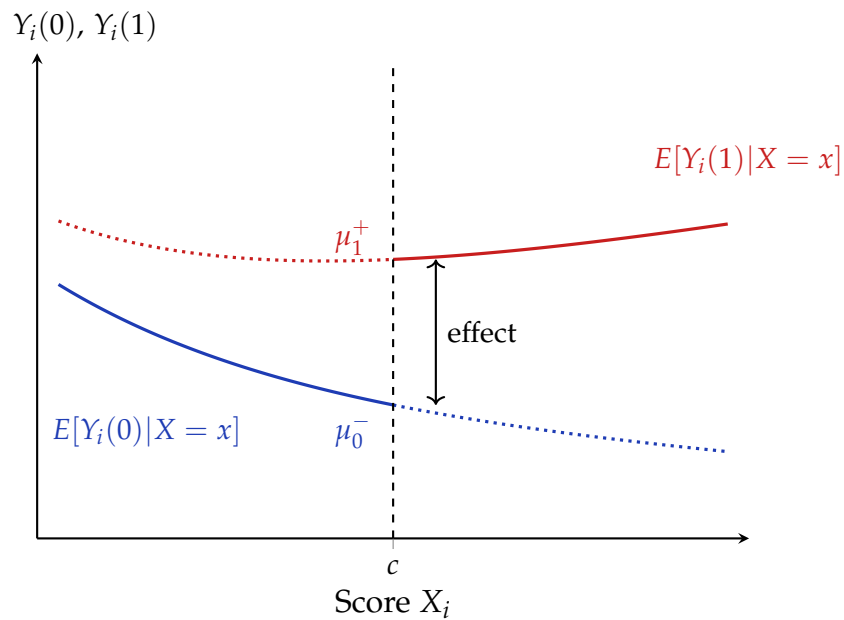
since $\mathbb{1}(x \geq c^+) = 1$ and $\mathbb{1}(x \geq c^-) = 0$.

The main assumption of RD Design is that the CEF of $Y_i(0)$ conditional on X_i is a continuous function (at least around c)

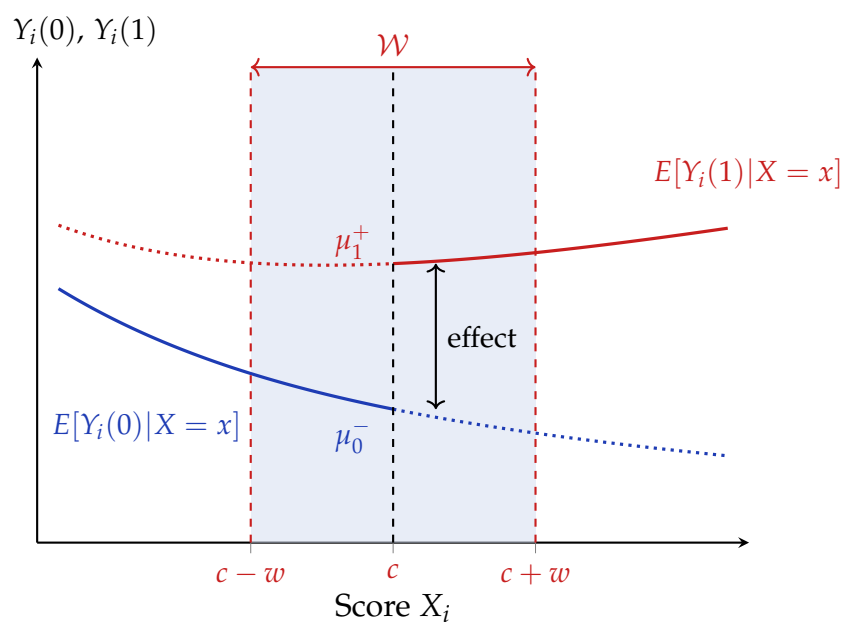
$$\mathbb{E}(Y_i(0)|X_i = c^-) = \mathbb{E}(Y_i(0)|X_i = c^+)$$

The idea to estimate a Sharp RD parameter is that in a counterfactual world without treatment, CEFs at the threshold c would present no statistical anomaly/discontinuity.

Outcomes



Outcomes



References

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A Appendix - Additional Results

A.1 Asymptotic Tools for i.n.i.d. Data

When the error covariance is non-spherical, the sequence of terms used in consistency proofs, such as $\mathbf{x}_i \varepsilon_i$, is often **independent but not identically distributed (i.n.i.d.)**. In this case, standard i.i.d. Laws of Large Numbers (LLN) cannot be directly applied.

To ensure the Weak Law of Large Numbers (WLLN) holds for a sequence of independent random variables $\{X_i\}_{i=1}^n$ (where $X_i = \mathbf{x}_i \varepsilon_i$ in the context of OLS with non-spherical errors), the key requirement is often the condition of **Uniform Integrability (UI)**.

Definition A.1: Uniform Integrability

A sequence of random variables $\{X_i, i \geq 1\}$ is uniformly integrable if

$$\lim_{\delta \rightarrow \infty} \sup_{i \geq 1} \mathbb{E}[|X_i| \cdot \mathbb{1}\{|X_i| > \delta\}] = 0$$

The core idea behind uniform integrability is controlling the probability mass in the tails of the distribution. For convergence theorems like the WLLN, we need to ensure that the tail behavior (where $|X_i|$ is large, i.e., beyond δ) does not become overwhelmingly large for any single variable X_i in the sequence.

The condition requires that the expected value of the tail (the portion of the distribution where $|X_i| > \delta$) goes to zero **uniformly** for all variables X_i in the sequence as δ increases. This ensures that no individual variable "carries too much weight" in the limit, which allows the Law of Large Numbers to hold even when the individual variances (second moments) are not bounded. We now introduce sufficient and necessary conditions for uniform integrability.

Theorem A.1: Necessary Conditions for UI

The following condition is necessary for a sequence of random variables $\{X_i, i \geq 1\}$ to be uniformly integrable (UI):

$$\max_{i \geq 1} \mathbb{E}(|X_i|) < \infty$$

Theorem A.2: Sufficient Conditions for UI

The following conditions are sufficient for a sequence of random variables $\{X_i, i \geq 1\}$ to be uniformly integrable (UI):

1. **Existence of a Moment Higher than 1:** If $\sup_{i \geq 1} \mathbb{E}\{|X_i|^{1+\eta}\} < \infty$ for some $\eta > 0$, then $\{X_i, i \geq 1\}$ is UI.
2. **Identically Distributed:** If $\{X_i, i \geq 1\}$ is identically distributed (I.D.) and

$\mathbb{E}|X_i| < \infty$, then $\{X_i, i \geq 1\}$ is UI.

The proof of these theorems are fairly easy, but are omitted. What is interesting in our application is mostly sufficient condition 1. Indeed, it is enough to have finiteness of a moment slightly above one to have uniform integrability, which we will see below will allow us to derive another version of the WLLN.

Theorem A.3: WLLN for Independent, Uniformly Integrable Sequences

Let $\{X_i\}_{i=1}^n$ be a sequence of independent and uniformly integrable random variables. Then

$$\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] \xrightarrow{p} 0.$$

Application to OLS Consistency: This WLLN is vital in showing the consistency of the OLS estimator ($\hat{\beta} \xrightarrow{p} \beta$) when the data is independent but not identically distributed (i.n.i.d.) due to heteroskedasticity. Specifically, it ensures that the critical term $\frac{1}{n} \sum_{i=1}^n x_i \varepsilon_i$ converges in probability to 0, provided the sequence $\{x_i \varepsilon_i\}$ is uniformly integrable.

A.2 Functional CLTs

A natural starting point for understanding functional central limit theorems is the classical CLT, which describes the limiting distribution of a single normalized sum of random variables (for iid data or not). However, in many problems, especially in time series, we are not interested only in the final sum, but in how the cumulative sum evolves over time. This leads us to consider the sequence of partial sums $S_t = \sum_{i=1}^t u_i$, which defines a stochastic process capturing the entire path of accumulation.

To relate discrete-time partial sums to a continuous-time object, we index time using a rescaled³⁷ parameter $r \in [0, 1]$ and define $t = \lfloor Tr \rfloor$, where $\lfloor \cdot \rfloor$ denotes the integer part. This means that for each r , we select the largest integer t such that $t \leq Tr$. In this way, the discrete index $t = 1, \dots, T$ is mapped onto the continuous interval $[0, 1]$. For example, when $r = 0.5$, we have $t \approx T/2$, so we are looking at the partial sum halfway through the sample. This construction allows us to represent the sequence of partial sums as a function of r , thereby embedding a discrete-time process into continuous time. We are basically turning a sequence $S_1, S_2, \dots, S_T$ into a function $X_T(r)$ defined for all $r \in [0, 1]$.

While the classical CLT describes the limiting behavior of the endpoint S_T , the functional CLT characterizes the limit of the entire trajectory of partial sums. In particular, it shows that a properly rescaled random walk converges to Brownian motion, which I define below.

³⁷Formally, we simple rescale time $r = \frac{t}{T} \implies t \approx Tr$. We need t to be an integer, so we take $t = \lfloor Tr \rfloor$

Definition A.2: Standard Brownian Motion

Standard Brownian Motion $\mathcal{B}(\cdot)$ is a continuous-time stochastic process, associating each date $t \in [0, 1]$ with the scalar $\mathcal{B}(t)$ such that:

- (a) $\mathcal{B}(0) = 0$;
- (b) For any dates $0 \leq t_1 < t_2 < \dots < t_k \leq 1$, the changes $[\mathcal{B}(t_2) - \mathcal{B}(t_1)], [\mathcal{B}(t_3) - \mathcal{B}(t_2)], \dots, [\mathcal{B}(t_k) - \mathcal{B}(t_{k-1})]$ are independent multivariate Gaussian with $[\mathcal{B}(s) - \mathcal{B}(t)] \sim N(0, s - t)$;
- (c) For any given realization, $\mathcal{B}(t)$ is continuous in t w.p.1.

This definition implies that for any $t, t_1, t_2 \in [0, 1]$, we have $\mathcal{B}(t) \sim N(0, t)$ and $\mathbb{E}[\mathcal{B}(t_1)\mathcal{B}(t_2)] = \min(t_1, t_2)$. We can now build towards the functional CLT.

Consider

$$X_T(r) = \frac{1}{T} \sum_{t=1}^{\lfloor Tr \rfloor} u_t.$$

To obtain a non-degenerate limit, we multiply by $\sqrt{T}$ and divide by σ :

$$\sqrt{T} \frac{X_T(r)}{\sigma} = \frac{1}{\sqrt{T}} \frac{1}{\sigma} \sum_{t=1}^{\lfloor Tr \rfloor} u_t.$$

Now rewrite this as

$$\sqrt{T} \frac{X_T(r)}{\sigma} = \frac{\sqrt{\lfloor Tr \rfloor}}{\sqrt{T}} \left(\frac{1}{\sqrt{\lfloor Tr \rfloor}} \frac{1}{\sigma} \sum_{t=1}^{\lfloor Tr \rfloor} u_t \right).$$

By the classical CLT,

$$\frac{1}{\sqrt{\lfloor Tr \rfloor}} \frac{1}{\sigma} \sum_{t=1}^{\lfloor Tr \rfloor} u_t \xrightarrow{d} N(0, 1),$$

and since

$$\frac{\sqrt{\lfloor Tr \rfloor}}{\sqrt{T}} \rightarrow \sqrt{r},$$

we obtain (using CMT)

$$\sqrt{T} \frac{X_T(r)}{\sigma} \xrightarrow{d} \sqrt{r} N(0, 1) = N(0, r).$$

The following observation is key: let $r_2 > r_1$ with corresponding partial sums $\lfloor Tr_2 \rfloor$ and $\lfloor Tr_1 \rfloor$, respectively. Then, asymptotically,

$$\sqrt{T} [X_T(r_2) - X_T(r_1)] / \sigma \xrightarrow{d} \mathcal{N}(0, r_2 - r_1)$$

Since the increments between any two points r_1 and r_2 are asymptotically independent and normally distributed with variance proportional to the time elapsed, it follows that the sequence of stochastic functions $\{\sqrt{T} \cdot X_T(\cdot)/\sigma\}_{T=1}^{\infty}$ has an asymptotic probability law described by standard Brownian Motion, $\mathcal{B}(\cdot)$. Specifically:

$$\sqrt{T} \cdot X_T(\cdot)/\sigma \xrightarrow{d} \mathcal{B}(\cdot)$$

We now formally state this result.

Theorem A.4: Functional Central Limit Theorem (Donsker's Theorem)

Let $S_T = \sum_{t=1}^T u_t$, where the sequence $\{u_t, t = 1, \dots, T\}$ is i.i.d. with zero mean and variance $\sigma^2 < \infty$. If we define the rescaled process as $X_T(r) = S_{\lfloor Tr \rfloor} / (\sqrt{T}\sigma)$, then as $T \rightarrow \infty$:

$$X_T(r) \xrightarrow{d} \mathcal{B}(r)$$

where $\mathcal{B}(r)$ is a standard Brownian Motion for $r \in [0, 1]$.

A version of the CMT also applies here, which we state below.

Theorem A.5: Functional Continuous Mapping Theorem

Let $h : D \mapsto \mathbb{R}$ be a measurable functional that is continuous except on a set $D_h \subset D$ with $P(\mathcal{B} \in D_h) = 0$. If $X_T(r) \xrightarrow{d} \mathcal{B}(r)$, then

$$h(X_T(r)) \xrightarrow{d} h(\mathcal{B}(r))$$

As an application, consider a pure random walk process starting at $y_0 = 0$:

$$y_t = y_{t-1} + u_t, \quad u_t \stackrel{i.i.d.}{\sim} (0, \sigma^2)$$

which implies that $y_t = \sum_{j=1}^t u_j$. To bridge the gap between this discrete sequence and the continuous Brownian Motion, we define the stochastic function $X_T(r)$ as a step function over $r \in [0, 1]$:

$$X_T(r) = \begin{cases} 0 & \text{for } 0 \leq r < 1/T \\ y_1/T & \text{for } 1/T \leq r < 2/T \\ y_2/T & \text{for } 2/T \leq r < 3/T \\ \vdots & \\ y_T/T & \text{for } r = 1. \end{cases}$$

$X_T(r)$ creates a sequence of rectangles. The area of a typical rectangle j is given by its height (y_j/T) times its width ($1/T$), which equals y_j/T^2 . The integral of the entire process is simply the sum of these areas:

$$\int_0^1 X_T(r) dr = y_1/T^2 + y_2/T^2 + \dots + y_{T-1}/T^2$$

By scaling both sides by $\sqrt{T}$, we establish a direct link between the continuous integral and the normalized sum of the random walk:

$$\int_0^1 \sqrt{T} \cdot X_T(r) dr = T^{-3/2} \sum_{t=1}^T y_{t-1}$$

Applying the Functional Continuous Mapping Theorem to the integral functional $h(\cdot) = \int_0^1 (\cdot) dr$, we know that if the underlying process converges to Brownian Motion, its integral must converge to the integral of that Brownian Motion. This leads us to the following result.

Proposition A.1: Asymptotic Distribution of the Sum of a Random Walk

Given a random walk y_t satisfying the assumptions above, the normalized sum converges in distribution to the integral of a scaled Brownian Motion:

$$T^{-3/2} \sum_{t=1}^T y_{t-1} \xrightarrow{d} \sigma \int_0^1 \mathcal{B}(r) dr$$

A.3 Additional derivations

A.3.1 Restricted Least Squares Derivation

Consider the linear regression model

$$y = X\beta + u$$

and the linear restrictions

$$R\beta = c.$$

The restricted least squares (RLS) estimator is obtained by minimizing

$$(y - X\beta)'(y - X\beta)$$

subject to the constraint $R\beta = c$.

The Lagrangian is

$$\mathcal{Q}(\beta, \lambda) = (y - X\beta)'(y - X\beta) - 2\lambda'(R\beta - c).$$

Differentiating with respect to β and λ gives

$$\frac{\partial \mathcal{Q}}{\partial \beta} = -2X'y + 2X'X\beta - 2R'\lambda = 0, \quad (\text{A.1})$$

$$\frac{\partial \mathcal{Q}}{\partial \lambda} = R\beta - c = 0. \quad (\text{A.2})$$

From (A.1) we obtain

$$X'X\beta - R'\lambda = X'y,$$

which implies

$$\beta = (X'X)^{-1}X'y + (X'X)^{-1}R'\lambda.$$

Define the unrestricted OLS estimator

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Then

$$\beta = \hat{\beta} + (X'X)^{-1}R'\lambda.$$

Substitute into (A.2):

$$R\hat{\beta} + R(X'X)^{-1}R'\lambda = c.$$

Rearranging,

$$R(X'X)^{-1}R'\lambda = c - R\hat{\beta}.$$

Assuming $R(X'X)^{-1}R'$ is nonsingular,

$$\lambda = [R(X'X)^{-1}R']^{-1}(c - R\hat{\beta}).$$

Substituting back,

$$\tilde{\beta} = \hat{\beta} + (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1}(c - R\hat{\beta}).$$

This is the restricted least squares estimator.

A.3.2 MLE asymptotics

As beforen, it is easy to write the MLE as an extremum estimator. Indeed, let

$$Q_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(z_i, \theta) \implies \hat{\theta} = \arg \max_{\theta \in \Theta} Q_n(\theta)$$

We can now write a set of primitive conditions that guarantee the General consistency Theorem (5.1).

Theorem A.6: Consistency of MLE

Suppose the following conditions hold

- (i) Θ is compact
- (ii) $\{z_i\}_{i=1}^n$ is iid with pdf $f(z, \theta_0)$. For any $\theta \neq \theta_0$ on Θ , $f(z, \theta) \neq f(z, \theta_0)$
- (iii) $\log f(z, \theta)$ is almost surely continuous at each $\theta \in \Theta$
- (iv) $\mathbb{E}(\sup_{\theta \in \Theta} |\log f(z, \theta)|) < \infty$

Then, we achieve consistency:

$$\hat{\theta} \xrightarrow{p} \theta_0$$

Proof. As before, it is sufficient to check conditions 1-4 from Theorem 5.1. Condition 1 is guaranteed by (i). We now work towards proving uniform convergence (condition 4). In this case,

$$Q_*(\theta) = -\mathbb{E}(\log f(z, \theta))$$

so condition 4 amounts to checking

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \log f(z, \theta) - \mathbb{E}(\log f(z, \theta)) \right| \xrightarrow{p} 0$$

Using the first assumption in (i) and conditions (ii) to (iv), the ULLN applies. ULLN also guarantees continuity of $Q_*(\theta)$, satisfying condition 3. It remains to check condition 2, that is identification of θ_0 . The condition to be checked is

$$Q_*(\theta) = \mathbb{E}(\log f(z, \theta)) \text{ is uniquely maximized at } \theta_0$$

Pick any $\theta \neq \theta_0$. We want to show that $Q_*(\theta_0) - Q_*(\theta) > 0$ for any such θ . We can write

$$Q_*(\theta_0) - Q_*(\theta) = \mathbb{E}(\log f(z, \theta_0)) - \mathbb{E}(\log f(z, \theta)) = \mathbb{E} \left[-\log \frac{f(z, \theta)}{f(z, \theta_0)} \right]$$

By (i) and Jensen's Inequality, we have

$$\mathbb{E} \left[-\log \frac{f(z, \theta)}{f(z, \theta_0)} \right] > -\log \mathbb{E} \left[\frac{f(z, \theta)}{f(z, \theta_0)} \right] = -\log \int \frac{f(z, \theta)}{f(z, \theta_0)} f(z, \theta_0) dz = -\log \int f(z, \theta) dz = 0$$

This yields the desired result and we are done. $\square$

Theorem A.7: Asymptotic Normality of MLE

Suppose the assumptions for MLE consistency are satisfied. Additionally, assume:

- (i) $\theta_0 \in \text{int } \Theta$
- (ii) $f(z, \theta)$ is twice continuously differentiable and $f(z, \theta) > 0$ in a neighborhood $\mathcal{N}$ of θ_0 with probability one
- (iii) $\mathbb{E}[|\frac{\partial \log f(z, \theta_0)}{\partial \theta}|^2] < \infty$, and $\mathbb{E} \left[\sup_{\theta \in \mathcal{N}} \left| \frac{\partial^2 \log f(z, \theta)}{\partial \theta \partial \theta'} \right| \right] < \infty$
- (iv) $J \equiv \mathbb{E} \left[\frac{\partial \log f(z, \theta_0)}{\partial \theta} \frac{\partial \log f(z, \theta_0)}{\partial \theta'} \right]$ is nonsingular

Then, the MLE is asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, J^{-1})$$

Proof. As for the other general asymptotic proofs, we provide a proof by construction highlighting where each primitive assumptions come to play. Consider $Q_n(\theta) := \frac{1}{n} \sum_{i=1}^n \log f(z_i, \theta)$. Assuming i) and ii) allows us to write the MVT:

$$\sqrt{n}(\hat{\theta} - \theta_0) = - \left(\frac{\partial^2 Q_n(\tilde{\theta})}{\partial \theta \partial \theta'} \right)^{-1} \sqrt{n} \frac{\partial Q_n(\theta_0)}{\partial \theta}$$

Let us focus on the first order term first. The FOC evaluated at θ_0 yields

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(z_i, \theta_0)}{\partial \theta}$$

Now, we can define the variance as follows

$$J := \mathbb{V} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(z_i, \theta_0)}{\partial \theta} \right) = \frac{1}{n} \sum_{i=1}^n \mathbb{V} \left(\frac{\partial \log f(z_i, \theta_0)}{\partial \theta} \right) \equiv V \left(\frac{\partial \log f(z, \theta_0)}{\partial \theta} \right)$$

To show that the score has mean zero at the true parameter, note that

$$\mathbb{E} \left[\frac{\partial \log f(z, \theta_0)}{\partial \theta} \right] = \int \frac{\partial \log f(z, \theta_0)}{\partial \theta} f(z, \theta_0) dz = \int \frac{\partial f(z, \theta_0)}{\partial \theta} dz.$$

Under standard regularity conditions (in particular, a domination condition allowing differentiation under the integral sign via the Dominated Convergence Theorem), we can write

$$\int \frac{\partial f(z, \theta_0)}{\partial \theta} dz = \frac{\partial}{\partial \theta} \int f(z, \theta) dz \Big|_{\theta=\theta_0} = 0.$$

Therefore,

$$\mathbb{E} \left[\frac{\partial \log f(z, \theta_0)}{\partial \theta} \right] = 0.$$

This implies that

$$J = \mathbb{E} \left[\frac{\partial \log f(z, \theta_0)}{\partial \theta} \frac{\partial \log f(z, \theta_0)}{\partial \theta'} \right] < \infty \text{ by assumption (iii)}$$

Standard CLT yields that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(z_i, \theta_0)}{\partial \theta} \xrightarrow{d} \mathcal{N}(0, J)$$

Now, the second order term can be written as

$$\frac{\partial^2 Q_n(\theta)}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f(z, \theta)}{\partial \theta \partial \theta'}$$

Assuming the second part of (iii) and by ULLN, we have that

$$\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f(z, \theta)}{\partial \theta \partial \theta'} \xrightarrow{p} \mathbb{E} \left(\frac{\partial^2 \log f(z, \theta)}{\partial \theta \partial \theta'} \right) \equiv H \text{ uniformly in } \theta$$

By CMT, assuming (iv) holds,

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, H^{-1} J H^{-1}) = \mathcal{N}(0, J^{-1} J J^{-1}) = \mathcal{N}(0, J^{-1})$$

where we used the information matrix identity, which concludes the proof. $\square$

A.4 Asymptotics at the Boundary

In many econometric applications, parameters are naturally constrained to lie in a subset of $\mathbb{R}^p$, such as non-negativity or probability restrictions. In such cases, it may occur that the true parameter value lies on the *boundary* of the parameter space. This seemingly innocuous feature has important consequences for asymptotic theory.

Under standard regularity conditions, extremum estimators admit a quadratic expansion around the true parameter and, when the true value lies in the interior of the parameter space, the estimator is asymptotically normal. However, when the true parameter lies on the boundary, the local parameter space becomes *asymmetric*: certain directions of perturbation are no longer admissible. As a result, the usual first-order conditions may fail to characterize the estimator, and the limiting distribution is no longer Gaussian.

The purpose of this appendix is to provide a rigorous derivation of the asymptotic distribution of an extremum estimator when the true parameter lies at the boundary. The key idea is that, after appropriate rescaling, the estimator can be represented as the maximizer of a stochastic process defined over a restricted domain. While the local objective function still converges to a quadratic (Gaussian) process, the constraint set induces a projection of the unconstrained optimizer onto the feasible region, leading to nonstandard asymptotic distributions.

To illustrate these ideas, we consider the simple linear regression model

$$y_i = \beta x_i + u_i, \quad i = 1, \dots, n,$$

where $\{u_i\}$ is an i.i.d. sequence with mean zero and finite variance, and the parameter space is given by $\Theta = [0, M]$ for some $M > 0$. The estimator of interest is the constrained least squares estimator

$$\hat{\beta}_n = \arg \min_{\beta \in \Theta} Q_n(\beta), \quad Q_n(\beta) = \frac{1}{n} \sum_{i=1}^n (y_i - \beta x_i)^2.$$

We focus on the case where the true parameter satisfies $\beta_0 = 0$, so that it lies at the boundary of Θ . The goal is to characterize the asymptotic behavior of $\hat{\beta}_n$ in this setting. In particular, we will show that while the estimator still converges at rate $\sqrt{n}$, its limiting distribution is given by the maximizer of a Gaussian process over a half-line, rather than over the entire real line.

The proof proceeds by first establishing consistency and the rate of convergence, and then studying the behavior of the objective function in a shrinking neighborhood of the true parameter. This leads to a sequence of stochastic processes indexed by a local parameter, whose weak convergence is established using standard tools from empirical process theory. The asymptotic distribution of the estimator then follows from an application of the continuous mapping theorem to the argmax functional. Let us start with a proposition.

Proposition A.2: Consistency of the Constrained Least Squares Estimator

Consider the model

$$y_i = \beta_0 x_i + u_i, \quad i = 1, \dots, n,$$

and suppose that the parameter space is

$$\Theta = [0, M], \quad M < \infty.$$

Let the estimator be defined by

$$\hat{\beta}_n = \arg \min_{\beta \in \Theta} Q_n(\beta), \quad Q_n(\beta) := \frac{1}{n} \sum_{i=1}^n (y_i - \beta x_i)^2.$$

Assume that:

1. $\{(x_i, u_i)\}_{i=1}^n$ is an i.i.d. sequence;
2. $\mathbb{E}[u_i \mid x_i] = 0$;
3. $\mathbb{E}[x_i^2] < \infty$;
4. $\mathbb{E}[x_i^2] > 0$.

Then

$$\hat{\beta}_n \xrightarrow{p} \beta_0.$$

Proof. Fix $\beta \in \Theta$. Using $y_i = \beta_0 x_i + u_i$, we may write

$$Q_n(\beta) = \frac{1}{n} \sum_{i=1}^n ((\beta_0 - \beta)x_i + u_i)^2,$$

and therefore

$$Q_n(\beta_0) = \frac{1}{n} \sum_{i=1}^n u_i^2.$$

Subtracting yields

$$\begin{aligned} Q_n(\beta_0) - Q_n(\beta) &= \frac{1}{n} \sum_{i=1}^n u_i^2 - \frac{1}{n} \sum_{i=1}^n ((\beta_0 - \beta)x_i + u_i)^2 \\ &= -(\beta - \beta_0)^2 \frac{1}{n} \sum_{i=1}^n x_i^2 + 2(\beta_0 - \beta) \frac{1}{n} \sum_{i=1}^n x_i u_i. \end{aligned}$$

We now study the two terms on the right-hand side. Since $\{x_i^2\}$ is i.i.d. with finite mean, the Law of Large Numbers gives

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \xrightarrow{p} \mathbb{E}[x_i^2].$$

Likewise, because $\mathbb{E}[x_i u_i] = \mathbb{E}[x_i \mathbb{E}[u_i | x_i]] = 0$ and $\mathbb{E}[|x_i u_i|] < \infty$, another application of the Law of Large Numbers yields

$$\frac{1}{n} \sum_{i=1}^n x_i u_i \xrightarrow{p} 0.$$

Hence, for each fixed $\beta \in \Theta$,

$$Q_n(\beta_0) - Q_n(\beta) \xrightarrow{p} -(\beta - \beta_0)^2 \mathbb{E}[x_i^2].$$

Since $\mathbb{E}[x_i^2] > 0$, the limit is strictly negative for every $\beta \neq \beta_0$.

Because $\Theta = [0, M]$ is compact and the right-hand side is continuous in β , the convergence above is in fact uniform over Θ . Thus,

$$\sup_{\beta \in \Theta} \left| Q_n(\beta_0) - Q_n(\beta) + (\beta - \beta_0)^2 \mathbb{E}[x_i^2] \right| \xrightarrow{p} 0.$$

Now fix $\varepsilon > 0$, and define

$$N(\beta_0) := \{\beta \in \Theta : |\beta - \beta_0| \leq \varepsilon\}.$$

For every $\beta \notin N(\beta_0)$, we have $|\beta - \beta_0| > \varepsilon$, so

$$-(\beta - \beta_0)^2 \mathbb{E}[x_i^2] \leq -\varepsilon^2 \mathbb{E}[x_i^2] < 0.$$

Therefore,

$$\sup_{\beta \notin N(\beta_0)} \{Q_n(\beta_0) - Q_n(\beta)\} \xrightarrow{p} \sup_{\beta \notin N(\beta_0)} \{-(\beta - \beta_0)^2 \mathbb{E}[x_i^2]\} \leq -\varepsilon^2 \mathbb{E}[x_i^2] < 0.$$

It follows that

$$\mathbb{P} \left(\sup_{\beta \notin N(\beta_0)} \{Q_n(\beta_0) - Q_n(\beta)\} > 0 \right) \rightarrow 0.$$

Equivalently, with probability approaching one, no value of β outside the ε -neighborhood of β_0 attains a lower criterion value than $Q_n(\beta_0)$. Hence the minimizer $\hat{\beta}_n$ must lie in $N(\beta_0)$ with probability approaching one:

$$\mathbb{P}(|\hat{\beta}_n - \beta_0| > \varepsilon) \rightarrow 0.$$

Since $\varepsilon > 0$ was arbitrary, we conclude that

$$\hat{\beta}_n \xrightarrow{p} \beta_0. \quad \square$$

Having established consistency, we now determine the rate at which $\hat{\beta}_n$ converges to β_0 . This step is crucial, since it identifies the appropriate local reparametrization under which the criterion admits a nondegenerate stochastic limit. As in the standard theory of extremum estimators, the relevant rate is obtained by asking how quickly the neighborhood around β_0 may shrink while still ensuring that the minimizer lies inside it with probability approaching one.

Proposition A.3: Rate of Convergence

For $\alpha > 0$ and $L > 0$, define the shrinking neighborhood

$$N_n(\alpha, L) := \{\beta \in \Theta : |\beta - \beta_0| \leq Ln^{-\alpha}\}.$$

Under the assumptions of the previous Proposition, the largest value of α for which

$$\Pr(\hat{\beta}_n \in N_n(\alpha, L)) \rightarrow 1$$

for L sufficiently large is $\alpha = \frac{1}{2}$. Equivalently,

$$\hat{\beta}_n - \beta_0 = O_p(n^{-1/2}).$$

Proof. By Proposition A.2, $\hat{\beta}_n \xrightarrow{p} \beta_0$. We now refine this result by asking how quickly the neighborhood around β_0 may shrink while still containing the minimizer with probability approaching one.

From the expansion established in the consistency proof,

$$Q_n(\beta_0) - Q_n(\beta) = -(\beta - \beta_0)^2 \frac{1}{n} \sum_{i=1}^n x_i^2 + 2(\beta_0 - \beta) \frac{1}{n} \sum_{i=1}^n x_i u_i.$$

Write

$$\tilde{\beta} := \beta_0 - \beta.$$

Then

$$Q_n(\beta_0) - Q_n(\beta) = -\tilde{\beta}^2 \frac{1}{n} \sum_{i=1}^n x_i^2 + 2\tilde{\beta} \frac{1}{n} \sum_{i=1}^n x_i u_i.$$

We want to show that, for suitable α , no value of β outside $N_n(\alpha, L)$ can yield a lower criterion value than β_0 with high probability. Thus consider

$$A_n(\alpha, L) := \Theta \setminus N_n(\alpha, L) = \{\beta \in \Theta : |\beta - \beta_0| > Ln^{-\alpha}\}.$$

It suffices to study

$$\mathbb{P}\left(\sup_{\beta \in A_n(\alpha, L)} \{Q_n(\beta_0) - Q_n(\beta)\} > 0\right).$$

Now, if $\beta \in A_n(\alpha, L)$, then $|\tilde{\beta}| > Ln^{-\alpha}$. Hence

$$\begin{aligned} & \mathbb{P} \left(\sup_{\beta \in A_n(\alpha, L)} \left\{ -\tilde{\beta}^2 \frac{1}{n} \sum_{i=1}^n x_i^2 + 2\tilde{\beta} \frac{1}{n} \sum_{i=1}^n x_i u_i \right\} > 0 \right) \\ & \leq \mathbb{P} \left(\sup_{\beta \in A_n(\alpha, L)} 2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{-1} \left| \frac{1}{n} \sum_{i=1}^n x_i u_i \right| > \inf_{\beta \in A_n(\alpha, L)} |\tilde{\beta}| \right) \\ & \leq \mathbb{P} \left(2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{-1} \left| \frac{1}{n} \sum_{i=1}^n x_i u_i \right| > Ln^{-\alpha} \right). \end{aligned}$$

Multiplying both sides of the inequality inside the probability by n^α , we obtain

$$\mathbb{P} \left(2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{-1} \left| n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i \right| > L \right).$$

Since

$$n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i = n^{\alpha-\frac{1}{2}} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i \right),$$

the stochastic order of this term depends on α . By the Law of Large Numbers,

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \xrightarrow{p} \mathbb{E}[x_i^2] > 0,$$

while by the Central Limit Theorem,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i = O_p(1),$$

since $\mathbb{E}[x_i u_i] = 0$ and $\mathbb{E}[(x_i u_i)^2] < \infty$. Therefore,

$$n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i = n^{\alpha-\frac{1}{2}} O_p(1).$$

We now distinguish cases.

Case 1: $\alpha < \frac{1}{2}$. In this case,

$$n^{\alpha-\frac{1}{2}} \rightarrow 0,$$

so

$$n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i = o_p(1).$$

Hence

$$2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{-1} \left| n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i \right| = o_p(1).$$

It follows that for any $L > 0$,

$$\mathbb{P} \left(\sup_{\beta \in A_n(\alpha, L)} \{Q_n(\beta_0) - Q_n(\beta)\} > 0 \right) \rightarrow 0.$$

Thus

$$\Pr(\hat{\beta}_n \in N_n(\alpha, L)) \rightarrow 1.$$

Case 2: $\alpha = \frac{1}{2}$. Now

$$n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i = \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i = O_p(1),$$

so

$$2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{-1} \left| n^\alpha \frac{1}{n} \sum_{i=1}^n x_i u_i \right| = O_p(1).$$

Therefore, for every $\eta > 0$, one can choose L sufficiently large so that

$$\mathbb{P} \left(\sup_{\beta \in A_n(\alpha, L)} \{Q_n(\beta_0) - Q_n(\beta)\} > 0 \right) < \eta$$

for all n sufficiently large. Hence

$$\mathbb{P}(\hat{\beta}_n \in N_n(1/2, L)) \rightarrow 1.$$

Case 3: $\alpha > \frac{1}{2}$. In this case,

$$n^{\alpha - \frac{1}{2}} \rightarrow \infty,$$

so the stochastic linear term is no longer asymptotically negligible relative to the radius $Ln^{-\alpha}$. Thus the previous argument cannot establish that

$$\mathbb{P}(\hat{\beta}_n \in N_n(\alpha, L)) \rightarrow 1.$$

In other words, the neighborhood shrinks too fast for consistency on that scale to be guaranteed. We conclude that the largest admissible exponent is $\alpha = \frac{1}{2}$. Hence

$$\hat{\beta}_n - \beta_0 = O_p(n^{-1/2}),$$

or equivalently,

$$\sqrt{n}(\hat{\beta}_n - \beta_0) = O_p(1).$$

□

Having established the rate of convergence, we now study the behavior of the estimator in a shrinking neighborhood of the true parameter. Since $\hat{\beta}_n - \beta_0 = O_p(n^{-1/2})$, we can introduce the local reparametrization

$$\beta = \beta_0 + \frac{v}{\sqrt{n}},$$

and define the rescaled estimator

$$\hat{v}_n := \sqrt{n}(\hat{\beta}_n - \beta_0).$$

Under this transformation, the original minimization problem can be rewritten as

$$\hat{v}_n = \arg \min_{v \in V_n} Q_n \left(\beta_0 + \frac{v}{\sqrt{n}} \right),$$

where V_n denotes the set of admissible local parameters induced by the constraint $\beta \in \Theta$. It is convenient to work instead with the criterion

$$\Lambda_n(v) := n \left(Q_n(\beta_0) - Q_n \left(\beta_0 + \frac{v}{\sqrt{n}} \right) \right),$$

so that $\hat{v}_n$ can be expressed as

$$\hat{v}_n = \arg \max_{v \in V_n} \Lambda_n(v).$$

This reformulation transforms the estimation problem into the maximization of a stochastic process indexed by v . The asymptotic behavior of the estimator is therefore governed by the limiting behavior of $\Lambda_n(v)$ and the geometry of the set V_n . In particular, when β_0 lies on the boundary of Θ , the set of admissible local parameters becomes asymmetric, which leads to nonstandard limiting distributions.

To build intuition, it is helpful to interpret $\Lambda_n(v)$ as a *random function* of v . For each realization of the data, $v \mapsto \Lambda_n(v)$ defines a curve, and the estimator $\hat{v}_n$ corresponds to the location at which this random curve attains its maximum. The analysis therefore proceeds by studying the behavior of these random curves as n grows. In particular, rather than analyzing the estimator directly, we study the convergence of the entire function $\Lambda_n(\cdot)$ to a limiting random function $\Lambda(\cdot)$, and then characterize the behavior of its maximizer.

Formally, $\{\Lambda_n(v) : v \in [-L, L]\}$ is a stochastic process indexed by v . Weak convergence of such processes is understood in the space $\mathbb{C}([-L, L])$ of continuous functions, equipped with the supremum norm. A key requirement for this convergence is *tightness*, which ensures that the sequence of random functions does not exhibit excessive oscillations. One convenient sufficient condition for tightness is the following: there exist constants $\eta > 0$, $\delta > 0$, and a continuous, non-decreasing function F such that for all $v_1, v_2 \in [-L, L]$,

$$\mathbb{E} \left[\left| \Lambda_n(v_2) - \Lambda_n(v_1) - \mathbb{E}[\Lambda_n(v_2) - \Lambda_n(v_1)] \right|^\eta \right] \leq K(F(v_2) - F(v_1))^{1+\delta},$$

for some finite constant K . Intuitively, this condition controls the fluctuations of the process and guarantees that its sample paths remain well-behaved. Together with convergence of finite-dimensional distributions, it implies weak convergence of $\Lambda_n(\cdot)$ to $\Lambda(\cdot)$. This is what the following proposition relies on.

Proposition A.4: Weak Convergence of the Local Criterion Process

Under the assumptions of Proposition A.3, fix $L > 0$ and define

$$\Lambda_n(v) := n \left[Q_n(\beta_0) - Q_n\left(\beta_0 + \frac{v}{\sqrt{n}}\right) \right], \quad v \in [-L, L].$$

Then, as a stochastic process in $\mathbb{C}([-L, L])$,

$$\Lambda_n(\cdot) \rightsquigarrow \Lambda(\cdot),$$

where

$$\Lambda(v) := -v^2\sigma_x^2 - 2vW, \quad \sigma_x^2 := \mathbb{E}[x_i^2], \quad W \sim N(0, \mathbb{V}(x_i u_i)).$$

If in addition $\mathbb{V}(u_i | x_i) = \sigma_u^2$ and u_i is independent of x_i , then

$$\mathbb{V}(x_i u_i) = \sigma_u^2 \sigma_x^2,$$

so that we may write

$$\Lambda(v) = -v^2\sigma_x^2 - 2v\sigma_u\sigma_x Z, \quad Z \sim N(0, 1).$$

Proof. By Proposition A.3, the relevant local parameterization is

$$\beta = \beta_0 + \frac{v}{\sqrt{n}}.$$

Substituting this into the criterion expansion obtained earlier gives

$$\begin{aligned} \Lambda_n(v) &= n \left[Q_n(\beta_0) - Q_n\left(\beta_0 + \frac{v}{\sqrt{n}}\right) \right] \\ &= n \left[-\left(\frac{v}{\sqrt{n}}\right)^2 \frac{1}{n} \sum_{i=1}^n x_i^2 + 2\left(\beta_0 - \beta_0 - \frac{v}{\sqrt{n}}\right) \frac{1}{n} \sum_{i=1}^n x_i u_i \right] \\ &= -v^2 \frac{1}{n} \sum_{i=1}^n x_i^2 - 2v \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i. \end{aligned}$$

We now prove weak convergence in $\mathbb{C}([-L, L])$. It suffices to verify:

1. convergence of finite-dimensional distributions;
2. tightness of the process.

Step 1: Finite-dimensional convergence. Fix $v_1, \dots, v_q \in [-L, L]$. Then

$$(\Lambda_n(v_1), \dots, \Lambda_n(v_q))$$

can be written as

$$-\left(v_1^2, \dots, v_q^2\right) \frac{1}{n} \sum_{i=1}^n x_i^2 - 2\left(v_1, \dots, v_q\right) \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i.$$

By the Law of Large Numbers,

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \xrightarrow{p} \sigma_x^2,$$

while by the Central Limit Theorem,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i \xrightarrow{d} W, \quad W \sim N(0, \mathbb{V}(x_i u_i)),$$

since $\mathbb{E}[x_i u_i] = 0$ and $\mathbb{E}[(x_i u_i)^2] < \infty$. Therefore, by Slutsky's theorem,

$$(\Lambda_n(v_1), \dots, \Lambda_n(v_q)) \xrightarrow{d} (\Lambda(v_1), \dots, \Lambda(v_q)),$$

where

$$\Lambda(v) = -v^2 \sigma_x^2 - 2vW.$$

Thus the finite-dimensional distributions converge.

Step 2: Tightness. We next verify tightness in $\mathbb{C}([-L, L])$. For $v_1, v_2 \in [-L, L]$ with $v_2 > v_1$,

$$\Lambda_n(v_2) - \Lambda_n(v_1) = -(v_2^2 - v_1^2) \frac{1}{n} \sum_{i=1}^n x_i^2 - 2(v_2 - v_1) \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i.$$

Subtracting expectations eliminates the deterministic quadratic component, since

$$\mathbb{E} \left[\frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i \right] = 0,$$

so

$$(\Lambda_n(v_2) - \Lambda_n(v_1)) - \mathbb{E}[\Lambda_n(v_2) - \Lambda_n(v_1)] = -2(v_2 - v_1) \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i.$$

Hence, for any $\eta > 2$,

$$\begin{aligned} \mathbb{E} \left| (\Lambda_n(v_2) - \Lambda_n(v_1)) - \mathbb{E}[\Lambda_n(v_2) - \Lambda_n(v_1)] \right|^\eta &= 2^\eta |v_2 - v_1|^\eta \mathbb{E} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i u_i \right|^\eta \\ &\leq K |v_2 - v_1|^\eta, \end{aligned}$$

for some finite constant K , provided $\mathbb{E}|x_i u_i|^\eta < \infty$. Therefore Billingsley's tightness criterion applies, and $\{\Lambda_n\}$ is tight in $\mathbb{C}([-L, L])$.

Combining finite-dimensional convergence and tightness, we conclude that

$$\Lambda_n(\cdot) \rightsquigarrow \Lambda(\cdot) \text{ in } \mathbb{C}([-L, L]).$$

Finally, if u_i is independent of x_i with $\mathbb{V}(u_i) = \sigma_u^2$, then

$$\mathbb{V}(x_i u_i) = \mathbb{E}[x_i^2 u_i^2] = \mathbb{E}[x_i^2] \mathbb{E}[u_i^2] = \sigma_x^2 \sigma_u^2,$$

so that $W = \sigma_u \sigma_x Z$ for $Z \sim N(0, 1)$, yielding

$$\Lambda(v) = -v^2 \sigma_x^2 - 2v \sigma_u \sigma_x Z. \quad \square$$

The final step consists in linking the behavior of the estimator to that of the limiting process. Since $\hat{v}_n$ can be expressed as the maximizer of $\Lambda_n(v)$ over the set V_n , its asymptotic distribution is determined by the maximizer of the limiting process $\Lambda(v)$ over the corresponding limit set V . Intuitively, once the random functions $\Lambda_n(\cdot)$ converge to $\Lambda(\cdot)$, the location of their maxima also converges, provided the maximizer is well-defined. This step is formalized using the continuous mapping theorem applied to the argmax functional, and highlights the key role of the constraint set in shaping the limiting distribution.

Proposition A.5: Asymptotic Distribution at the Boundary

Under the assumptions of Proposition A.4, define

$$\hat{v}_n := \sqrt{n}(\hat{\beta}_n - \beta_0).$$

Then:

1. If $\beta_0 > 0$, so that the true value is an interior point of $\Theta = [0, M]$, then

$$\hat{v}_n \xrightarrow{d} \frac{W}{\sigma_x^2},$$

where $W \sim N(0, \mathbb{V}(x_i u_i))$. In particular, if u_i is independent of x_i with variance σ_u^2 , then

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} \frac{\sigma_u}{\sigma_x} Z, \quad Z \sim N(0, 1).$$

2. If $\beta_0 = 0$, so that the true value lies on the boundary of Θ , then

$$\hat{v}_n \xrightarrow{d} \max \left\{ 0, \frac{W}{\sigma_x^2} \right\}.$$

Under the additional independence assumption,

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} \max \left\{ 0, \frac{\sigma_u}{\sigma_x} Z \right\}.$$

Proof. By construction,

$$\hat{\beta}_n = \arg \min_{\beta \in \Theta} Q_n(\beta).$$

Using the local reparametrization

$$\beta = \beta_0 + \frac{v}{\sqrt{n}},$$

this may be rewritten as

$$\hat{v}_n = \arg \min_{v \in V_n} Q_n\left(\beta_0 + \frac{v}{\sqrt{n}}\right),$$

where

$$V_n := \left\{ v : \beta_0 + \frac{v}{\sqrt{n}} \in \Theta \right\}.$$

Equivalently, since maximizing $\Lambda_n(v)$ is the same as minimizing $Q_n(\beta_0 + v/\sqrt{n})$,

$$\hat{v}_n = \arg \max_{v \in V_n} \Lambda_n(v).$$

By Proposition A.4,

$$\Lambda_n(\cdot) \rightsquigarrow \Lambda(\cdot) \quad \text{in } C([-L, L]),$$

where

$$\Lambda(v) = -v^2\sigma_x^2 - 2vW.$$

The limit process is a continuous, strictly concave parabola in v , so it has a unique maximizer. Hence the argmax map is continuous at Λ , and the continuous mapping theorem yields

$$\hat{v}_n \xrightarrow{d} v^* := \arg \max_{v \in V} \Lambda(v),$$

where V denotes the limiting local parameter space. We now distinguish the interior and boundary cases.

Interior case: $\beta_0 > 0$. If β_0 is an interior point of $\Theta = [0, M]$, then for every fixed $v \in \mathbb{R}$,

$$\beta_0 + \frac{v}{\sqrt{n}} \in \Theta$$

for all sufficiently large n . Thus the limiting local parameter space is

$$V = \mathbb{R}.$$

The maximizer of

$$\Lambda(v) = -v^2\sigma_x^2 - 2vW$$

over $v \in \mathbb{R}$ is obtained from the first-order condition

$$-2v\sigma_x^2 - 2W = 0,$$

so that

$$v^* = -\frac{W}{\sigma_x^2}.$$

Therefore,

$$\hat{v}_n \xrightarrow{d} -\frac{W}{\sigma_x^2}.$$

Since $\hat{v}_n = \sqrt{n}(\hat{\beta}_n - \beta_0)$, this gives the usual asymptotic normality result. Under the additional assumption $W = \sigma_u \sigma_x Z$,

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} -\frac{\sigma_u}{\sigma_x} Z.$$

Since $Z \sim N(0, 1)$ implies $-Z \sim N(0, 1)$, this is equivalent to

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} \frac{\sigma_u}{\sigma_x} Z.$$

Boundary case: $\beta_0 = 0$. If $\beta_0 = 0$, the constraint $\beta \in [0, M]$ becomes

$$0 + \frac{v}{\sqrt{n}} \geq 0,$$

so that the admissible local parameters satisfy

$$v \geq 0.$$

Hence the limiting local parameter space is

$$V = [0, \infty).$$

We must therefore maximize

$$\Lambda(v) = -v^2 \sigma_x^2 - 2vW \quad \text{over } v \geq 0.$$

The unconstrained maximizer is again

$$-\frac{W}{\sigma_x^2}.$$

Imposing the constraint $v \geq 0$, the constrained maximizer is the projection of the unconstrained one onto $[0, \infty)$:

$$v^* = \max \left\{ 0, -\frac{W}{\sigma_x^2} \right\}.$$

Thus

$$\hat{v}_n \xrightarrow{d} \max \left\{ 0, -\frac{W}{\sigma_x^2} \right\}.$$

Under the additional assumption $W = \sigma_u \sigma_x Z$,

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} \max\left\{0, -\frac{\sigma_u}{\sigma_x} Z\right\}.$$

Since $-Z \sim N(0, 1)$, this is equivalent to

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} \max\left\{0, \frac{\sigma_u}{\sigma_x} Z\right\}.$$

□

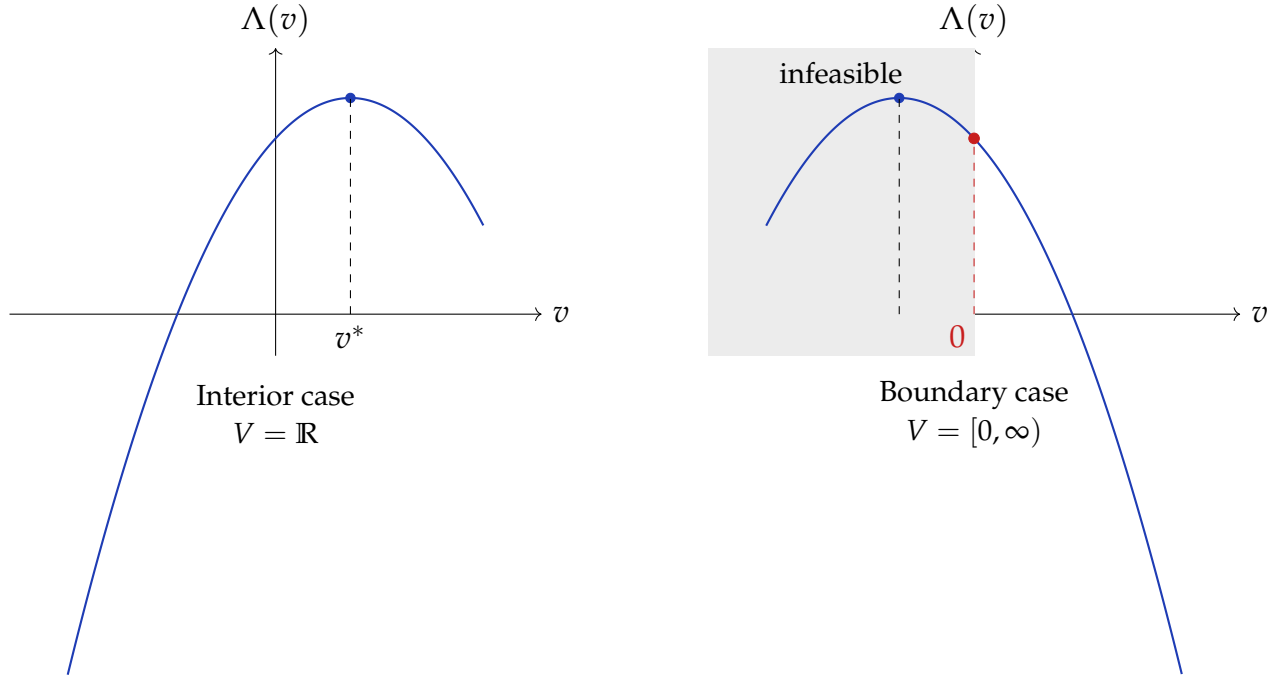


Figure 15: Local criterion in the interior and at the boundary. The objective function is the same in both cases, but the feasible set differs. At the boundary, infeasible directions lead the estimator to select the constrained maximizer.

B Appendix - The imputation estimator

In this section, I discuss a recent paper more in details since I had to present it during class (Borusyak et al., 2024).

B.1 Research Question

The central research question of this paper asks: How can we estimate dynamic treatment effects in event studies with staggered treatment adoption and heterogeneous causal effects in an efficient manner? This question is particularly relevant given the increasing prevalence of settings where treatment is rolled out to different units at different times, and where the effect of treatment may vary across units and over time.

B.2 Framework

The paper uses a household-level example to illustrate the framework. Each household i is treated (that is, has received a tax rebate) from period E_i onwards. We define $K_{it} = t - E_i$ as the time (in weeks) relative to treatment.

To study the average treatment effect, researchers typically use the canonical regression specification: $Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}$, where $D_{it} = \mathbb{1}\{K_{it} > 0\}$ is an indicator for whether unit i is treated at time t .

However, to study the dynamics of the treatment effect, we need a fully dynamic specification: $Y_{it} = \alpha_i + \beta_t + \sum_{k=-\infty}^{\infty} \tau_k \mathbb{1}\{K_{it} = k\} + \varepsilon_{it}$, where α_i represents individual fixed effects, β_t represents week fixed effects, and τ_k captures the dynamic causal effects at each relative time period k .

The estimation target in this framework is $\tau_w = \sum_{it \in \Omega_1} w_{it} \tau_{it} \equiv w' \tau$, where $\tau_{it} = Y_{it} - Y_{it}(0)$ represents the individual treatment effect for unit i at time t .

B.3 Assumptions

The paper relies on four key assumptions. Assumption 1 is the parallel trend assumption, which states that for all $(i, t) \in \Omega$, the expected value of the untreated potential outcome can be written as $\mathbb{E}[Y_{it}(0)] = \alpha_i + \beta_t$, where $Y_{it}(0)$ is the period t stochastic potential outcome of unit i if it is never treated.

Assumption 2 is the no anticipation assumption, which requires that for all non-treated units $(i, t) \in \Omega_0$, we have $Y_{it} = Y_{it}(0)$. This means that units do not change their behavior in anticipation of treatment.

Assumption 3' is an optional parametric model of causal effects, which posits that $\tau = \Gamma \theta$, where θ is an $(N_1 - M) \times 1$ vector of unknown parameters and Γ is a known $N_1 \times (N_1 - M)$ matrix of full column rank. This assumption imposes a linear structure on treatment effects so that they can be efficiently estimated and related to each other.

Assumption 4 concerns spherical errors and states that $\mathbb{E}[\varepsilon \varepsilon'] = \sigma^2 I_N$. While this assumption is useful for the main theorem, it can be relaxed in practice.

B.4 The Underidentification Problem

One of the key insights of the paper is the formal identification of an underidentification problem in fully dynamic specifications. The paper presents the following proposition.

Proposition B.1: Underidentification

If there are no never-treated units, the path of $\{\tau_h\}_{h \neq -1}$ coefficients is not point-identified in the fully dynamic specification. In particular, for any $\kappa \in \mathbb{R}$, the path $\{\tau_h + \kappa(h+1)\}$ fits the data equally well, with the fixed-effect coefficients appropriately modified.

Proof. In the absence of never-treated units and defining $\tau_{-1} = 0$, we can write

$$\sum_{h \neq -1} \tau_h \mathbb{1}\{K_{it} = h\} = \tau_{K_{it}}.$$

Now consider some collection of $\{\tau_h\}$ (with $\tau_{-1} = 0$) and fixed effects $\tilde{\alpha}_i$ and $\tilde{\beta}_t$. For any $\kappa \in \mathbb{R}$, let

$$\tau_h^* = \tau_h + \kappa(h+1), \quad \tilde{\alpha}_i^* = \tilde{\alpha}_i + \kappa(E_i - 1), \quad \tilde{\beta}_t^* = \tilde{\beta}_t - \kappa t.$$

Then for any observation (i, t) ,

$$\begin{aligned} \tilde{\alpha}_i^* + \tilde{\beta}_t^* + \tau_{K_{it}}^* &= \tilde{\alpha}_i + \kappa(E_i - 1) + \tilde{\beta}_t - \kappa t + \tau_{K_{it}} + \kappa(K_{it} + 1) \\ &= \tilde{\alpha}_i + \tilde{\beta}_t + \tau_{K_{it}} + \kappa(E_i - 1 - t + K_{it} + 1). \end{aligned}$$

Using $K_{it} = t - E_i$, we have $E_i - 1 - t + K_{it} + 1 = 0$, so

$$\tilde{\alpha}_i^* + \tilde{\beta}_t^* + \tau_{K_{it}}^* = \tilde{\alpha}_i + \tilde{\beta}_t + \tau_{K_{it}}.$$

Thus equation (2) has exactly the same fit under the original and modified fixed effects and $\{\tau_h\}$ coefficients, indicating perfect collinearity. $\square$

The fundamental identity underlying this problem is that $K_{it} = t - E_i$, which implies that event time is perfectly collinear with unit and time fixed effects. The conclusion is striking: without restrictions, the model cannot distinguish τ_k from $\tau_k + h \cdot k$. Dynamic treatment effects are identified only up to an arbitrary linear trend.

To understand this result graphically, consider a simple example with two units where one appears to have a trend and level difference relative to the other. There are two possible interpretations of this pattern. First, the treatment might have no impact on the outcome, with trends merely reflecting features of the environment and the level difference arising from unit fixed effects. Second, the outcome could be entirely driven by causal effects plus anticipation. Without additional assumptions, these interpretations are observationally equivalent.

The paper proposes two solutions to this underidentification problem. The first solution is to impose a no-anticipation effect and estimate a semi-dynamic model. However, this requires a strong assumption on anticipation, which must be motivated by an a priori argument. This approach separates estimation and pre-trend testing through an F-test on treatment leads, arbitrarily dropping two leads. The second solution is to introduce a control group of units that are never treated whenever possible, though one should not allow the control group to be on its own trend.

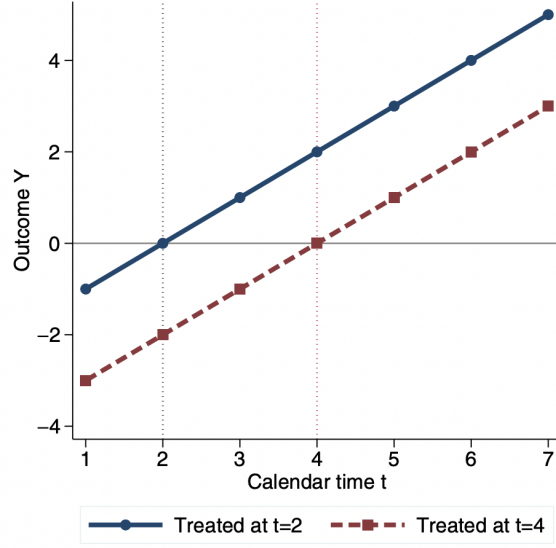


Figure 16: Underidentification graphically

B.5 Negative Weights Problem

The paper also highlights the problem of negative weights that can arise in standard two-way fixed effects (TWFE) estimation. To illustrate this, consider a simple example with two units (A and B) and three periods. Unit A is treated starting in period 2 ($E_A = 2$), while unit B is treated starting in period 3 ($E_B = 3$), as described in the Table B.5.

$\mathbb{E}[Y_{it}]$	$i = A$	$i = B$
$t = 1$	α_A	α_B
$t = 2$	$\alpha_A + \beta_2 + \tau_{A2}$	$\alpha_B + \beta_2$
$t = 3$	$\alpha_A + \beta_3 + \tau_{A3}$	$\alpha_B + \beta_3 + \tau_{B3}$
Event date	$E_i = 2$	$E_i = 3$

Table 7: Two-Unit, Three-Period Example

Suppose we estimate the TWFE model:

$$Y_{it} = \alpha_i + \beta_t + \tau^{static} D_{it} + \varepsilon_{it}$$

An admissible comparison would be

$$(Y_{A2} - Y_{A1}) - (Y_{B2} - Y_{B1}) = \tau_{A2}$$

which uses unit B as a control for unit A. However, a forbidden comparison would be

$$(Y_{B3} - Y_{B2}) - (Y_{A3} - Y_{A2}) = \tau_{B3} + \tau_{A2} - \tau_{A3}$$

which compares already-treated unit A to newly-treated unit B.

The key conclusion is that if τ_{A3} goes up, the apparent treatment effect on B at time 3 goes down. This means that τ_{A3} receives a negative weight because OLS imposes treatment effect homogeneity. For large enough relative time periods, when there are no never-control units, only forbidden comparisons exist, making estimation of long-run causal effects incorrect.

B.6 The Efficient Estimator

B.6.1 The general theorem

The paper's main theoretical contribution is an efficient estimator theorem.

Theorem B.1: Efficient Estimator

Suppose Assumptions 1, 2, 3', and 4 hold. Among all linear unbiased estimators of τ_w , the unique efficient estimator $\hat{\tau}_w^*$ is obtained via:

1. **Estimate:** Within the untreated observations only $(i, t) \in \Omega_0$, estimate the α_i and β_t by OLS in

$$Y_{it}(0) = \alpha_i + \beta_t + \varepsilon_{it}.$$

2. **Extrapolate:** Set $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\beta}_t$ and

$$\hat{\tau}_{it}^* = Y_{it} - \hat{Y}_{it}(0) \text{ for each treated observation } (i, t) \in \Omega_1$$

3. **Take Averages:** Estimate the target τ_w by a weighted sum,

$$\hat{\tau}_w^* = \sum_{(i,t) \in \Omega_1} w_{it} \hat{\tau}_{it}^*.$$

This is a powerful result because it shows that efficiency can be achieved while relaxing some of the more stringent assumptions.

B.6.2 The Imputation Estimator

The paper provides an imputation representation of the efficient estimator that is particularly useful for practitioners.

Theorem B.2: Imputation representation of the efficient estimator

With a null Assumption 3' (that is, if $\Gamma = I_{N_1}$), the unique efficient linear unbiased estimator $\hat{\tau}_w^*$ of τ_w from Theorem 0.1 can be obtained via an imputation procedure:

1. **Estimate:** Within the untreated observations only $(i, t) \in \Omega_0$, estimate the α_i and

β_t by OLS in

$$Y_{it}(0) = \alpha_i + \beta_t + \varepsilon_{it}.$$

2. **Extrapolate:** Set $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\beta}_t$ and

$$\hat{\tau}_{it}^* = Y_{it} - \hat{Y}_{it}(0) \text{ for each treated observation } (i, t) \in \Omega_1$$

3. **Take Averages:** Estimate the target τ_w by a weighted sum,

$$\hat{\tau}_w^* = \sum_{(i,t) \in \Omega_1} w_{it} \hat{\tau}_{it}^*.$$

B.7 Pre-trend Testing

One of the key insights of the paper is the separation of estimation of treatment effects from pre-trends testing. To test for pre-trends in the imputation framework, use the sample of untreated or not yet treated observations to estimate

$$Y_{it} = \alpha_i + \beta_t + \sum_{k=E_i-R}^{E_i-1} \tau_k \mathbb{1}\{K_{it} = k\} + \varepsilon_{it}$$

for $(i, t) \in \Omega_0$ Then perform an F-test of joint significance, testing

$$\mathcal{H}_0 : \tau_{E_i-R} = \dots = \tau_{E_i-1} = 0$$

This approach ensures that the pre-trend test is not contaminated by the estimation of treatment effects.

B.8 Application: Marginal Propensity to Consume Out of Tax Rebates

The paper provides an application studying the Economic Stimulus Act of 2008, a 100 billion dollar programme that sent tax rebates to approximately 130 million tax filers. The goal is to estimate the marginal propensity to consume (MPC) out of the tax rebate, following Broda and Parker, 2014.

The timing of the tax rebate (measured in weeks) was determined by the last two digits of the taxpayer's Social Security number, providing plausibly exogenous variation in treatment timing. The researchers use the following specification:

$$Y_{it} = \alpha_i + \beta_t + \sum_{h=-a}^b \tau_h \mathbb{1}\{K_{it} = h\} + \varepsilon_{it}$$

where Y_{it} is the dollar amount of spending in calendar week t for household i .

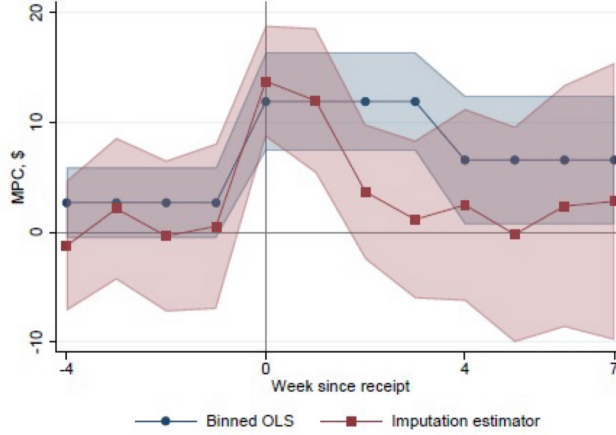


Figure 17: Binned OLS vs. Imputation Estimator

Figure B.8 shows the difference between OLS and the imputation estimator. The results comparing binned OLS versus the imputation estimator are striking. The imputation estimator shows much less persistence than binned OLS. The two-month MPCs are very different: \$73.8 for binned OLS versus \$38 for the imputation estimator. This difference arises because binned OLS uses forbidden comparisons that inflate long-run effects.

The reason for this discrepancy can be understood by considering how OLS estimates treatment effects at longer horizons. Tax rebates were sent out in a seven-week window. Information on treatment effects eight weeks after treatment comes from late periods in the sample and from individuals who were treated early. Crucially, we can never observe τ_8 and τ_{-1} at the same time: if anyone is observed eight weeks after treatment, everyone else must already be treated. Therefore, OLS performs extrapolations by observing τ_8 and τ_3 at the same time, as well as τ_3 and τ_{-1} , and inferring something about the difference between τ_8 and τ_{-1} . This works if treatment effects are homogeneous, but not otherwise, especially if people treated earlier have larger treatment effects, which would be a forbidden extrapolation.

In this specific example, a cohort that is treated late will be compared with another cohort that has been treated early and who is therefore on a decreasing consumption trend (the marginal propensity to consume is not increasing systematically after a one-off tax rebate). Because the control group is on a negative trend, the relative difference makes the treatment effect for the treatment group look like it is increasing more than it actually is, creating an upward bias because of the forbidden comparison.

B.9 Asymptotic Properties

The paper also establishes the asymptotic properties of the imputation estimator. Assumption 5 concerns clustered error terms and requires that errors ε_{it} are independent across units i and have bounded variance, with $\text{Var}(\varepsilon_{it}) \leq \bar{\sigma}^2$ for all $(i, t) \in \Omega$. Assumption 6 is the Herfindahl condition, which requires that along the asymptotic sequence, the

Herfindahl norm of the weights

$$\|v\|_H^2 = \sum_i \left(\sum_{t:(i,t) \in \Omega} |v_{it}| \right)^2$$

converges to zero, where v_{it} are the weights in the unbiased linear estimator

$$\hat{\tau}_w = \sum_{(i,t) \in \Omega} v_{it} Y_{it}$$

Under Assumptions 1', 2, 3', 5, and 6, the paper establishes consistency:

$$\hat{\tau}_w - \tau_w \xrightarrow{L^2} 0$$

for any unbiased linear estimator $\hat{\tau}_w$ of τ_w , such as $\hat{\tau}_w^*$ from the main theorem.

The paper also establishes asymptotic normality. If the assumptions of the consistency proposition hold, and there exists $\kappa > 0$ such that $\mathbb{E}[|\varepsilon_{it}|^{2+\kappa}]$ is uniformly bounded, the weights are not overly concentrated (formally, $\sum_i \left(\sqrt{n_H} \sum_{t:(i,t) \in \Omega} |v_{it}| \right)^{2+\kappa} \rightarrow 0$), and the variance does not vanish ($\liminf n_H \sigma_w^2 > 0$), then with $\sigma_w^2 = \text{Var}[\hat{\tau}_w]$, we have $\sigma_w^{-1}(\hat{\tau}_w - \tau_w) \xrightarrow{d} \mathcal{N}(0, 1)$.

B.10 Key Takeaways

The paper provides several important lessons for practitioners. First, researchers should use difference-in-differences if they have an ex-ante reason to believe that the parallel trends and no anticipation assumptions hold. Second, it is crucial to separate pre-trends testing from the estimation of treatment effects to avoid contamination. Third, researchers should use a valid control group even in the absence of never-treated units, and the imputation estimator provides the efficient way to do so.